LABG SUITE · OPEN SCIENCE IN THE BROWSER



SigmaPro

User's Manual

Statistics for agronomic and biological research

Luis Ángel Barrera-Guzmán

VERSION 1.0 · 2026

SigmaPro

User's Manual · Statistics for agronomic and biological research

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SOFTWARE	SigmaPro, version 1.0.0 (November 2026)
MANUAL	First English edition, 2026. It describes version 1.0 of the program.
SOFTWARE LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL COPYRIGHT	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.
INTERFACE LANGUAGE	Spanish and English; the ES · EN selector on the top bar switches the language. This manual describes the English interface and writes the names of buttons, fields and menus as they appear on screen, for example Add to report.

HOW TO CITE THE SOFTWARE

Barrera-Guzmán, L. Á. (2026). *SigmaPro: Statistics for agronomic and biological research* (Version 1.0.0) [Computer software]. LABG Suite. https://doi.org/10.5281/zenodo.23935865

HOW TO CITE THIS MANUAL

Barrera-Guzmán, L. Á. (2026). *SigmaPro. User's manual* (1st ed.). Distributed with SigmaPro, version 1.0.

The methods paragraph that the report builds (section 5.4) cites the program with the version you have open, together with R, and the reference list of the exported report includes the reference of each method you used.

> **PRIVACY**
>
> SigmaPro runs entirely inside your browser. The data you open, the analyses and the reports are computed on your computer and **are never sent to any server**, not even for the models that run in the R engine, which also works inside the browser. You can analyze the unpublished data of a thesis or a trial without them ever leaving your machine.

The screenshots, figures and numbers in this manual come from the program itself, with the 27 example trials it includes: six are published data, cited where they are used; the rest are **fictitious**, built so that each analysis has a clear case, and must not be cited as data.

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How to read this manual

The manual follows the order of the app: ten numbered blocks, walked from top to bottom in the sidebar. Each chapter explains one block and carries its number: chapter 4 is Block 4, and its sections, figures and tables are numbered 4.1, 4.2… The introduction, before the blocks, is numbered I.1, I.2…

One color for each block

The color of a block appears on the opening page of its chapter, in the number of each section, on the tab at the edge of the page and in the table of contents.

Block	Name in the app	What it does
1	Home	The front page: what the app is for, the path of an analysis, the list of what you can analyze, how to cite it, the opening screen and the notice for recovering saved work
2	Data	Importing, pasting or opening an example; the worksheet, the variable types, computed variables, transformations, filters, wide ⇄ long and what SigmaPro sees in the trial
3	Explore	Descriptive statistics, frequencies, proportions and goodness of fit, with their intervals
4	Analysis	The assistant, the catalog and the model builder; its 74 analyses (the other 4 are in Explore), from experimental designs and mean comparisons to mixed models, generalized models and the rest of the families
5	Results	The LABG Report: the saved analyses, with their tables, figures and assumptions, the paragraph for the thesis and the R code
6	Figures	The report figures, their themes, their downloads, the ✎ editor and the Figure Studio
7	Export	The report in .docx, PDF or HTML, the tables, the figures, the .labg project and autosave
8	Validation	The comparison of each result against R, case by case
9	Fundamentals	One fact sheet per analysis: what it is for, the model, the assumptions, the data it needs, how to read it and its references, with example data and templates
10	LABG Suite	The sister apps that continue the work, the suggestions within the analysis and how to take the data from one app to another

Types of boxes

NOTE

Clarifications and practical tips that save time.

IMPORTANT

What can change your conclusions if it is overlooked.

THEORY IN BRIEF

The idea behind a calculation, without unnecessary formulas.

EXAMPLE

A text or a result exactly as the app produces it.

Writing conventions

Written like this	Meaning
Add to report	A button, check box, field or menu of the app, with its text exactly as it appears on screen.
`rend`, `hibrido * dosis + (1 | bloque)`	A column name, a formula or a value that you type exactly as shown.
"rend", "Germinó"	The names of the variables and of their levels, in quotation marks, as the app writes them in its conclusions. The example datasets keep their original Spanish column and level names (for example, "rend" is yield).
5.938 · F = 60.63 · p < 0.001	Numbers with a decimal point, as in the app; the letters of the statistics, in italics.
Figure 4.28 · Table 4.3	The figures and tables of the manual. The tables that the app numbers within a result ("Table 1. Analysis of covariance") are referred to by their title.

THE NUMBERS IN THIS MANUAL

All the numbers in the text are the ones the app gives with its examples (Data block → Examples ▾) and the default values, unless stated otherwise. The main example is **"Maize: hybrids × nitrogen"**, a factorial of 3 hybrids by 4 nitrogen rates in 4 blocks; when a chapter uses another example, it names it before giving its results. With the same example you will see the same numbers.

In

INTRODUCTION · BEFORE THE BLOCKS

Introduction to SigmaPro

What the program is, which questions about a trial it answers, what data it accepts and what it delivers, how to open it and work through it, which language it works in and what it does not do yet.

I.1 What is SigmaPro?

I.2 Questions it answers

I.3 Five ideas behind the manual

I.4 Data it accepts and what it delivers

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The R engine inside the browser

I.8 How to work through a study

I.9 The examples

I.10 What the program does not do

I.1 What is SigmaPro?

SigmaPro is a statistics program for agronomic and biological research that runs inside the browser, with nothing to install and no internet connection. It takes the data of a trial along a single path, from the field file to the published table and figure.

That path is the path of the blocks: load the data, explore them, choose the analysis that matches the design, read the conclusion, check the assumptions and export the report. It is made for people who analyze their own experiments—undergraduate and graduate students, teachers, field and greenhouse researchers—and want correct results without writing code. That is why each result begins with a conclusion in words, checks its assumptions, cites its methods and gives the R code that repeats the calculation. Its 78 analyses, in 18 families, range from the analysis of variance of an experimental design to mixed models, heritability, multi-environment trials, survival and time series. Each one has its **Fundamentals fact sheet**, which explains what it does, what it assumes, what data it needs and how to read it (chapter 9).

And every number is checked against R: the program's computing core is its own, but R is the judge. The comparison, case by case, comes inside the app (chapter 8). SigmaPro is part of the LABG Suite, a set of free apps that also run in the browser; chapter 10 explains which of them continue the work of an analysis.

I.2 Questions it answers

The question	Where
What are my data like? Are there odd values, missing values, groups with few observations?	Explore (chapter 3) and what SigmaPro sees in the trial (section 2.13)
Do my treatments differ? Which ones?	Experimental designs and mean comparisons (sections 4.10 and 4.11)
Are the assumptions of the analysis met? If not, what do I do?	The traffic light of each result and the model assumptions (section 4.7)
What is the optimum rate? How does the crop respond to a quantitative factor?	Polynomials, regression, nonlinear models and response surfaces (sections 4.11, 4.12, 4.18 and 4.19)
How closely are two variables related? Which variables explain yield?	Correlation and multiple regression (sections 4.12 and 4.13)
Are the blocks, locations or years random? How do I analyze missing plots or repeated measures?	Linear mixed models (section 4.15)
How much of the variation is genetic? Which genotypes should I select?	Heritability, BLUP and multi-environment trials (sections 4.16 and 4.20)
How do I analyze counts, proportions or times to an event?	Generalized models and survival (sections 4.17 and 4.23)
How many replications or how many plants do I need?	Power, replications and sampling (section 4.22)
How do I present it in the thesis or in the paper?	The report, the figures and the exports (chapters 5 to 7)
What does an analysis do, what does it assume and how must my data be arranged to use it?	Its fact sheet in Fundamentals (chapter 9)
Which other tool continues the work: randomizing the design, exploring many variables, forming groups?	LABG Suite (chapter 10)

I.3 Five ideas behind the manual

1. The design comes first

The analysis of an experiment depends on how it was carried out: whether there were blocks, whether a factor was applied to large plots, whether the measurements were repeated on the same plant. SigmaPro reads the structure of the trial from the data (section 2.13), and the assistant asks about what cannot be read, so that it proposes the analysis that matches the design, not the best-known one.

2. The conclusion first, in words

Each result begins by saying what it found ("hibrido: there are differences ($F = 60.635$, $p < 0.001$)") and then gives the tables. The numbers are there for anyone who wants to check them; the conclusion is there so that nobody misreads them.

3. Assumptions are not taken for granted

The analyses check their own assumptions: most of them show the check with a traffic light in the result, and the experimental designs check them with `Check the model assumptions` (section 4.7). If something fails, the app says what to do: transform, change the test or use a model that does not make that assumption.

4. Every number is validated against R

A method does not appear in the app until all its validation cases agree with R to six digits (chapter 8).

5. Everything can be reproduced

Each result comes with its R code, the report cites the methods with their versions, and the .labg project saves the data and the analyses with their fingerprints, so that another person arrives at the same numbers.

I.4 Data it accepts and what it delivers

SigmaPro reads tables in delimited text (CSV, TSV), spreadsheet workbooks (.xlsx, .ods) and pasted tables, with one row per experimental unit and one column per variable (section 2.2). It recognizes factors, numeric variables, dates and identifiers, and lets you correct any of them. It works comfortably with about 100,000 rows.

It delivers the complete report in .docx, PDF or HTML; each table in CSV, and all of them together in a spreadsheet or in LaTeX; the figures in SVG, vector PDF, PNG or TIFF at 600 dpi; and the .labg project with the data, the analyses and the R code. In addition, each Fundamentals fact sheet gives a template in CSV or XLSX for preparing your data (section 9.5). Appendix A describes each file in detail.

I.5 What you need and how to open it

- **A recent browser**, from the last two or three years, on a computer; it also works on a tablet or a phone, with the screen rearranged (section I.6).
- **Nothing to install.** SigmaPro opens with a double-click on its `index.html` file, from the program folder, or from its web page. The first time the web page is opened, the browser stores the app so that it can be used offline afterward.
- **Internet only for the R engine**, the first time you use an analysis that needs it (section I.7). Nothing else uses the internet.

The opening screen

When you open SigmaPro, its opening screen appears: on a black background, the logo and the name are drawn in about six seconds, and the scene stays still until you press Start (or Enter). Skip or Esc dismisses it at any time.

- The first time, it is shown in full; on later visits, a short version of a second and a half, also with Start. Reloading the same tab does not show it again.
- If the system asks for reduced motion, the scene appears still.
- It can be turned off, replayed or projected in a loop from Home → About SigmaPro (chapter 1).

Spanish or English

The ES · EN selector on the top bar changes the language of the whole app: the menus, the conclusions, the tables, the report, the methods paragraph and the Fundamentals fact sheets. When you change it, the app saves the work and reloads in the other language, with the same data and the same analyses; the chosen language is remembered the next time. The comments in the equivalent R code stay in Spanish. This manual describes the English interface.

In

I.6 A tour of the interface

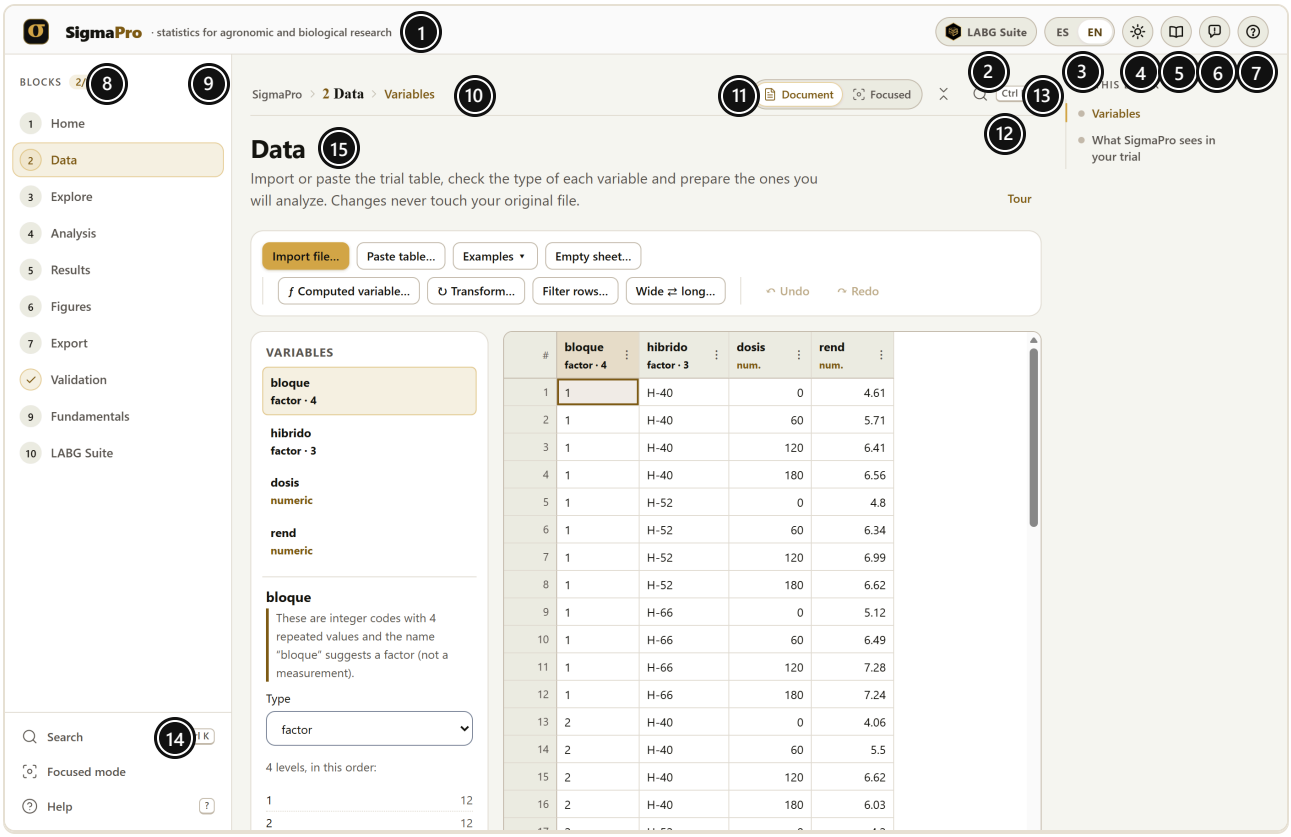


Figure I.1. The SigmaPro screen, here in the Data block with the maize example. The marked parts are common to all blocks.

1. The logo and the name of the program.
2. LABG Suite: the other apps of the suite.
3. ES · EN: the language of the app.
4. The light or dark theme.
5. The glossary (Ctrl G), described below.
6. Report a problem or suggest something.
7. Help and keyboard shortcuts.
8. The bar of the ten blocks, with the active one highlighted.
9. Collapse the bar to gain space.
10. The breadcrumb: where you are within the block.
11. Document or Focused: the whole block or one section at a time.
12. The search (Ctrl K).
13. In this block: the list of its sections.
14. Search, Focused mode and Help.
15. The title and the explanation of the block, with the Tour button on the right (described below).

The **block bar** takes you to any block at any time; a block that needs something that does not exist yet (data, an analysis in the report) says so and offers the button to go and get it. Alt + ← and → move to the previous or the next block.

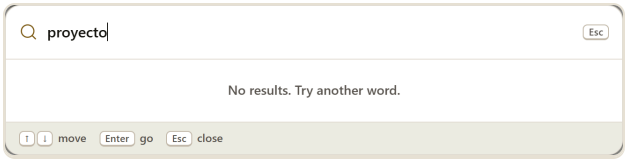


Figure I.2. The search (Ctrl K) finds blocks, sections, buttons and fields by name and takes you to them.

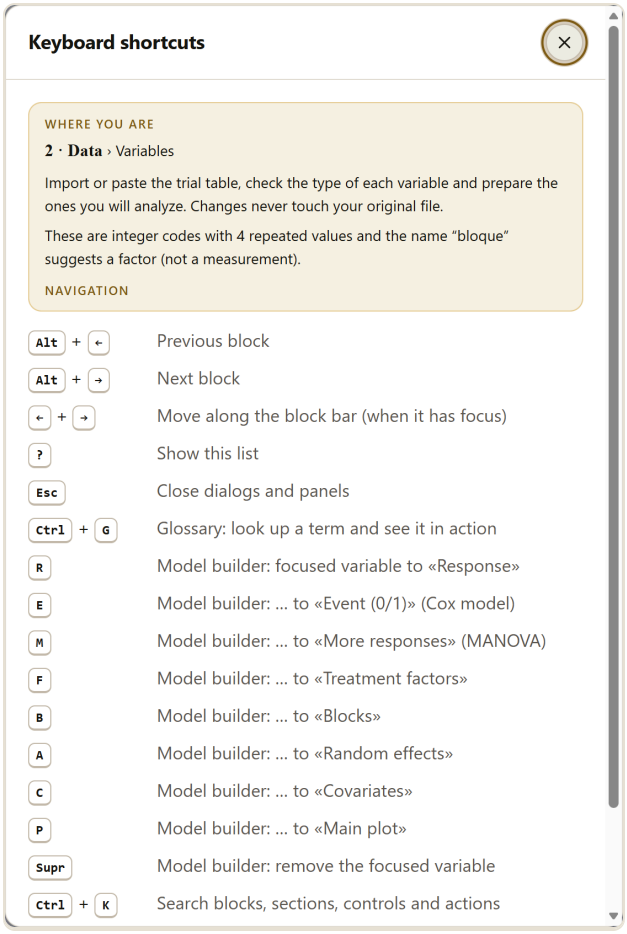


Figure I.3. The ? key shows where you are and the keyboard shortcuts, including those of the model builder.

The sun button switches to the **dark theme**, more comfortable at night or on a projector; figures are always downloaded with the colors of the light theme (and with the chart theme you choose, chapter 6). On a narrow screen (a phone, a tablet held upright), the block bar moves to the bottom of the screen, as a row of tabs that slides sideways, and the buttons of the top bar drop to a second line.

The glossary

The book button, or Ctrl G from any block, opens the **glossary**: 146 terms, namely the 121 statistics that appear in the app's tables (*F*, *p*, R^2, CV, BLUP…) and 25 concepts of design and analysis (block, interaction, error stratum, REML, overdispersion…). Ctrl G again, or Esc, closes it.

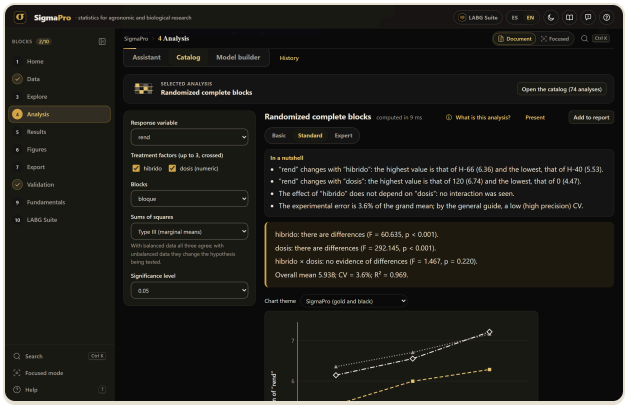


Figure I.4. The dark theme.

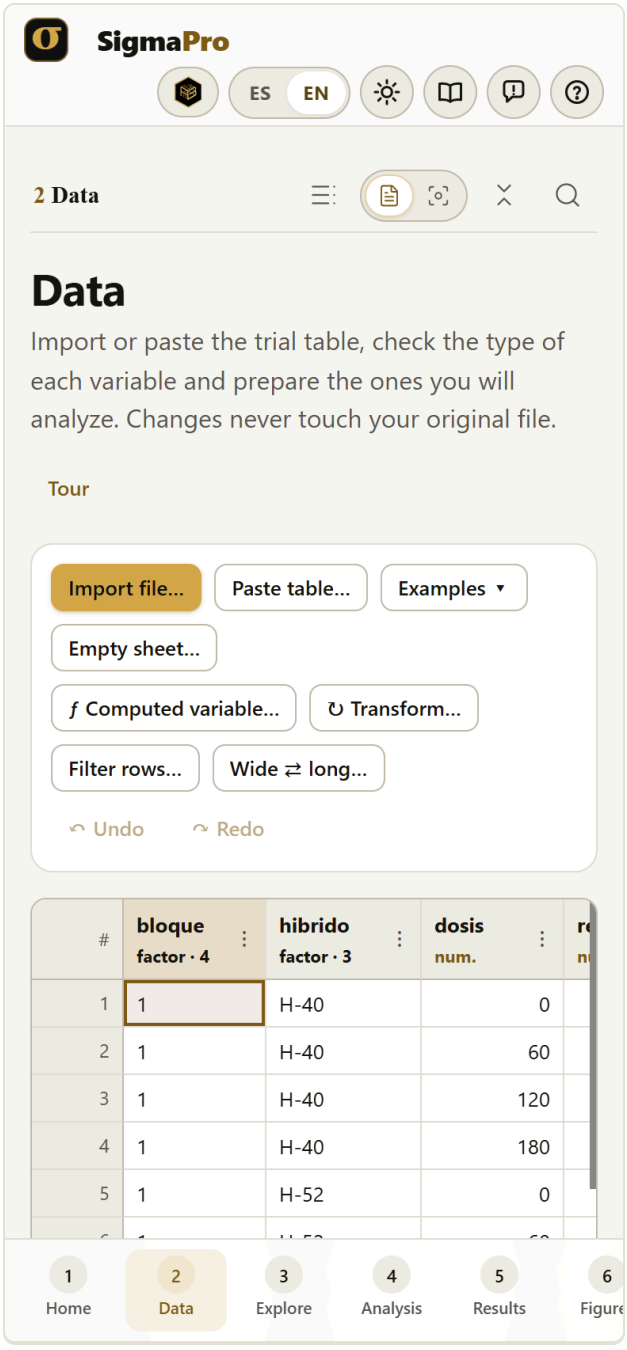


Figure I.5. SigmaPro on a phone.

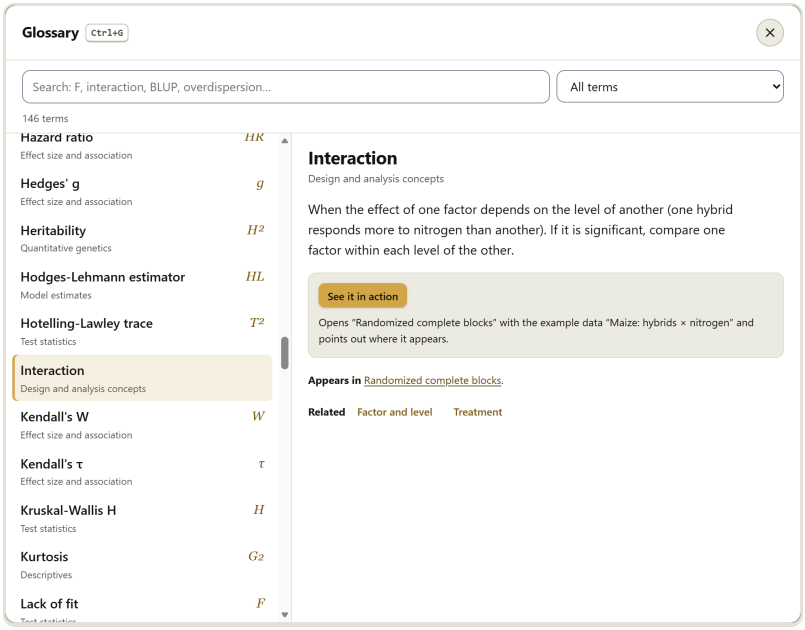


Figure I.6. The glossary at the term "Interaction": what it is, "See it in action", where it appears and its related terms.

- **Search.** The search box ignores accents and capitalization and recognizes symbols, abbreviations and alternative names: "R²", "chi", "odds ratio", "marginal mean". The selector on the right shows a single group (the concepts, the test statistics, the effect sizes, quantitative genetics...). ↓ moves to the list and Enter opens the first term; if nothing matches, a button searches for the same text in Fundamentals (chapter 9).
- **The entry.** It says what the term is, in which analyses it appears (each name opens its fact sheet in Fundamentals) and which terms are related to it.
- **See it in action.** It loads the example data of the analysis where the term appears, runs it and points out the table, the column or the row in the result with a gold frame (figure I.7). With *F* it points out the F column of the maize analysis of variance; with "Interaction", the hibrido × dosis row. Some terms offer other analyses under Also in.

Table 1. Analysis of variance: Randomized complete blocks

Source of variation	df	SS	MS	F	p
bloque	3	1.4052	0.4684	10.40	< 0.001
hibrido	2	5.4594	2.7297	60.63	< 0.001
dosis	3	39.4562	13.1521	292.15	< 0.001
hibrido × dosis	6	0.3962	0.0660	1.47	0.220
Error	33	1.4856	0.0450	—	—
Total	47	48.2026	—	—	—

Figure I.7. "See it in action" for "Interaction": the maize analysis of variance with the interaction row pointed out.

The example data take the place of the loaded table, which is not lost: Back to the previous table, in Data, brings it back (section 2.12). At the Basic level, which hides the tables, "See it in action" switches to Standard (section 4.5).

Tours and help

Each block has the Tour button next to its title: it takes you step by step through the parts of the block, highlighting each one (the rest of the screen is dimmed) with two lines that say what it is for (figure I.8). Previous and Next, or the ← and → arrows, change the step; Finish or Esc closes the tour. In Home, First time? SigmaPro tour takes you around the app: the paths, the block bar, the search, the glossary, the shortcuts and the language. The tours of Data, Explore and Analysis offer to load the maize example if there are no data.

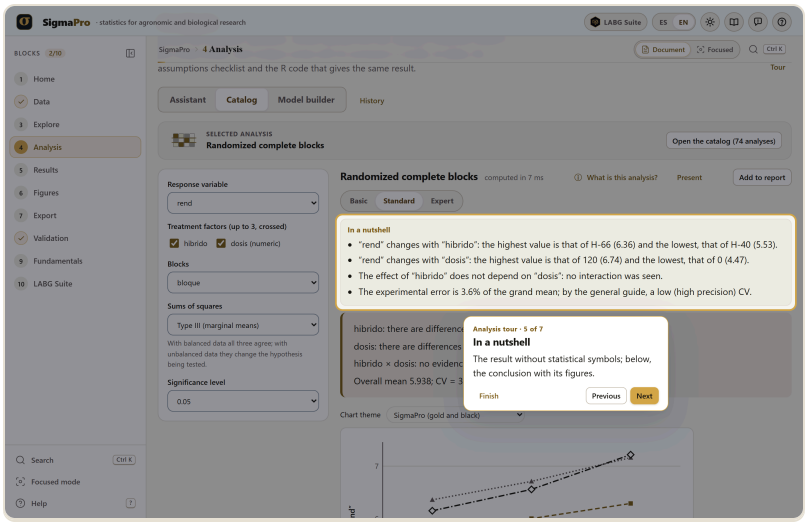


Figure I.8. A step of the Analysis tour: "In a nutshell", highlighted over the maize result.

Tooltips appear when you rest the pointer on a control for a moment, or when you reach it with the keyboard, and they close with Esc. The table headers say what their statistic is (figure I.9), and the fields for the significance level, the sums of squares and the *p*-value adjustment carry their definition from the glossary.

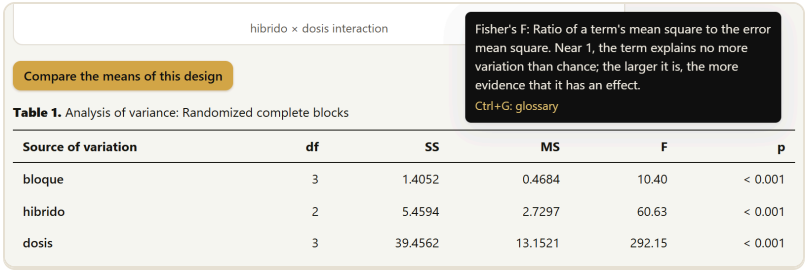


Figure I.9. The help for the *F* header of the analysis of variance.

I.7 The R engine inside the browser

Almost all of SigmaPro runs on its own computing core, written for the program. Some methods—linear mixed models, repeated measures and generalized linear mixed models—need algorithms that R handles better than anything else, and for them SigmaPro includes an **R engine**: R compiled to run inside the browser (webR), with the packages `lme4`, `lmerTest`, `emmeans`, `nlme` and `car`.

- The first time you use one of those methods, the app tells you and downloads the engine (about 25 MB). After that, it stays stored in the browser and works without internet.
- The data **do not leave your machine**: the engine runs on your own computer, like the rest of the app.
- When the app is published with its `motor-r/` folder and opened from a server, the engine is loaded from that same folder, and no internet is needed, not even the first time.

Section 4.6 explains the notice and which methods use the engine.

I.8 How to work through a study

The order of the blocks is the order in which a trial is analyzed. You do not have to go through all of them: whatever a block needs from another, it takes on its own.

1. **Block 2.** Load the data and check the type of each variable; read what SigmaPro sees in the trial: the design, the missing values, the empty cells.
2. **Block 3.** Explore: means, spread and odd values by treatment, before testing anything.
3. **Block 4.** Choose the analysis with the assistant (or in the catalog, or build the model in the model builder); read the conclusion and the traffic light; compare the means; add to the report what you want to keep.
4. **Block 5.** Review the report: the order, the titles, the notes and the paragraph for the thesis.
5. **Block 6.** Adjust the figures: the error bars, the width and, if needed, the ✎ editor or the Figure Studio.
6. **Block 7.** Export the report, the tables and the figures, and save the .labg project.

At any step, the ⓘ What is this analysis? button opens the analysis's fact sheet in Fundamentals (Block 9), with what it assumes and the data it needs, and at the foot of some results the Continue in the LABG suite strip suggests the app that continues the work (Block 10).

I.9 The examples

The Data block includes **27 example trials** (table 2.2), one or more for each family of analyses, embedded in the program so that they work with a double-click. Six are published data—Mendel's segregation, the poisons of Box and Cox (1964), the oats in split plots of Yates (1935), an alpha-lattice and an oat trial in 14 environments from Olivoto and Lúcio (2020), and the elongation of a yarn made from three polymers, from Cornell (2002)—and the rest are fictitious, built so that each analysis has a clear case. The main example of the manual is "Maize: hybrids × nitrogen": 3 hybrids with 4 nitrogen rates in 4 randomized complete blocks, 48 plots.

I.10 What the program does not do

SigmaPro does what it does well, and it says what it does not do yet. These are its known limits.

- The manual and the interface are available only in Spanish and English.
- The PDF of the report comes from the browser's print dialog; in the .docx, the figures are embedded as PNG images at 300 dpi. The LaTeX is written ready to compile, but the app does not compile it.
- The Data block does not average or sort rows: if you need the mean of each plot or the dates in order, prepare them in your spreadsheet before importing.
- With two SigmaPro tabs open at the same time, autosave keeps the work of the last one that changed; the worksheet's undo history is not saved.
- With many rows, each scatter plot draws at most 2,500 points (the calculations use all of them).
- Sampling: only simple random and stratified sampling (not cluster or systematic sampling).
- Generalized linear mixed models: they use the Laplace approximation; Gauss-Hermite quadrature only with a single random intercept, in binomial or Poisson models.
- Heritability: it does not handle fixed controls within a trial of random genotypes, or relationship matrices.
- Multivariate: it includes MANOVA and discriminant analysis; principal components and cluster analysis are done in PCAPro and ClusteringPro, from the same suite (chapter 10).

1

BLOCK 1 · HOME

Home: what SigmaPro promises and where it stands

The first block is the front door of the app. It says in one sentence what the app is for, offers three ways to begin (your data, an example, or finding out which analysis to use), remembers what was analyzed last, sums up its three commitments (accuracy verified against R, reproducible code and complete privacy) and shows where its development stands. Further down it lists everything that can be analyzed, with a link to the fact sheet of each analysis, shows the program's details and the opening-screen settings, and introduces the sister apps of the suite. If the tab was closed last time with work in progress, the notice to recover it appears here.

1.1 What you see when you open the app

1.2 The three commitments

1.3 Development status

1.4 If there is saved work

1.5 What you can analyze

1.6 About SigmaPro and the opening screen

1.7 Explore the LABG suite

1.1 What you see when you open the app

SigmaPro always opens in this block, after the opening screen (section 1.6): when you press Start, the logo and the name travel to their place in the top bar and Home appears. The title sums up the idea of the whole app: **from experiment to conclusion, with numbers you can verify**. Below it, a sentence says what it does: experimental designs, mean comparisons, regression and mixed models, with an assistant that understands your trial, publication-ready reports and the R code for every analysis.

The illustration on the right is a miniature summary of a typical result: treatment means joined by a curve, each with the letter that shows which means differ from each other, and below them the formula of the analysis that produced them (`rend ~ bloque + híbrido × N`), with the σ of the logo in the background. It is decorative; the real numbers appear when data are loaded.

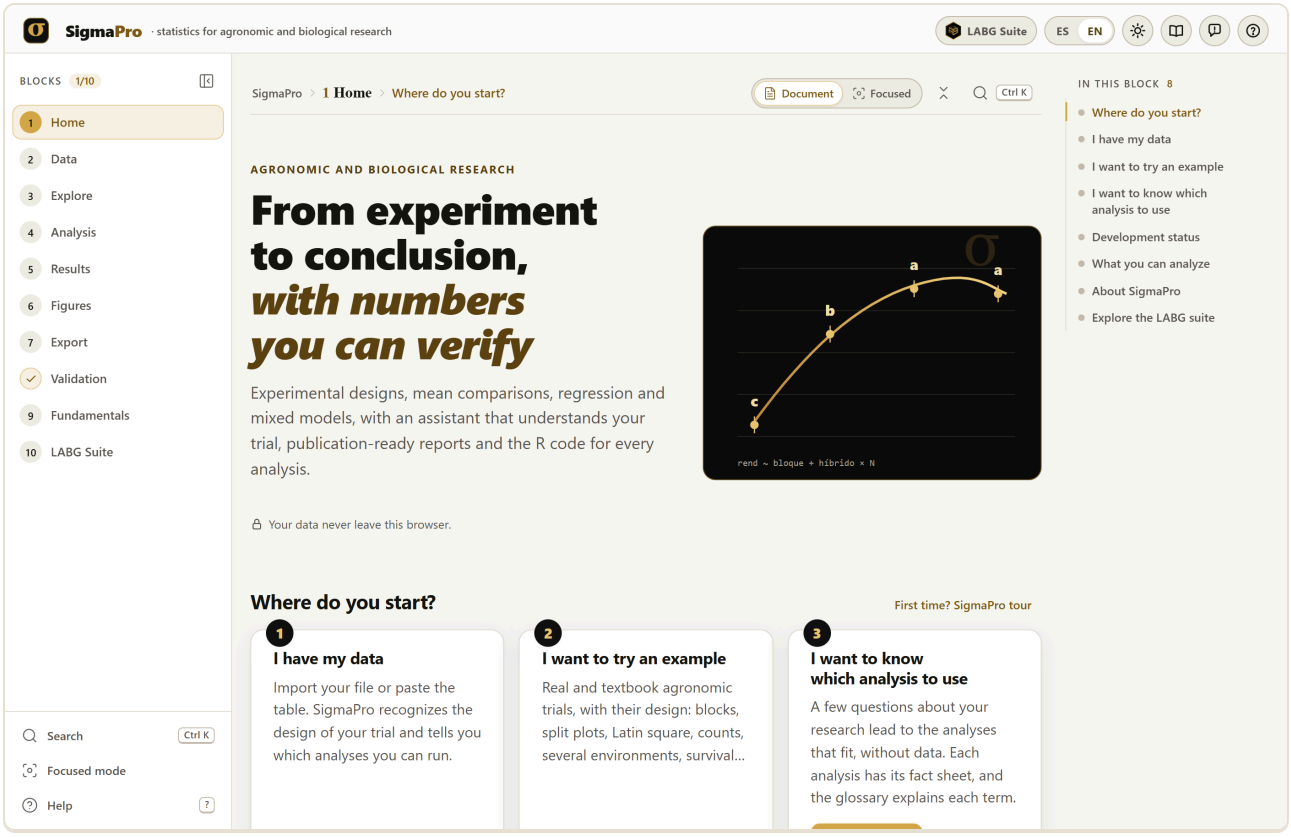


Figure 1.1. The Home block, as it appears when you open SigmaPro. The bar on the left leads to the ten blocks; the content of the block fills the center.

The note *Your data never leave this browser* is a reminder that everything is computed on the computer of whoever is using the app (section 1.2).

B1

Where do you start?

Below the title, three numbered cards offer the three ways to begin (figure 1.2):

Path	What it does
1 · I have my data	Import a file and Paste a table open the same windows as in the Data block (sections 2.3 and 2.4); once the table is loaded, the app moves to Data, which says what it sees in the trial (section 2.13).
2 · I want to try an example	Choose an example opens the list of the 27 examples (section 2.5); when you choose one, the app loads it and moves to Data.
3 · I want to know which analysis to use	Decision tree leads you to the analysis through a few questions about your research, without data (section 9.3); See the analyses opens the Fundamentals catalog, with the fact sheet of each analysis (chapter 9), and Open the glossary, the app's glossary (section I.6).

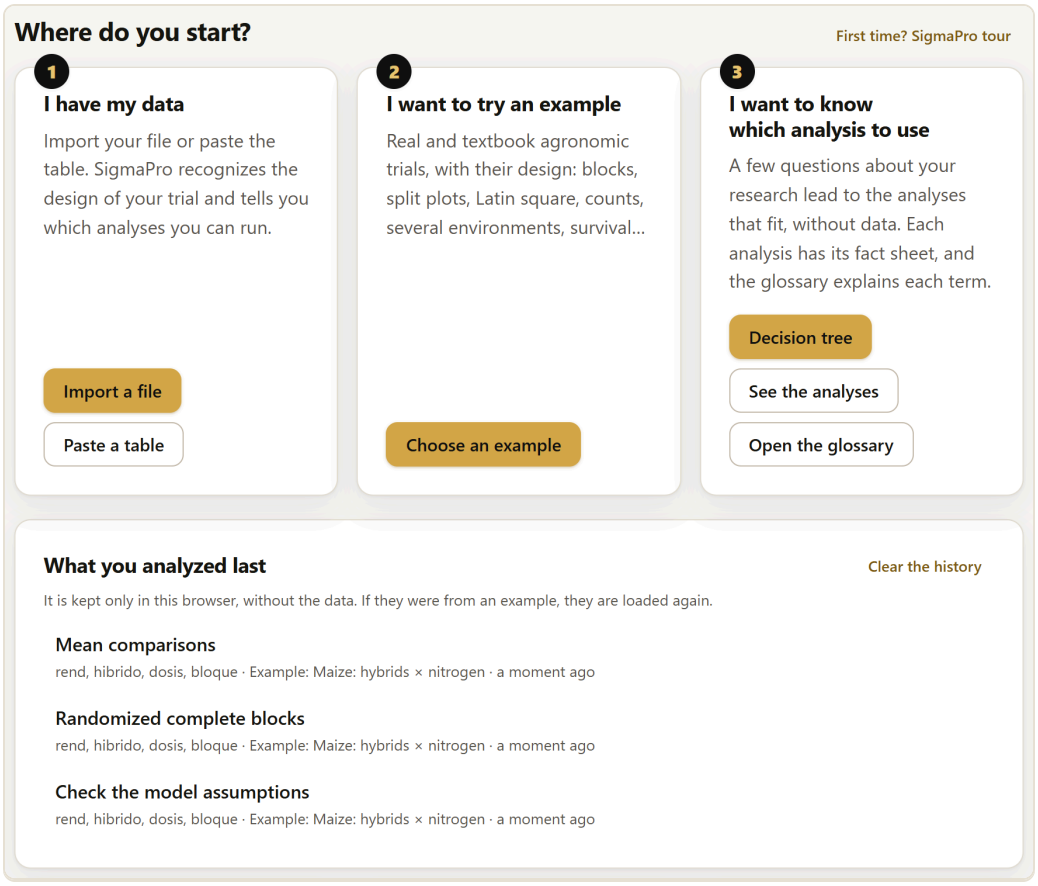


Figure 1.2. The three paths and, below them, what was analyzed last with the maize example.

What you analyzed last

SigmaPro remembers every analysis computed in Explore or in Analysis: the method, the variables and options, the data it was run on, and when; the same analysis with the same data takes up a single row, that of the latest computation. The **What you analyzed last** card shows the five most recent ones, and a click on one opens it again with its variables and options. If the loaded table does not have its columns and the data came from an example, the app loads that example again (the previous table remains available under Back to the previous table, section 2.12); if they came from a file, a notice says which columns are missing. The same history, with up to fifteen analyses, is under the History button of the Analysis block (section 4.1).

The history is kept only in that browser and contains no data values: only the names of the columns and of the data. Clear the history empties it.

At the end of each block there are two buttons, ← Previous and Next →, with the name of the block they lead to; in Home only the second one appears, and in LABG Suite, the last block, only the first. The natural route follows the bar: from the data to the analysis, to the report, to the figures and to the export. But any block can be opened at any time.

1.2 The three commitments

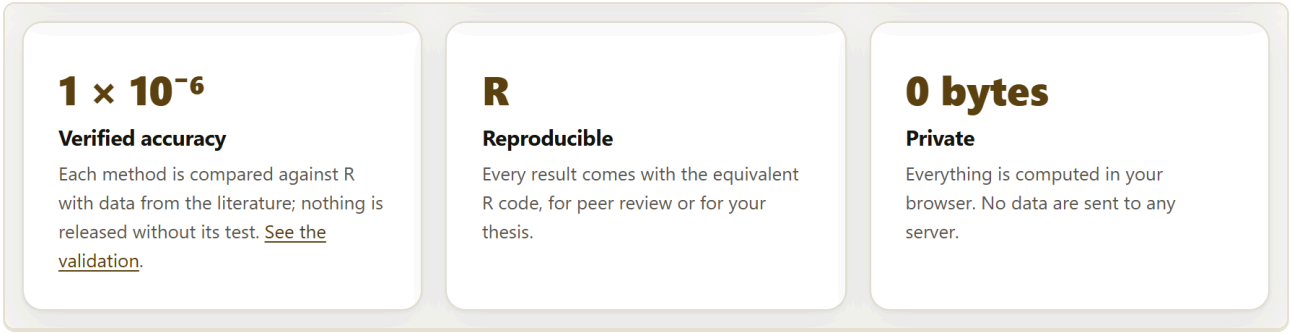


Figure 1.3. The three cards of Home. Each one is a promise that the app keeps in a way you can check.

Below the paths, three cards sum up what sets SigmaPro apart. They are not slogans: each one can be checked; in the first, See the validation leads to Block 8 (chapter 8).

10^{-6}

VERIFIED ACCURACY

Each method is compared against R, with data from the literature and with cases built on purpose, before it is offered in the catalog. The rule is that **no number may differ from R's by more than one millionth** (relative to its size). A method whose cases have not all passed does not appear in the app. Block 8 shows the outcome of those comparisons.

B1

R REPRODUCIBLE

Every result comes with the **R code** that gives the same result, with the data, the model and the options that were used. It lets a reviewer, a thesis advisor or you yourself, years later, repeat the analysis without SigmaPro. The `.labg` project also stores a script that repeats all the analyses in the report (chapter 7).

0 PRIVATE

Everything is computed **in the browser**: the data are not sent to any server, not even for the models that run in the R engine, which also works inside the browser (section I.7). Opened with a double-click, the app makes no request outside the computer, except for the first download of the R engine.

A difference of one millionth means that, in practice, SigmaPro's figures and R's are the same in every decimal place that gets published. An F of 60.635 in SigmaPro is an F of 60.635 in R; a p value of 0.0034 is 0.0034. When a method cannot reach that accuracy because R itself introduces larger numerical noise (for example, in certain iterative fits), the case carries its own tolerance with its reason written down, and that is how it appears in the validation (chapter 8).

1.3 Development status

Further down, the **Development status** card explains how SigmaPro is built: in **milestones**, and each milestone is closed only when all its comparisons against R and all its automated tests pass. The list marks finished milestones with ✓, the one in progress with • and pending ones with ○; the number of each milestone tells which phase it belongs to. All of them are finished, up to the last one on the list: the release of version 1.0.0.

Phase	Milestones	What it brought
Phase 1	1.0 to 1.12	The foundations and what is used most in the field: the worksheet, descriptive statistics, assumptions with a traffic light, t tests and nonparametric tests, the analysis of variance of experimental designs, mean comparisons, ANCOVA, regression, correlation, contingency tables, the assistant, the model builder, the report, the figures and the exports.
Phase 2	2.0 to 2.10	The models that demand more computation: mixed models (in the R engine inside the browser), repeated measures, variance components and heritability, generalized linear models and their mixed versions, nonlinear regression and dose-response, response surfaces and multi-environment trials with their stability.
Phase 3	3.0 to 3.9	Sample size and power, sampling, survival, time series, multivariate analysis (MANOVA and discriminant analysis) and, as an extension, strata in survival, Box's M test, ARIMA models, Cox and MANOVA in the model builder, and autosave of the work.

Phase	Milestones	What it brought
Phase 4	4.0 to 4.5	The complete calculations: nonlinear regression with hormesis, Michaelis-Menten and the economic optimum rate; multi-environment analysis with cell imputation, WAAS and two-stage analysis; the response surface design generator, mixtures, desirability and the 3D surface; generalized models with profile likelihood intervals and Firth's correction; the assistant that recognizes split plots in the data; and the English interface.
Phase 5	5.0 to 5.11	The app that explains what it computes: the Fundamentals fact sheets of the 78 analyses, the LABG Suite block with suggestions from the sister apps, the opening screen, the catalog illustrations and the chart themes; fast loading; "How to read this table", with gauges, curves and interpretation cards; the Basic, Standard and Expert levels; the traffic light with icons and "Try the alternative"; "In a nutshell" and "How to report it"; the glossary; what guides you (the Home paths, the history, the decision tree, the tours, the help, presentation mode and errors in plain words); the redesigned report, with its methodological appendix, Markdown and the class worksheet; and the quality tests of the wording, the links, accessibility and speed.

The number at the foot of every page of the app (*SigmaPro v1.0.0*) is the version. This manual describes version 1.0.0.

1.4 If there is saved work

While you work, SigmaPro saves on its own, in the browser itself, the data, the analyses in the report and the figure settings (**autosave**, section 7.7). If the tab was closed without saving the project, or the computer shut down, this notice appears at the top of Home and of Data when you open the app again:

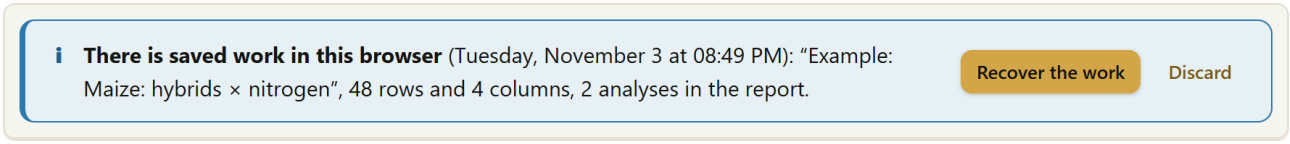


Figure 1.4. The autosave notice: when it was saved, which data, how many rows and columns, and how many analyses the report had.

1. Recover the work reloads the data exactly as they were (with their types, their computed columns and their filter), the analyses in the report and the figure settings. The results are recomputed with those data, so they are exactly the same as before. The app opens in Results, or in Data if the report was empty.
2. Discard deletes what was saved and leaves the app empty.
3. If you choose neither and start other work, the earlier work is not lost: the notice stays in view and the work can be recovered while the tab is open. Recovering it when other data are already loaded asks for confirmation, because it replaces them.

B1

> **!** AUTOSAVE DOES NOT REPLACE THE PROJECT
>
> What is saved lives in **that browser on that computer**. It does not travel to another machine, it is lost if the browser data are cleared and, in a private window, it is deleted when the window is closed. To keep your work, share it or take it to another computer, save the **.labg project** in the Export block (section 7.6).

1.5 What you can analyze

The **What you can analyze** card lists the 78 analyses in SigmaPro, grouped into their 18 families, from descriptive statistics to survival and time series. It lets you see at a glance what the program does before loading any data.

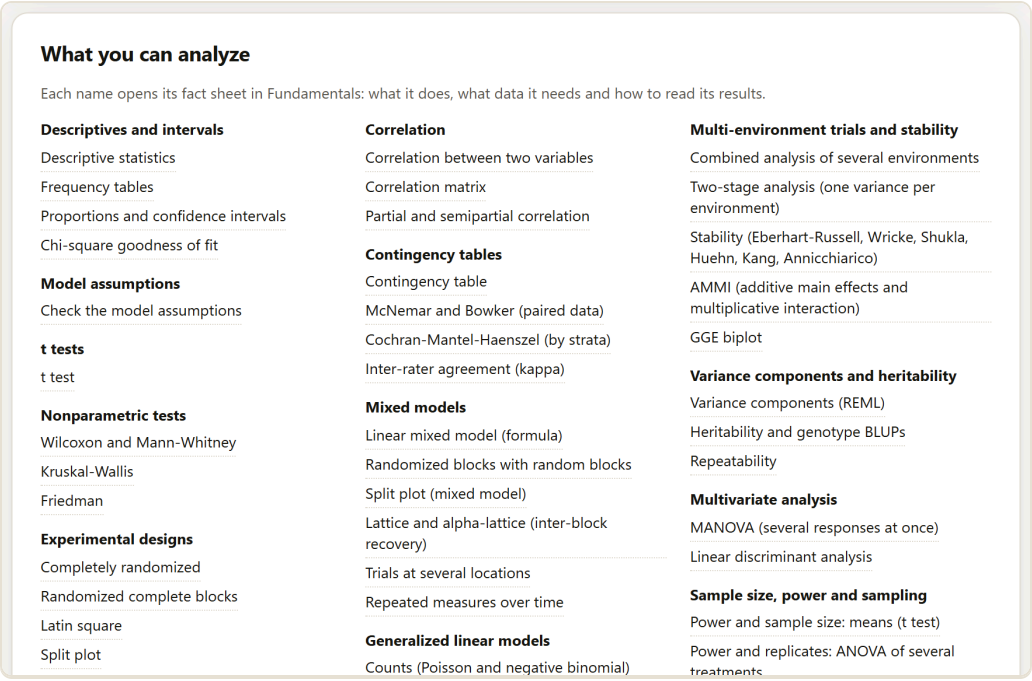


Figure 1.5. The beginning of the What you can analyze card. Each name is a link to the fact sheet of that analysis.

Each name opens the **fact sheet of the analysis** in Block 9, Fundamentals: what it is for, the model and its assumptions, the data it needs, how to read the result, and its references (chapter 9). From the fact sheet, Open this analysis takes you to the analysis and Load the example data opens it with its example already loaded, so this list is also a shortcut for trying out any method.

1.6 About SigmaPro and the opening screen

The **About SigmaPro** card says who made the program and under which license, and brings together the controls of the opening screen.

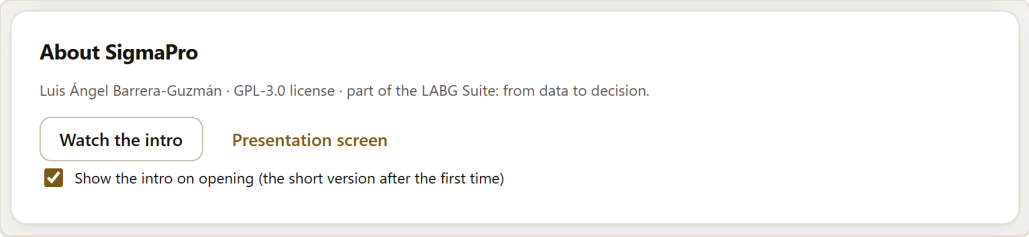


Figure 1.6. The About SigmaPro card.

Control	What it does
Watch the intro	Shows the full opening screen again, up to Start.
Presentation screen	The opening scene in a loop, without a button, to project in a class or at a booth. Any key or a click exits it.
Show the intro on opening	When checked (the default), the opening appears when the app is opened: in full the first time and in a short version (a second and a half) after that. When unchecked, SigmaPro opens directly in Home. The choice is remembered in the browser.

> **NOTE**
>
> The opening never gets in the way of your work: Skip or Esc dismisses it at any time, it does not repeat when the same tab is reloaded and, if the system asks for reduced motion, it appears without animation. While it is showing, the app finishes loading behind it.

1.7 Explore the LABG suite

At the end of Home, the black and gold box **Explore the LABG suite** introduces four sister apps of SigmaPro, the ones that most often continue an agronomic analysis: PCAPro (principal components), ClusteringPro (cluster analysis), AgriDesign (design and randomization of experiments) and CorrelationPro (correlation). Each card shows its logo, its version, its features and the Open app, Manual and Cite buttons; it is the same box as in Block 10, which shows the 22 apps of the suite by area; chapter 10 explains it in full (figure 10.1 and table 10.1).

2

BLOCK 2 · DATA

Data: the trial table

Every analysis starts with a well-built table. In this block you import a file, paste a table or open an example; check the type of each variable, which decides how it is analyzed; create computed and transformed variables; filter rows without deleting them; change the shape of the table; and read what SigmaPro recognizes in the design of the trial. At the bottom, depending on what the table contains, the block suggests the suite app that can continue the work. None of this touches the original file.

2.1 Four ways to begin

2.2 What the table should look like

2.3 Importing a file

2.4 Pasting a table and the empty sheet

2.5 The 27 examples

2.6 The worksheet

2.7 The five variable types

2.8 Computed variables

2.9 Transforming a variable

2.10 Filtering rows

2.11 Wide and long tables

2.12 Undo, redo and go back

2.13 What SigmaPro sees in your trial

2.14 Continue in the LABG suite

2.1 Four ways to begin

While there are no data, the block shows its toolbar and, below it, a box with the three most common ways to begin. The toolbar buttons do the same and stay in view once a table is loaded.

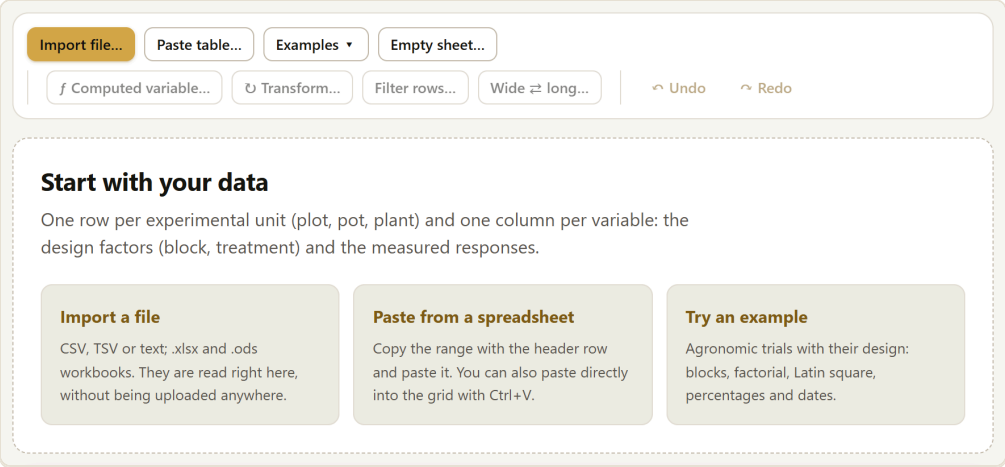


Figure 2.1. The Data block without data. The tools in the second row become active when a table is loaded.

Button	What it does	Section
Import file...	Opens a text file (CSV, TSV, TXT) or a spreadsheet workbook (.xlsx, .xlsm, .ods). The file is read on your computer; it is not uploaded anywhere.	2.3
Paste table...	Takes a range copied from a spreadsheet, with its row of names.	2.4
Examples ▾	Opens one of the 27 example trials, each one designed for a type of analysis.	2.5
Empty sheet...	Creates a blank table with the column names and the number of rows you ask for, for entering data by hand.	2.4

Loading new data replaces the previous table. If the previous one had changes, the report and the open analyses are recomputed with the new data; that is why it is a good idea to save the project (chapter 7) before switching to another trial.

2.2 What the table should look like

SigmaPro reads the table layout that almost all statistics programs require: **one row per experimental unit** (the plot, the pot, the plant, the animal) and **one column per variable**. The columns are of two kinds: those that describe the design (block, treatment, variety, rate, date) and those that were measured (yield, height, number of fruits).

Table 2.1. The first rows of the maize example: one plot per row.

bloque	hibrido	dosis	rend
1	H-40	0	4.61
1	H-40	60	5.71
1	H-40	120	6.41
1	H-40	180	6.56
1	H-52	0	4.80

B2

GOOD PRACTICES FOR THE TABLE

One row of names, at the top	Short names, with no repeats: `rend`, `altura_30dds`, `peso_seco`. Spaces and accents are allowed, but simple names read better in formulas and in R code.
Nothing else on the sheet	No titles above the table, no totals below it, no merged cells and no notes at the side.
Missing values, left empty	An empty cell is a missing value. `NA`, `.` and `-` are also accepted, or the codes you specify when importing. Never a zero in place of a lost value.
One type per column	A yield column with numbers only; if there is a stray text ("s/d", "lost"), it is taken as missing and the app tells you so.
Clear group codes	If the block or the treatment was written with numbers (1, 2, 3), check that the app has taken it as a factor (section 2.7).
Long format	If the same variable was measured several times (dates, years, harvests), one row per measurement. If the table comes in wide format (one column per date), the block converts it (section 2.11).

2.3 Importing a file

Import file… opens the system's file picker. SigmaPro accepts:

- **Delimited text** (.csv, .tsv, .txt): separated by commas, semicolons, tabs or vertical bars. The app detects on its own the column separator, the decimal separator and the text encoding (UTF-8, UTF-16 or the one used by Windows in Spanish, Windows-1252), so accents and the ñ come through correctly even if the file comes from an old spreadsheet.
- **Spreadsheet workbooks** (.xlsx, .xlsm and .ods): if the workbook has several sheets, you choose which one to import. Workbooks in the old format (.xls) cannot be opened; save them first as .xlsx or .csv.

Before anything is loaded, the **Import data** window shows how the table will look:

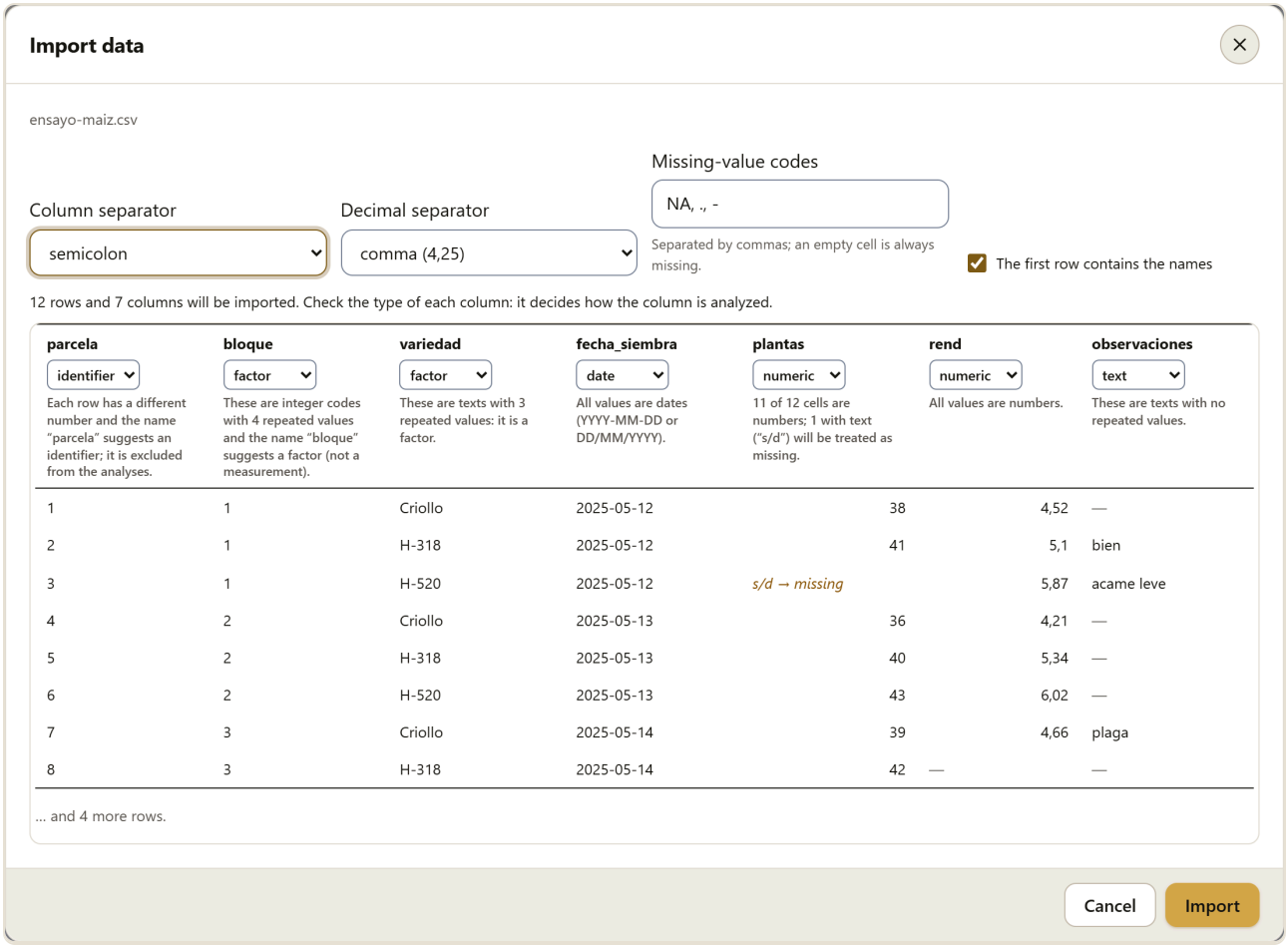


Figure 2.2. The import window with a file separated by semicolons and with a decimal comma. Each column shows the type the app proposes and the reason; the "s/d" cell in *plantas* is highlighted because it will become missing.

Option	What it is for
Column separator	Comma, semicolon, tab or vertical bar (text files only). If the preview shows everything in a single column, the separator is not the right one.
Sheet	In a workbook, the sheet to import.
Decimal separator	Point (4.25) or comma (4,25). In Mexico the point is the usual one; files from other countries, or from spreadsheets set to Spanish (Spain), use a comma.
Missing-value codes	The texts that mean "no data", separated by commas (by default `NA, ., -`). An empty cell is always missing.
The first row contains the names	If you uncheck it, the columns are named V1, V2… and the first row is taken as data.
Type of each column	The selector above each column. What you choose here prevails; the app only proposes (section 2.7).

The preview shows the first eight rows **as they will be**: numbers with the chosen decimal separator, dates in the form YYYY-MM-DD and, in another color, each cell that cannot be converted and will become missing (*s/d → missing*). When you press Import, the table is loaded and the block moves to the worksheet. If any numeric column had stray texts, a notice says so, and the explanation of that variable tells how many there were.

B2

> **THE ORIGINAL FILE DOES NOT CHANGE**
> SigmaPro works with a copy in memory. No change made in the block (types, new columns, filters, edited cells) is written to the file that was imported. To save the table with its changes, use the .labg project (section 7.6), which also includes the data as CSV.

2.4 Pasting a table and the empty sheet

Paste table... opens a text box where you paste, with Ctrl+V, a range copied from a spreadsheet (with its row of names). When you press Continue, the same import window from the previous section appears, with tab as the separator. It is the quickest way to bring in some data without saving a file first. You can also paste directly onto the empty grid: the app recognizes it as a new table and opens the same window.

Empty sheet... asks for the **column names**, separated by commas (by default `bloque, tratamiento, rend`), and the **number of rows** (up to 100,000). Columns whose name suggests a design factor (bloque, tratamiento, variedad, repetición, lote...) start as factors; the rest, as numeric. Then you type in the grid as in any spreadsheet (section 2.6).

2.5 The 27 examples

Examples ▾ drops down the list of example trials, with a line that says what each one is for. Six are published data (Mendel, Box and Cox, Yates, Cornell and two from an R package for multi-environment trials); the rest are fictitious, built so that each analysis has a clear case. All the results in this manual come from them, so any number can be reproduced with a click.

Table 2.2. The SigmaPro examples and what each one is for.

Example	Rows	Design and variables	For
Maize: hybrids × nitrogen	48	3 × 4 factorial in 4 blocks; yield (t/ha)	ANOVA, comparisons, polynomials, figures
Bean: germination (%)	20	5 treatments, 4 boxes of 50 seeds	Arcsine, binomial model
Wheat: 5 × 5 Latin square	25	5 varieties, rows and columns	Latin square
Tomato: height in the greenhouse	18	3 substrates; height at 15, 30, 45 and 60 days, in wide format	Wide → long, dates, missing values
Avocado: thrips per shoot	30	5 treatments in 6 blocks; counts with zeros	√(x + 0.5), count models

Table 2.2. The SigmaPro examples and what each one is for. *(continued)*

Example	Rows	Design and variables	For
Poisons × treatments	48	Box and Cox (1964): 3 poisons × 4 treatments	Box-Cox transformation
Pea: Mendel's segregation	4	Mendel (1866): 556 seeds already counted	9:3:3:1 goodness of fit
Oats in a split plot	72	Yates (1935): 3 varieties × 4 N rates, 6 blocks	Split plots
Sorghum: varieties and plants	24	6 varieties in 4 blocks; harvested plants	ANCOVA
Coffee: yield and site	40	40 farms with 7 site variables	Multiple regression, variable selection
Bean: treated seed in four lots	16	Germination by lot and treatment, already counted	Contingency, Cochran-Mantel-Haenszel
Rust: two raters	80	Severity 0 to 3 scored by two technicians	Kappa, Bowker
Alpha-lattice: 24 genotypes	72	Olivoto and Lúcio (2020): 3 replicates, blocks of 4	Mixed lattice, heritability
Oats in 14 environments	420	Olivoto and Lúcio (2020): 10 genotypes, 3 replicates	Multi-environment, stability, AMMI, GGE
Maize: height on five dates	80	4 rates in 4 blocks; the same plant on 5 dates	Repeated measures
Guava: 20 trees, four harvests	80	Kilograms per tree and harvest, with 5 missing	Repeatability
Herbicides: dry weight by dose	56	7 doses of 2 herbicides, 4 replicates	Dose-response (ED50)
Tomato: nitrogen × potassium	13	Rotatable central composite with 5 center points	Response surface
Avocado: sampling in three orchards	38	Random trees from 3 orchards; weight and thrips (0/1)	Stratified sampling
Bean: days to germination	120	3 treatments; days until germination, with censoring	Kaplan-Meier, log-rank, Cox
Weather station: rainfall and temperature	96	Monthly values from 2017 to 2024	Time series, ARIMA
Chili pepper: fruit measurements	60	4 varieties in 3 blocks, 5 fruits per plot	MANOVA, discriminant analysis
Yarn: elongation of three polymers	15	Cornell (2002): {3, 2} simplex-lattice with replicates	Mixtures (Scheffé models)
Fruit: yield, cost and pH	13	Rotatable central composite: nitrogen × irrigation, three responses	Surfaces, desirability
Wheat: irrigation × variety	48	3 irrigation depths (main plots) × 4 varieties, 4 blocks	Split plots, the assistant
Rice: irrigation × variety × nitrogen	36	Split-split plots in 3 blocks	Split-split plots
Sorghum: seven lines in incomplete blocks	21	7 blocks of 3 plots; each pair of lines occurs together once (λ = 1)	Balanced incomplete blocks

2.6 The worksheet

With a table loaded, the block is divided into three parts: the list of variables on the left, the grid with the data in the center and, below the list, the details of the selected variable. Figure 2.3 points them out.

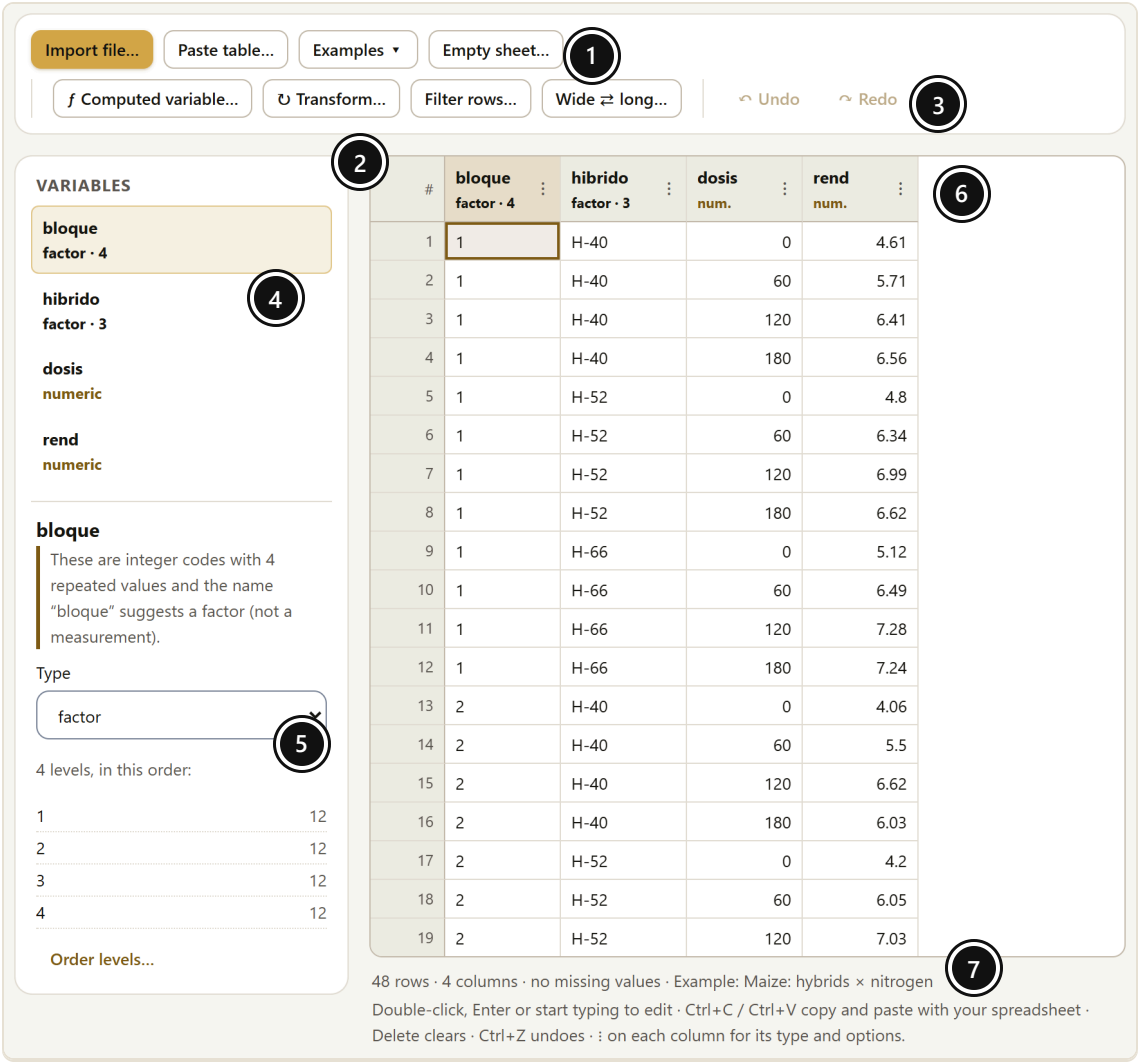


Figure 2.3. The worksheet with the maize example.

1. **Load**: import, paste, examples and empty sheet (2.1).
2. **Prepare**: computed variable, transform, filter and wide ⇄ long (2.8–2.11).
3. **Undo and redo** (2.12).
4. **Variables**: each column with its type, its levels and its missing values. A click selects it.
5. **Details** of the selected variable: why it has that type, the selector to change it, and its summary.
6. **The grid**: the data, with the type under each name and the ⋮ menu of each column.
7. **Status**: rows, columns, missing cells and where the data come from.

The list of variables and the details

Each variable in the list shows its type in color (*factor · 4* means a factor with four levels), its missing cells if there are any (*2 miss.*) and a mark if it is computed (*ƒ*) or transformed (↻). When you select one, the **details** show:

- the **reason** for the type, in one sentence ("These are integer codes with 4 repeated values and the name 'bloque' suggests a factor"), and, if it is computed or transformed, where it came from (*ƒ = rend / 1000*);
- the Type selector, which changes the type just like the ⋮ menu;
- for a numeric variable, n, missing values, mean, standard deviation, minimum and maximum, and the Transform… button; for a factor, its levels in order with how many rows each one has, and the Order levels… button; for a date, the range of dates.

If a filter is active (section 2.10), these summaries are computed only with the rows that pass the filter.

The grid

The grid works like a spreadsheet and is operated with the keyboard. It draws only the visible rows, so a table of 100,000 rows scrolls just as fast as one of 100.

Keys	What they do
Arrows, Home, End, Page Up, Page Down	Move around the table; Ctrl+Home and Ctrl+End go to the first and to the last cell. With Shift, you select a range.
Double-click, Enter, F2 or start typing	Edit the cell. Enter saves; Esc cancels.
Tab and Shift+Tab	Next and previous cell.
Delete or Backspace	Clear the selected cells (they become missing).
Ctrl+C and Ctrl+V	Copy and paste ranges to and from a spreadsheet, separated by tabs.
Ctrl+Z and Ctrl+Y	Undo and redo.
Ctrl+A	Select the whole table.

What you type in a cell is interpreted according to the type of the column: in a numeric column, a text that is not a number is not accepted and the app explains why (for a lost value, leave the cell empty); in a date column, YYYY-MM-DD and DD/MM/YYYY are accepted. When you paste a range, the cells that do not match the type of their column are left as they were and a notice says how many there were. When the table has unsaved changes, the browser asks before closing the tab.

B2

2.7 The five variable types

The type of a column decides how it enters the analyses. It is the most important decision in the block: a block written with numbers (1, 2, 3, 4) and taken as numeric turns a block analysis into a meaningless regression.

Type	What it is	How it enters the analyses
numeric	A measurement: yield, height, weight, count, percentage. Also a rate that you want to study as a curve.	As a response or as a covariate. A numeric rate with few repeated values can also be used as a design factor (the analysis offers this).
factor	A group: block, treatment, variety, lot, site, year of the trial.	As a design factor; its levels have an order that decides the order of the means in tables and figures.
date	A date (sowing, transplanting, harvest), in YYYY-MM-DD or DD/MM/YYYY.	To order time series and to compute days between dates; never as a design factor.
text	Free notes, with no repeated values.	It does not enter, unless it is used explicitly; then it counts as a factor.
identifier	The key of each unit: plot number, record number, plant.	It does not enter the analyses. It serves to recognize each row and, in the long table, as the subject of the repeated measures.

How the app proposes the type

When importing, SigmaPro proposes a type for each column and **always explains why**. The rules, in order:

1. If all the values are numbers, the column is **numeric**, except in two cases: if they are integers, each row has a different one and the name suggests a key (parcela, id, folio, número, planta…), it is an **identifier**; if they are integers with up to 30 values that repeat and the name suggests a design factor (bloque, rep, tratamiento, variedad, lote, sitio, año…), it is a **factor**.
2. If at least eight out of ten values are numbers and the rest are stray texts ("s/d", "lost"), it is **numeric** and those texts become missing, with a warning.
3. If all of them are dates, it is a **date**.
4. If they are texts that repeat, it is a **factor**. If each row has a different one, it is an **identifier** when they look like codes (P01, P02…), when the name suggests a key or when there are more than 30 rows; otherwise, it is **text**.

> **!** ALWAYS CHECK NUMERIC CODES
>
> If a column of integers with few values does not have a recognizable name (for example `T` or `grupo2`), the app leaves it as numeric and warns you: "These are numbers, but there are only 4 distinct repeated values: if they are group codes, change it to a factor". It is worth reading the reason for each variable once, when you load the data.

The ⋮ menu of each column

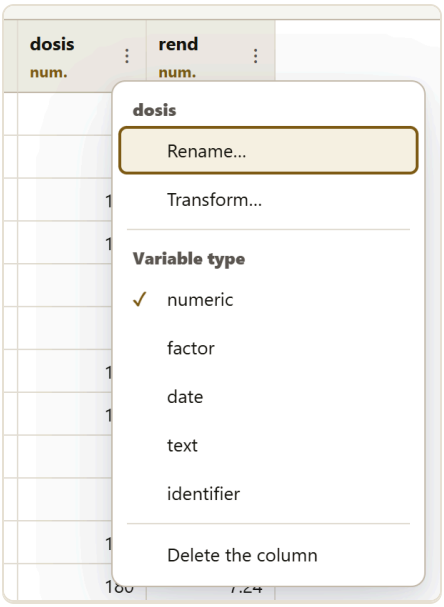


Figure 2.4. The ⋮ menu of a numeric column.

The ⋮ button next to the name of each column opens its menu:

- Rename... changes the name (it cannot repeat an existing one).
- Transform... (for numeric columns) opens the window of section 2.9.
- Order levels... (for factors) changes the order of the levels.
- Variable type: the five types, with the current one checked.
- Delete the column. Ctrl+Z brings it back.

When you change a column to numeric or to date, the values that cannot be converted become missing and a notice says how many; the change can be undone.

The order of the levels

The order of the levels decides the order of the means in tables and figures, and which one is the **reference level**: the first one, against which the others are compared in a Dunnett test, in Cox hazard ratios or in the coefficients of a model. That is why it is a good idea to put the control first.

The Numeric, Alphabetical and As they appear buttons sort them all at once; the ↑ and ↓ arrows move one level. Apply saves the order (and it can be undone).

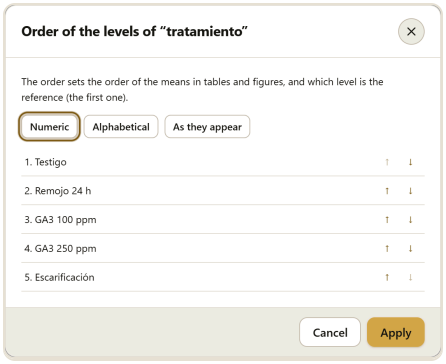


Figure 2.5. The order of the levels of a factor. The first one is the reference level.

2.8 Computed variables

ƒ Computed variable... creates a new column from the existing ones, with an expression written in the same notation as R. As you type, the window shows whether the expression is valid, what type of column it will give and its first values; a click on the name of a column or of a function inserts it at the cursor.

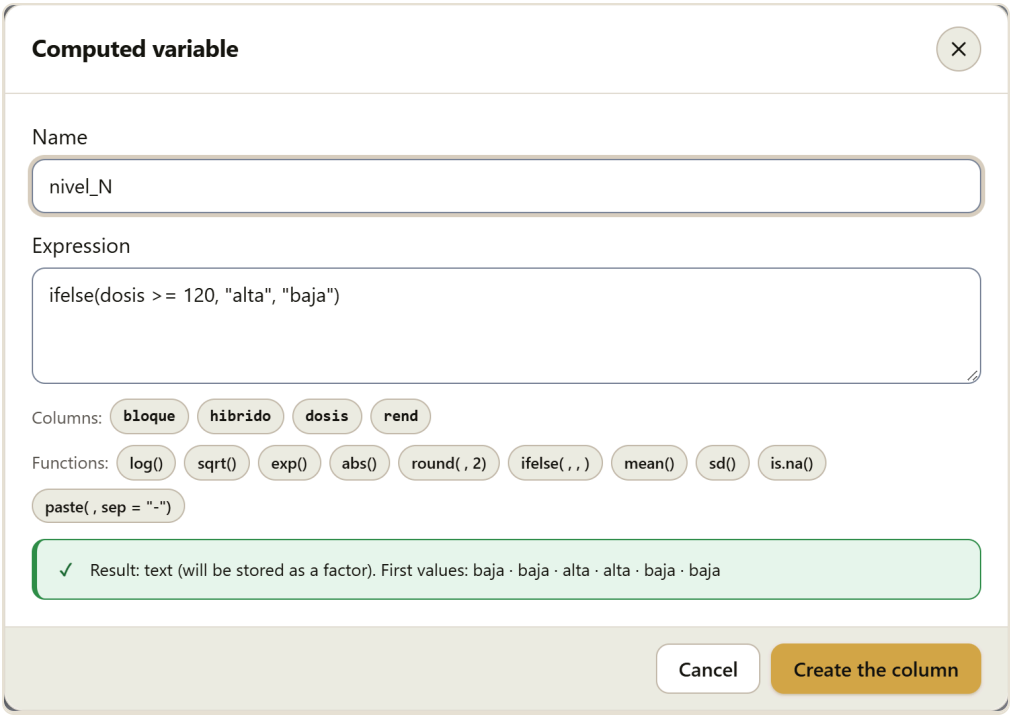


Figure 2.6. A computed variable with a condition: a rate of 120 kg/ha or more, "alta" (high); less, "baja" (low).

You write	It means
`+ - * / ^` and `%%`	Addition, subtraction, multiplication, division, power and remainder of the division.
`== != < <= > >=`	Comparisons (they give true or false).
`& | !`	And, or, not.
`x %in% c("a", "b")`	Membership in a list.
`` `peso seco` ``	A name with spaces, between backticks.
`log(x)`, `log(x, 10)`, `log10`, `log2`, `log1p`, `exp`, `sqrt`, `abs`	Logarithms (natural by default), exponential, square root and absolute value.
`round(x, 2)`, `floor`, `ceiling`, `trunc`, `sign`	Rounding and sign.
`sin`, `cos`, `tan`, `asin`, `acos`, `atan`	Trigonometric functions (in radians).
`pmin(a, b)`, `pmax(a, b)`	The smaller or the larger, row by row.
`ifelse(cond, yes, no)`	One value if the condition holds and another if it does not.
`is.na(x)`, `as.numeric(x)`	Is the value missing? · Convert to a number.
`paste(a, b, sep = "-")`, `paste0`	Join texts (for example, to combine two factors).
`mean`, `sd`, `var`, `median`, `min`, `max`, `sum`, `length`	Summaries of the whole column: they give a single number, which is repeated in every row.

Some functions also have a Spanish name: `raiz`, `si` (= ifelse), `media`, `de` (= sd), `mediana`, `minimo`, `maximo`, `suma`, `varianza`, `redondear`, `exponencial`, `absoluto` and `faltante` (= is.na).

EXAMPLES

`rend * 1000 / 10.5`	Kilograms per hectare from a plot of 10.5 m² (if *rend* were in kg/plot).
`(rend - mean(rend)) / sd(rend)`	Standardized yield.
`germinadas / sembradas * 100`	Germination percentage.
`ifelse(trips == 0, "sin", "con")`	Presence or absence (stored as a factor).
`paste(hibrido, dosis, sep = "_")`	A factor with the 12 combinations of hybrid and rate.

B2

The result is stored according to what the expression gives: a number, as a **numeric** column; a text, as a **factor**; a true or false, as a factor with the levels VERDADERO and FALSO (Spanish for TRUE and FALSE). Missing values follow R's rules: any operation with a missing value gives a missing value, and `NA & FALSE` is FALSE. One difference from R: summaries (`mean`, `sd`...) **ignore** missing values unless you write `na.rm = FALSE`, because that is almost always what you want.

2.9 Transforming a variable

When the residuals of an analysis do not have constant variance or are not normal (chapter 4, the assumptions), a transformation of the response often solves the problem. ↻ Transform... creates a **new column** with the transformed variable; the original does not change, so the two analyses can be compared.

Transformation	When it helps
ln(x)	Positive data whose standard deviation grows in proportion to the mean (weights, biomass, pathogen loads).
$\log_{10}$(x)	The same, in base 10 (easier to read in orders of magnitude).
ln(x + c)	When there are zeros. You choose the constant *c* (1 by default).
√x	Counts whose variance is proportional to the mean (Poisson distribution).
√(x + 0.5)	Counts with zeros or with small values (insects, weeds, lesions).
asin(√p)	Proportions and percentages (germination, incidence, mortality). The window detects whether the data are in percent (0 to 100) and lets you choose the result in degrees or in radians.
1/x	Times (days to flowering, to germination) whose variance grows strongly with the mean.
Box-Cox	When you do not know which one to use: it estimates the power that best stabilizes the variance and normalizes the residuals.
(x − media)/DE	Convert to standard deviation units (to compare variables on different scales).

B2

Box-Cox

The Box and Cox (1964) transformation is a family of powers, x^{λ}, that includes the previous ones as special cases: λ = 1 leaves the data as they are, λ = 0.5 is the square root, λ = 0 the logarithm and λ = −1 the reciprocal. The value of λ is estimated by maximum likelihood from the **residuals of the experiment's model**, not from the raw data: what must be stabilized is the variation within treatments, not the difference between them.

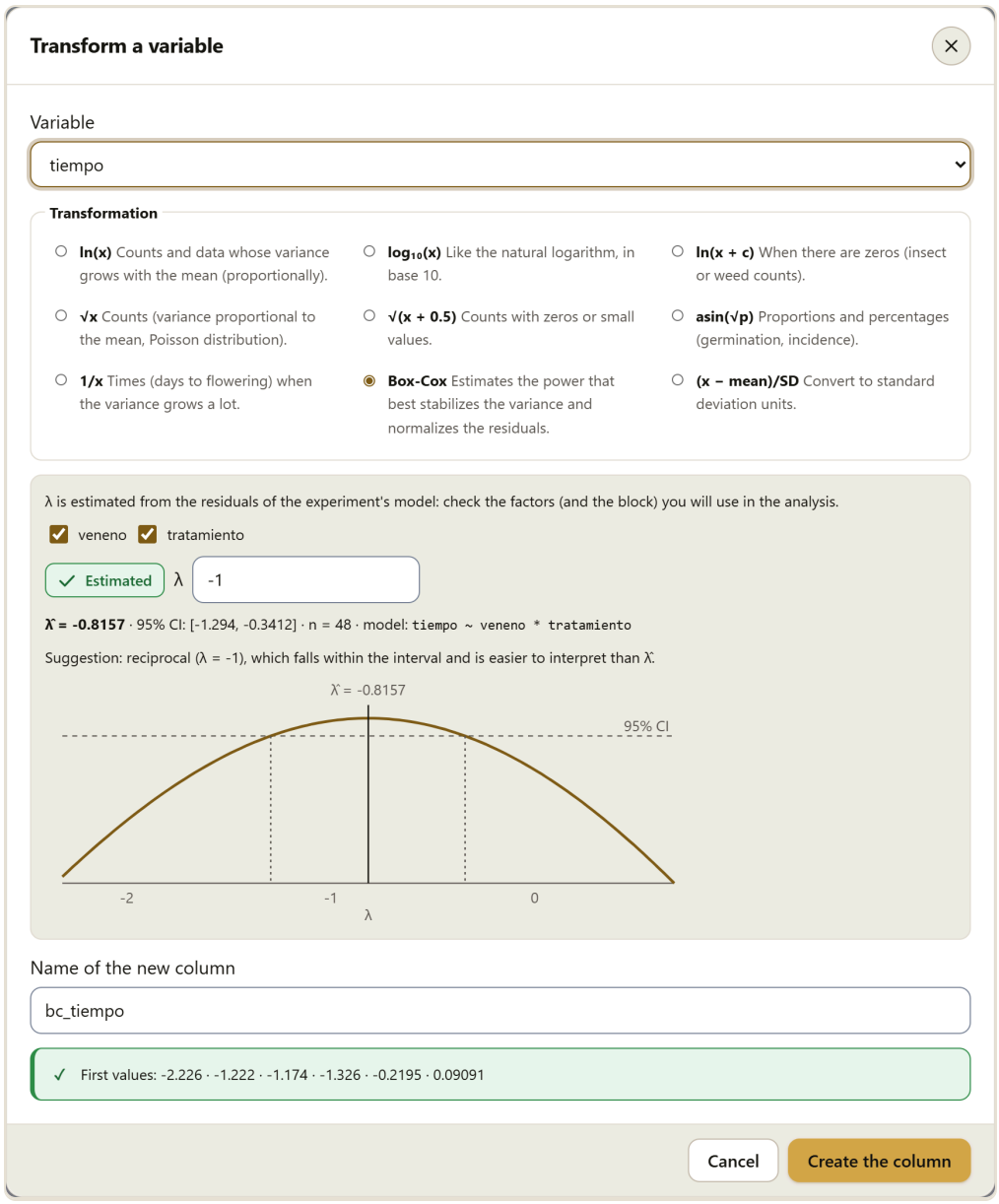


Figure 2.7. Box-Cox for the survival time in the poisons example. The maximum is at $\hat{\lambda} = -0.82$; the 95% interval includes −1, so the app proposes the reciprocal.

1. Choose the variable and select **Box-Cox**.

2. Check the factors (and the block) that will be used in the analysis. The app builds the model: blocks, rows and columns are added; treatments are crossed. In the example, `tiempo ~ veneno * tratamiento`.

3. Estimate λ computes $\hat{\lambda}$, its 95% confidence interval (the values of λ with $2[L(\hat{\lambda}) - L(\lambda)] \le \chi^2_{0.95;\,1} = 3.84$) and the likelihood curve, with the maximum and the interval marked.

4. The app **suggests a simple power** (none, square root, logarithm, reciprocal square root, reciprocal or square) if it falls within the interval, because it is easier to interpret than $\hat{\lambda}$ and almost as good. The value to be used goes in the **λ** box, which you can change by hand.

5. Create the column stores $(x^{\lambda} - 1)/\lambda$ (or $\ln x$ if $\lambda = 0$) under the name you choose; the details of the variable record the λ used.

> **THE MEANS OF A TRANSFORMED VARIABLE**
>
> The means and their differences are on the transformed scale. To present them in the original units, apply the inverse transformation to the means (for example, square the means of √x), bearing in mind that the result is a median, not a mean. An alternative that avoids the problem is generalized linear models (section 4.17), which model counts and proportions on their own scale.

2.10 Filtering rows

Filter rows… leaves out of the analyses the rows that do not meet a condition, **without deleting them**. It is useful for analyzing a single site of a multi-site trial, removing a damaged block or excluding plots with a recorded problem. The condition is written in the notation of section 2.8, and the window tells you instantly how many rows pass.

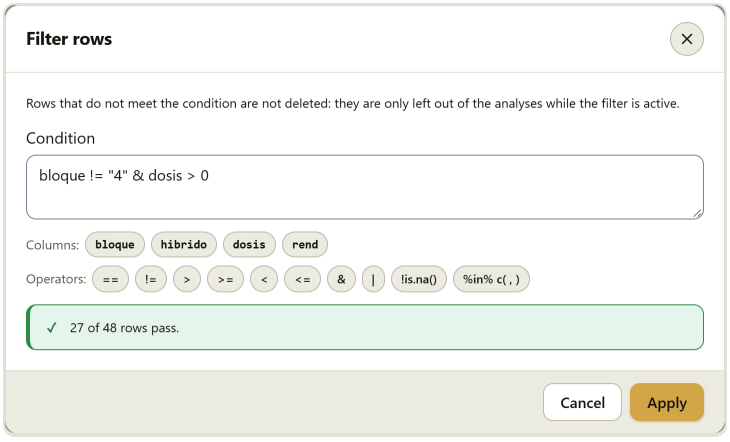


Figure 2.8. A filter that removes block 4 and the control without nitrogen: 27 of the 48 rows pass.

With the filter active, a notice above the grid reminds you of it, with the condition and how many rows pass, and the Remove the filter button. The rows that do not pass remain in the grid, hatched and with their number struck through; you can still see and correct them.

Everything that comes afterward uses only the rows that pass: the summaries in the details, the diagnosis in section 2.13, Explore, Analysis and the report. The .labg project saves the filter condition.

Some useful conditions:

- `sitio == "Norte"`
- `bloque != "4"`
- `!is.na(rend)`
- `hibrido %in% c("H-52", "H-66")`
- `dosis > 0 & dosis < 180`

A row whose result is missing (for example, `rend > 5` with *rend* empty) does not pass. A filter that leaves no rows is not applied.



Figure 2.9. The active filter: the rows that do not pass are hatched.

2.11 Wide and long tables

A variable measured several times on the same unit is often entered in **wide format**: one row per plant and one column per date (`altura_15dds`, `altura_30dds`...). The analyses require the **long format**: one row per measurement, with one column that gives the date and another with the value. Wide ⇄ long... makes the change in both directions.

1. In **Wide → long**, check the **columns that identify each unit** (those that will be repeated in each new row: here sustrato and planta) and the **columns to stack** (the measurements).
2. Name the new column that will hold the names of the stacked columns (here `dds`, days after transplanting) and the values column (`altura`).
3. If each stacked column has a number in its name, the **Keep the number in each name** checkbox keeps only that number (altura_15dds → 15), and the new column comes out **numeric**, ready for a curve over time; without the checkbox, it keeps the full name as a factor.

4 **Reshape** loads the new table. The identifiers keep their type; a column that was an identifier (one plant per row) becomes a factor, because in the long table each plant is repeated: it is the subject of the repeated measures.

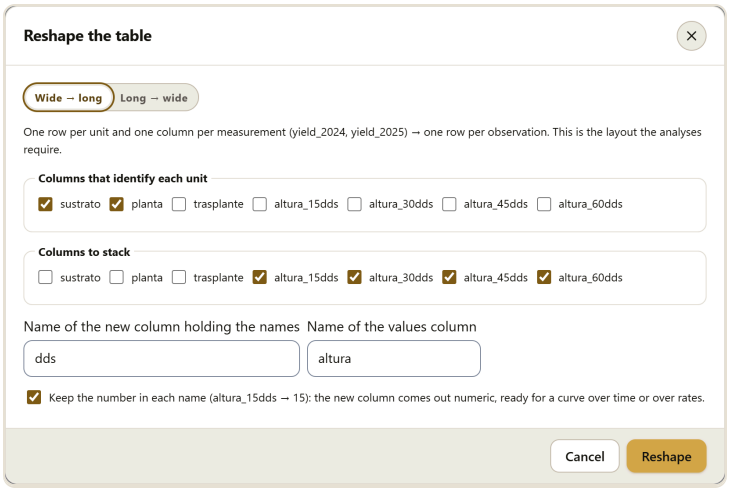
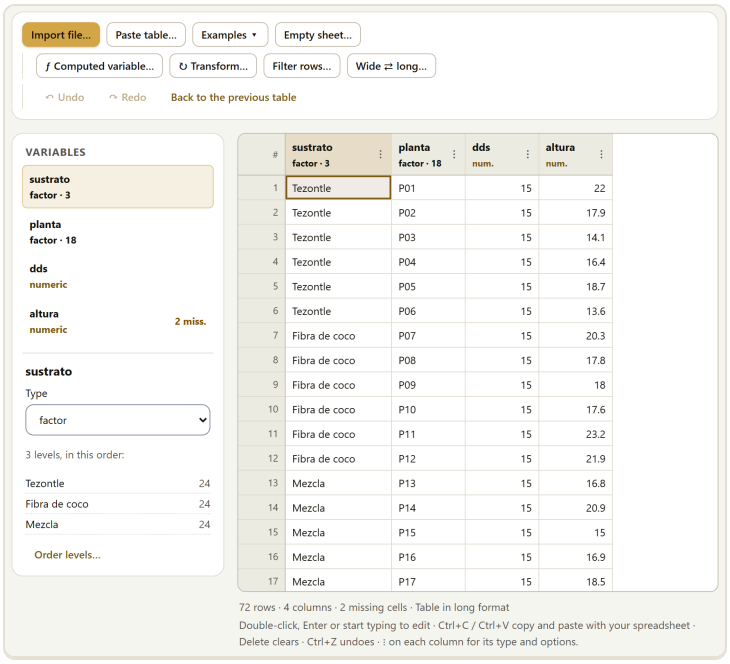


Figure 2.10. The tomato heights, from wide to long: each plant will now have four rows, one per date.



#	sustrato factor · 3	planta factor · 18	dds num.	altura num.
1	Tezontle	P01	15	22
2	Tezontle	P02	15	17.9
3	Tezontle	P03	15	14.1
4	Tezontle	P04	15	16.4
5	Tezontle	P05	15	18.7
6	Tezontle	P06	15	13.6
7	Fibra de coco	P07	15	20.3
8	Fibra de coco	P08	15	17.8
9	Fibra de coco	P09	15	18
10	Fibra de coco	P10	15	17.6
11	Fibra de coco	P11	15	23.2
12	Fibra de coco	P12	15	21.9
13	Mezcla	P13	15	16.8
14	Mezcla	P14	15	20.9
15	Mezcla	P15	15	15
16	Mezcla	P16	15	16.9
17	Mezcla	P17	15	18.5

Figure 2.11. The result: 72 rows (18 plants × 4 dates). The **Back to the previous table** button brings back the wide table.

In **Long → wide**, choose the columns that identify each unit, the column whose values will name the new columns, and the column with the values. If a combination of identifiers is repeated, the app stops and explains why: the wide table would not know which value to use, and another column is needed to tell those rows apart.

2.12 Undo, redo and go back

Every change to the sheet (a cell, a paste, a type, a new or deleted column, an order of levels) can be undone with ↶ Undo or Ctrl+Z, and redone with ↷ Redo or Ctrl+Y. Hovering the pointer over the buttons shows what is about to be undone ("Undo: Change 'dosis' to factor").

Reshaping the table creates a new table, and so does loading the example data with "See it in action" from the glossary (section I.6). To return to the previous table there is the Back to the previous table button, which appears in the toolbar for as long as you can go back.

2.13 What SigmaPro sees in your trial

At the end of the block, the **What SigmaPro sees in your trial** card summarizes the structure that the app recognizes in the table. It is the same reading that the assistant and the model builder (chapter 4) use to propose the analysis, and it is worth checking before you analyze: if something does not match the actual design, now is the time to correct a type or a column.

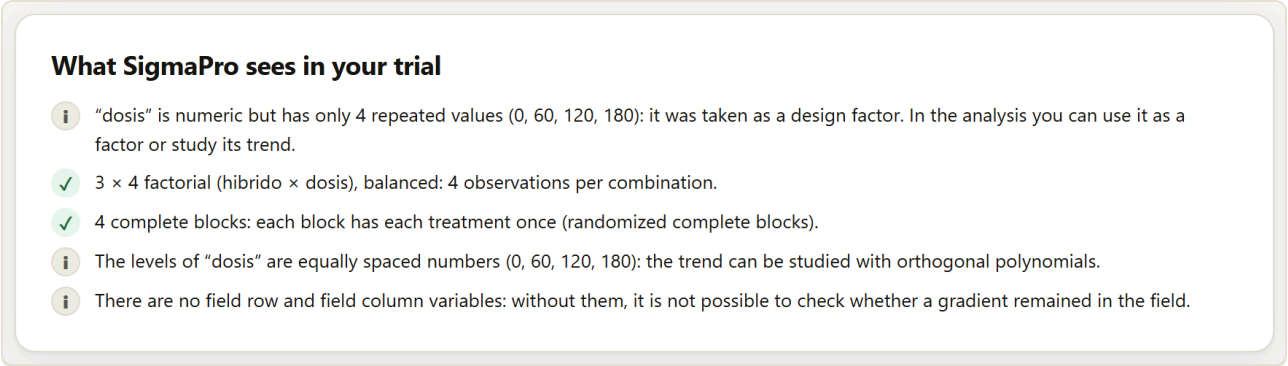


Figure 2.12. What SigmaPro sees in the maize trial: a balanced 3 × 4 factorial in four randomized complete blocks.

Each line carries a symbol: ✓ for what is fine, ! for what deserves attention and i for a note. What is checked:

Check	What it may say
The response	Which column was taken as the response and whether it has missing values (the analysis will use the complete observations).
Columns that are not factors	An identifier (a different value in each row), subsamples (the number of the fruit or plant within each plot, 1, 2, ..., which is not a treatment) or a numeric column with few repeated values taken as a design factor (a rate).
Combinations and balance	Whether it is a factorial and whether each combination has the same number of observations. If not, it warns that type I, II and III sums of squares will give different results (section 4.10); if there are empty combinations, that type III is not defined.

Check	What it may say
Blocks	Whether each block has all the treatments (randomized complete blocks), how many blocks are incomplete and whether a treatment appears more than once in a block.
Latin square	With row and column variables, whether each treatment appears once in each row and in each column.
Rates	If the levels of a numeric factor are equally spaced, that the trend can be studied with orthogonal polynomials (section 4.11).
The field	Without field row and field column variables, it is not possible to check whether a gradient remained in the field.

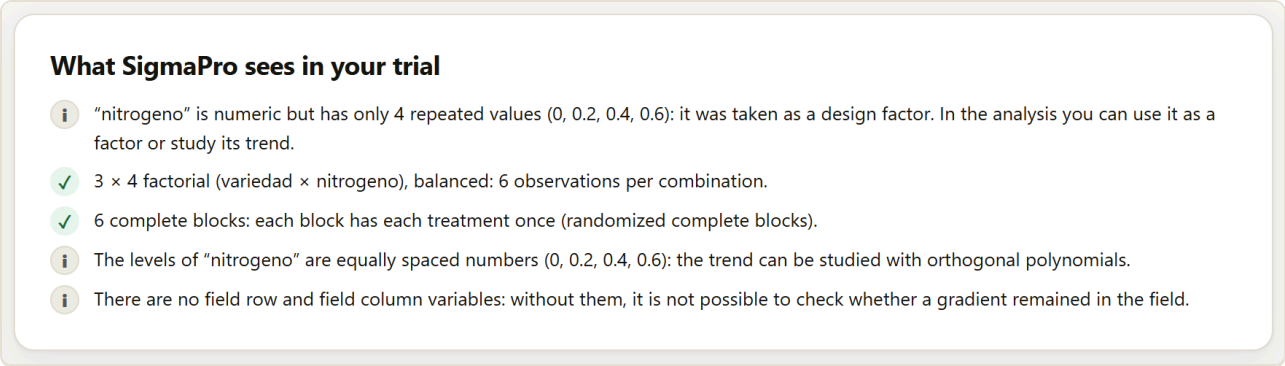


Figure 2.13. Yates's oats trial: the card reads it as a factorial in complete blocks. That the varieties are on main plots cannot be seen in these data; you have to specify it (or add the plot column, see below).

WHAT THE DATA DO NOT SAY

The card reads the table, not the field layout. A **split-plot** trial has the same columns as a factorial in blocks (figure 2.13): the difference lies in how it was randomized, and that is not recorded in the data. The same happens with repeated measures when there is no column that says which unit was measured. In those cases, you specify the design in the assistant, in the catalog or in the model builder (chapter 4); analyzing a split plot as a factorial in blocks would give wrong F tests for the main-plot factor.

A COLUMN THAT NUMBERS THE PLOTS

If the table has a column that numbers the main plots (or the strips), with a name such as `parcela`, `parcela_grande`, `pg`, `subparcela` or `franja`, SigmaPro does not take it as a treatment and the **assistant** uses it to recognize split plots, split-split plots or strip plots on its own: if a factor does not change within each plot, it is the main-plot factor (section 4.2). The "Wheat: irrigation × variety" example comes this way. The card in this section does not say so; the assistant does, when it proposes the analysis.

2.14 Continue in the LABG suite

Below the trial card, the Continue in the LABG suite strip may appear, with the sister app that makes use of the structure of the table. Each card says what it saw in the data and why that app is useful.

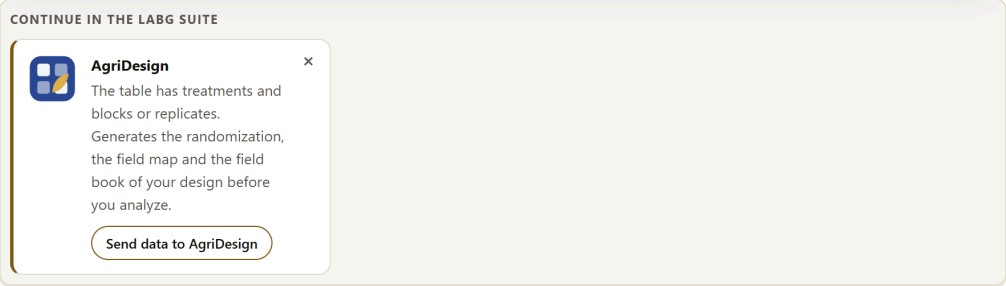


Figure 2.14. With the maize example, which has treatments and blocks, Data suggests AgriDesign.

If the table has…	It suggests
treatments and blocks or replicates	AgriDesign: the randomization, the field map and the field book of the design.
8 or more numeric variables	PCAPro, CorrelationPro and ClusteringPro: reduce dimensions, see the relationships among many variables or form groups.
latitude and longitude columns	BioModellingPro: map the sites, extract climate and soil variables and model the distribution of a species.
dates	PhenologyPro: with dates and daily weather, degree days, evapotranspiration, phenological stages and frost risk.

Send data to … saves the table in a CSV file (with the active filter, if there is one) and opens the app in a new tab; the data do not travel over the internet. The × button hides that suggestion in this place. Chapter 10 explains all this in detail.

3

BLOCK 3 · EXPLORE

Explore: look at the data before analyzing them

Before testing hypotheses, it pays to get to know the data: where the center is, how much they vary, whether there are odd values, how many observations each group has. Explore offers four summaries (descriptives, frequencies, proportions and goodness of fit), each with its confidence interval, its conclusion in words, the R code that gives the same result and its fact sheet in Fundamentals. Everything computed here can also go to the report.

3.1 The four summaries

3.2 How to read a result

3.3 Descriptive statistics

3.4 Frequency tables

3.5 Proportions and intervals

3.6 Goodness of fit

3.1 The four summaries

At the top of the block, four buttons choose the summary. Its options are on the left and the result on the right, and the result is recomputed whenever you change any option.

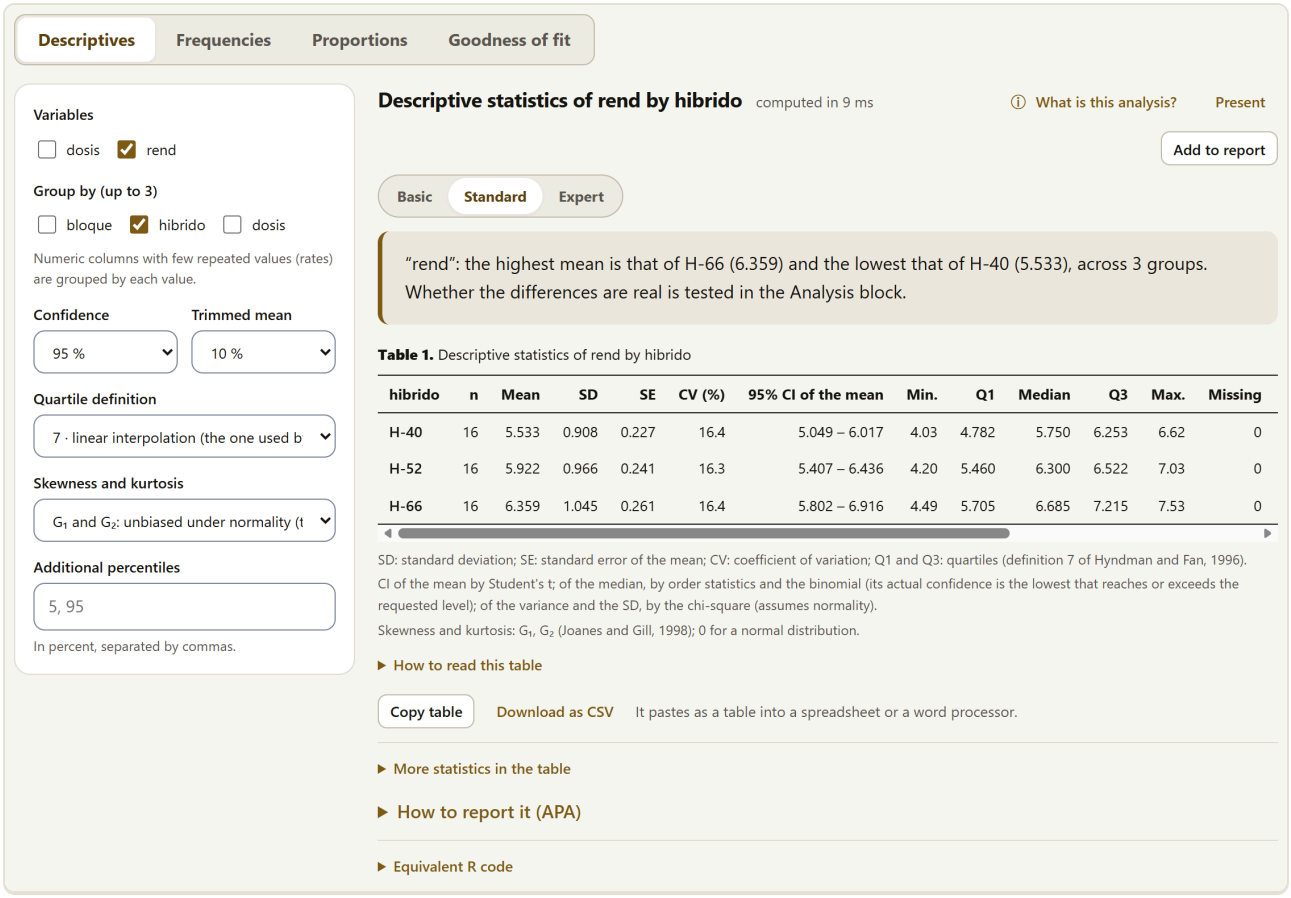


Figure 3.1. Descriptives of maize yield by hybrid. When data are loaded, the block proposes the first numeric response grouped by the first treatment factor.

Summary	What it is for	Section
Descriptives	Center, spread and shape of numeric variables, overall or by group, with the interval of the mean, of the median and of the variance.	3.3
Frequencies	How many observations there are at each level of a factor, or at each combination of two (a cross-tabulation), as counts or percentages.	3.4
Proportions	A proportion (germination, incidence, mortality) with its confidence interval, by group, with six interval methods.	3.5
Goodness of fit	Whether some observed frequencies follow an expected ratio (for example, the 9:3:3:1 segregation).	3.6

If a filter is active in the Data block, a note above the options reminds you of it: "With the filter from the Data block: 27 of 48 rows are used." With no data loaded, the block offers to load them right there (Import a file, Paste a table or Try an example, and the summary appears as soon as they arrive) or to go to Data and, for those planning a trial, the Calculate a sample size or power button, which needs no data (section 4.22).

> **EXPLORE DESCRIBES; ANALYSIS DECIDES**
>
> The summaries in this block say what the data look like, not whether the differences between groups are real. That hybrid H-66 has the highest mean is a description; whether it is significantly higher than H-52 is shown by an analysis of variance with its mean comparison, in the Analysis block (chapter 4). That is why, when there are groups, the conclusions in Explore end with "Whether the differences are real is tested in the Analysis block."

3.2 How to read a result

Every SigmaPro result, here and in Analysis, has the same parts in the same order. It is worth getting to know them once:

Part	What it is
Title	What was computed and on which variables. Next to it, the computing time in milliseconds.
Change the options ↓	Takes you to the summary's options; useful on a narrow screen, where they sit below the result.
ⓘ What is this analysis?	Opens the summary's fact sheet in Fundamentals: what it is for, what it assumes, what data it needs and how to read it (chapter 9).
Add to report	Sends the result to the Results block (chapter 5). The button changes to In the report ✓; the report stores the analysis, not the numbers, and recomputes it if the data change.
Present	Shows the result as full-screen slides, for projecting it; ← and → move through them and Esc exits (section 4.5).
Basic · Standard · Expert	The level of detail for the whole app (section 4.5). In Explore the table is shown at all three levels; Basic hides the R code and Expert adds the technical notes.
Conclusion	The result in words, with the figures that support it. It is the first thing to read.
Table	Numbered and with its footnotes, in journal format: rules only at the top, at the bottom and below the headers; aligned figures; decimals according to the precision of the data. If it is wider than the screen, it scrolls sideways; with the keyboard, Tab focuses it and the arrow keys scroll it.
View buttons	When a result has several tables, some buttons choose which one is shown (for example, frequencies or percentages).
Copy table	Copies the table with all its figures, ready to paste into a spreadsheet or a word processor.

Part	What it is
Download as CSV	Saves the table as a CSV file, with unrounded values, to open in another program. The file is UTF-8, so accented characters reach a spreadsheet intact.
More statistics in the table	(In descriptives.) Checkboxes that add columns to the table.
How to read this table	Collapsed below the table when it includes a statistic with a reading guide: a scale with the zones of the guide, a mark at each value and a card that interprets it with the numbers of the case. In Explore it appears below the descriptives, for the skewness, and below the goodness of fit, for the standardized residuals; in Analysis, below almost every table (section 4.5).
How to report it (APA)	Collapsed: the methods paragraph and the results paragraph of the summary in APA 7 style, in Spanish and in English, each with Copy (section 4.5).
Equivalent R code	The code that gives the same numbers in R, with the Copy the code button. It starts with a comment on how to read in the data and which packages it requires.

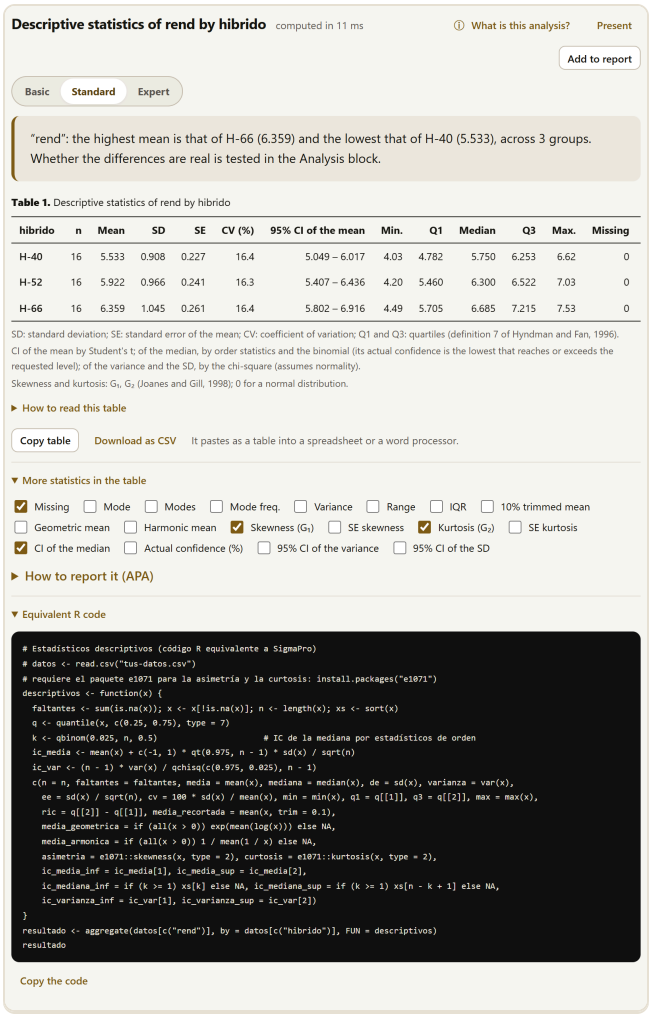


Figure 3.2. A result with its two collapsible sections open: more statistics and the R code.

3.3 Descriptive statistics

They describe one or more numeric variables, overall or by group. The options:

Option	What it decides
Variables	The numeric columns to describe; one row of the table per variable and group.
Group by (up to 3)	The factors that form the groups. A numeric column with a few repeated values (a rate) is grouped by each value. With two or three factors, the groups are their combinations present in the data.
Confidence	The level of the intervals (90, 95 or 99%).
Trimmed mean	The fraction removed from each end for the trimmed mean (5, 10, 20 or 25%).
Quartile definition	The nine definitions of Hyndman and Fan (1996). Type 7 (linear interpolation) is the one used by R and by spreadsheets; type 6 is the one in many textbooks. With large samples they agree; with few observations they may differ slightly.
Skewness and kurtosis	Three definitions (Joanes and Gill, 1998): G_1 and G_2, unbiased under normality and the ones in common use; g_1 and g_2, moment-based; b_1 and b_2, corrected by $(n-1)/n$.
Additional percentiles	Other percentiles added to the table, in percent and separated by commas (for example `5, 95`).

What the table includes

The main table gives, for each variable and group: **n**, **mean**, **standard deviation** (SD), **standard error of the mean** (SE), **coefficient of variation** (CV, in %), the **confidence interval of the mean**, the **minimum**, the **quartiles** Q1 and Q3, the **median** and the **maximum**. In More statistics in the table you can add: missing values, mode (and how many modes there are and their frequency), variance, range, interquartile range (IQR), trimmed mean, geometric mean, harmonic mean, skewness and kurtosis with their standard errors, the interval of the median with its actual confidence, and the intervals of the variance and of the SD.

Σ THE FORMULAS

With n observations of mean $\bar{x}$ and standard deviation s:

$$SE = s / \sqrt{n} \cdot CV = 100\ s / \bar{x} \cdot CI\ of\ the\ mean: \bar{x} \pm t_{1-\alpha/2;\ n-1} \cdot SE$$

The interval of the **median** is formed by two ordered observations, $x_{(k)}$ and $x_{(n-k+1)}$, with k taken from the binomial distribution $(n, ½)$; that is why its actual confidence ("Actual confidence (%)") is the first one that reaches or exceeds the requested level, not exactly that level. The interval of the **variance** comes from the chi-square, $(n-1)s^2 / \chi^2$; it assumes normality, and the interval of the SD is its square root.

The conclusions

Without groups, the conclusion summarizes the variable: "'rend' (n = 48): mean 5.938 (95% CI: 5.644 to 6.232), standard deviation 1.013 and coefficient of variation 17.1%." With groups, it says which one has the highest mean and which the lowest. It also warns in two cases: when, in a group of eight or more observations, the **skewness is greater than twice its standard error** (a sign of outliers or that a transformation would help, section 2.9), and when some rows were left out because they have no level in a grouping factor.

> **THE CV IN AGRICULTURAL EXPERIMENTS**
>
> The CV of a group describes the variation of the raw data; it mixes the variation among blocks with the error variation. The CV published for a trial is a different one: that of the analysis of variance, computed with the error mean square (in the maize example, 3.6% against the 16% of the raw data by hybrid). That one is given by the analysis of the design in the Analysis block.

3.4 Frequency tables

They count how many observations there are at each level of a factor (Variable) or, with Cross with, at each combination of two. The Count missing values as a category checkbox adds a row and a column for the empty cells. Numeric columns with few values appear in the lists marked "(numeric)".

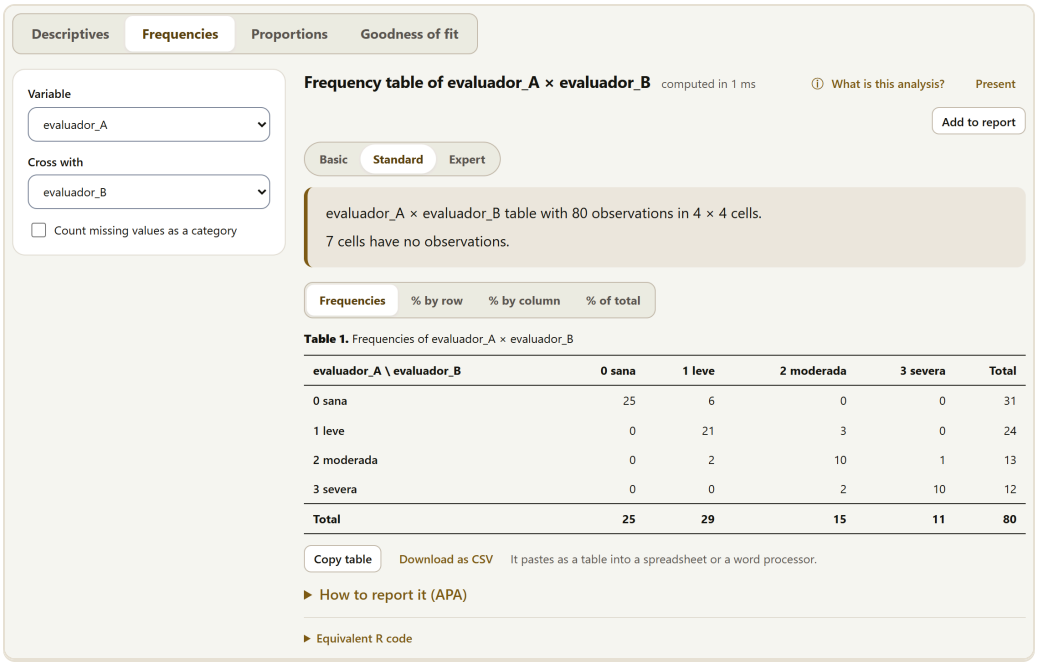


Figure 3.3. The rust scores of the two raters, cross-tabulated. The cells on the diagonal are the leaves on which they agreed.

A cross-tabulation has four views, chosen with its buttons: Frequencies, % by row (each row adds up to 100), % by column and % of total. The conclusion gives the size of the table and how many cells have no observations at all. A one-factor table gives, besides the count, the percentage and the cumulative values.

The frequency table describes; to **test** whether two factors are associated (for example, whether germination depends on the treatment), or whether two raters agree, there are the contingency table, the McNemar and Bowker test and Cohen's kappa, in the Analysis block (section 4.14).

3.5 Proportions and intervals

A proportion can be requested in two ways, chosen with the buttons above the options:

- Categories of a factor: the proportion of each level of a categorical variable (for example, what fraction of the leaves has each rust score).
- Successes out of trials: two numeric columns, one with the successes (germinated seeds, dead plants, damaged fruits) and one with the trials (seeds sown, plants observed). The rows are summed within each group.

By group computes one proportion per level of a factor, and Confidence sets the level of the intervals. Under Intervals you check the methods you want to see, one column each.

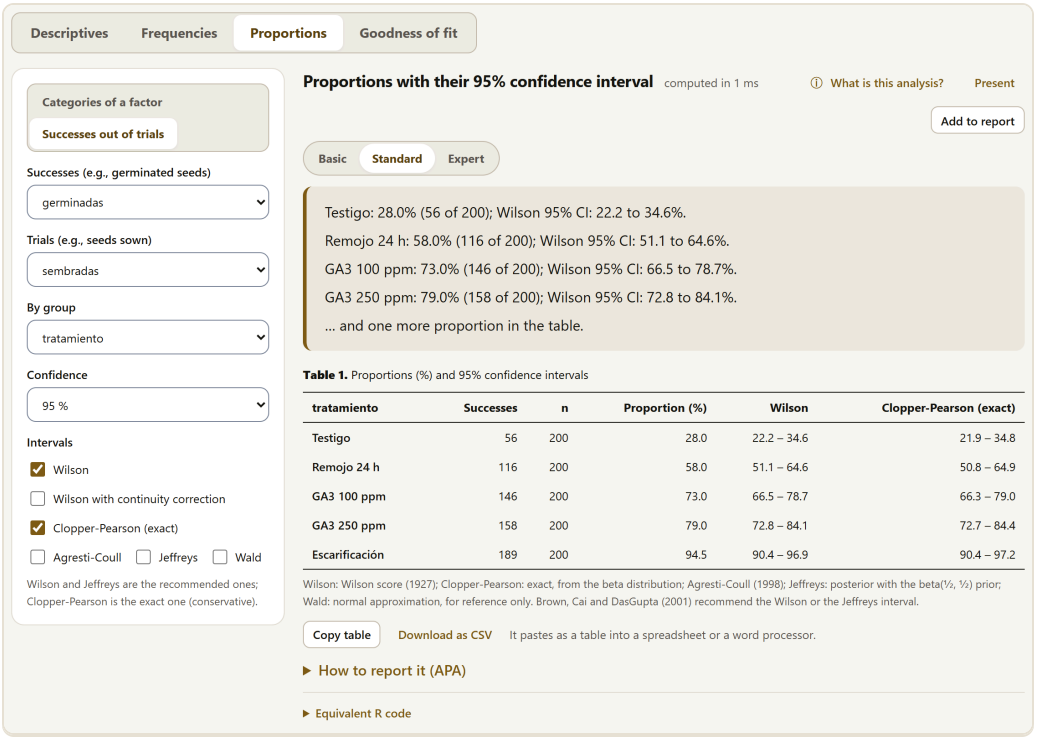


Figure 3.4. The germination of five treatments (200 seeds each), with the Wilson and Clopper-Pearson intervals.

Interval	What it is like
Wilson	Inverts the score test (Wilson, 1927). Its actual coverage stays very close to the requested level even with few trials or extreme proportions. **Recommended.**
Wilson with continuity correction	Wilson's interval with a continuity correction: wider and conservative.
Clopper-Pearson (exact)	Based on the beta distribution. It guarantees at least the requested confidence, at the price of being wider than necessary.
Agresti-Coull	The Wald interval with two successes and two failures added (Agresti and Coull, 1998); easy to compute by hand.
Jeffreys	The Bayesian interval with the beta(½, ½) prior. **Recommended**, with properties similar to those of Wilson's.
Wald	The textbook normal approximation, $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$. For reference only: with fewer than 30 trials or proportions close to 0 or 1 it fails, and the app warns about it.

The recommendation of Wilson and Jeffreys comes from the review by Brown, Cai and DasGupta (2001), who compared the actual coverage of these intervals. The conclusion shows the first four proportions in words, with the Wilson interval (if Wilson was unchecked, with that of the first checked method), and says how many more there are in the table.

3.6 Goodness of fit

The chi-square goodness-of-fit test tells whether some observed frequencies are consistent with an expected ratio. The classic case in agronomy is Mendelian segregation: in the F_2 of a dihybrid, the phenotypes are expected in a 9:3:3:1 ratio.

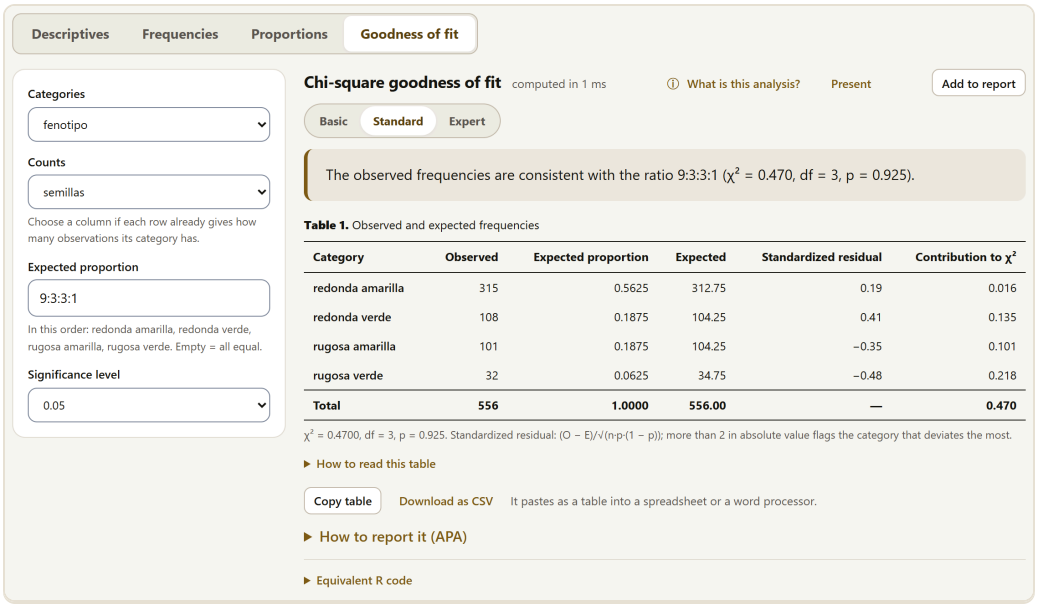


Figure 3.5. Mendel's (1866) 556 seeds against the 9:3:3:1 ratio: $\chi^2 = 0.470$ with 3 degrees of freedom, $p = 0.925$.

Option	What it decides
Categories	The variable with the categories (the phenotypes).
Counts	If each row already gives how many observations its category has (as in Mendel's example, with four rows), the column with the counts. If each row is one observation, leave it at "one row per observation".
Expected proportion	One proportion per category, in the order of the levels shown in the help text below, separated by colons: `9:3:3:1`, `3:1`, `1:2:1`. Decimals are also accepted (`0.5625:0.1875:0.1875:0.0625`). If left empty, equal proportions are tested.
Significance level	The α the conclusion uses to decide (0.05, 0.01 or 0.10).

The table gives, for each category, the **observed** count, the **expected proportion**, the **expected** count ($n \times$ proportion), the **standardized residual** and the **contribution to χ^2**:

$$\chi^2 = \Sigma\,(O - E)^2 / E,\ df = k - 1,\ residual = (O - E) / \sqrt{(n\,p\,(1 - p))}$$

A residual greater than 2 in absolute value flags the category that deviates most from what was expected; when the test rejects, the conclusion names it. If any category expects fewer than five observations, the chi-square approximation may not be good, and the app warns about it: it is better to merge categories or to use an exact test.

> **"CONSISTENT" IS NOT "PROVEN"**
>
> A high p value says that the data do not contradict the expected ratio, not that they prove it. With few observations, the test has no power to reject almost anything. Mendel's 556 seeds are enough to tell 9:3:3:1 apart from very different ratios, not from similar ones.

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4

BLOCK 4 · ANALYSIS

Analysis: from the design to the conclusion

The heart of SigmaPro: 74 analyses in 17 families, from the analysis of variance of an experimental design to mixed models, survival and time series (the other 4 summaries are in Explore, chapter 3). You choose them with the assistant, in the catalog or in the model builder, and they all respond the same way: a conclusion in words, tables, charts, assumptions, the R code that gives the same numbers and their fact sheet in Fundamentals.

4.1 Three ways to choose an analysis

4.2 The assistant

4.3 The catalog

4.4 The model builder

4.5 How to read a result

4.6 The R engine inside the browser

4.7–4.11 Assumptions, tests, designs and means

4.12–4.17 Regression, contingency, mixed and generalized models

4.18–4.24 Nonlinear, multi-environment, multivariate, sampling, survival and time series

4.1 Three ways to choose an analysis

At the top of the block, three buttons choose how you get to the analysis. All three lead to the same place (the same method with the same options), and you can switch from one to another at any time.

Way	Who it is for	Section
Assistant	For those who know what they want to learn from their trial but not which analysis to use. A few questions in field language (what did you measure?, did you group the units into blocks?, was any factor applied to main plots?) and a recommendation with its reasons. It is the mode the block opens in.	4.2
Catalog	For those who already know which analysis they want: the complete list, grouped by families, with the form of each method.	4.3
Model builder	For those who think in terms of a model: you drag the variables to their roles (response, factors, blocks, random effects, covariates…) and the app writes the formula and chooses the analysis. The formula can also be typed by hand.	4.4

Next to the three buttons, History lists the most recently computed analyses, with their variables and their data; choosing one reopens it (section 1.1). With no data loaded, the block offers to load them right there (Import a file, Paste a table or Try an example) or to go to Data; the calculations that need no data (sample size, power, replicates) open with Calculate a sample size or power, here or in Explore, which selects Power and sample size: means (t test) (section 4.22). If a filter is active in Data, the whole block works with the rows that pass it.

4.2 The assistant

The assistant asks the questions a statistician would ask on seeing a trial for the first time, in order, and only the ones that are needed: each answer decides the next question. On the right, the **recommendation** is updated with each answer. The answers come already proposed from the data (what the card in section 2.13 recognizes), so often you only need to review them.

The questions

The first question, **What do you want to know?**, opens one of ten paths:

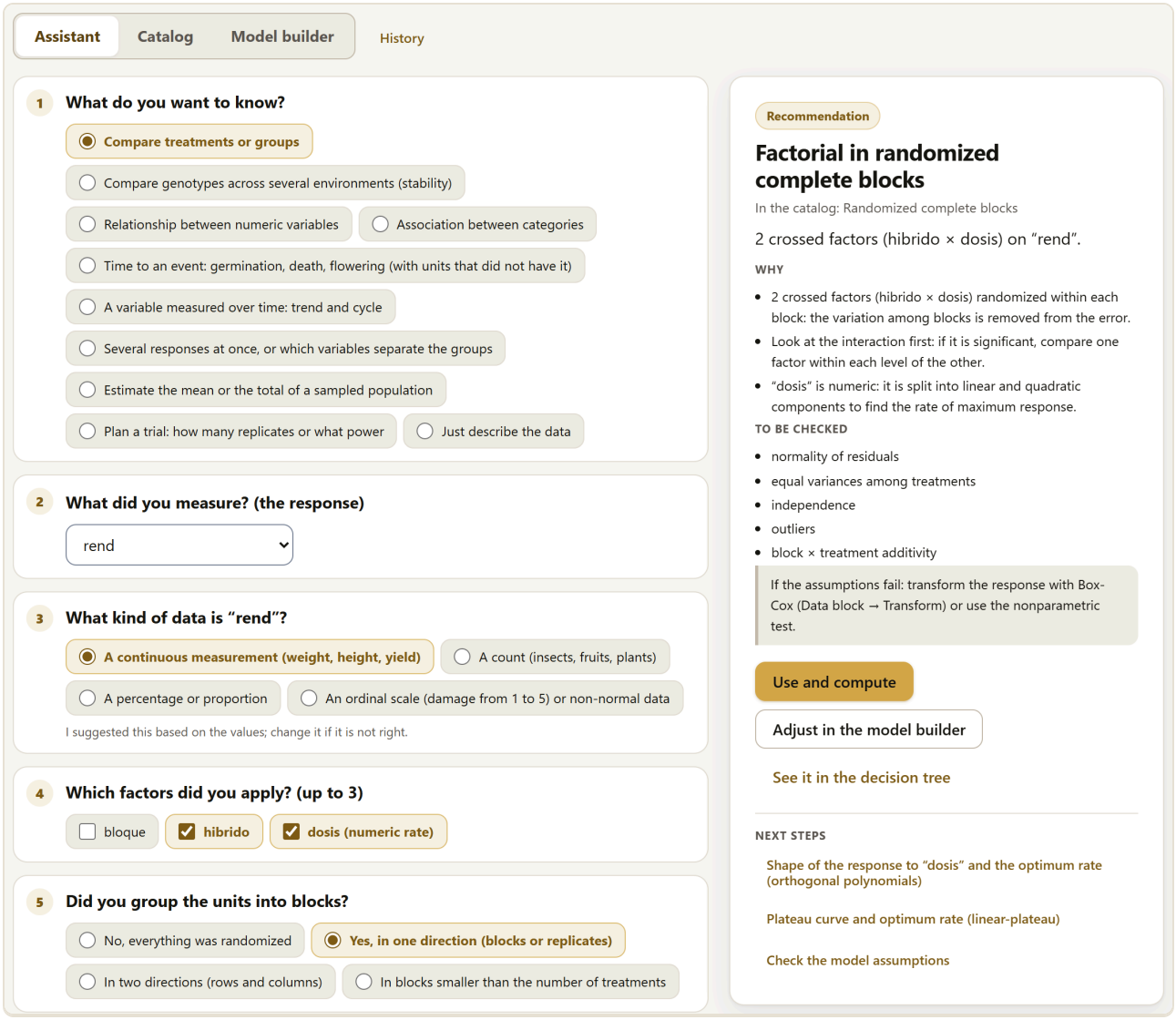


Figure 4.1. The assistant with the maize trial. With the proposed answers, it recommends a factorial in randomized complete blocks.

Goal	What it asks next and what it usually recommends
Compare treatments or groups	The response and its kind (continuous, count, percentage, ordinal scale), the factors (up to 3), the blocks (none, in one direction, in two, incomplete), whether there were main plots or strips, whether there is a covariate measured before the treatment, whether the same unit was measured on several dates, whether the blocks are random and how to compare (all pairs, against a control, contrasts, or the curve of a rate). It recommends the corresponding design (sections 4.10 and 4.11), an ANCOVA, a repeated-measures analysis, a mixed model, a generalized model or a nonparametric test.
Compare genotypes across several environments	Response, genotype, environment, replicates and what you are interested in: the combined analysis, stability, AMMI or GGE (section 4.20).

Goal	What it asks next and what it usually recommends
Relationship between numeric variables	Whether there is a response to explain: with one predictor, the shape (straight line, curve, growth, dose-response, plateau); with several, all of them, stepwise, all combinations or a response surface. Without a response: a correlation, partial if other variables are controlled for (sections 4.12, 4.13, 4.18 and 4.19).
Association between categories	Whether they are different cases, the same case classified twice, two raters or an association across several lots, and whether the data come already counted (section 4.14).
Time to an event	The time, the event (1 yes, 0 censored), whether groups are compared and whether there are covariates: Kaplan-Meier or Cox (section 4.23).
A variable measured over time	The series, the time and what you are interested in: trend, cycle or autocorrelation (section 4.24).
Several responses at once	MANOVA or discriminant analysis (section 4.21).
Estimate the mean or the total of a sampled population	The variable and whether there were strata (section 4.22).
Plan a trial	Which variable will be measured, to take its variation from the loaded data: number of replicates or power (section 4.22).
Just describe the data	Takes you to the Explore block (chapter 3).

The recommendation

The card on the right gives the recommended analysis and everything you need to decide whether to accept it:

- the **title** and, below it, the name under which it appears in the catalog;
- a sentence with what is going to be analyzed (*2 crossed factors (hibrido × dosis) on "rend"*);
- **Why**: the reasons, in terms of the trial (*the variation among blocks is removed from the error*; *"dosis" is numeric: it is split into linear and quadratic components*);
- **To be checked**: the assumptions the analysis puts to the test, and a box with **what to do if they fail**;
- warnings, when something in the data deserves attention;
- the Use and compute button, which opens the analysis in the catalog with its variables already set, and, when applicable, Adjust in the model builder, which takes it to the model builder (section 4.4), and See it in the decision tree, which shows the path that leads to that analysis among all the others (section 9.3);
- **Next steps**: the analyses that usually follow (the mean comparison, the rate curve, the check of the assumptions), each as a button that opens it directly.

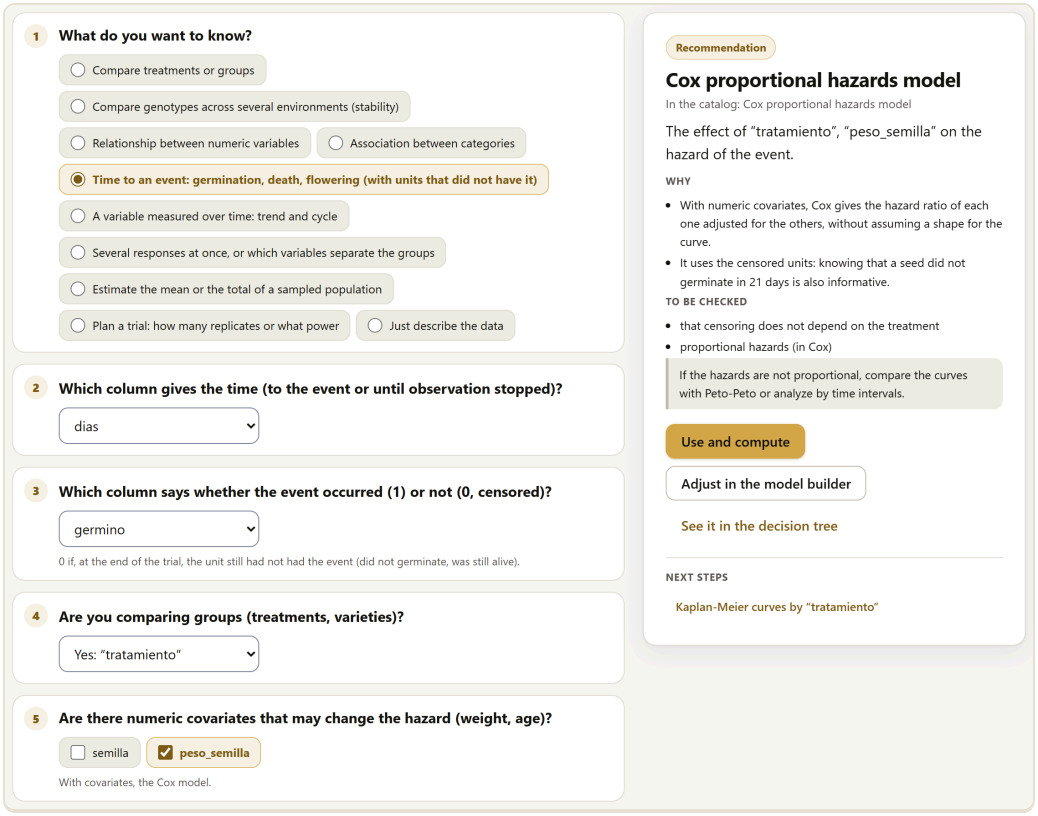


Figure 4.2. Another path: time to an event. With a covariate (seed weight), the assistant recommends the Cox model and leaves the Kaplan-Meier curves for later.

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I THE QUESTIONS THAT CHANGE THE ANALYSIS THE MOST

Two questions deserve special care, because the data almost never answer them (section 2.13): whether **any factor was applied to main plots** (irrigation, tillage, sowing date: then it is a split plot or a strip plot, and each factor has its own error) and whether **the same unit was measured several times** (repeated measures). Answering them wrongly does not produce a visible error, but wrong F tests.

The exception is plots: if the table has a column that numbers them (`parcela`, `subparcela`, `franja`...), the assistant recognizes in the data split plots (a factor that does not change within each plot of the block), split-split plots and strip plots, proposes them in its question and, in the recommendation, says what it saw. Without blocks, the plot is taken as the replicate of each main plot. What you answer in the question prevails. The "Wheat: irrigation × variety" example comes this way.

4.3 The catalog

The catalog shows the block's 74 analyses, grouped in their 17 families, and tells you which ones can be run with the loaded table.

At the top, two cards read the data:

- **With your data**: what is in the table (rows, numeric columns, factors with their levels, blocks, plots, dates, counts, proportions and 0/1 columns) and how many analyses can be run with it (with the maize example, "You can run 57 of the 74 analyses in the catalog with this table.").
- **Recommended for your data**: the analysis the assistant proposes from what it sees in the table, with its why and, under **Next:**, the analyses that usually follow, each as a button. The assistant's questions refine the choice (section 4.2).

Below, each family shows its analyses on **cards**, with the analysis's illustration, its name, what it is for and its status: Ready with your data; Missing, with what is missing (with the maize example, the MANOVA says "Missing: at least two numeric responses."); or No data needed. Those that run in the R engine say so (R engine, section 4.6). The ⓘ icon on each card opens the analysis's fact sheet in Fundamentals (chapter 9). Next to the name of each family you can read how many of its analyses can be run (with the maize example, Experimental designs says "8 of 9 with your data").

Three controls narrow down the list: the text search (Search: Tukey, blocks, counts, regression...), the Only the ones I can run with my data checkbox, which is remembered, and the family selector (All families).

Clicking a card opens the analysis: the catalog collapses into a bar with the **Selected analysis** and the Open the catalog (74 analyses) button to open it again, and below it appear, on the left, its form (the **experiment model**: which column is the response, which are the treatments, the block and the options) and, on the right, the result, which is recomputed whenever you change any field. With no data loaded, the catalog only offers the calculations that do not need them: sample size and power (section 4.22).

When you open a method, the app proposes the variables of its form from the data (the response, the factors, the block it recognizes), just as the assistant does. With another table, the proposals are redone; with changes to the same table, the choices already made are respected.

> **!** ONLY WHAT IS VALIDATED IS OFFERED
>
> A method appears in the catalog only if all its validation cases against R passed (chapter 8). If for some reason one were not validated, in its place you would read "This analysis does not yet have an approved validation against R." It is the accuracy rule of section 1.2 applied to the interface.

Sections 4.7 to 4.24 go through the families in catalog order. For each method they explain what it is for, what the form asks for, what it delivers and what its R equivalent is.

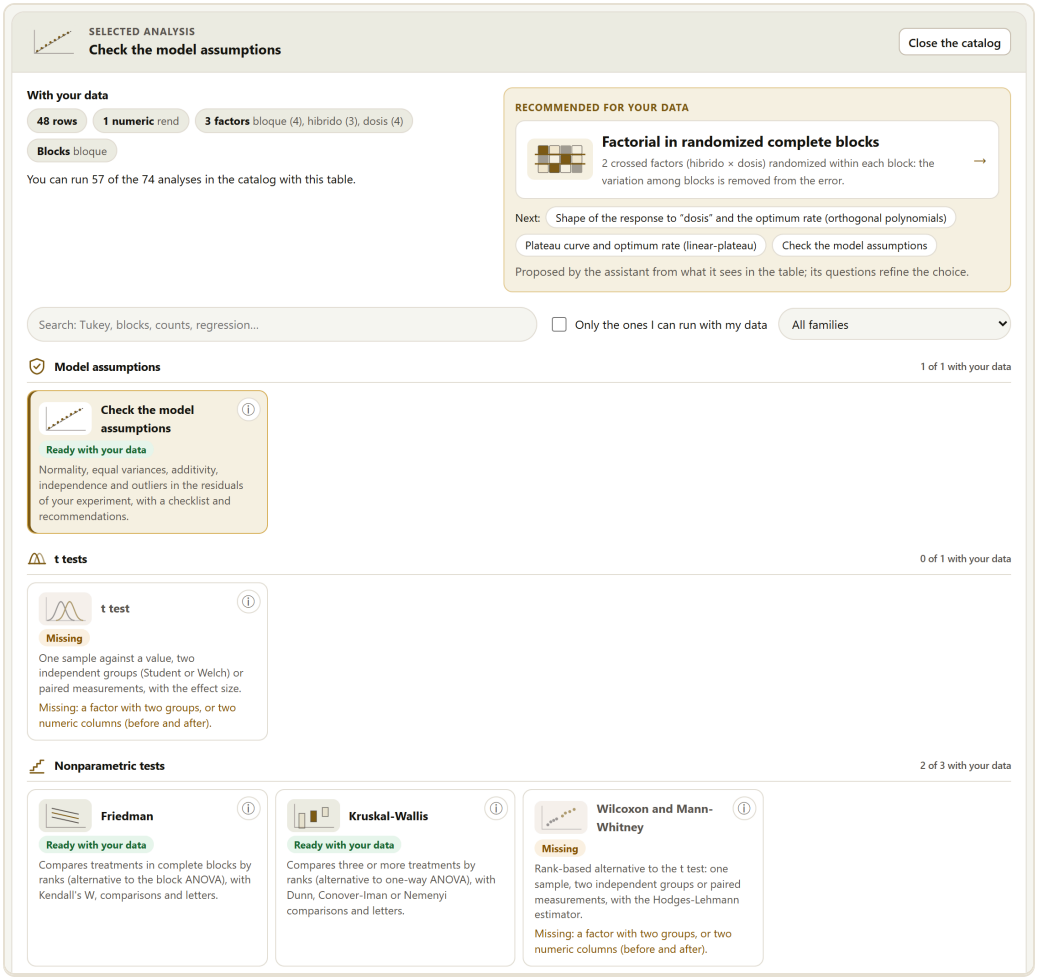


Figure 4.3. The catalog open with the maize example: at the top, what is in the data and the recommendation; below, the families with their cards. Descriptives, frequencies, proportions and goodness of fit are in Explore (chapter 3).

4.4 The model builder

The builder starts from a simple idea: an analysis is a model, and a model says what role each variable plays. On the left are the table's **variables**; on the right, eight **zones**, one per role. Below, the resulting **formula** and the **analysis** the app chose for it.

Putting each variable in its zone

There are three ways, so that it works with a mouse, with a keyboard and on a phone:

- **drag** the variable from the list to the zone;
- choose it in the Add... menu of each zone;
- with the variable focused (with Tab or a click), press the zone's **key**: R, E, M, B, A, F, P or C. Delete removes it from its zone.

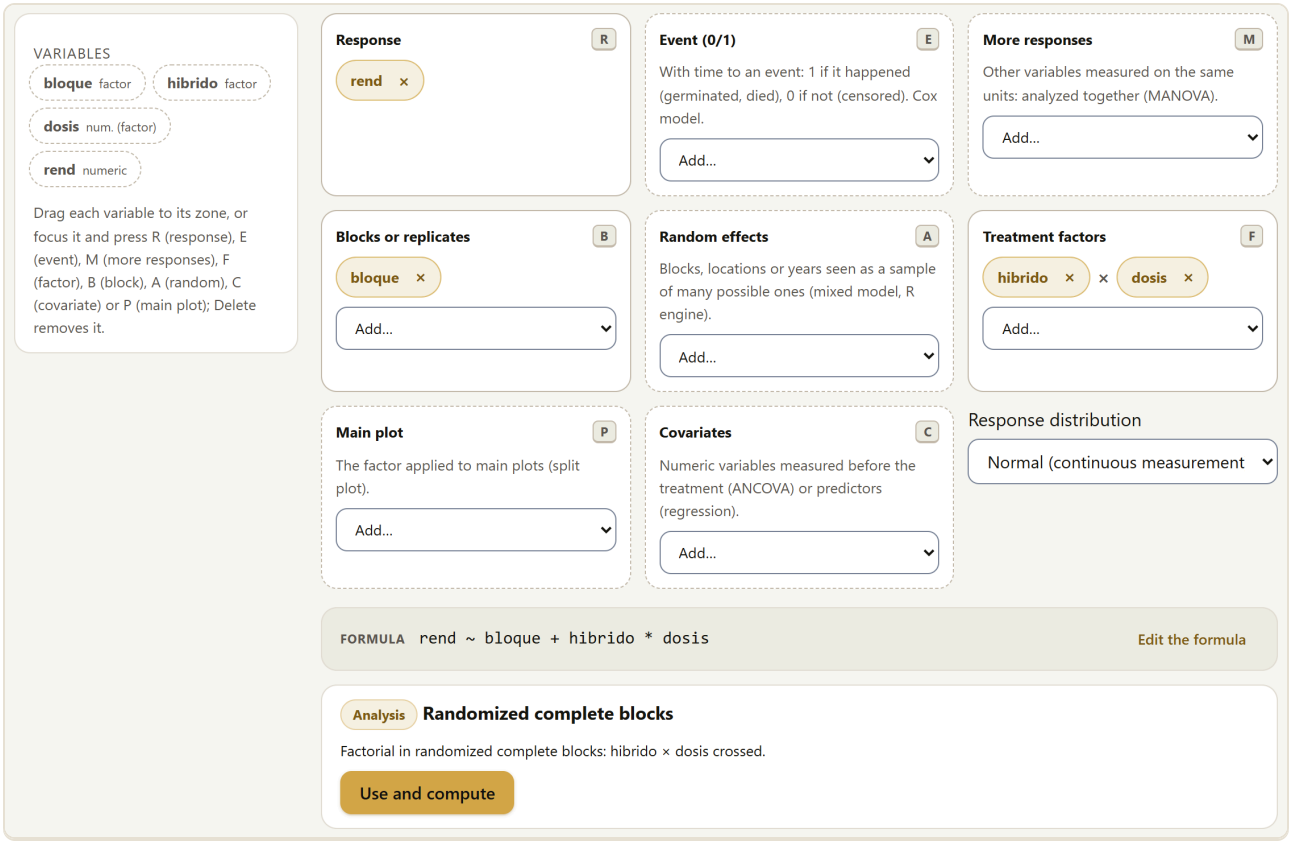


Figure 4.4. The builder with the maize example. The app has already put each variable in its zone: the formula is `rend ~ bloque + hibrido * dosis` and the analysis, randomized complete blocks.

A variable already in use is dimmed in the list; the × on each chip returns it. The zones that accept only numeric variables (response, event, more responses, covariates) offer only those in their menu.

Zone	Key	What it is for
Response	R	The variable that was measured (or the time to an event).
Event (0/1)	E	With time to an event: 1 if it happened (germinated, died), 0 if not (censored). It builds the Cox model.
More responses	M	Other variables measured on the same units, to analyze them together (MANOVA).
Blocks or replicates	B	One (complete blocks), or two (rows and columns: Latin square).
Random effects	A	Blocks, locations or years seen as a sample of many possible ones: mixed model, in the R engine.
Treatment factors	F	The factors that were applied; several are crossed (×).
Main plot	P	The factor that was applied to main plots (split plot); it must also be among the factors.
Covariates	C	Numeric variables measured before the treatment (ANCOVA) or predictors (regression).

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The Response distribution selector says what the response is like: **Normal** (a continuous measurement), **Count** (Poisson) or **Positive and skewed** (gamma). The last two lead to generalized linear models (section 4.17).

Which analysis it chooses

The app reads the zones and chooses the corresponding analysis from the catalog, with the same logic a statistician would use:

If the zones have…	…the analysis is
Factors, no blocks	Completely randomized (factorial if there are several)
Factors and one block	Randomized complete blocks
Factors and two blocks (row and column)	Latin square (up to two factors)
Two factors, one on a main plot, and one block	Split plot (with two blocks, in a Latin square)
Three factors, one on a main plot, and one block	Split-split plot
Only covariates	Linear regression (with blocks, the general linear model)
Factors and covariates (up to 2 and 2, with at most one block)	ANCOVA
Some random effect	Linear mixed model (R engine); with a main plot, also its error `(1 | bloque:parcela)`
Count or gamma distribution	Generalized linear model; with random effects, the generalized linear mixed model (Poisson or gamma, in the R engine)
An event	Cox model (the block as a stratum); without factors or covariates, Kaplan-Meier
More responses	MANOVA (with the block and the factors crossed)
Any other combination (more than three factors, covariates with plots…)	The general linear model, with a warning

Below the formula, the analysis box names it, explains the decision ("Factorial in randomized complete blocks: hibrido × dosis crossed.") and offers Use and compute, which opens it in the catalog with everything already set.

Writing the formula by hand

Edit the formula turns the formula into a text field with R notation. As you type, the app explains each term: which variable it is, whether it is a factor or a covariate, whether it is an interaction or a nested effect.

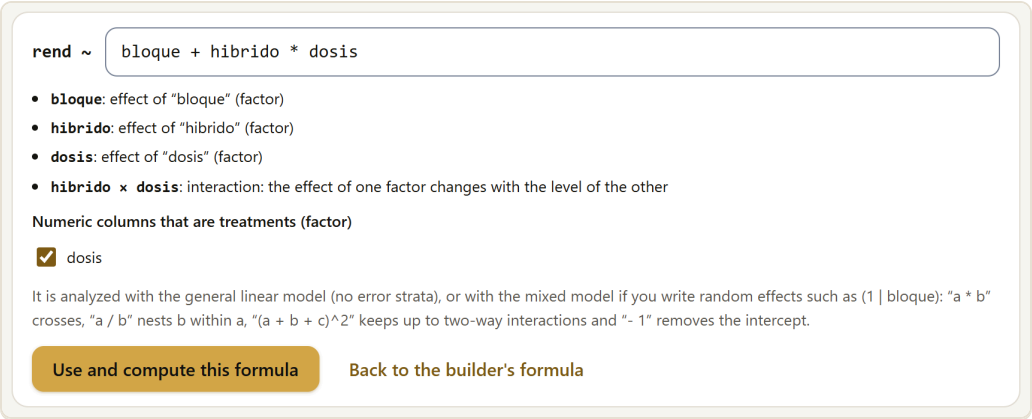


Figure 4.5. The formula typed by hand, with the reading of each term.

You type	It means
`a + b`	The main effects of *a* and *b*.
`a * b`	Crosses: *a*, *b* and their interaction *a* × *b*.
`a:b`	Only the interaction.
`a / b`	Nests *b* within *a*.
`(a + b + c)^2`	The main effects and all two-way interactions.
`- 1`	Removes the intercept.
`(1 | bloque)`	A random effect: the formula is fitted with the mixed model in the R engine.

The **Numeric columns that are treatments (factor)** checkboxes say which numeric columns are treated as a factor (a rate with four levels) and which as a covariate. Use and compute this formula analyzes it with the general linear model (without error strata: for a split plot, the dedicated method is the better choice) or, if it has random effects, with the mixed model; with a count or gamma distribution, with the generalized model. Back to the builder's formula undoes what you typed. With an event or with several responses, the formula comes from the zones and cannot be edited by hand.

When the model cannot be built

If the zones ask for something that no SigmaPro analysis does, the box offers not an analysis but the **problem and its fix**:

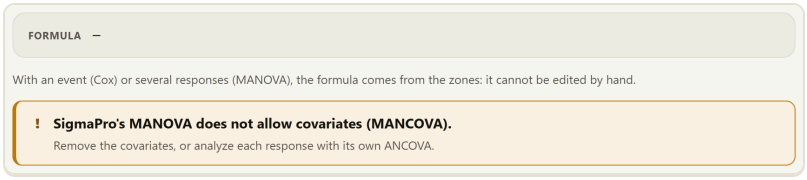


Figure 4.6. Several responses with a covariate: the app explains that it does not do MANCOVA and proposes the fix.

4.5 How to read a result

The results of the Analysis block have the parts of those in Explore (section 3.2) and a few more. Figure 4.7 shows the top part of the maize analysis in randomized complete blocks, and figure 4.8, its first two tables.

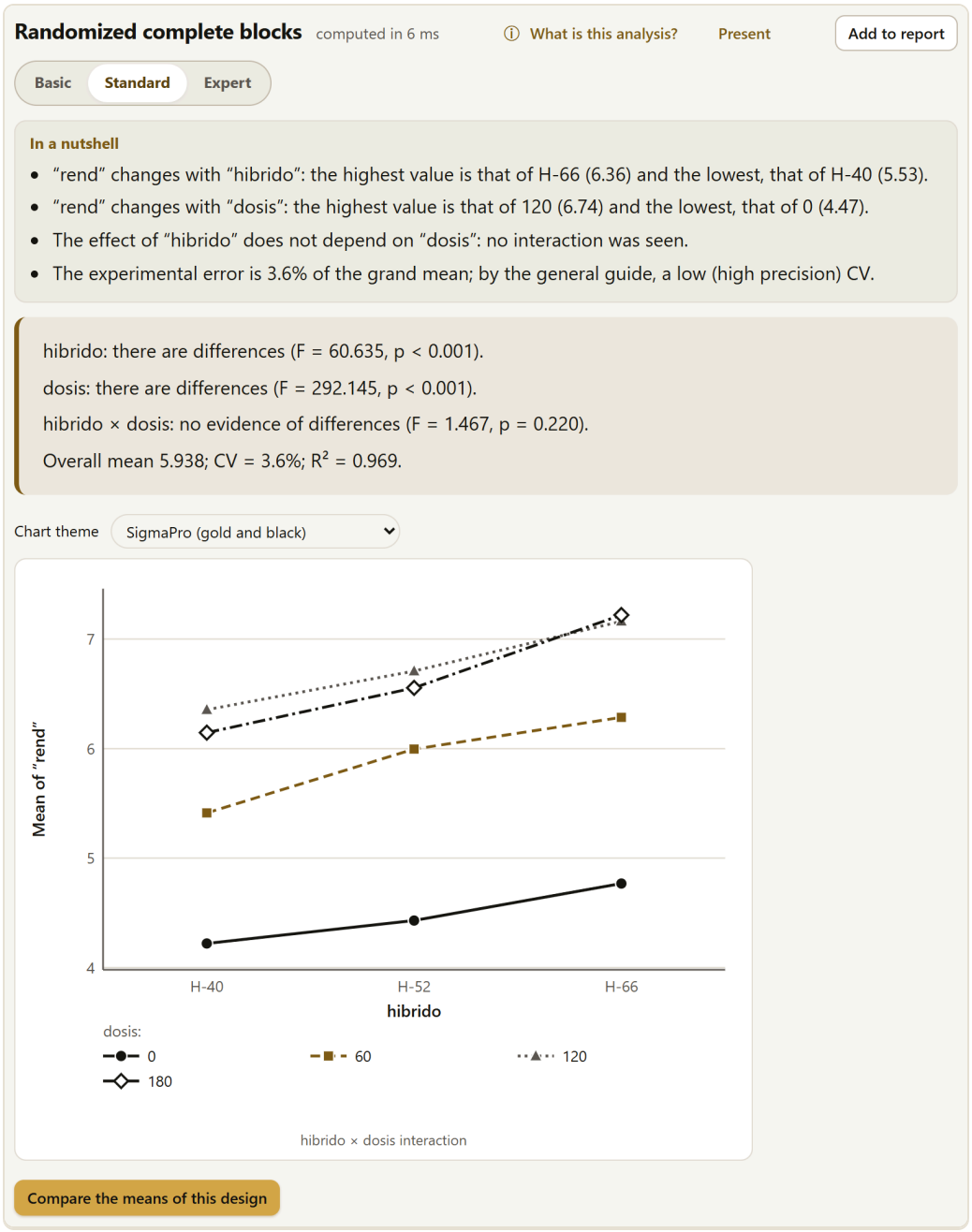


Figure 4.7. The top part of a result: title, level of detail, In a nutshell, conclusion, chart and the next step.

Table 1. Analysis of variance: Randomized complete blocks

Source of variation	df	SS	MS	F	p
bloque	3	1.4052	0.4684	10.40	< 0.001
hibrido	2	5.4594	2.7297	60.63	< 0.001
dosis	3	39.4562	13.1521	292.15	< 0.001
hibrido × dosis	6	0.3962	0.0660	1.47	0.220
Error	33	1.4856	0.0450	—	—
Total	47	48.2026	—	—	—

Balanced data: type I, II and III sums of squares coincide.

▶ How to read this table

Copy table Download as CSV

Table 2. Trial summary

Measure	Value
Grand mean	5.938
Observations	48
Coefficient of variation, %	3.6
R^2	0.969
Relative efficiency vs. completely randomized, %	160.0

Figure 4.8. The first two tables of the same analysis: the ANOVA, with How to read this table collapsed below it, and the trial summary. The marginal means of each factor and of their combinations follow.

Part	What it is
Title	The analysis, with the computing time, ⓘ What is this analysis?, which opens its fact sheet in Fundamentals (chapter 9), Present, which shows it as slides (below), and Add to report.
Level of detail	Below the title, Basic, Standard and Expert: how much of the result is shown (below, "Three levels of detail").
In a nutshell	A box with the result without statistical symbols, in up to four sentences (below, "In a nutshell").
Conclusion	One sentence per effect or per key result, with its statistic and its *p* value: "hibrido: there are differences (F = 60.635, p < 0.001)." At the end, the fit indicators (grand mean, CV, R^2).
Assumption traffic light	In the tests that depend on assumptions (t tests, regression, ANCOVA, generalized models…), one line per assumption in green, amber or red, with its icon and the test that decides it (section 4.7); if any of them fails, Try the alternative offers the analysis that addresses it. In the designs, the analysis of variance has no traffic light: its assumptions are checked with Check the model assumptions (section 4.7); in the report, those of completely randomized and randomized complete blocks designs are checked automatically.

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Part	What it is
Charts	The ones that correspond to the analysis: the interaction of two factors, the means with their standard error, the regression line with its bands, the residual diagnostics, the survival curves… At the top, the Chart theme selector changes the style of all of them at once (the six themes are in table 6.1). Each chart has its legend and the ✎ button of the figure editor (chapter 6).
Next step	Buttons that lead to the analysis that usually follows: Compare the means of this design in the designs, Analyze the chosen model after a variable selection, the CSV for other apps in the multivariate analyses.
Tables	Numbered, each with Copy table and Download as CSV (with the full numbers, without the on-screen rounding): the analysis of variance, the trial summary, the means, the comparisons… A table wider than the screen scrolls sideways; with the keyboard, Tab focuses it and the arrow keys scroll it.
How to read this table	Collapsed below each table that includes a statistic with a reading guide: one gauge per statistic with its interpretation card and the reference curve of each test (figures 4.9 and 4.10, below).
How to report it (APA)	Collapsed after the tables: the methods and results paragraphs in APA 7 style, in Spanish and in English, ready to copy (below).
Equivalent R code	Collapsed at the end, with Copy the code.
Continue in the LABG suite	At the foot of some analyses, a strip with the suite app that continues the work: AgriDesign after an experimental design, CorrelationPro and PCAPro after a correlation or a regression, PCAPro and ClusteringPro after the multivariate or multi-environment analyses. Each card says why it is worthwhile and can be hidden (chapter 10).

B4

How to read this table

Below each table that includes a statistic with a published reading guide there is a closed drop-down, How to read this table. It is drawn when you open it, so it does not slow down the result. It contains two things:

- **One gauge per statistic.** A scale divided into the zones of its guide, with a triangle at the value of each row of the table (with more than twelve values, dots and the count per zone). Below are the value with its zone and the guide that defines it, with a note when the guide has nuances. The magnitude zones (the *p*, the effect sizes, the R^2) are in golds, from light to dark; the quality zones (the CV, the collinearity, the residuals), in green, amber and red, like the assumption traffic light. Table 4.1 brings the guides together.
- **The interpretation card of each gauge.** A few sentences that translate the value with the numbers of the case: what part of the variation the R^2 explains, what the error standard deviation amounts to according to the CV and the grand mean, how many times the number of replicates the relative efficiency stands for, how many standard deviations separate two means, how often a result like this would occur if there were no effect (the *p*)… With several values, it says which ones fall in each zone: the terms with $p < \alpha$, the residuals beyond ±2 and ±3, the observations beyond $4/n$ in Cook's distance. Collapsed, What it is gives the definition of the statistic, and In the glossary opens its entry in the glossary, with "See it in action" (section I.6).

- **The reference curve of each test.** For each F, t, z or χ^2 in the table (up to eight), the density with the degrees of freedom of that test: the dashed line is the critical value at the analysis α, the colored line is the observed value and the shaded area is the *p*. If the value lies beyond the chart, an arrow announces it ("60.6 →"). In designs with several errors (split plots, strip plots), each F is drawn with the degrees of freedom of its own error. If the table gives an adjusted or exact *p*, a note says so: the curve shows the one from the reference distribution. Below each curve, a sentence says whether the value passes the critical value ("Before the critical value 2.39: $p = 0.220 \geq \alpha = 0.05$.").



Figure 4.9. How to read this table open below the maize trial summary: the CV, the R^2 and the relative efficiency, each with its scale, its card and its guide. According to the CV card, the error standard deviation, 0.212, is 3.57% of the grand mean, 5.94.

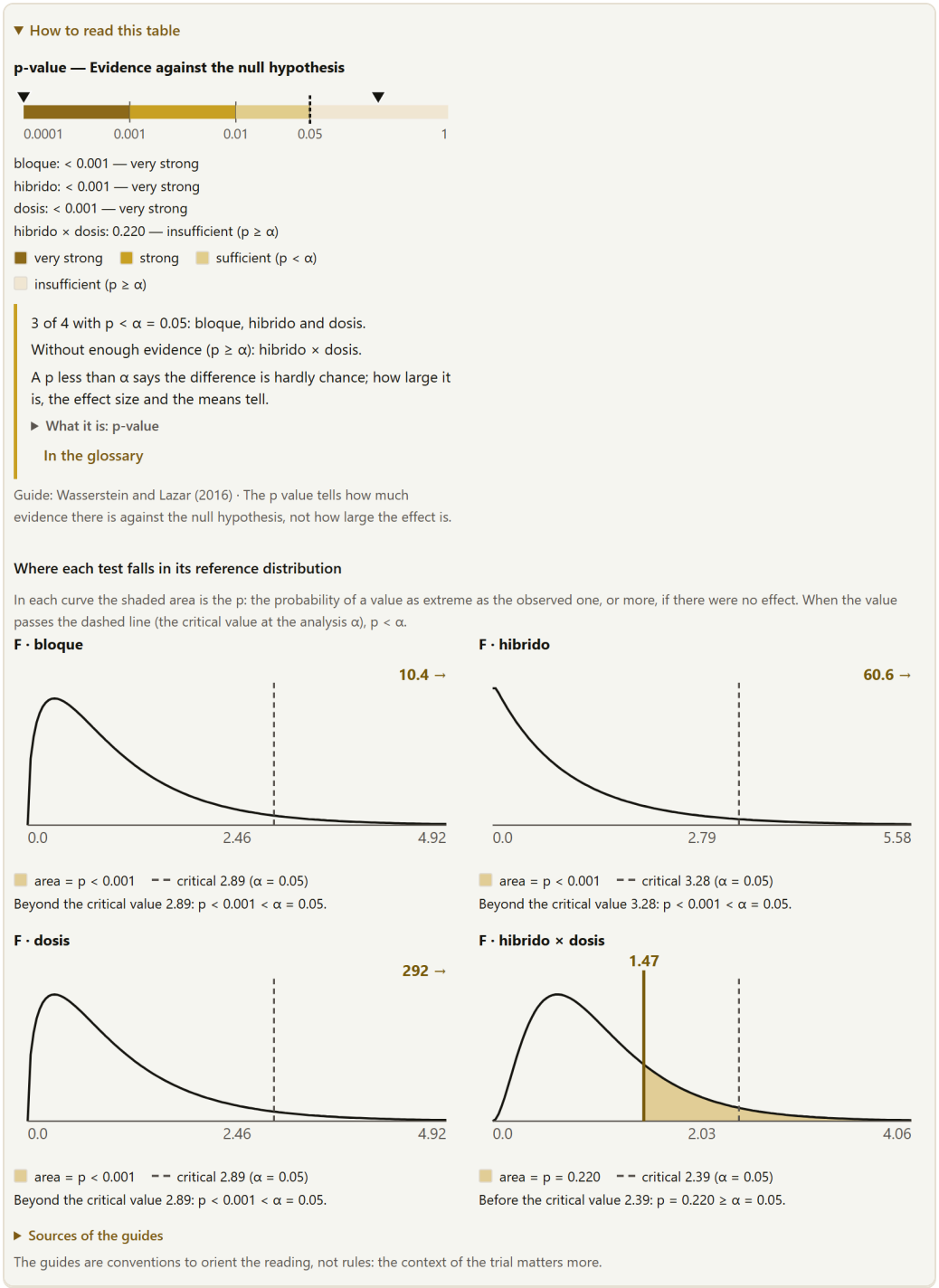


Figure 4.10. The same drop-down below the ANOVA: the *p* gauge for the four terms with its card (3 of 4 with $p < \alpha$) and the F curve of each one with its sentence. Only the interaction (F = 1.47) falls before the critical value, 2.39.

At the foot, Sources of the guides gives the full references (they are also in appendix D). The guides are conventions that orient the reading, not rules: the crop, the variable and the trial weigh more. A CV of 25% can be good in a pest count and bad in grain yield; that is why the CV guide can be changed (below). Some gauges depend on the context: the CV gauge does not appear in the descriptives of Explore (there the CV is that of the data, not that of the error of a trial), the gauge for Cramér's *V* adjusts its limits to the size of the table and the gauge for Cook's distance uses $4/n$ with the number of observations in the analysis.

Table 4.1. The guides of the gauges. Each limit is the start of the next zone.

Statistic	Zones	Guide
p value (also the adjusted one)	very strong < 0.001; strong < 0.01; sufficient < α; insufficient from α. A line marks the analysis α.	Wasserstein and Lazar (2016)
Cohen's *d* and *h*, Hedges' *g* (absolute value)	small from 0.2; medium from 0.5; large from 0.8	Cohen (1988)
Cohen's *f*	small from 0.10; medium from 0.25; large from 0.40	Cohen (1988)
Pearson's *r*, Spearman's ρ, φ (absolute value)	small from 0.1; medium from 0.3; large from 0.5	Cohen (1988)
Cramér's *V*	those of φ divided by √(min(rows, columns) − 1)	Cohen (1988)
R^2 and adjusted R^2	small from 0.02; medium from 0.13; large from 0.26	Cohen (1988)
Power	adequate from 0.80	Cohen (1988)
η^2 of *H* (Kruskal-Wallis)	small from 0.01; medium from 0.06; large from 0.14	Tomczak and Tomczak (2014)
Kendall's *W*	small from 0.1; medium from 0.3; large from 0.5	Tomczak and Tomczak (2014)
Odds ratio	small from 1.68; medium from 3.47; large from 6.71. A ratio below 1 is read by its inverse.	Chen, Cohen and Chen (2010)
Cohen's κ	worse than chance < 0; slight up to 0.20; fair up to 0.40; moderate up to 0.60; substantial up to 0.80; almost perfect	Landis and Koch (1977)
Coefficient of variation of the trial (general guide)	low < 10%; medium < 20%; high < 30%; very high from 30%	Pimentel-Gomes (2009)
Relative efficiency, %	precision gained from 100%	Gomez and Gomez (1984)
Heritability	low < 0.3; medium < 0.6; high from 0.6	Johnson, Robinson and Comstock (1955)
Variance inflation factor	no worrying collinearity < 5; moderate < 10; severe from 10	James et al. (2013)
Condition index	weak < 10; moderate < 30; strong from 30	Belsley, Kuh and Welsch (1980)
Durbin-Watson	no apparent autocorrelation from 1.5 to 2.5; possible from 1 to 1.5 and from 2.5 to 3; strong outside 1 to 3	Durbin and Watson (1950)
Studentized residual (absolute value)	common < 2; uncommon < 3; possible outlier from 3	Montgomery, Peck and Vining (2012)
Cook's distance	check from 4/*n*; influential from 1	Bollen and Jackman (1990); Cook and Weisberg (1982)

Table 4.1. The guides of the gauges. Each limit is the start of the next zone. *(continued)*

Statistic	Zones	Guide
Skewness (absolute value)	approximately symmetric < 0.5; moderate < 1; marked from 1	Bulmer (1979)
AIC difference	substantial support < 2; considerably less support < 7; very little support < 10; essentially no support from 10	Burnham and Anderson (2002)

The CV guide for your crop and variable

Below the CV gauge, Set the CV guide for your crop and variable changes the guide used to read the CV in all open tables, in Analysis and in the report; the choice is remembered in the browser. There are three options under Guide:

- Field trials in general: that of Pimentel-Gomes (2009), the one the app comes with (table 4.1).
- **Maize**: seven variables with the limits of Fritsche-Neto et al. (2012), derived from the CVs of the trials published in Brazil from 2005 to 2010 (table 4.2). Two rows of the original table (lodging and days to flowering) do not agree with the mean and the standard deviation that the same article publishes and were left out.
- Own limits: the crop, the variable and the three limits (Low up to, Medium up to, High up to), typed by hand or computed with Compute them from the CVs of similar trials (Garcia, 1989): low up to the mean of those CVs minus one standard deviation, medium up to the mean plus one, high up to the mean plus two and very high beyond (figure 4.11). They are applied with Apply my limits.

Table 4.2. CV limits for maize (Fritsche-Neto et al., 2012), in %. Each limit is included in its zone; anything beyond the last one is very high.

Variable	Low up to	Medium up to	High up to
Grain yield	8.24	15.49	19.11
Plant height	3.55	8.25	10.60
Ear height	4.58	10.76	13.85
Ear diameter	3.82	13.78	18.75
Ears per plant	6.13	19.26	25.81
Weight of commercial ears	10.46	17.37	20.82
Weight of 100 grains	4.81	20.04	27.66

Three levels of detail

Below the title of each result, and in the report header, three buttons choose how much is shown. The level applies to the whole app (Analysis, Explore and the report) and is remembered in the browser. It does not change any number or what is exported.

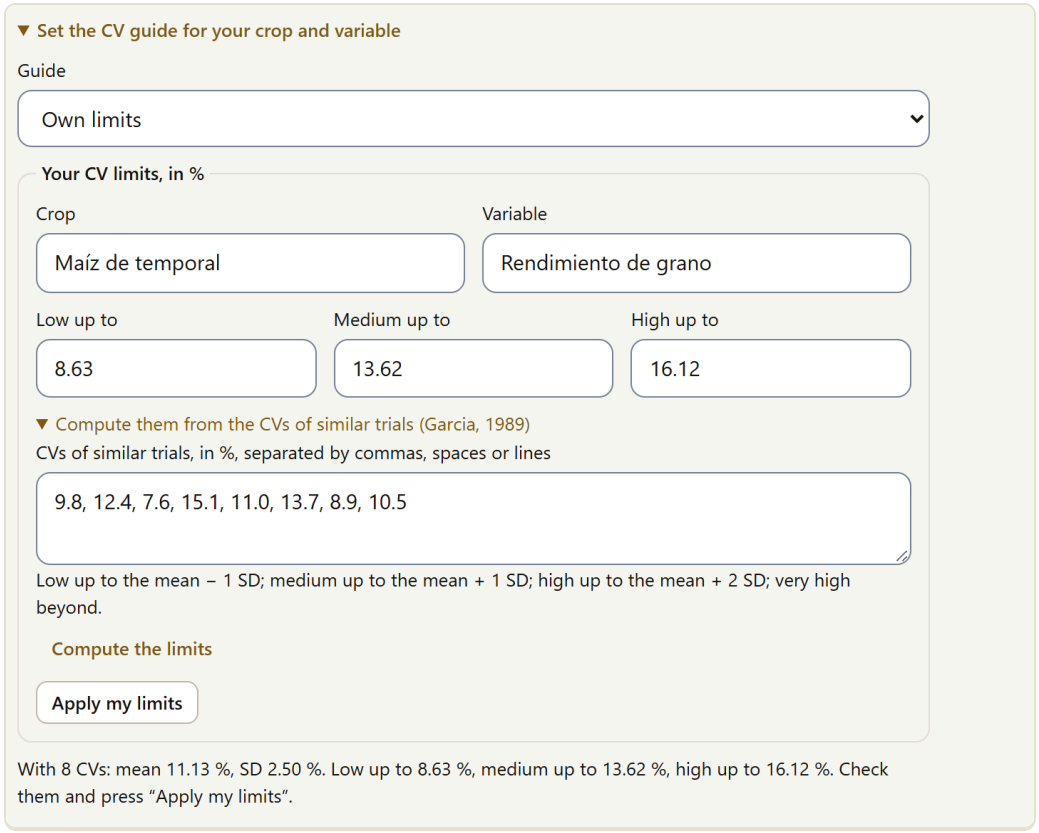


Figure 4.11. The form below the CV gauge: own limits computed by Garcia's method from eight CVs typed as an example: mean 11.13% and standard deviation 2.50%; low up to 8.63%, medium up to 13.62% and high up to 16.12%. With Apply my limits, the gauge and its card are read with them.

Level	What is shown
Basic	In a nutshell instead of the conclusion with the statistics (if the analysis has no such box, the conclusion), the charts, the traffic light, How to report it (APA) and, open below the name of each table ("Reading · Table 1. …"), its reading: gauges, cards and curves (figure 4.12). The tables with their notes and buttons, the R code and the technical details are hidden; a table without a reading guide does not appear. In Explore the table is the summary itself and is always shown.
Standard	The whole result, as described in this section. It is the level the app opens with.
Expert	Everything in Standard plus the Technical notes: method, inputs, options, data, rows used, software and computing time, with the model and its hypotheses taken from the fact sheet in Fundamentals (chapter 9). The notes, How to report it (APA) and the R code appear open (figure 4.13).

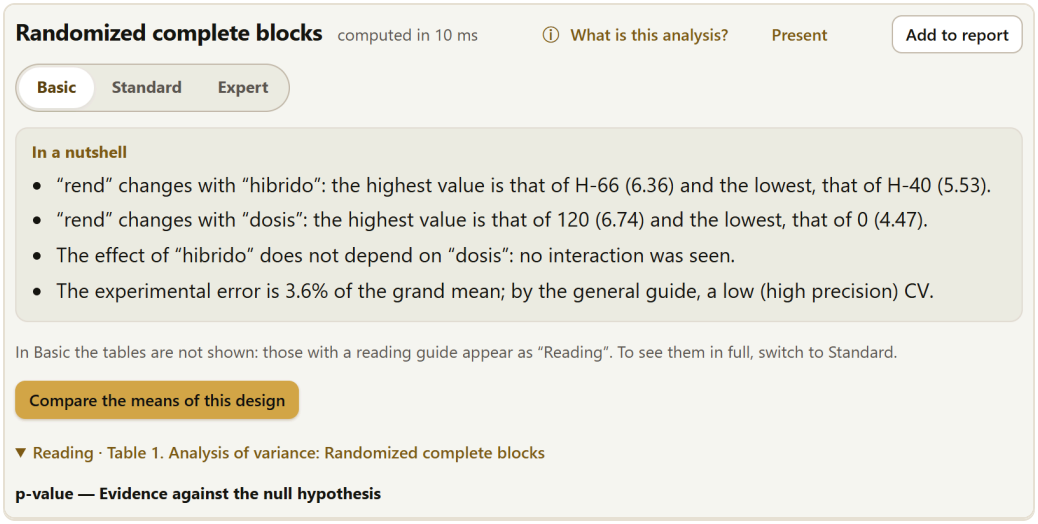


Figure 4.12. The Basic level: instead of the ANOVA table, its reading, open (it continues as in figure 4.10). The interaction plot, which in the app sits between the conclusion and the reading, was removed from the screenshot.

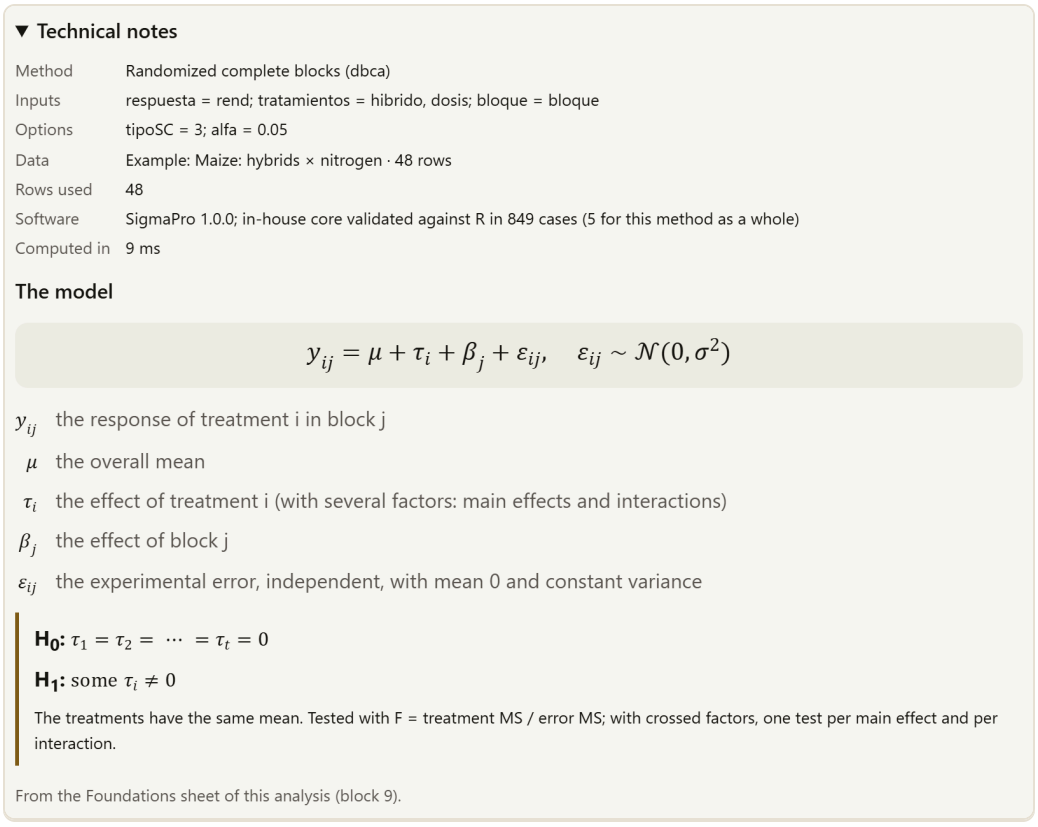


Figure 4.13. The Expert level adds the technical notes, with the model of the block design and its hypotheses from the fact sheet in Fundamentals.

In a nutshell

Above the conclusion, a box states the result without statistical symbols, in up to four sentences, for those who do not read an analysis of variance table (figure 4.7). It computes nothing new: it reads what the analysis already gave.

- **In experimental designs** it is built from the ANOVA table and the means of each factor: which factor changes the response, with its highest and lowest levels and their means; whether there is an interaction; and the precision of the trial, with the CV and its guide. A factor with no effect at the analysis α is stated like this: "With these data, … does not appear to change…"; with two levels, which one is higher. If there is an interaction, the box advises comparing the means of one factor within each level of the other.
- **In the other analyses** they are the sentences of the conclusion without the parentheses or the parts with statistics (F, t, p, degrees of freedom, intervals). Sentences that would say nothing without the statistics are dropped, as are technical ones, such as the one about letters in common. With Within each level of (section 4.11), each sentence starts with its level: "'hibrido' H-40: 3 groups of means…".

THE T TEST OF THE POISONS (SECTION 4.8)

Conclusion: The mean of "A" (0.314) differs from that of "B" (0.677): difference −0.362 (95% CI: −0.572 to −0.153); t = −3.729, df = 13.21, p = 0.002. Effect size: large (Hedges' g = −1.470; 95% CI: −2.342 to −0.572).

In a nutshell: The mean of "A" (0.314) differs from that of "B" (0.677). Effect size: large.

At the Basic level the box is shown and the technical conclusion is hidden; at Standard and Expert, both are shown. The box appears in Analysis and in the report; in Explore it is not needed, because there the conclusion is already in words.

How to report it (APA)

After the tables, the How to report it (APA) drop-down gives the **Methods** paragraph and the **Results** paragraph of the analysis, the same ones as in the report (section 5.4), with the conventions of APA 7 style (figure 4.14):

- statistical symbols in italics (*F*, *t*, *p*, *r*, *n*, *M*…); Greek letters, not;
- the *p* and the statistics that cannot exceed 1 (*r*, R^2, η^2, Cramér's *V*, κ, Kendall's *W*) without the leading zero: $F(2, 33) = 60.63$, $p < .001$; $R^2 = .969$;
- commas between the statistics within the same parentheses, confidence intervals in square brackets (in the t test of the poisons, 95% CI [−0.5722, −0.1528]) and the mean as *M*;
- in English, the percent sign attached to the number (3.6%); in Spanish, separated from it (3.6 %).

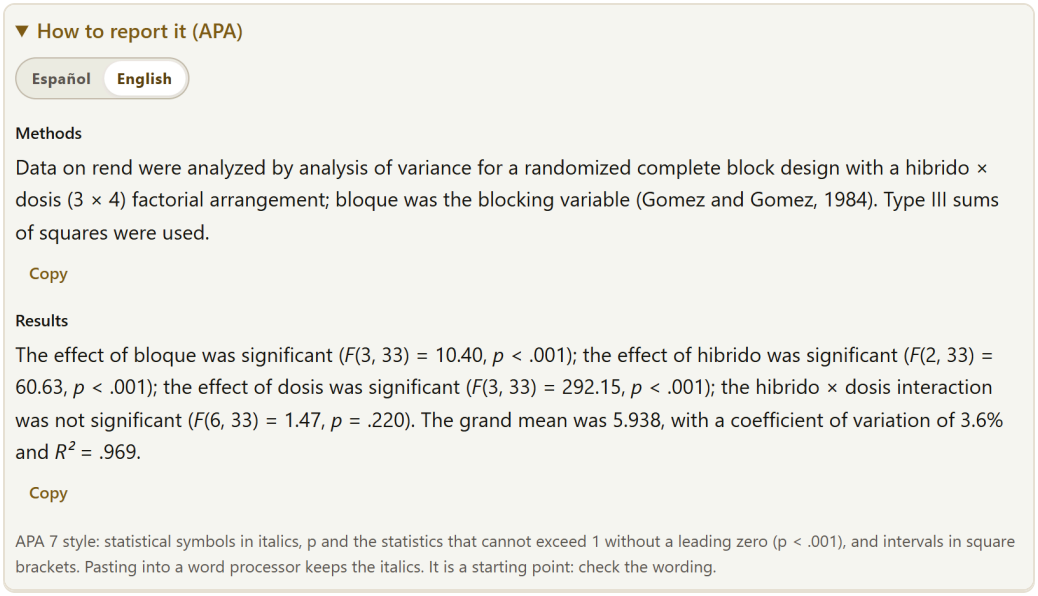


Figure 4.14. How to report it (APA) for the maize example on the English tab: the same analysis, written in English with the APA 7 conventions, without changing the app language.

The Español and English tabs change the language of the text, not that of the app: the paragraph is built with the same figures and with the names of the methods, the designs and the citations in that language (when the app is in Spanish, the first time English is requested the app loads it in an instant). Copy, below each paragraph, copies it with its italics, which a word processor keeps when you paste it. As in the report, it is a starting point: review the wording, replace the column names with full names and add the agronomic interpretation.

The drop-down is in Analysis, in the assumption check (section 4.7), in Explore and in each analysis of the report. It is built when you open it, so it does not slow down the result; at the Expert level it appears open.

> **THE REPORT BRINGS IT ALL TOGETHER**
>
> In the Analysis block the result is explored: options are changed, models are tried. When you add it to the report (chapter 5), SigmaPro adds what a document requires: the paragraphs of all the analyses in a single text, with the citation of each test and that of SigmaPro, the technical details and the numbering of tables and figures across the whole report.

Presenting the result

The Present button, in the title of the result, shows it as full-screen **slides** with large type, for projecting it in a class or a meeting without putting together a separate presentation. Each part of the result is a slide, in the order in which it appears: the title, In a nutshell, the conclusion, each chart and each table with its notes (figure 4.15). What is presented is what is shown at the chosen level of detail: at Basic there are no tables, and whatever is collapsed is left out.

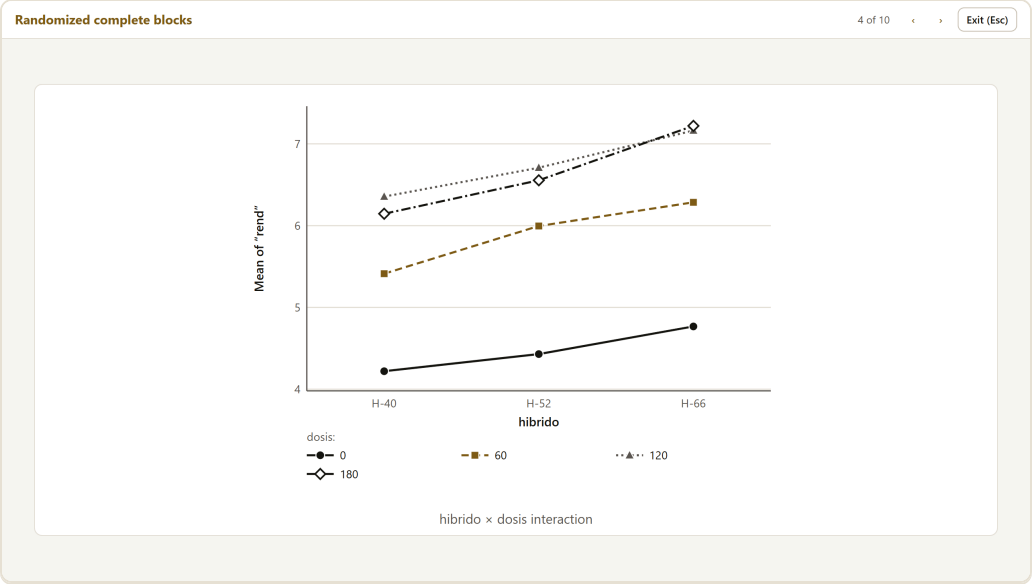


Figure 4.15. The fourth slide of the maize example: the interaction plot, across the width of the screen. At the top, the analysis, the slide count and the buttons.

The ← and → arrow keys (or Page Up, Page Down and the spacebar) change slides; Home and End go to the first and the last one; Esc exits and leaves the focus on Present. The slides are copies without buttons: they do not change the result. The same button is in Explore (section 3.2) and in the report header, where it presents all the analyses one after another, with the name of each one in the bar (section 5.2).

When it cannot be computed

If an analysis cannot be computed with the chosen data and options, instead of the result a box appears with **what happened**, in words, and What to do:, the fix (figure 4.16). It is the same in Analysis, in Explore and in the report.



Figure 4.16. A t test on a column with a single value: what happened and what to do.

- **A data problem** (a column that should be numeric, one that no longer exists, one in two roles, too few data, a factor left with a single level) comes with its fix. It is almost always in Data: the column type, the missing values or the filter.
- **An R engine error** (section 4.6) is explained in words when it is one of the known mixed-model errors: a random effect with a single level or with as many levels as observations, a response with the same value in every row, a model that does not converge or that is too complex for the data. The original R text remains below, in Technical detail, collapsed and with Copy the detail.

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- **A program error** is stated as such: it is not a data problem. The box suggests changing some option or reloading the page and, if it happens again, reporting it with the Report a problem icon in the header (number 6 in figure I.1), pasting the technical detail.

4.6 The R engine inside the browser

Almost all SigmaPro analyses are programmed in the app's own core, in JavaScript, and validated against R. A few, those that require the fitting machinery of mixed models, run in a **full R that works inside the browser** (webR, R compiled to WebAssembly): linear mixed models and their designs (section 4.15), repeated measures and generalized linear mixed models (section 4.17). The data do not leave the computer here either: R runs in the same tab.

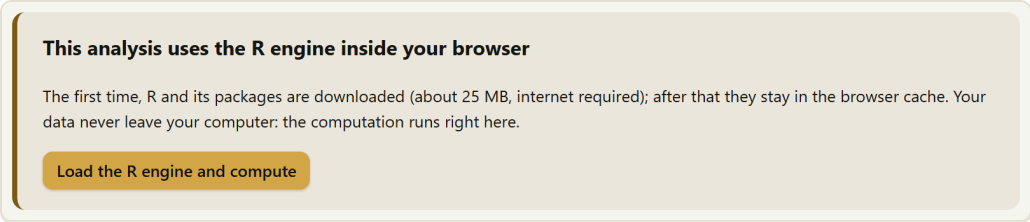


Figure 4.17. The first time an R engine analysis is used, the app asks for permission before downloading it.

1. The first time, the notice in figure 4.17 appears instead of the result. Load the R engine and compute downloads R and its packages (about 25 MB) and computes.
2. Meanwhile, above the result you can read each step ("Downloading the R engine (the first time takes longer)…", "Installing lme4 in the R engine (first time only)…", "Computing in the R engine…") and a Cancel button. The first load takes from a few seconds to a couple of minutes, depending on the connection; a computation, a few seconds.
3. Afterwards, the engine stays in the browser cache: internet is no longer needed and it does not ask again.

When SigmaPro is used published on a web page (not by double-clicking), it brings its own R engine repository and works offline from the first visit (section I.7).

4.7 Model assumptions

Analysis of variance, mean comparisons and regression rest on assumptions about the **residuals** (what the model does not explain): that they follow a normal distribution, that they have the same variance in all treatments, that they are independent of each other and that none of them is extreme. Check the model assumptions, the first method in the catalog, tests them all at once and tells you what to do if any of them fails.

Option	What it decides
Response variable	The response of the model.
Treatments (up to 3, crossed)	The factors; the variances are compared among their combinations.
Blocks	The block, if there is one (— no blocks — if not). With blocks, block × treatment additivity is also checked.
Order for the independence check	The order in which similar neighboring residuals are looked for: — row order — or the order of a numeric or date column (the position in the field, the measurement date).
Significance level	The α of the tests (0.05, 0.01 or 0.10), which also sets the cutoffs of the traffic-light colors.

The check fits the model `respuesta ~ bloque + A * B * C` with your choices; the header of the result writes it out, with the number of residuals (n), and a line warns how many rows with a missing value are left out.

> **WHICH MODELS IT CHECKS**
>
> The check accepts treatments and a single blocking factor: it properly checks completely randomized designs, block designs and their factorials. In a Latin square (put one of the two directions in Blocks), a lattice or a split plot the check is approximate, because it does not separate the other blocking factors or the main-plot error; it does not accept covariates either. When you add a completely randomized or randomized complete block design to the report (chapter 5), the report adds the traffic light of this check on its own, with the same model. The fact sheet of each analysis in Fundamentals (chapter 9) says, assumption by assumption, how to check it here and which alternative to use if it fails.

The traffic light

At the top of the result there is one row per assumption, each with its color, its icon (✓ OK, ! check, × problem, – N/A), its status and a sentence with the test that decides it. The icon and the word say the same as the color, so you do not need to tell the colors apart. The overall color is that of the worst row: **The assumptions are met**, **Some assumptions need checking** or **Some assumptions are not met**.

Assumption check model rend ~ bloque + hibrido * dosis · n = 48 · computed in 14 ms ⓘ What is this analysis?

Present Add to report

Basic Standard Expert

The assumptions are met

Normality of residuals OK
The residuals are consistent with a normal distribution (Shapiro-Wilk, p = 0.366).

Equal variances among treatments OK
The treatment variances are homogeneous (Brown-Forsythe, p = 0.538; the largest is 8.3 times the smallest).

Block × treatment additivity OK
Block and treatment effects are additive (Tukey, p = 0.671).

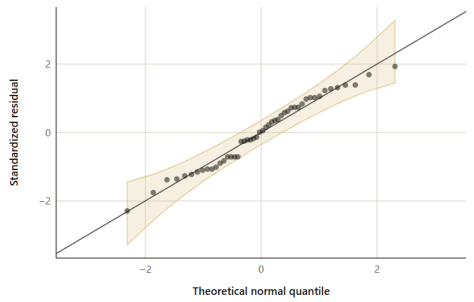
Independence of residuals OK
No autocorrelation of the residuals in the order of the rows (Durbin-Watson = 1.981, p = 0.331).

Outliers OK
No residual is an outlier (the largest studentized residual, -2.459 in row 26, has Bonferroni p = 0.937).

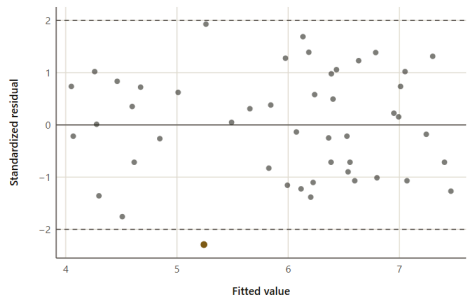
What to do

- The assumptions are reasonably met: the analysis of variance and the mean comparisons will be reliable.

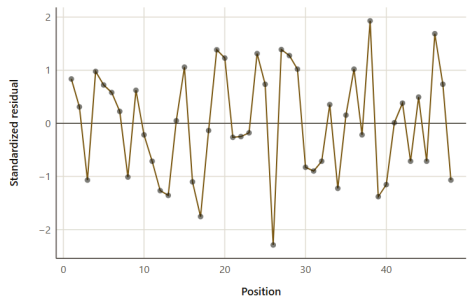
Chart theme SigmaPro (gold and black)



Normal Q-Q of the standardized residuals



Residuals vs. fitted values



Residuals in row order

Figure 4.18. The assumptions of the maize trial are met: all five rows of the traffic light are green.

Assumption	Deciding test	Also computed
Normality of residuals	Shapiro-Wilk (from 3 to 5,000 residuals; with more, Anderson-Darling)	Lilliefors, Anderson-Darling; the skewness G_1 and the kurtosis G_2
Equal variances among treatments	Brown-Forsythe (Levene with the median)	Levene, Bartlett (sensitive to non-normality), Fligner-Killeen (rank-based); the ratio of the largest variance to the smallest
Block × treatment additivity	Tukey's non-additivity (with blocks)	—
Independence of residuals	Durbin-Watson, against positive autocorrelation; exact *p* with fewer than 100 residuals	Runs above and below the median (exact *p* with fewer than 30 residuals)
Outliers	The largest studentized residual, with a Bonferroni *p*	—

The colors follow the *p*-value of the main test and the chosen α: **green** if $p \geq \alpha$ (OK), **amber** if it falls between α/5 and α (CHECK) and **red** if $p < \alpha/5$ (PROBLEM). With $\alpha = 0.05$, the cutoffs are 0.05 and 0.01. A gray row says N/A (additivity without blocks; variances without treatments or with a treatment that has a single observation) or that there are too few observations for the test.

Normality has one exception: with 50 residuals or more and a nearly normal shape ($|G_1| < 0.5$ and $|G_2| < 1$), the row goes no further than amber even if the test rejects, and its sentence says that the departure is small. With so many data any departure turns out significant, and analysis of variance tolerates it.

What to do

Below the traffic light, the **What to do** box turns the result into concrete actions. Each recommendation appears only when its condition holds:

When	What it proposes
Everything green	That the analysis of variance and the mean comparisons will be reliable.
Normality or variances fail and the response is positive	The Box-Cox transformation estimated with the same model (λ with its 95% interval) and the simple transformation that corresponds to it (square root, logarithm, reciprocal square root, reciprocal…), to be done in the Data block (section 2.9) before checking again.
Normality or variances fail and the response is counts (non-negative integers)	$\sqrt{(x + 0.5)}$, or a Poisson or negative binomial model (section 4.17).
Normality or variances fail and the response is a percentage (the column name says so and the values range from 0 to 100)	The arcsine square root, or a binomial model with the total of each row (section 4.17).
Unequal variances with a single factor and no blocks	Compare the means with Games-Howell (section 4.11) or use Kruskal-Wallis (section 4.9): neither assumes equal variances.

When	What it proposes
Unequal variances with several factors or blocks, with no Box-Cox possible	Transform the response; a model with a different variance for each treatment is not yet in SigmaPro.
Only normality fails	With 30 residuals or more, that analysis of variance is robust to that lack of normality; with fewer, compare with Kruskal-Wallis or, with blocks, Friedman (section 4.9).
Additivity fails	The logarithm (it makes a multiplicative effect additive) or, with replicates within each block, including the interaction in the model.
Independence fails	Check the order of the rows; smaller blocks (spatial analysis is not yet in SigmaPro).
There is an outlier	Check that row: first, that it is not a data-entry error; if the value is real, analyze with and without it and report it.

B4

With Box and Cox's poisons, for example, normality and variances fail and there is an outlying residual (figure 4.19): the app estimates the Box-Cox transformation with that same model ($\lambda = -0.816$, interval from -1.294 to -0.341), suggests the reciprocal and reminds you that the original column does not change; for row 24, it proposes checking first that it is not a data-entry error. If you check the poison alone, without the treatment, the model becomes one-way and the variances come out amber (Brown-Forsythe, $p = 0.021$): the box then adds Games-Howell and Kruskal-Wallis.

Trying the alternative

When a row of the traffic light fails, the Try the alternative box appears below it (in this check, below **What to do**), with one button per remedy. Each button runs the analysis that addresses the assumption, with the same variables; the numbers of the previous result do not change, and you go back to it by choosing it in the catalog. The buttons depend on the analysis:

Analysis	When	What the button runs
Assumption check	Normality or variances fail and Box-Cox proposes a transformation	Creates the transformed column in Data (`ln_`, `raiz_`, `inv_` or `bc_` plus the name of the response) and checks again with it.
	… and the response is counts	The count models, with the family chosen according to the overdispersion (section 4.17).
	With a single factor: unequal variances, or normality with fewer than 30 residuals	Kruskal-Wallis, or Friedman if there are blocks (section 4.9); with unequal variances and no blocks, also Games-Howell (section 4.11).
t test	Unequal variances with Student's t	Welch's t.
	Normality fails	Wilcoxon: Mann-Whitney with two groups, signed-rank with one sample or paired data (section 4.9).

Analysis	When	What the button runs
Linear and polynomial regression	Normality or constant variance fails	With a positive response, its logarithm in a new column and the fit again; with counts, the count model with the predictors as covariates (up to two).
	The shape fails, with one predictor	Polynomial regression of degree 2.
	Collinearity, with three or more predictors	Best subsets (section 4.12).
Generalized models	Overdispersion with the Poisson or the binomial	The negative binomial or the quasibinomial.
Mixed models and repeated measures	Normality or variances fail and the response is positive	Its logarithm in a new column and the fit again.

The transformed column is created only once: if it already exists with the same transformation, that one is used. A notice gives its name, and the original column does not change (section 2.9). With the poisons, Transform with the reciprocal and check again creates `inv_tiempo`, and the check with it comes out with all rows green, except additivity, which does not apply without blocks. Outliers, additivity and independence have no button: they call for checking the data or the design.

The charts and the table

- **Normal Q-Q of the standardized residuals**: if the residuals are normal, the points follow the diagonal. The shaded band is the confidence region; points that fall outside it are marked and the farthest ones carry their row number.
- **Residuals vs. fitted values**: you should see a shapeless cloud. A funnel (more scatter at larger values) reveals unequal variances; the lines at ±2 flag the large residuals.
- **Residuals in the order** of the rows or of the chosen column: long runs of the same sign reveal dependence (similar neighboring plots).

Above the three charts is the Chart theme selector, the same as in the other results (table 6.1). This result does not carry the LABG Suite band at the bottom: no app in the suite continues the assumption check.

Table 1 lists all the tests, each with its statistic, value, degrees of freedom, *p* and what it checks; the main test of each assumption is marked "(main)", and the note recalls the traffic-light cutoffs. Residual variance by treatment (collapsed) holds the variance of each combination, so you can see which one stands apart.

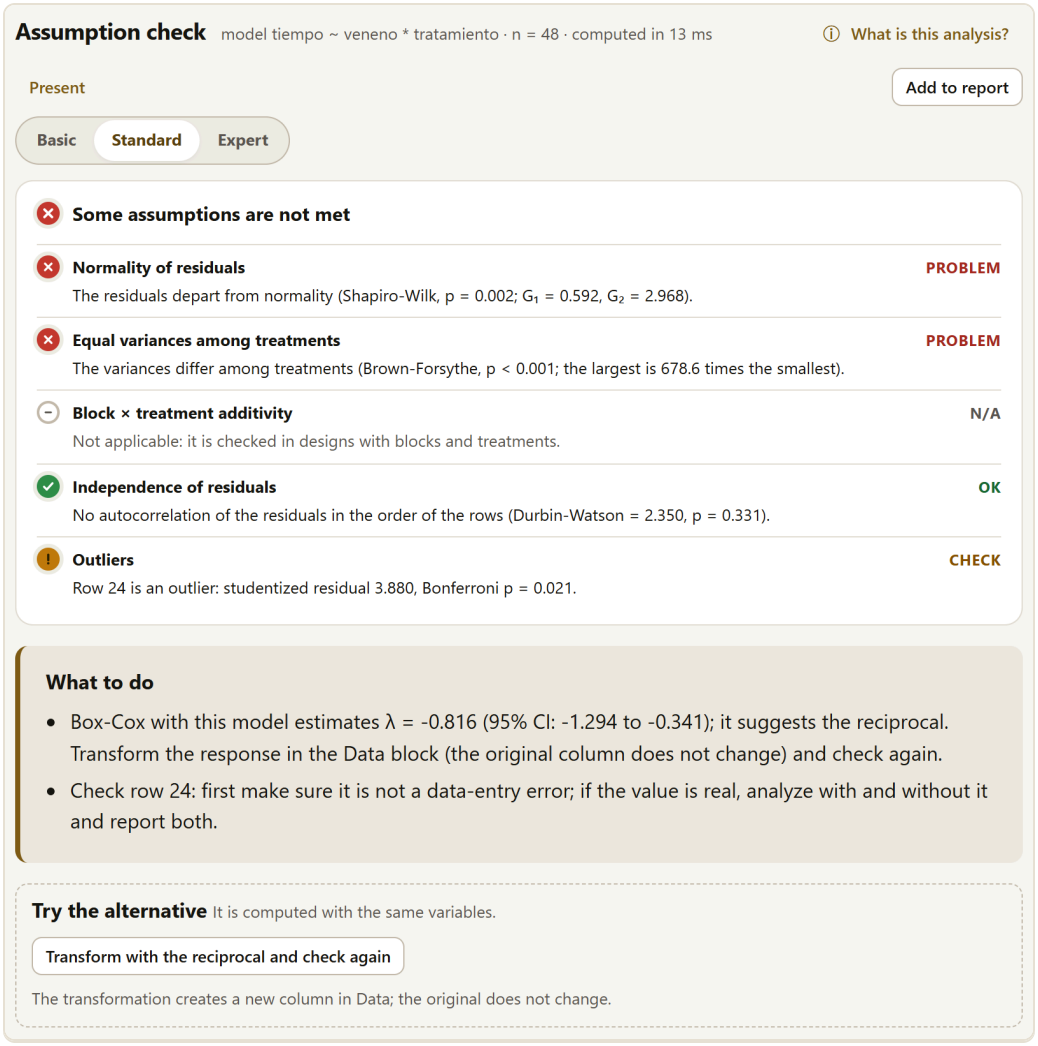


Figure 4.19. The opposite case: the survival times of the poisons do not meet the assumptions; the app proposes the reciprocal and offers it in Try the alternative.

WHY BROWN-FORSYTHE AND NOT BARTLETT

Bartlett's test is very sensitive to non-normality: with heavy-tailed residuals it rejects equality of variances even when the variances are equal. The Brown-Forsythe test (Levene's test with the median instead of the mean) is robust to this, which is why it decides the row; Bartlett stays in the table as a reference. For a similar reason, with very large data sets you should look at the charts as well as the *p*-values: with thousands of residuals, tiny departures from normality already turn out significant (hence the normality exception with 50 residuals or more).

Table 1. Assumption tests on the residuals of tiempo ~ veneno * tratamiento

Test	Statistic	Value	df	p	Checks
Shapiro-Wilk	W	0.9123		0.002	normality (main)
Lilliefors	D	0.1634		0.003	normality
Anderson-Darling	A	1.5923		< 0.001	normality
Brown-Forsythe	F	4.1323	11, 36	< 0.001	variances (main)
Levene	F	4.8535	11, 36	< 0.001	variances
Bartlett	K^2	45.1369	11	< 0.001	variances; sensitive to non-normality
Fligner-Killeen	X^2	31.0520	11	0.001	variances; rank-based
Durbin-Watson	DW	2.3501		0.331	independence; exact p
Runs test	R	29		0.136	independence; normal approx.
Studentized residual (Bonferroni)	t	3.8801	35	0.021	outlier: row 24

Figure 4.20. The table of tests, with the main test of each assumption marked.

B4

4.8 t tests

The t test compares means in three situations, chosen in Comparison type:

Type	What it asks for	Example
Two independent groups	The Response variable, the Group column and, if that column has more than two levels, the ones compared in Group 1 and Group 2.	Survival time with treatments A and B.
Paired measurements	The First measurement and the Second measurement, two columns measured on the same units (each row, one unit); first − second is compared.	Height of the same plant at 15 and at 30 days.
One sample vs. a value	The Variable and the Reference value (μ_0).	Does mean germination differ from 50%?

The other options: the Difference under H_0 (for two groups and paired data; usually 0), the Alternative hypothesis (Two-sided (≠), Less (<) or Greater (>)), the Confidence level of the interval (95, 99 or 90%), the Significance level and, with two groups, the Assume equal variances (Student's t) checkbox. Left unchecked, **Welch's t** is used, which does not assume equal variances and is the recommended one: with equal variances it gives almost the same as Student's, and with unequal variances it remains valid.

The result includes:

- the **conclusion** with the two means, the difference and its interval, and *t*, df and *p*;
- the **effect size** in words, based on Hedges' *g* and Cohen's guide: very small (below 0.2), small (from 0.2), medium (from 0.5) or large (from 0.8). For paired data it is computed on the differences (d_z) and for one sample, against the reference value;

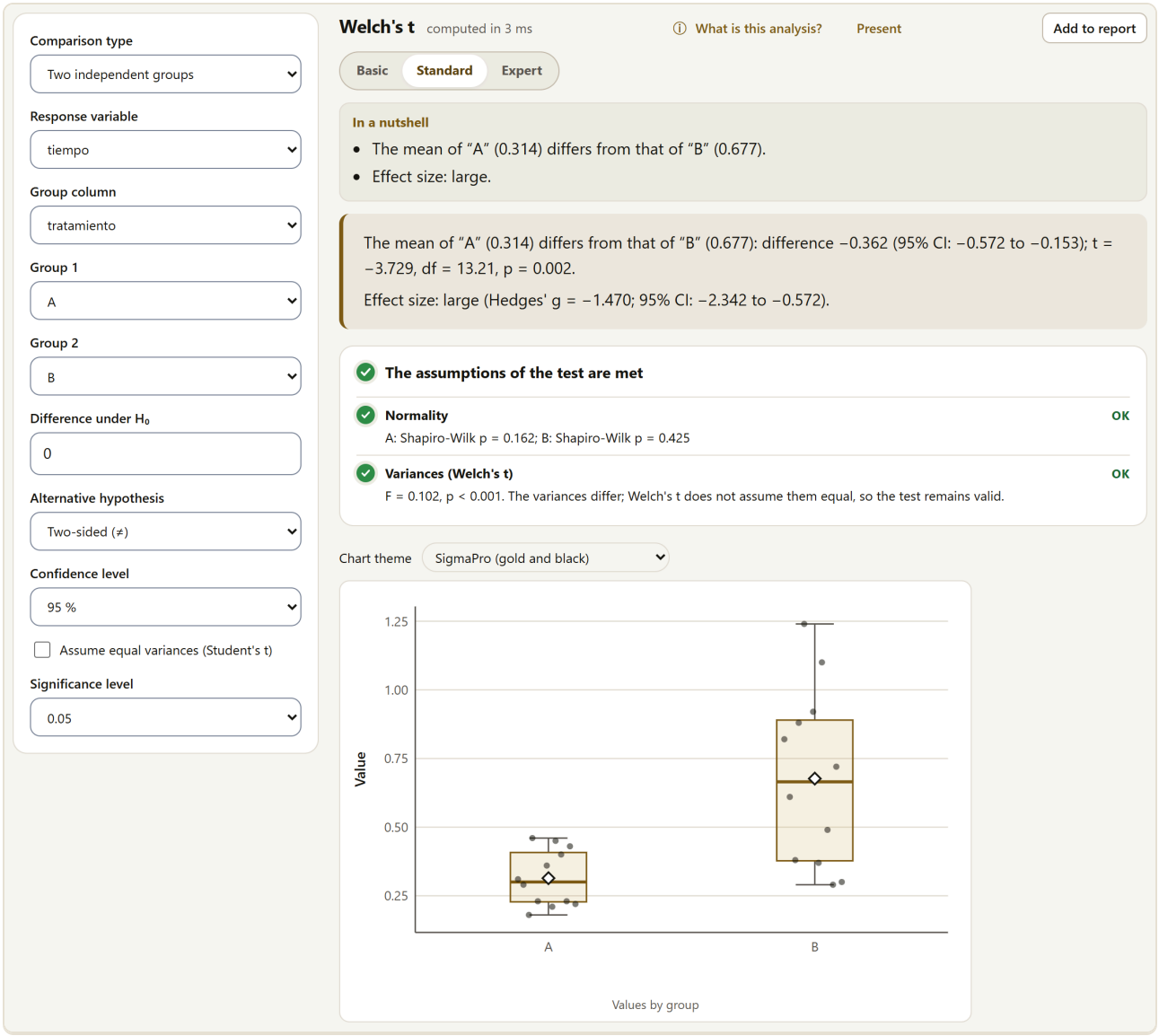


Figure 4.21. Welch's t: treatment B prolongs survival compared with A. The chart shows each value, the quartile box and the mean (diamond).

- the **traffic light**: the normality of each group (or of the differences, for paired data) by Shapiro-Wilk and, with two groups, the comparison of the variances with the *F* test. With Welch's t, that row (Variances (Welch's t)) stays green even if the variances differ, because Welch does not assume them equal; with Student's (Equal variances) it is colored according to the *F* and, if they differ, Try the alternative runs Welch's (if normality fails, Wilcoxon);
- the **chart** of the values of each group (for paired data, of the differences), with the quartile box and the mean; for paired data and one sample, a dashed line marks the reference value;
- the **tables**: the chosen test (with two groups, the other t below as a reference), the means and standard deviations, and the effect size: Cohen's *d* and Hedges' *g* (*d* corrected for small-sample bias), with intervals based on the noncentral t (Steiger and Fouladi, 1997).

With treatments A and B of the poisons, the variances differ greatly (F = 0.102, p < 0.001). With Welch, the test uses 13.21 df (p = 0.002) and the variances row says that it remains valid; if you check the Student box, the test uses 22 df (p = 0.001) and the same row comes out red with "Welch's t, which does not assume them equal, is preferable".

> **A T TEST IS NOT FOR MORE THAN TWO TREATMENTS**
> With three or more treatments, comparing them two at a time with t tests inflates the probability of finding differences that do not exist. The way to go is the analysis of variance of the design (section 4.10) followed by a mean comparison that controls that error (section 4.11).

4.9 Nonparametric tests

B4

When the data do not meet normality and a transformation does not achieve it, or when the response is an ordered scale (damage from 1 to 5, a score), rank tests compare groups without assuming a distribution. They sort all the values and work with their position. SigmaPro computes their **exact distributions** when the size allows it, as R does. None of the three has a traffic light.

Test	Equivalent to	When
Wilcoxon and Mann-Whitney	The t test	Two independent groups (rank sum, Mann-Whitney), two paired measurements or one sample against a value (signed ranks).
Kruskal-Wallis	The completely randomized ANOVA	Three or more treatments without blocks.
Friedman	The block ANOVA	Three or more treatments with one value of each in each block (plot, judge, date).

Wilcoxon and Mann-Whitney

The form is that of the t test (without the variances checkbox), with two more options: p-value (Exact if possible, Exact or Normal approximation) and Continuity correction, checked by default, for the normal approximation. With Exact if possible, the exact p is used with fewer than 50 values per group and no ties (nor differences equal to the reference value, in the signed ranks), as in R; with ties the app uses the normal approximation even if the exact one is requested, and the table note says which was used.

The result gives the statistic (W for two groups, V for paired data and one sample), the p and, as the effect measure, the **Hodges-Lehmann estimator** of the shift between the groups (the median of all the differences between one value from each group; for paired data and one sample, the pseudomedian) with its interval, plus a table of medians and quartiles and a box plot. If, with so few data, the interval cannot reach the requested confidence, a note says which one it reaches. With treatments A and B of the poisons, which have ties: W = 20.5, p = 0.003, and a shift of −0.360 (95% CI: −0.610 to −0.090), with the normal approximation corrected for continuity and for ties.

Kruskal-Wallis

Kruskal-Wallis computed in 4 ms ⓘ What is this analysis? Present Add to report

Basic Standard Expert

In a nutshell

- At least one treatment differs from the others in "tiempo".
- Dunn comparisons (Holm): 2 of 3 pairs differ.

At least one treatment differs from the others in "tiempo" ($H = 25.584$, $df = 2$, $p < 0.001$).

Effect size: η^2 from $H = 0.524$.

Dunn comparisons (Holm): 2 of 3 pairs differ at $\alpha = 0.05$. Treatments sharing a letter do not differ.

Chart theme SigmaPro (gold and black)

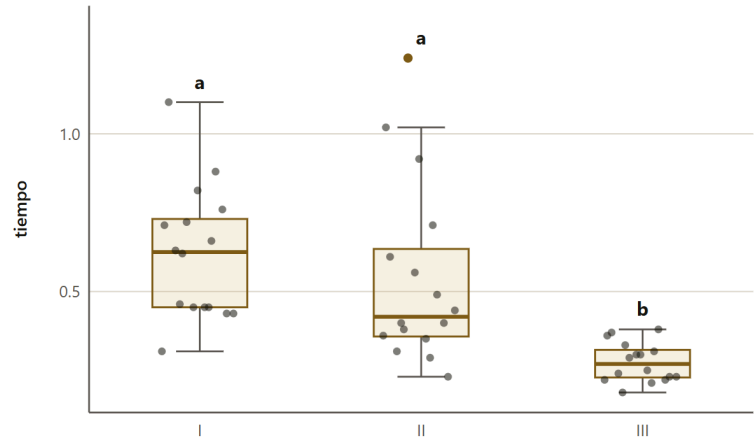


Table 1. Treatments sorted by mean rank

Treatment	n	Median	Q1	Q3	Mean rank	Group
I	16	0.63	0.45	0.73	34.84	a
II	16	0.42	0.36	0.64	28.06	a
III	16	0.27	0.23	0.32	10.59	b

Figure 4.22. Kruskal-Wallis with the three poisons: poison III shortens survival compared with I and II (different letters).

You choose the Response variable, the Treatments, the Pairwise comparisons method (Dunn, Conover-Iman or Nemenyi) and the p-value adjustment for multiple comparisons: Holm (the default), Bonferroni, Benjamini-Hochberg (FDR) or No adjustment; with Conover-Iman, also Tukey (single-step). Nemenyi needs no adjustment: its p already comes from the studentized range. The result gives H with its degrees of freedom and its p (corrected for ties), the effect size η^2 from H, and the comparisons: the Treatments sorted by mean rank table, with the median, the quartiles and their **letters** (treatments that share a letter do not differ), which also appear on the chart, and the table for each pair, with the difference in mean ranks, the statistic, the unadjusted p and the adjusted one.

Friedman

It also asks for the Blocks column, with one value per treatment in each block. The pairwise comparisons can be Conover, Nemenyi or paired Wilcoxon, with the same adjustments; Tukey (single-step) only with Conover. Besides χ^2 and its p, it gives **Kendall's W**, the agreement among blocks (from 0, none, to 1, all blocks rank the treatments the same way), and the table of treatments sorted by their rank sum, with their letters.

The data must form a complete table of blocks by treatments:

- A block with an empty response in some treatment is left out, as in R, and a note says how many. With the avocado thrips, two of the six blocks have a plot with no value: the test uses the other four ($\chi^2 = 8.308$, df = 4, $p = 0.081$; $W = 0.519$) and the chart shows only those blocks.
- If the row of some block and treatment combination is missing, the app stops: "Friedman needs the same treatments in all blocks, and some combinations are missing".
- If a treatment has more than one value in a block (replicates, or a factorial in blocks), it also stops, and proposes averaging the replicates of each treatment in each block in the spreadsheet, importing that table of means and trying again. That is what happens with the maize if you ask for the hybrids in blocks: each block has four plots of each hybrid, one per rate.

> **WHAT YOU LOSE WITH RANKS**
>
> Rank tests say whether one group tends to have higher values than another, not how much their means differ, and with normal data they have somewhat less power than ANOVA. Before turning to them, it is worth trying the transformation suggested by the assumption check (section 4.7), or a generalized linear model if the response is a count or a proportion (section 4.17).

4.10 Experimental designs

The analysis of variance of an experimental design splits the variation in the response into the part due to the treatments, the part due to the way the trial was laid out (blocks, rows, columns, main plots) and the part left unexplained (the error). Each design has its own model and, in some, more than one error. SigmaPro includes the eight designs most used in agricultural research and, for whatever does not fit them, the general linear model.

Design	Model	When
Completely randomized	`y ~ A * B * C`	Homogeneous units (pots in a growth chamber, germination boxes); up to three crossed factors.
Randomized complete blocks	`y ~ bloque + A * B * C`	The most common in the field: each block contains all the treatments and groups similar plots.
Latin square	`y ~ fila + columna + A * B`	Variation in two directions (slope and fertility); each treatment once in each row and in each column.
Split plots	`Error(bloque/A)`	One factor that can only be applied to large plots (irrigation, tillage, sowing date) and another within them.
Split-split plots	`Error(bloque/A/B)`	Three factors, each one on plots smaller than the previous one.
Strip plots	`Error(bloque/(A + B))`	Two factors applied in strips that cross each other (tillage in one direction, irrigation in the other).
Balanced incomplete blocks	`y ~ bloque + trat`	More treatments than plots per block, with each pair of treatments together the same number of times.
Lattice and alpha-lattice	`y ~ rep + rep:bloque + trat`	Many genotypes (plant breeding): incomplete blocks within complete replicates.
General linear model	The formula you write	Whatever does not fit the designs above: nested factors, covariates with interactions, extra factors.

The form

They all ask for the Response variable and the roles of the design. Each field is named as on screen:

Design	Roles
Completely randomized, blocks	Treatment factors (up to 3, crossed); in blocks, also Blocks.
Latin square	Treatment factors (up to 2), Rows and Columns.
Split plots	Main plots are arranged in (randomized complete blocks, a completely randomized design or a Latin square), Main-plot factor, Subplot factor and, depending on how the main plots are arranged, Blocks, the Main-plot replicate (the number of each main plot within its treatment) or Rows and Columns.
Split-split plots	Main-plot factor, Subplot factor, Sub-subplot factor and Blocks.
Strip plots	Horizontal strip factor, Vertical strip factor and Blocks.
Balanced incomplete blocks	Treatments (one factor) and Blocks.
Lattice and alpha-lattice	Treatments (one factor), Replicates and Incomplete blocks (within each replicate).

A numeric column with few values (a rate) appears marked "(numeric)" and is used as a factor. The completely randomized, block and Latin square designs also offer the Sums of squares (Type III (marginal means), Type II (marginality) or Type I (sequential)); incomplete blocks always use the sequential ones (see below), and all of them ask for the Significance level. When a column identifies the plots, the assistant (section 4.2) recognizes split plots, split-split plots and strip plots in the data and proposes them.

> **Σ THE THREE TYPES OF SUMS OF SQUARES**
>
> With **balanced** data (the same number of observations in each combination), the three types give the same result, and the table note says so. With unbalanced data (missing plots, treatments with different numbers of replicates), each type tests a different hypothesis:
>
> - **Type I** (sequential): each term adjusted for those that come before it; the order matters.
> - **Type II** (marginality): each term adjusted for the other terms that do not contain it; it is the most powerful if there is no interaction.
> - **Type III** (marginal means, the default option): each term adjusted for all the others, with sum-to-zero contrasts; it tests the marginal means with equal weights, which is almost always what you want in an experiment.

What the result includes

- The **conclusion** for each effect ("hibrido: there are differences (F = 60.635, p < 0.001)") and the trial indicators: grand mean, coefficient of variation (one per error in designs with strata: CV(a), CV(b)...) and, with a single error, R^2.
- The **chart**: with one factor, boxes with the points of each treatment and its mean; with two or more, the interaction plot of the first two, one line per level of the second (figure 4.7). Above it is the Chart theme selector (section 4.5).
- The Compare the means of this design button, which opens the mean comparison (section 4.11) with the same design and the same variables.

- The **tables**: the analysis of variance; the trial summary (grand mean, observations, CV, R^2 and the relative efficiency, where it applies); the **means** of each factor and of the combinations of the first two; and, with two or more factors in a single-error design, the **simple effects**: the *F* test of each factor within each level of the other. With a single error the means are the estimated marginal means, with their standard error (a mean that cannot be estimated appears as "—"); in designs with strata they are the averages of each level, with their number of values.
- At the bottom of the eight designs (not of the general linear model), the **Continue in the LABG suite** band suggests AgriDesign, which generates the randomization, the field layout and the field sheet of the design before sowing; with data loaded, its Send data to AgriDesign button downloads the table and opens the app (chapter 10).

With two or more factors, if the interaction is significant the conclusion warns about it. In single-error designs it refers you to the simple effects and to the means of the combinations; in designs with strata, which do not have the simple-effects table, it says to compare one factor within each level of the other in Mean comparisons, with Within each level of (section 4.11).

B4

Relative efficiency

The trial summary says how much the trial gained by blocking (Gomez and Gomez, 1984), as a percentage and with the degrees-of-freedom adjustment factor when the error has fewer than 20:

- In **randomized complete blocks**, relative to completely randomized. In the maize it is 160%: a completely randomized design would have needed 60% more replicates for the same precision. It is not computed with missing plots or with generalized blocks.
- In the **Latin square**, three: relative to completely randomized, relative to blocks with the rows as blocks and relative to blocks with the columns as blocks. With the rows as blocks, the variation among columns would go into the error: that is why that efficiency depends on the mean square of the columns (and the other one, on that of the rows). In the wheat, with "hilera" (the planting rows) as the rows: 128.4% relative to completely randomized, 111.1% with the rows as blocks and 123.7% with the columns as blocks; blocking by columns alone would have lost more. It is computed with the complete square (t^2 plots).

Some situations are recognized automatically and explained in a notice: more than one plot per treatment in each block (generalized blocks: the block × treatment interaction is estimated), missing block and treatment combinations (the means are adjusted), empty cells (the type III sum of squares does not exist and type II is used) or a term confounded with another.

Split plots, split-split plots and strip plots

In these designs each factor was applied to units of different sizes, so each one has its own error: the main-plot factor is tested against **error a**, and the subplot factor and the interaction against **error b** (the variation within main plots). Error a is block × main plot when the main plots are in blocks, main plot × replicate if they are completely randomized, and row × column if they are in a Latin square. The table note says which error each term is tested against, and the blocks are not tested because their stratum has no error of its own. Analyzing these trials as an ordinary factorial would use a single error and give false *F* tests, almost always in favor of the main-plot factor.

Split-split plots add a third stratum (`Error(bloque/A/B)`, with errors a, b and c), and **strip plots** have three errors: a for the horizontal strip factor (block × A), b for the vertical one (block × B) and c for their interaction. With the maize laid out in strips, for example, errors a, b and c have 6, 9 and 18 degrees of freedom. In split-split plots, the table of means of the combinations and the interaction plot show only the first two factors (A × B); the other interactions are studied in Mean comparisons.

The example Rice: irrigation × variety × nitrogen (split-split plot), fictitious and for practice, shows this (figure below): irrigation, in the main plots, is tested with only 2 degrees of freedom of error a (F = 95.839, p = 0.010); the irrigation × nitrogen interaction is significant (F = 23.273, p < 0.001), and the conclusion asks you to compare one factor within each level of the other; the chart and the table of combinations, on the other hand, show irrigation × variety. The coefficients of variation decrease from error a to error c: 8.5, 3.1 and 2.2%. For split plots there is another practice example, Wheat: irrigation × variety (split plot).

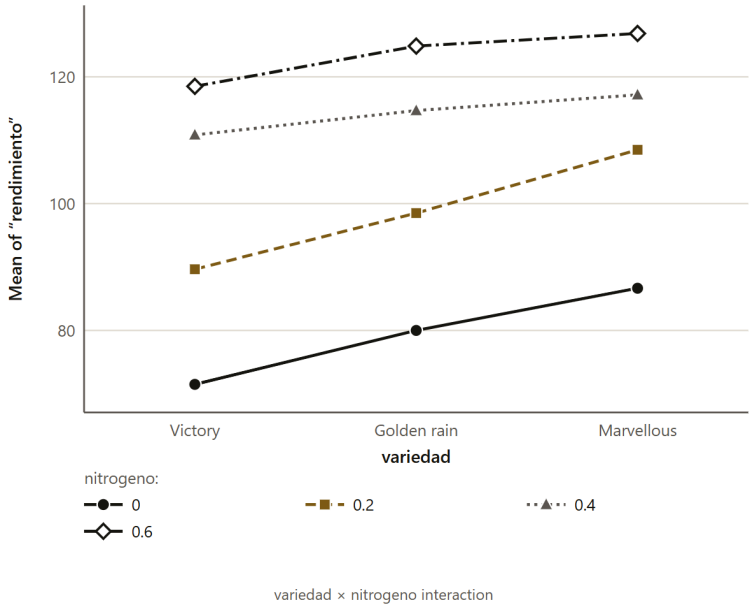
variedad: no evidence of differences (F = 1.485, p = 0.272).

nitrogeno: there are differences (F = 37.686, p < 0.001).

variedad × nitrogeno: no evidence of differences (F = 0.303, p = 0.932).

Overall mean 103.972; CV(a) = 23.6%, CV(b) = 12.8%.

Chart theme SigmaPro (gold and black)



Compare the means of this design

Table 1. Analysis of variance: Split plot

Source of variation	df	SS	MS	F	p
bloque	5	15875.2778	3175.0556	—	—
variedad	2	1786.3611	893.1806	1.49	0.272
Error a	10	6013.3056	601.3306	—	—
nitrogeno	3	20020.5000	6673.5000	37.69	< 0.001
variedad × nitrogeno	6	321.7500	53.6250	0.30	0.932
Error b	45	7968.7500	177.0833	—	—
Total	71	51985.9444	—	—	—

Figure 4.23. Yates's (1935) oats in split plots: nitrogen, in the subplots, has a clear effect; the varieties, in the main plots, are tested against error a, with only 10 degrees of freedom.

With unbalanced data (missing or extra plots), the *F* tests of the strata are no longer exact and the app warns about it; in that case the mixed model of section 4.15, which handles the imbalance, is the better choice. If an error term is confounded with another, a notice says that the error model is singular.

riego: there are differences (F = 95.839, p = 0.010).

variedad: there are differences (F = 36.254, p < 0.001).

riego × variedad: no evidence of differences (F = 4.068, p = 0.060).

nitrogeno: there are differences (F = 459.841, p < 0.001).

riego × nitrogeno: there are differences (F = 23.273, p < 0.001).

variedad × nitrogeno: no evidence of differences (F = 0.496, p = 0.621).

riego × variedad × nitrogeno: no evidence of differences (F = 0.806, p = 0.469).

The riego × nitrogeno interaction is significant: the effect of one factor depends on the level of the other; compare one factor within each level of the other (Mean comparisons, "Within each level of"), not the main effects separately.

Overall mean 5.938; CV(a) = 8.5%, CV(b) = 3.1%, CV(c) = 2.2%.

Compare the means of this design

Table 1. Analysis of variance: Split-split plot

Source of variation	df	SS	MS	F	p
bloque	2	12.1080	6.0540	—	—
riego	1	24.5685	24.5685	95.84	0.010
Error a	2	0.5127	0.2564	—	—
variedad	2	2.4122	1.2061	36.25	< 0.001
riego × variedad	2	0.2707	0.1353	4.07	0.060
Error b	8	0.2661	0.0333	—	—
nitrogeno	1	8.0089	8.0089	459.84	< 0.001
riego × nitrogeno	1	0.4053	0.4053	23.27	< 0.001
variedad × nitrogeno	2	0.0173	0.0086	0.50	0.621
riego × variedad × nitrogeno	2	0.0281	0.0140	0.81	0.469
Error c	12	0.2090	0.0174	—	—
Total	35	48.8069	—	—	—

Figure 4.24. The split-split plots of the rice (fictitious data): one error per stratum. The irrigation × nitrogen interaction is significant and the conclusion directs you to compare them within each level. The interaction plot, which the app shows between the conclusion and the table, is omitted here.

Incomplete blocks: BIB, lattice and alpha-lattice

When there are more treatments than plots fit in a homogeneous block, each block receives only some of them. The **intra-block** analysis compares the treatments within the blocks: their sum of squares is sequential (type I), with the treatments last, **adjusted for the blocks**, and their means are adjusted means. The table note recalls this. The *F* of the blocks, on the other hand, is not adjusted for the treatments and is only an indication.

In balanced incomplete blocks, the trial summary also includes the **efficiency factor** $E = \lambda t/(rk)$ and λ, the number of times each pair of treatments occurs together in a block. With the example Sorghum: seven lines in incomplete blocks (BIB), fictitious and for practice (7 lines in 7 blocks of 3), the lines differ ($F = 8.604$, $p = 0.004$), with $E = 0.778$ and $\lambda = 1$. If the design is not an exact BIB (blocks of different sizes or pairs that do not occur together the same number of times), the app says so and does not compute the efficiency; the intra-block analysis remains valid.

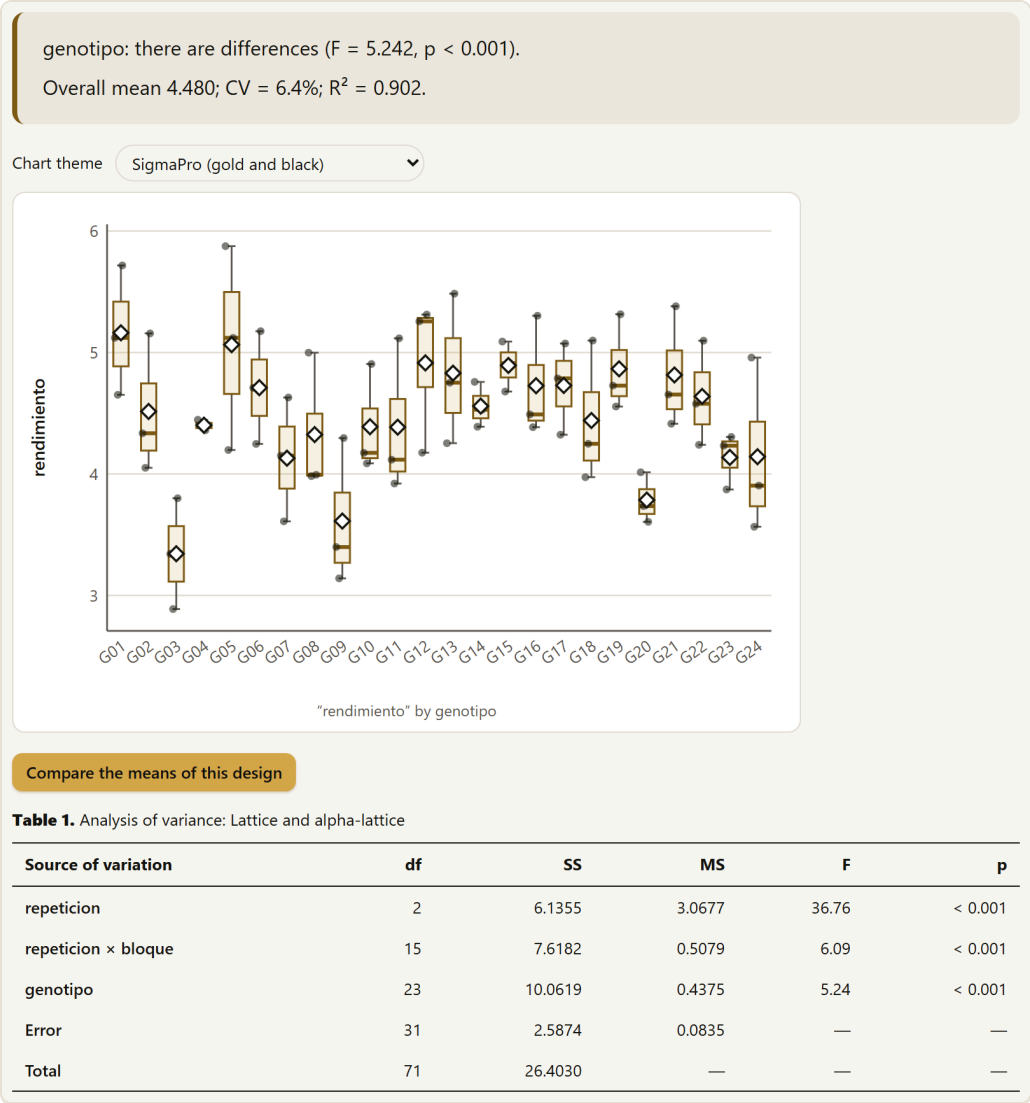


Table 1. Analysis of variance: Lattice and alpha-lattice

Source of variation	df	SS	MS	F	p
repeticion	2	6.1355	3.0677	36.76	< 0.001
repeticion × bloque	15	7.6182	0.5079	6.09	< 0.001
genotipo	23	10.0619	0.4375	5.24	< 0.001
Error	31	2.5874	0.0835	—	—
Total	71	26.4030	—	—	—

Figure 4.25. The alpha-lattice of 24 genotypes (Olivoto and Lúcio, 2020). Genotype goes last in the table, adjusted for the incomplete blocks.

In these two designs, the box plot shows the raw data of each treatment, not adjusted for the blocks; the means you should publish are the adjusted ones in the table, with their standard error. The intra-block analysis wastes the information there is between blocks. To recover it, with blocks as a random effect, there is Lattice and alpha-lattice (inter-block recovery), among the mixed models (section 4.15); for the heritability of the genotypes, see section 4.16.

The general linear model

For models that are not a design in the catalog, General linear model (formula) asks for the Response variable and the Model terms (what goes after "~"), with the notation of section 4.4; below, the app explains how it read each term. It also asks for the Numeric columns that are treatments (factor), the Sums of squares, the Confidence level and the Significance level. It does not accept error strata (`Error(…)`: that is what the split-plot designs are for) or functions inside the formula (the transformation is done in the Data block); without an intercept, it uses type I sums of squares.

It delivers the analysis of variance, the **coefficients** (with treatment contrasts: each level compared with the first), a chart of the marginal means (of the first interaction between two factors or, if there is none, of the factor with the largest *F*; with no factors, observed vs. fitted values) and the means of each factor with their confidence interval. It does not include the trial summary, the efficiency or the suite band. It is also what the model builder uses when the formula is written by hand.

4.11 Mean comparisons and contrasts

An analysis of variance tells you whether the treatments differ, not which ones. That is what the four methods of this family are for; they work on the model of any of the eight designs in the previous section: the form starts with the Experimental design and its roles, and then asks what to compare.

Method	What for
Mean comparisons	All the treatments with one another (or against a control), with the group letters.
Contrasts	Comparisons planned in advance (the control against the rest, one group against another).
Orthogonal polynomials	The shape of the response to a quantitative factor (a rate): linear, quadratic, cubic, and the optimum rate.
Adjusted means	The estimated marginal means with their confidence interval, without comparing them.

Two fields are common to all of them. The first is what is compared: in mean comparisons and adjusted means, Means of, with one checkbox per factor (one or two at a time; with two, their combinations are compared, such as "H-66 : 180"); in contrasts, Factor, and in polynomials, Quantitative factor (rate). The second is Within each level of, which by default says — all together — and accepts a single factor. If the interaction is significant, comparing the means of one factor averaged over the other can be misleading; you then compare one factor **within each level** of the other (the rates within each hybrid), and the result includes a conclusion, its tables and a chart for each level (up to six charts).

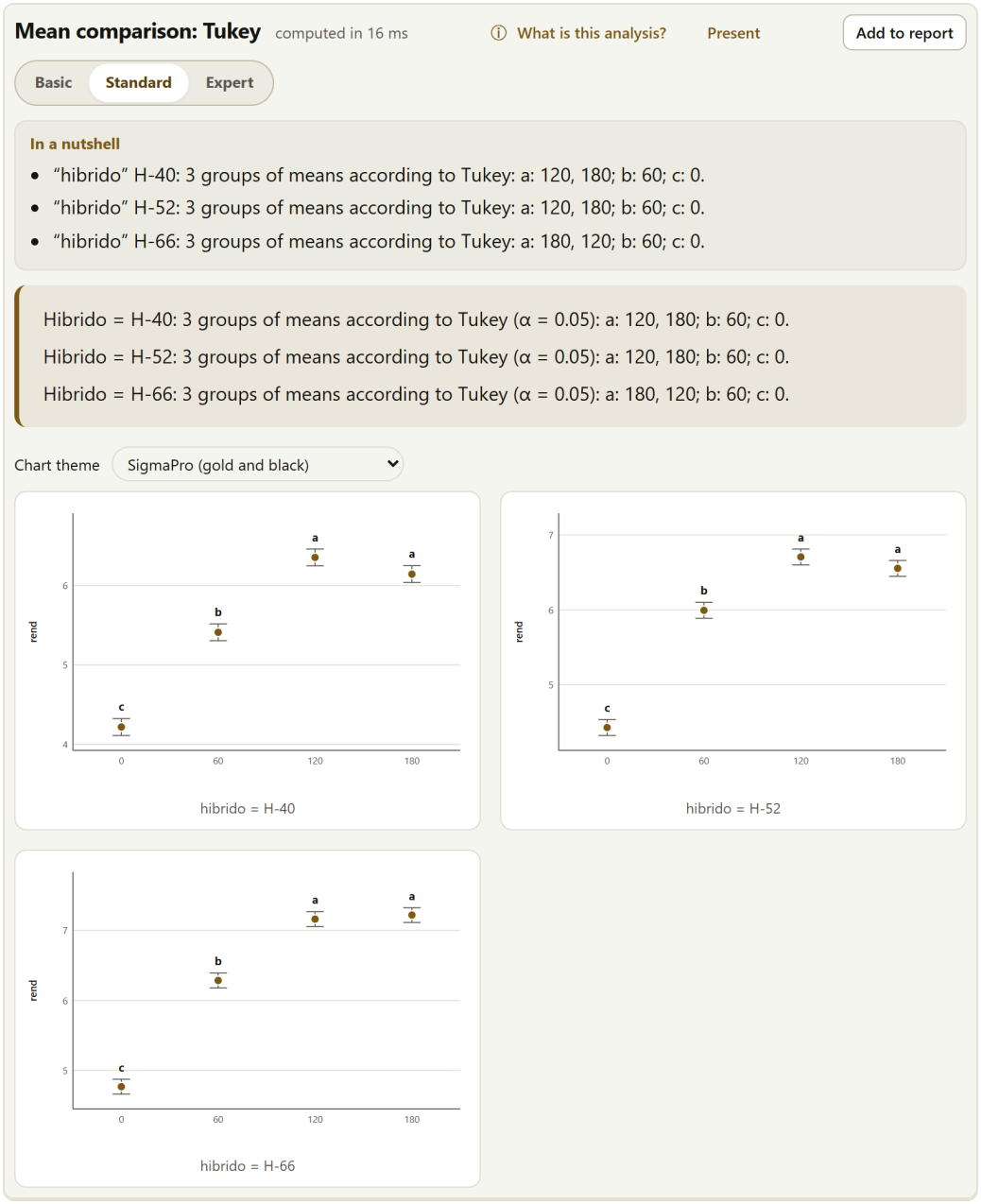


Figure 4.26. "Within each level of": the maize rates compared within each hybrid. In all three, 120 and 180 kg/ha do not differ from each other and outperform 60 and 0.

Mean comparisons

The Test chooses how the error of making many comparisons at once is controlled:

Experimental design: Randomized complete blocks

Response variable: rend

Treatment factors (up to 3, crossed): hibrido, dosis (numeric)

Blocks: bloque

Means of: hibrido

Within each level of: — all together —

With interaction, compares one factor within each level of the other.

Test: Tukey

Significance level: 0.05

Mean comparison: Tukey computed in 8 ms ⓘ What is this analysis? Present Add to report

Basic Standard Expert

In a nutshell

- 3 groups of means according to Tukey: a: H-66; b: H-52; c: H-40.

3 groups of means according to Tukey (α = 0.05): a: H-66; b: H-52; c: H-40.

Chart theme: SigmaPro (gold and black)

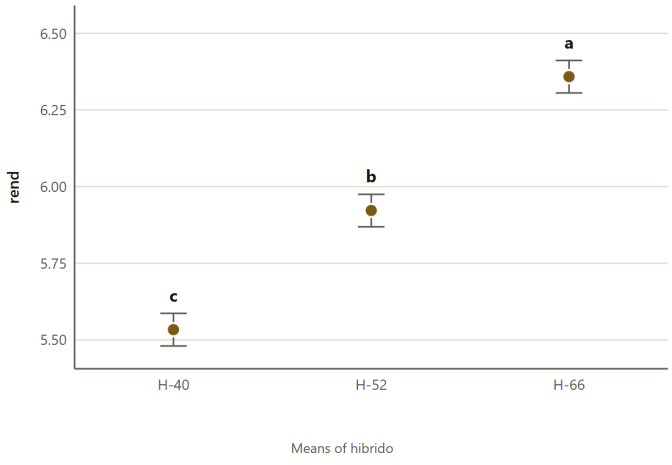


Table 1. Means of hibrido with their group (Tukey)

hibrido	Mean	SE	Group
H-66	6.359	0.053	a
H-52	5.922	0.053	b
H-40	5.533	0.053	c

Means sharing a letter do not differ (α = 0.05).

Copy table Download as CSV

Table 2. Pairwise comparisons: Tukey

Pair	Difference	SE	df	t	adjusted p	Differ?
H-40 – H-52	−0.389	0.075	33	−5.18	< 0.001	yes
H-40 – H-66	−0.826	0.075	33	−11.01	< 0.001	yes
H-52 – H-66	−0.437	0.075	33	−5.82	< 0.001	yes

Figure 4.27. Tukey with the maize hybrids: all three differ from one another. The letters appear on the chart and in the table; means that share a letter do not differ.

Test	How it works
Tukey	All pairs, with the error of the whole family of comparisons controlled (Tukey-Kramer if the number of replicates differs). The default option and the most recommended for comparing everything against everything.
Fisher's LSD and Protected LSD	The least significant difference, without adjustment for multiplicity (it finds more differences, including false ones). The protected version only separates means if the *F* of that term was significant; if not, all the means get the same letter and a note says so. That *F* comes from the analysis with sequential (type I) sums of squares, which in an unbalanced factorial may not be the type III one shown by the design.
Duncan and Student-Newman-Keuls	Stepwise tests, traditional in agronomy; they control the error less well than Tukey. They use a common standard error (with the harmonic mean of the replicates) and the *q* statistic.
Scheffé	Tests the pairs with Scheffé's criterion (*F* with $k - 1$ df), which protects any contrast; that is why it is very conservative.
Bonferroni, Šidák and Holm	General adjustments of the LSD *p*-value; Holm is the most powerful of the three.
Dunnett (vs. a control)	Each treatment against a **control**, chosen in Control; the Hypothesis can be Different from control, Greater than control or Less than control. More powerful than Tukey for that question.
Games-Howell (unequal variances)	All pairs without assuming equal variances: each pair with its own error and its own degrees of freedom. Only with one factor in a completely randomized design.

The result includes the conclusion with the groups ("a: H-66; b: H-52; c: H-40"), the chart of the means with their standard error and their letters, the table of means sorted from largest to smallest with their **group**, and the table of **pairwise comparisons** (difference, SE, df, the *t* statistic —or *q* in Duncan, SNK and Games-Howell—, adjusted *p* and whether they differ). The means are the estimated marginal means of the design's model and the error is that of the design: in a split plot, each comparison uses the error that corresponds to it and Satterthwaite degrees of freedom, and the table note says so. In designs with strata, the table of means does not include their standard error and the chart has no bars.

With the rice of the previous section, for example, Means of nitrogen ("nitrogeno") Within each level of irrigation ("riego") shows the interaction: the N80 rate outperforms N0 by 0.731 under "Temporal" (rainfed) and by 1.156 under "Riego" (irrigated) (both with $p < 0.001$, with the 12 df of error c).

Dunnett

Dunnett's *p*-values are exact and single-step: they come from the multivariate *t* distribution of the comparisons against the control and are computed in a few seconds, also in balanced incomplete blocks. With the avocado thrips (complete blocks with two missing plots), the hypothesis Less than control finds that only Spinosad reduces thrips relative to the control, "Testigo" (−6.333 thrips per shoot, $p = 0.016$).

WHEN A TEST DOES NOT RUN

- **Dunnett** needs a single error. In split plots, split-split plots or strip plots, the app stops and explains that Bonferroni and Holm, which adjust all pairs, can be used instead, although against the control they are conservative. It also stops when the comparisons against the control do not have the correlation structure that the exact calculation requires, as in the alpha-lattice of 24 genotypes or with certain missing plots: the message proposes the same.
- **Duncan** and **Student-Newman-Keuls** also need a single error; in split plots, the app proposes Tukey, LSD or Bonferroni.
- **Games-Howell** is only for one factor in a completely randomized design, without Within each level of; in other designs, the app proposes Tukey or Bonferroni.
- **Adjusted means** needs a single error; in split plots, split-split plots or strip plots, the means are compared with Mean comparisons.

Contrasts

A contrast compares groups of treatments with coefficients that add up to zero. They are written in the Contrasts (one per line) field, in the form `nombre: coeficientes` (name: coefficients), in the order of the levels that the help shows below it (the coefficients are separated by spaces, commas or semicolons; a line without a name is called "Contrast 1", "Contrast 2"...). With the field empty, the app shows the first level against the rest as an example:

```
I y II vs III: 1 1 -2
I vs II: 1 -1 0
```

Table 1. Contrasts of "veneno"

Contrast	Coefficients	Estimation	SE	df	t	p	SS
I y II vs III	1 1 −2	0.609	0.091	36	6.67	< 0.001	0.9902
I vs II	1 −1 0	0.073	0.053	36	1.39	0.174	0.0428

Figure 4.28. Two orthogonal contrasts among the poisons: III differs from the average of I and II; I and II do not differ.

Each contrast includes its estimate, SE, df, *t*, *p* and, in single-error designs, its sum of squares. If the contrasts are **orthogonal** to each other (with the same number of replicates per level, their coefficients multiplied term by term add up to zero; the app checks this with the actual replicates of each level), their sums of squares partition that of the treatments without overlapping, and the note says so; if not, it warns that they overlap. A line that does not add up to zero is also computed, with the note that it is a linear combination and not a contrast. Contrasts are planned before looking at the data; chosen afterwards, they have the problem of multiple comparisons.

Orthogonal polynomials

With a quantitative factor (nitrogen rates, plant densities, temperatures), what matters is not the differences between rates but the **shape of the response**. Orthogonal polynomials split the variation among rates into a linear component, a quadratic one, a cubic one… (as many as the number of rates minus one), each with its own test. You choose the Quantitative factor (rate), whose levels must be numbers (at least three), and the Curve degree (Straight line, Quadratic (with the optimum rate) or Cubic), which decides the fitted curve; the table always includes all the components.

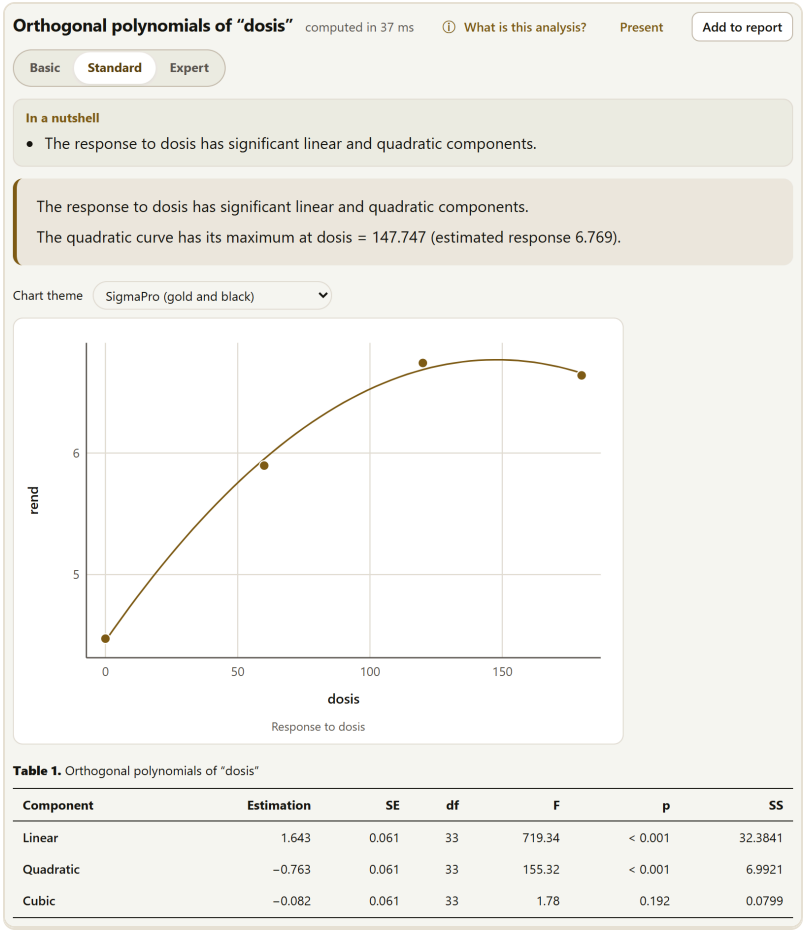


Figure 4.29. The response of the maize to nitrogen is quadratic; the maximum is at 147.7 kg/ha, with an estimated 6.77 t/ha.

The orthogonal coefficients are computed on the actual levels of the rate, even if they are not equally spaced. With the quadratic curve, the app fits the equation to the means (y = 4.454 + 0.03133·x − 1.060·10^{-4}·x^2 in the maize, with R^2 = 0.998) and gives the rate at the maximum or minimum; if it falls outside the rates tested, it warns about it, because extrapolating is not advisable. With Within each level of there is one curve per level: in the maize, the maximum of each hybrid is at 147.5 (H-40), 139.5 (H-52) and 157.7 kg/ha (H-66). For a response that levels off at a plateau, the linear-plateau model (section 4.18) is the better choice.

Adjusted means

The **estimated marginal means** are the averages that the model predicts for each level, with equal weights over the other factors. With balanced data they are the usual means; with missing plots, incomplete blocks or covariates, they are the ones you should publish. For the chosen factor (and within each level of another, if requested), the method gives the mean, its SE, the degrees of freedom and its confidence interval at the chosen Confidence level, with a chart of the means and their intervals. In the alpha-lattice, for example, the 24 adjusted means have standard errors from 0.194 to 0.195, with 31 df.

4.12 Analysis of covariance and regression

When the response depends on a numeric variable measured on each plot (the number of plants, the altitude of the site, the rate), the question changes: it is no longer enough to know whether the treatments differ; you want to know how much the response changes per unit of that variable. In the catalog, the ANCOVA and regression family has five methods.

Method	What for
Analysis of covariance (ANCOVA)	Compare treatments after discounting one or two covariates that the treatments do not control (plants harvested, initial weight, the previous year's yield).
Simple and multiple linear regression	Explain or predict the response with one or several numeric predictors, with full diagnostics.
Polynomial regression	A curve of degree 2 to 4 on a single variable (the response to the rate), with its maximum or minimum.
Stepwise selection	Keep the predictors that matter among many candidates, by AIC or by *p*-value.
Best subsets	Review every combination of predictors and compare the best models of each size.

Analysis of covariance

The form asks for the Response variable, the Treatment factors (up to 2, crossed), the Blocks (with — no blocks (completely randomized) — if the trial has none) and the Covariates (up to 2). The options are the Sums of squares (Type III (marginal means) by default, Type II (marginality) or Type I (treatments adjusted for the covariate)), the Comparison of adjusted means (Tukey, Fisher's LSD, Bonferroni, Šidák, Holm, Scheffé or None), the Confidence level and the Significance level. With empty cells, the type III sum of squares does not exist: type II is used, and the note under the table says so.

The sorghum example compares six varieties (five lines and a control) in four blocks; the harvested plants were counted in each plot, and the count was not the same in all of them. A plot with more plants yields more, regardless of the variety: the covariate separates the two things.

Σ THE TWO ASSUMPTIONS OF COVARIANCE ANALYSIS

The traffic light of the result checks the two assumptions specific to ANCOVA:

- **Equal slopes.** The covariate must weigh the same in all treatments: the model with a common slope is compared against one with a slope per treatment (in the sorghum, $F = 0.649$ with 5 and 9 df, $p = 0.670$). If the slopes differ, the effect of the variety depends on the number of plants, and a single adjusted mean does not summarize it.
- **The covariate does not depend on the treatments.** The covariate is analyzed with the same design ($F = 1.391$ with 5 and 15 df, $p = 0.283$). If the treatments changed it (a variety that emerges poorly), adjusting for it would take away part of the variety's real effect.

The traffic light does not test the normality or the constant variance of the residuals, and Check the model assumptions (section 4.7) does not accept covariates. As an approximation, check the residuals of the same design without the covariate there, and look at the plot to see that the points of each treatment follow their line with an even spread; if the relationship with the covariate is curved, transform the covariate in the Data block (section 2.9).

The **adjusted means** are the values each treatment would have if all its plots had the same number of plants (the overall mean of the covariate). They are the ones you publish, and they can change the order of the unadjusted means:

Table 4.3. Sorghum means, unadjusted and adjusted for the number of plants, with Tukey letters (α = 0.05), as SigmaPro gives them (table 4 of the result).

variedad	Unadjusted mean	Mean of plantas	Adjusted mean	SE	95% CI	Group
Línea 1	4.125	122.3	4.127	0.06143	3.995 – 4.259	a
Línea 2	3.442	110.0	3.770	0.07099	3.617 – 3.922	bc
Línea 3	3.520	126.8	3.403	0.06274	3.268 – 3.537	de
Línea 4	3.833	122.5	3.828	0.06143	3.696 – 3.960	b
Línea 5	3.593	124.3	3.542	0.06168	3.409 – 3.674	cd
Testigo	3.280	128.3	3.123	0.06376	2.986 – 3.260	e

Línea 2 had fewer plants (110 on average): unadjusted, it looked like one of the worst (3.442); adjusted, it falls in the second group (3.770). Línea 3, with more plants than average, drops from 3.520 to 3.403.

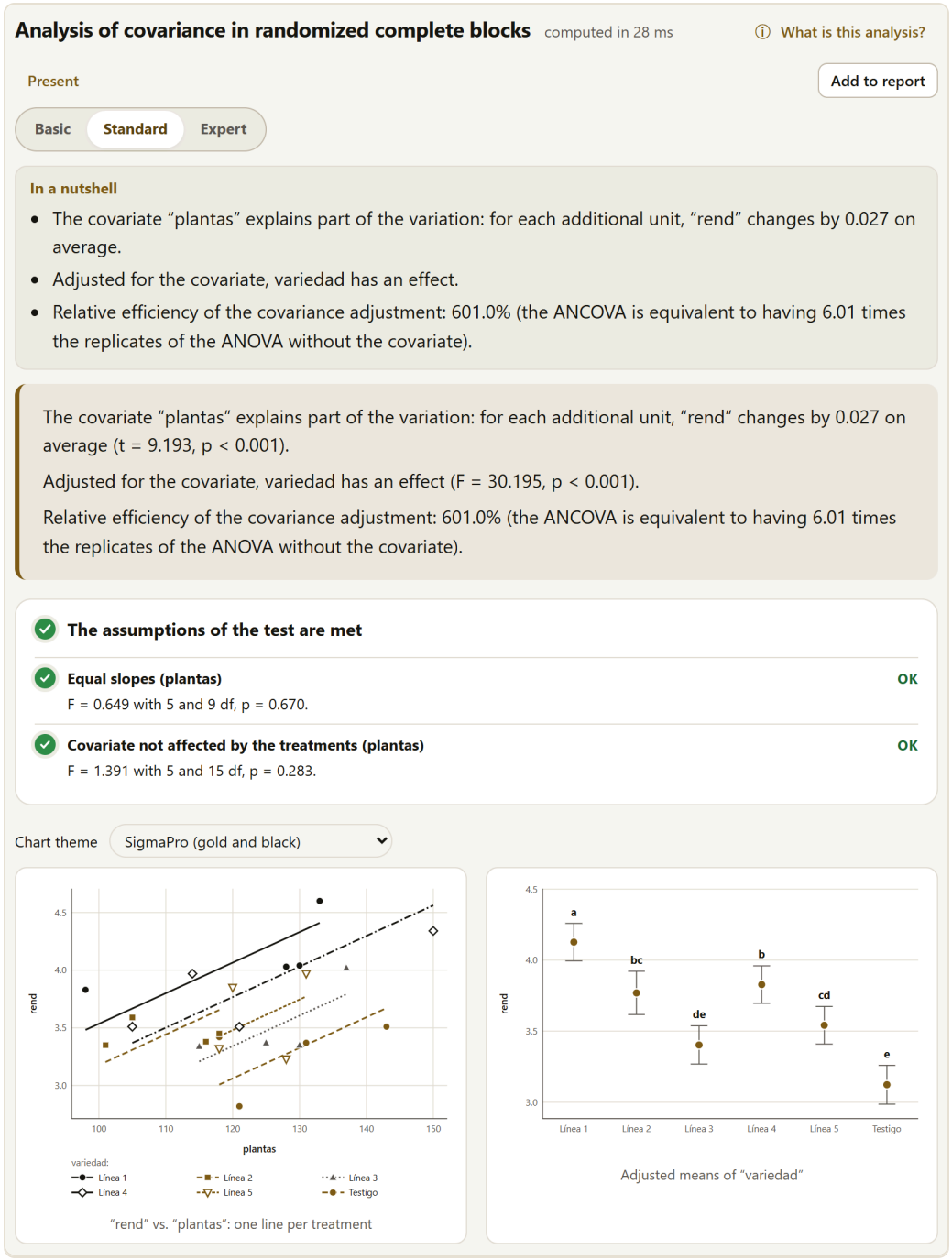


Figure 4.30. The sorghum covariance analysis: each additional plant adds 0.027 to yield; adjusted for plants, the varieties differ (F = 30.195, $p < 0.001$).

There are five tables. The first is the **analysis of covariance**, with the treatment effect already adjusted (variety, $F = 30.19$ with 5 and 14 df). The second, the **common slope of each covariate** (0.02652 per plant, $t = 9.19$, 95% CI: 0.02033 to 0.03270). The third, the **covariance assumptions**, with the two tests of the traffic light. The fourth, the adjusted means (table 4.3). The last, the **trial summary**: overall mean, CV, R^2 and the two error mean squares that give the **relative efficiency** of the covariance adjustment. In the sorghum, the error mean square drops from 0.0991 without the covariate to 0.0151 with it, and the ANCOVA is equivalent to having 6.01 times the replicates of the ANOVA without the covariate (Cochran and Cox, 1957). The efficiency is calculated only with one covariate; with two, the summary does not include it.

The two plots show the response against the covariate (the first one, if there are two), with one line per variety, all with the same slope, and the adjusted means with their letters. With two treatment factors, the analysis of covariance includes their interaction and, if it is significant, an additional table compares the first factor within each level of the second, with the adjusted means.

B4

Simple and multiple linear regression

Choose the Response variable (y) and the Predictors (up to 15); with one predictor it is simple regression, and with more, multiple regression. The predictors enter the model in the order in which they appear in the list, which is the order of the columns in the table, not the order in which you checked them. The Predict at (optional) field takes one line per point, with the values of the predictors in that same order, separated by spaces; the help under the field names them in order. The Confidence level and the Significance level complete the form. In the coffee example, the yield of 40 farms is explained by their altitude, the soil organic matter and the percentage of shade.

The conclusion says whether the model is significant as a whole, which predictors have an effect of their own (adjusted for the others) and how much it explains (R^2 and adjusted R^2). In simple regression, it says instead how much the response rises or falls per unit of the predictor, with its interval. The regression **traffic light** has six rows (if one fails, Try the alternative offers the corresponding remedy: section 4.7):

- **Normality of residuals**: Shapiro-Wilk.
- **Constant variance**: studentized Breusch-Pagan (Koenker).
- **Model form**: the RESET test (with $\hat{y}^2$ and $\hat{y}^3$), and lack of fit when the predictors have repeated values. In the coffee, RESET gives $F = 3.576$ and $p = 0.039$: perhaps a curve or an interaction is missing.

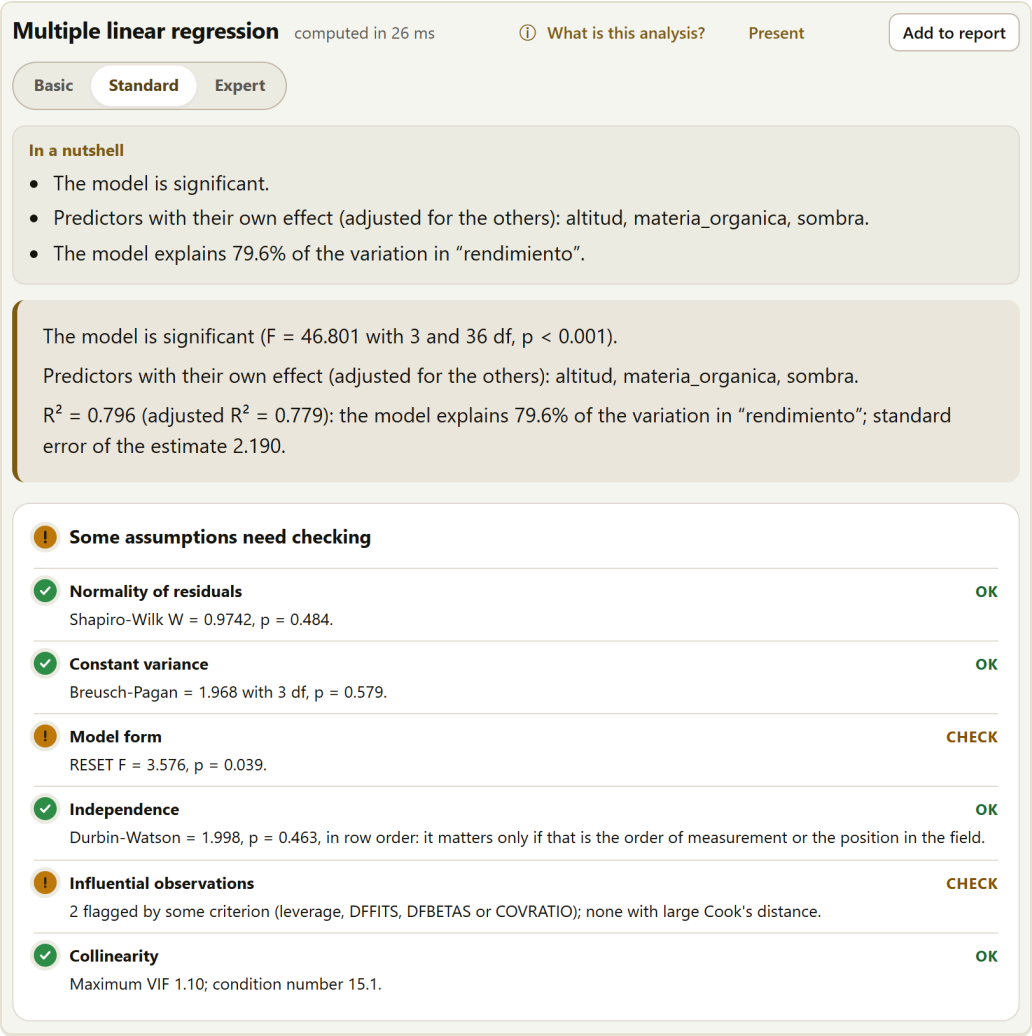


Figure 4.31. The coffee regression explains 79.6% of the variation in yield; the traffic light asks you to check the model form and two observations.

- **Independence**: Durbin-Watson, with an exact *p* if there are fewer than 100 observations. It measures autocorrelation *in the order of the rows*, so it only matters if that is the order of measurement or the position in the field; that is why this row never goes beyond amber.
- **Influential observations**: R's criteria (leverage, DFFITS, DFBETAS, COVRATIO and Cook's distance). The row turns red if any observation has a large Cook's distance, and amber if only the other criteria flag it. In the coffee, two farms, 17 and 6, are flagged by COVRATIO and none by Cook's distance.
- **Collinearity**, with two or more predictors: the VIF of each predictor and the condition number; the row turns amber with a VIF greater than 5 and red with one greater than 10 (maximum VIF 1.10 and condition number 15.1 in the coffee: no problem).

With a single predictor, the first plot is not observed against fitted but the **fitted line** over the data, with the confidence band of the mean (shaded) and the prediction band (dashed border); the other three are the same.

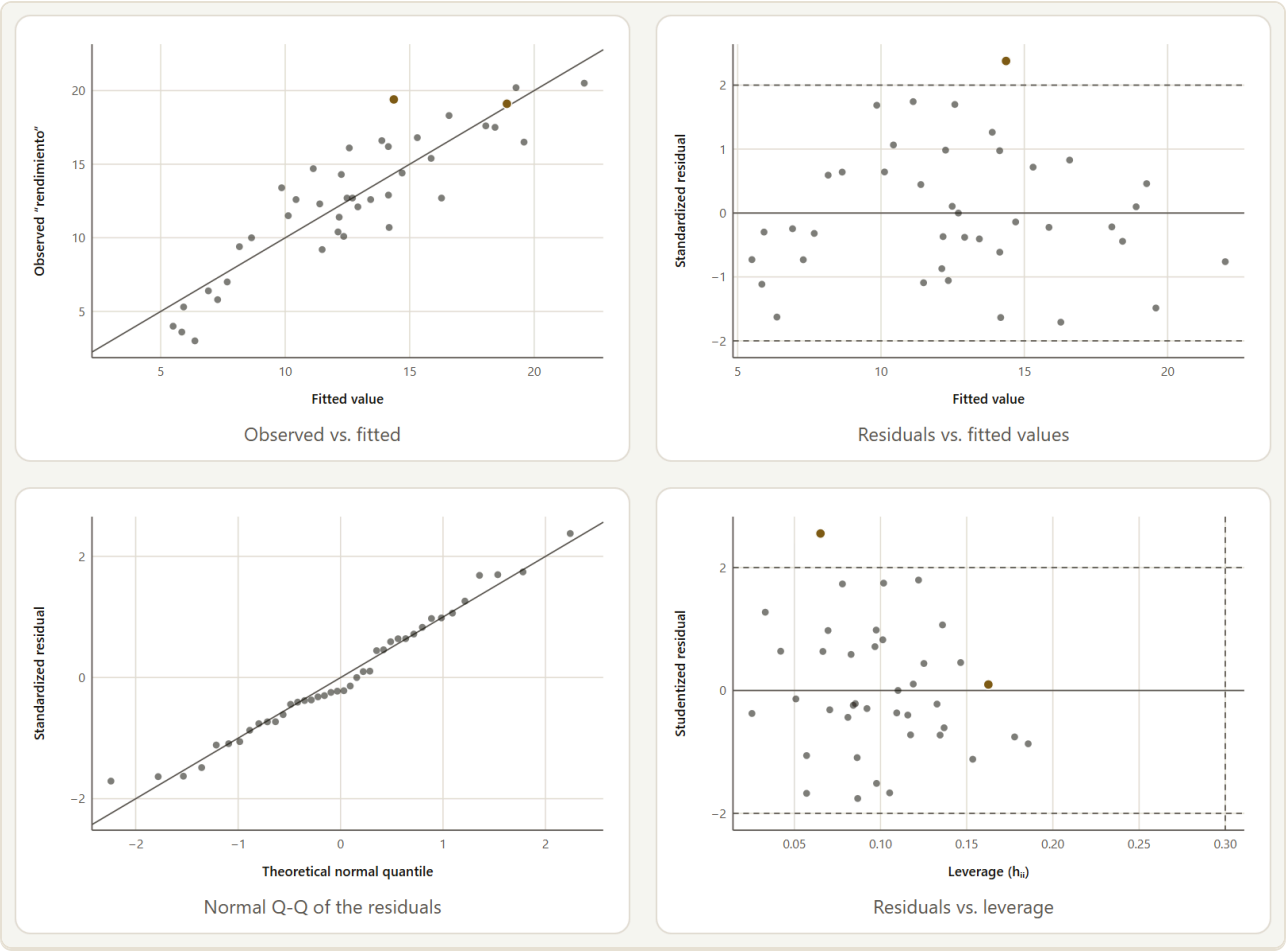


Figure 4.32. The four diagnostic plots. The two flagged farms appear in gold: 17, for its studentized residual of 2.55, and 6, with an almost null residual (0.10) but a leverage of 0.16; both are flagged by COVRATIO and neither by Cook's distance.

There are seven tables: the **coefficients**, with their interval, the standardized coefficient (β) and the VIF; the **analysis of variance** of the regression; the **goodness of fit** (R^2, adjusted R^2, standard error of the estimate, CV, AIC, BIC and the number of observations); the **contribution of each predictor**, with the sequential sum of squares (type I, in model order) and the partial one (type II); the **assumption tests**; the **influential observations**, with the criterion that flagged them (only if there are any); and the **condition indices** with the variance proportions of Belsley, Kuh and Welsch (1980), with two or more predictors. With values in Predict at (optional), the **predictions** table is added, with the confidence interval of the mean and the prediction interval of a new observation.

Table 4.4. The coefficients of the coffee regression (table 1 of the result).

Term	Estimate	SE	t	p	95% CI	β	VIF
(Intercept)	−7.324	2.164	−3.38	0.002	−11.71 – −2.934	—	—
altitud	0.01118	0.001467	7.62	< 0.001	0.008209 – 0.01416	0.59	1.07

Table 4.4. The coefficients of the coffee regression (table 1 of the result). *(continued)*

Term	Estimate	SE	t	p	95% CI	β	VIF
materia_organica	1.841	0.2157	8.53	< 0.001	1.403 – 2.278	0.65	1.02
sombra	−0.04835	0.02369	−2.04	0.049	−0.09640 – −0.0003028	−0.16	1.10

Each coefficient is the change in yield (in quintals of green coffee per hectare) per unit of the predictor, with the others held fixed: 1.841 per unit of organic matter, 0.01118 per meter of altitude and −0.04835 per unit of shade. The β values let you compare predictors measured on different scales: organic matter (0.65) and altitude (0.59) weigh much more than shade (−0.16).

Polynomial regression

It fits a curve to a single x variable, with the Polynomial degree you choose (2 (quadratic), 3 (cubic) or 4 (quartic)), and tests each degree: the last one that enters must be significant for the curve to need it. With the data of each maize plot, the quadratic curve of yield against rate explains 81.7% of the variation (F = 100.377 with 2 and 45 df) and has its maximum at 147.747 kg/ha, with an estimated 6.769 t/ha: the same maximum given by the orthogonal polynomials on the means (section 4.11), because the data are balanced. With degree 3, the conclusion says that the cubic term is not significant (p = 0.529) and that one degree less is enough. The maximum or minimum is sought in curves of degree 2 and 3 when their last term is significant; if it falls outside the range of the data, the conclusion says so and asks you not to extrapolate it.

Because the rate is repeated (four rates, twelve plots each), the app splits the error into **lack of fit** and **pure error**: the lack of fit is not significant (F = 0.402, p = 0.529), so the parabola describes the means of the four rates well. The first plot shows the curve with its confidence band and its prediction band, and the Predict at (optional) field takes x values separated by spaces. Everything else is as in linear regression: the traffic light (without the collinearity row), the other three diagnostic plots and the same tables, except for the β values, the VIF and the condition indices, which make no sense among the powers of a single variable.

> **WHEN DURBIN-WATSON LIGHTS UP FOR NO REASON**
>
> In the maize, the traffic light of the polynomial regression marks independence in amber (Durbin-Watson = 0.982, p < 0.001). It is not a flaw of the trial: the rows of the example are sorted by block, hybrid and rate, and the residuals of the same block and hybrid resemble each other because the model does not include the hybrid. The test only makes sense if the order of the rows is the order of measurement or the position in the field; that is why the app warns about it in the row itself.

Stepwise selection

With many Candidate predictors (2 to 15), stepwise selection adds or removes one per step until no change improves the model. The Criterion can be AIC (like R's step), the default, which at each step makes the change that lowers the AIC the most, or Partial F p-value, with a Level to enter (0.05 by default) and a Level to remove (0.10). The options of the Direction change with the criterion:

Criterion	Direction	How it proceeds
AIC	Both directions, from the full model (the default)	Starts from all the candidates; at each step it can remove one or bring back one that has already left.
AIC	Backward	Starts from all the candidates and only removes.
AIC	Forward	Starts from the model with only the mean and only adds.
p-value	Stepwise (enter and remove) (the default)	Starts from the mean alone: the candidate with the smallest *p* enters if it is below the level to enter and, after each entry, the one with the largest *p* leaves if it exceeds the level to remove. The level to enter cannot be greater than the level to remove.
p-value	Forward	Starts from the mean alone and adds while the smallest *p* stays below the level to enter.
p-value	Backward	Starts from all the candidates and removes while the largest *p* exceeds the level to remove.

In the coffee, temperature leaves first: it is almost determined by altitude ($r = -0.967$, section 4.13) and, with altitude in the model, it adds nothing. The table of steps says what changed at each one, with the error degrees of freedom and sum of squares, the AIC, the adjusted R^2 and the predictors that remain (with the *p*-value criterion, also the *F* and the *p* of each change); the second table gives the coefficients of the final model. The Analyze the chosen model button opens linear regression with those predictors, so you can check its diagnostics.

The path matters. With the same data, going forward by AIC, organic matter enters first and then temperature, not altitude, and the selection ends with five predictors (materia_organica, temperatura, edad, ph and lluvia; adjusted R^2 0.843); by *p*-value, entering and removing, it ends with four (adjusted R^2 0.831). That is why it is worth comparing it with best subsets, which review every combination.

> **THE AIC OF THE SELECTION IS NOT THAT OF THE REGRESSION**
>
> The selection uses the AIC of R's `step` function, $n \cdot \ln(\mathrm{SSE}/n) + 2k$; the goodness-of-fit table of linear regression gives the one derived from the log-likelihood, like R's `AIC()`. They differ by a constant that depends only on n: the six-predictor model chosen by the selection has AIC = 52.735 there and 168.250 in linear regression. Compare AIC values only within the same analysis.

Stepwise variable selection computed in 6 ms ⓘ What is this analysis? Present Add to report

Basic Standard Expert

In a nutshell

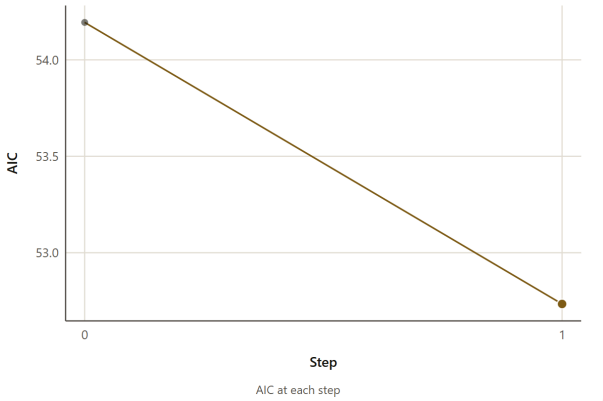
- Selection by AIC, in both directions from the full model, keeps 6 of 7 predictors: altitud, lluvia, materia_organica, ph, sombra, edad.
- Automatic selection is exploratory: confirm the model with knowledge of the crop and check its diagnostics.

Selection by AIC, in both directions from the full model, keeps 6 of 7 predictors: altitud, lluvia, materia_organica, ph, sombra, edad.

Final model: R^2 = 0.875, adjusted R^2 = 0.853, AIC = 52.735.

Automatic selection is exploratory: confirm the model with knowledge of the crop and check its diagnostics.

Chart theme SigmaPro (gold and black)



Analyze the chosen model

Table 1. Selection steps

Step	Change	error df	Error SS	AIC	Adjusted R^2	Predictors in the model
0	start	32	103.9307	54.19	0.85	altitud, temperatura, lluvia, materia_organica, ph, sombra, edad
1	− temperatura	33	105.3457	52.73	0.85	altitud, lluvia, materia_organica, ph, sombra, edad

Figure 4.33. With the seven coffee predictors, selection by AIC removes temperature in one step and stops.

! AUTOMATIC SELECTION IS EXPLORATORY

The *p*-values of the chosen model are optimistic, because the same data served both to choose and to test. The conclusion says so: the model is confirmed with knowledge of the crop, and you should check its diagnostics before using it. The selection does not check the assumptions: the traffic light is in the linear regression of the chosen model.

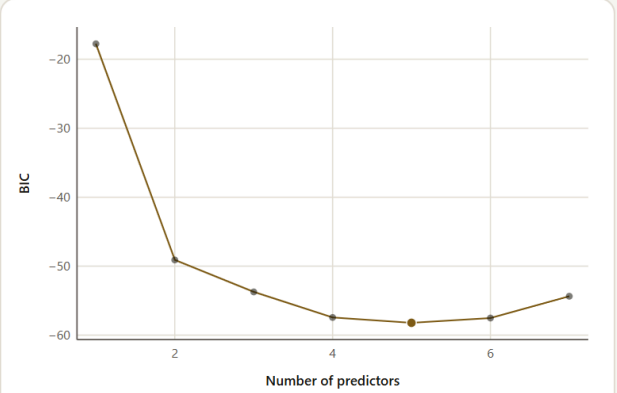
Best subsets

Instead of moving one predictor at a time, it reviews **all** combinations of the Candidate predictors (2 to 15) (like `leaps::regsubsets`) and gives the best models of each size: one, two or three per size, according to Models by size (with more than one, the table adds the rank column). Each model comes with its R^2, the adjusted R^2 (higher is better), Mallows' (1973) C_p and the BIC (lower is better).

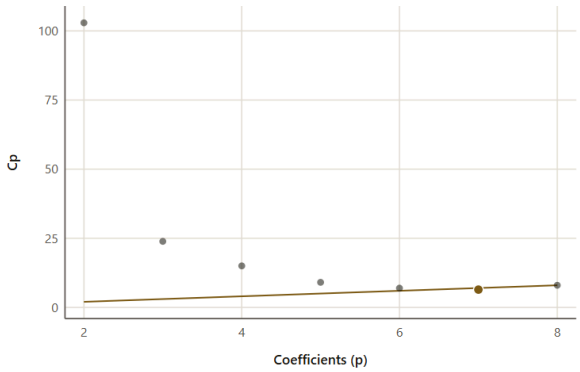
Table 4.5. The best model of each size with the seven coffee predictors (table 1 of the result).

Predictors	Variables	R^2	Adjusted R^2	C_p	BIC	Chosen by
1	materia_organica	0.47	0.45	102.94	−17.76	
2	temperatura, materia_organica	0.78	0.77	23.89	−49.10	
3	temperatura, materia_organica, edad	0.82	0.80	15.02	−53.73	
4	altitud, materia_organica, sombra, edad	0.85	0.83	9.08	−57.43	
5	altitud, materia_organica, ph, sombra, edad	0.87	0.85	6.95	−58.21	BIC
6	altitud, lluvia, materia_organica, ph, sombra, edad	0.88	0.85	6.44	−57.51	Cp, Adj. R^2
7	altitud, temperatura, lluvia, materia_organica, ph, sombra, edad	0.88	0.85	8.00	−54.36	

The **Chosen by** column marks the size that wins with each criterion: the highest adjusted R^2, the lowest C_p and the lowest BIC. BIC penalizes predictors more and keeps five; C_p and adjusted R^2 keep six (the same ones chosen by stepwise selection). When they do not agree, the BIC model is the simplest.



BIC of the best model of each size



Mallows' Cp vs. coefficients

Analyze the model chosen by BIC (5 predictors) Analyze the model chosen by Cp (6 predictors)

Figure 4.34. The best subsets of the coffee: BIC reaches its minimum with five predictors; in the C_p plot, the six-predictor model lies below the line $C_p = p$. The buttons take each model to linear regression.

Σ HOW TO READ C_p

C_p compares the error of each model with that of the full model, which is taken as the true variance. An unbiased model has a C_p close to its number of coefficients p (the predictors plus the intercept), and the full model always has $C_p = p$ (8.00 with the seven coffee predictors). SigmaPro marks the lowest C_p; the classic rule, a C_p close to p or below it, is read in the plot on the right. In the coffee, the six-predictor model ($p = 7$) lies below the line, with 6.44, and the five-predictor one ($p = 6$) just above it, with 6.95.

The buttons, Analyze the model chosen by BIC (5 predictors) and Analyze the model chosen by Cp (6 predictors) in the coffee, open linear regression with those predictors and their full diagnostics. When adjusted R² chooses a size different from the other two, a third button appears for its model.

I CONTINUE IN THE LABG SUITE

At the foot of linear regression, stepwise selection and best subsets, the Continue in the LABG suite strip suggests CorrelationPro, to explore and visualize the relationships among many variables at once, and PCAPro, which summarizes highly correlated predictors (high VIF) into independent principal components. Analysis of covariance and polynomial regression have no strip (chapter 10).

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4.13 Correlation

Correlation measures how much two variables move together, without saying which depends on which. The family includes the correlation between two variables, the correlation matrix of many variables and partial correlation, which discounts the effect of others.

Correlation between two variables

Choose the x variable, the y variable, the Coefficient (Pearson (linear relationship), the default; Spearman (ranks) or Kendall (ranks), for a relationship that only rises or only falls) and the Alternative hypothesis (Correlation different from zero, Positive correlation or Negative correlation). With Pearson, you also choose the Confidence level of the interval; with Spearman or Kendall, the p-value (Exact if possible or Approximate) and the Continuity correction (in the approximate p) checkbox.

In the coffee, altitude and yield have a Pearson correlation $r = 0.565$ (95% CI: 0.307 to 0.745; $t = 4.217$ with 38 df, $p < 0.001$): they share 31.9% of their variation ($r^2 = 0.319$). The conclusion rates it as *strong* using Cohen's (1988) cutoffs: less than 0.1, negligible; up to 0.3, weak; up to 0.5, moderate; beyond that, strong.

The result includes the scatter plot with the least-squares line and two tables: the test (the coefficient, its interval with Pearson, the statistic, the *p*-value, how the *p* was calculated and the number of pairs) and **the three coefficients** at once (Pearson 0.565, Spearman 0.574 and Kendall 0.406 in the coffee). If Pearson and Spearman differ a lot, the relationship is not linear or some values weigh too much. Without ties, the *p*-values of Spearman (up to 1,290 observations) and Kendall (fewer than 50) are exact, as in R. With ties, as in the coffee, the *p* is approximate and the conclusion says so: Spearman calculates it with Student's *t*, and Kendall with the normal approximation corrected for ties ($z = 3.683$).

This analysis has no traffic light: the shape of the relationship and the extreme values can be seen in the plot, and linear regression gives its full diagnostics.

Correlation matrix

With the Variables (2 to 30) you check, it gives all the pairwise correlations with the Coefficient you choose (Pearson, Spearman or Kendall). Because there are many tests at once, the *p*-values are adjusted (p-value adjustment: Holm, the default, Bonferroni, Benjamini-Hochberg (FDR) or No adjustment). In Rows with missing values, you choose between Only rows complete in all variables (the default) and Each pair with its complete rows.

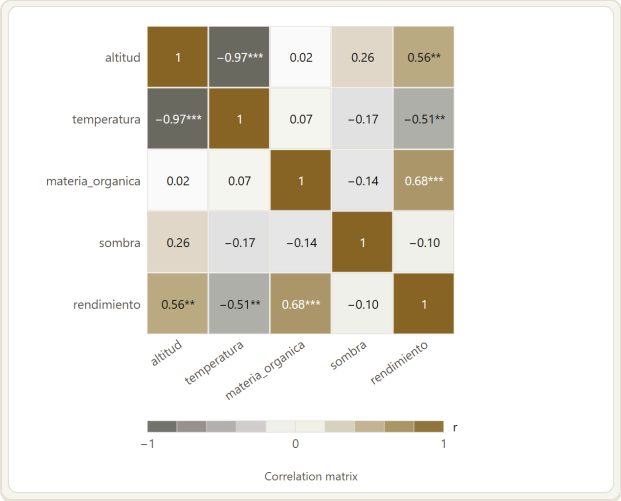


Figure 4.35. The heat map of the coffee correlations: gold, positive; gray, negative; the stars come from the Holm-adjusted *p*.

The conclusion says how many pairs have a significant correlation (in the coffee, 4 of 10) and which one is the strongest.

The first table is the matrix, with stars according to the adjusted *p* (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$); the second gives each pair with its *p*, its adjusted *p* and its number of pairs, and with Pearson also its confidence interval. The tests are two-sided. In the coffee, altitude and temperature lie almost on a straight line ($r = -0.967$): that is why stepwise selection does not want both.

Partial and semipartial correlation

Choose the x variable, the y variable, the Control variables (1 to 10) and the Coefficient (Pearson or Spearman; with Spearman, everything is calculated on the ranks). The **partial correlation** is that of x with y after removing from both what the controls explain; the **semipartial** removes the controls from only one of them, and its square is the part of the variation in y that x explains on its own. The tests use $n - 2 - k$ degrees of freedom (k controls), like R's `ppcor` package.

Table 4.6. Temperature and coffee yield, controlling for altitude and shade.

Correlation	Value	t	df	p
Simple	−0.512	−3.68	38	< 0.001
Partial (controls removed from both)	0.279	1.74	36	0.090
Semipartial: "rendimiento" with "temperatura" without the controls	0.219	1.35	36	0.186
Semipartial: "temperatura" with "rendimiento" without the controls	0.068	0.41	36	0.686

Temperature seemed to lower yield (−0.512), but after discounting altitude and shade, the two are no longer correlated (partial 0.279, $p = 0.090$): the cooler farms yield more because they are higher up, not because of the temperature itself. The conclusion adds the squared semipartial (0.048): the part of the variation in yield that temperature explains on its own, beyond altitude and shade. The added-variable plot shows the residuals of the two variables without the controls; their correlation is the partial correlation.

> **CONTINUE IN THE LABG SUITE**
>
> At the foot of the three correlation analyses, the Continue in the LABG suite strip suggests CorrelationPro, to explore and visualize the relationships among many variables at once, and PCAPro, to reduce the dimensions of the variables and examine their relationships in biplots (chapter 10).

4.14 Contingency tables

When both variables are categories (germinated or not, healthy or diseased, treated or control), the data are summarized in a table of counts. The four methods of the family accept two forms of data: **one row per case** (each seed, each leaf), with — each row is one case — in Frequencies, or the table **already counted**, with a column of counts chosen in that same field.

The Frequencies field only offers numeric columns of non-negative integers, and it proposes on its own the one named like a count (n, frecuencia, conteo, casos, semillas…). The categorical variables must have 2 to 50 categories. Contingency tables have neither a traffic light nor a suite strip.

Contingency table

It tests whether two variables are independent. Choose the Row variable, the Column variable, the Frequencies, the Yates correction (2 × 2 tables) checkbox (checked by default, as in R), the Confidence level and the Significance level. The treated-seed example counts how many seeds germinated with and without fungicide in four lots; here, the lots are pooled.

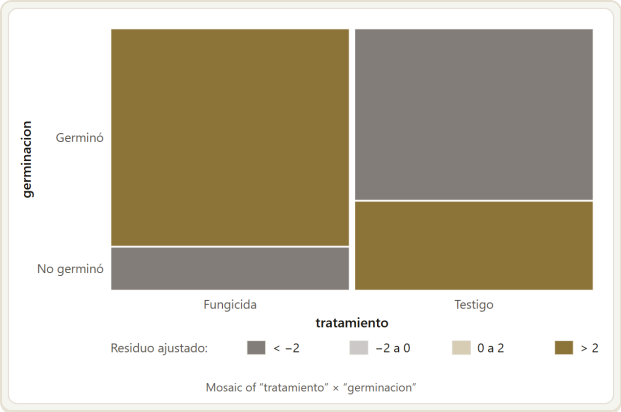


Figure 4.36. The mosaic of the treated seed: gold marks the cells with more cases than expected under independence.

Treated seed germinates more ($\chi^2 = 31.326$, $p < 0.001$). The conclusion says whether there is an association, how strong it is (**Cramér's V**: 0.201, weak) and which cell departs most from independence, by its **adjusted residual** (beyond ±1.96, the cell departs at $\alpha = 0.05$; here, fungicide with germinated, 5.678). In 2 × 2 tables, it adds the **odds ratio** (2.589, 95% CI: 1.854 to 3.615: treated seeds have 2.6 times the odds of germinating) and the **relative risk** of the first column (1.266, 95% CI: 1.165 to 1.376).

The five tables are the observed frequencies; the expected ones with their adjusted residual; the independence tests (Pearson's chi-square, the likelihood ratio G^2 and Fisher's exact test, with its conditional odds ratio in 2 × 2 tables); the measures of association (φ, Cramér's V, the contingency coefficient, Goodman and Kruskal's λ and the uncertainty coefficient); and the measures for ordered categories (γ, Kendall's τ-b, Stuart's τ-c and Somers' *d*). SigmaPro calculates the latter in any table, but they are only meaningful if both variables have a natural order (doses, damage grades).

Fisher's exact test is always calculated in 2 × 2 tables; in larger ones, only with up to 5,000 cases and if some expected frequency is less than 5 or there are 300 cases or fewer (if it would require examining too many tables, the note says so and recommends merging categories). When some expected frequency is less than 5, the note warns that the chi-square is approximate, and the conclusion decides with Fisher's exact test.

McNemar and Bowker: paired data

When each case is classified twice (before and after a treatment, or with two methods), the two classifications are not independent and the chi-square does not apply. Choose the First classification (before) and the Second classification (after), with the same categories, the Frequencies and the Continuity correction checkbox (checked by default). With two categories, it is the **McNemar** test, with its exact binomial version; with more, **Bowker's symmetry** test, which asks whether changes in one direction are as frequent as in the other.

Table 4.7. Rust severity on 80 leaves, scored by technicians A (rows) and B (columns): 0 healthy, 1 mild, 2 moderate, 3 severe.

A \ B	0	1	2	3	Total
0	25	6	0	0	31
1	0	21	3	0	24
2	0	2	10	1	13
3	0	0	2	10	12
Total	25	29	15	11	80

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In 2 × 2 tables, the test table gives McNemar's chi-square (with the correction, if it is checked and the two types of change are not equal) and the exact binomial test, which uses only the cases that changed; the conclusion decides with the exact binomial test, whether or not the correction is checked.

With the rust, Bowker gives $\chi^2 = 6.533$ with 3 df and $p = 0.088$: there is no evidence that one technician systematically scores higher than the other. The pairs of categories with no change at all (healthy–moderate, healthy–severe, mild–severe) do not enter the test; R would give NaN, while SigmaPro lowers the degrees of freedom to 3 and says so in the note. The third table gives the proportion of each category in each classification: technician A scored 0.3875 of the leaves as healthy and B, 0.3125, because B saw as mild six of the leaves that A saw as healthy. The plot is a heat map of the table, with the diagonal of the cases that did not change.

Cochran-Mantel-Haenszel: association by strata

When the data come from several lots, sites or years, pooling them can create or hide an association. Choose the Row variable, the Column variable, the Strata (lots, sites, years) and the Frequencies, with the Continuity correction checkbox (checked by default) and the Confidence level. The **Cochran-Mantel-Haenszel** test measures the association within each stratum and combines it, with the **common odds ratio**. The **Breslow-Day** test says whether the odds ratio is similar in all strata; the table gives it without and with Tarone's correction, and the conclusion uses the corrected one. With tables larger than 2 × 2 in each stratum, the test is the generalized one, with $(I - 1)(J - 1)$ df, without a common odds ratio, without Breslow-Day and without a plot.

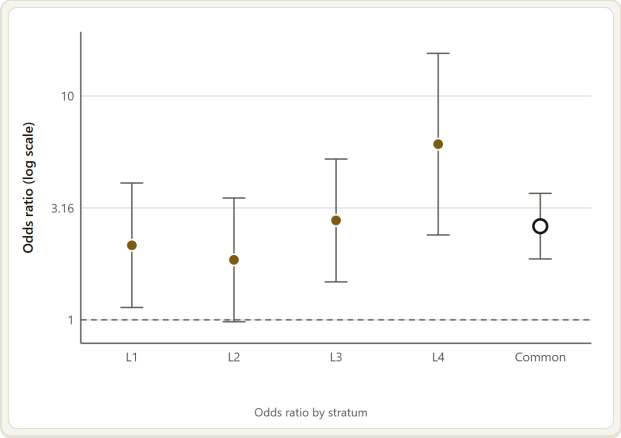


Figure 4.37. The odds ratio of each lot and the common one (2.621), with their intervals, on a logarithmic scale.

Controlling for lot, treatment and germination remain associated (CMH $\chi^2 = 31.600$, $p < 0.001$), with a common odds ratio of 2.621 (95% CI: 1.871 to 3.671). Lot 4 has the highest ratio (6.093), but Breslow-Day finds no differences between lots ($p = 0.186$), so the common ratio summarizes them well.

The table for each stratum gives its table (a b / c d), its odds ratio and its interval; the dashed line in the plot marks 1, where there would be no association.

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Inter-rater agreement: kappa

It measures how much two raters (or two methods) agree beyond what they would agree by chance. Choose Rater 1, Rater 2, the Frequencies, the Weighted kappa weights (Quadratic, the default, or Linear), the Confidence level and the Significance level. The weights count a disagreement between neighboring categories as less serious than one between extreme categories: with quadratic weights, a disagreement of two grades weighs four times as much as one of a single grade; with linear weights, twice as much. The weighted kappa appears when there are more than two categories.

With the rust (table 4.7), the technicians agree on 82.5% of the leaves, when by chance they would agree on 28.1%. Cohen's kappa is 0.757 (95% CI: 0.642 to 0.871), a *substantial* agreement according to Landis and Koch (1977). The weighted kappa rises to 0.920 with quadratic weights and to 0.846 with linear weights, because almost all the disagreements are between neighboring categories. The agreement table gives each kappa with its standard error (Fleiss, Cohen and Everitt, 1969), its interval and its *z* test, which uses the standard error of chance agreement. The plot is the heat map of the table, with the agreements on the diagonal.

Table 4.8. The agreement of the two technicians on rust, with the default quadratic weights (table 2 of the result).

Measure	Value	SE	95% CI	z	p
Cohen's kappa	0.757	0.059	0.6419 – 0.8713	11.27	< 0.001
Weighted kappa (quadratic)	0.920	0.022	0.8763 – 0.9630	8.26	< 0.001

The weighted kappa depends on the order of the categories. SigmaPro builds a square table with the categories of both raters in the order of those of rater 1, which is the order of the column in Data (the order of appearance, or the one set with Order levels..., section 2.7). A category used only by rater 2 is inserted right after the one that precedes it in that rater's column, and if all the categories are numbers, they are sorted from smallest to largest. Thus, if rater 1 scored only "mild", "medium" and "high" and rater 2 also "healthy", the table becomes healthy, mild, medium, high. Before reading the weighted kappa, check that order in the first table of the result.

> **WEIGHTED KAPPAS FROM EARLIER VERSIONS**
>
> In SigmaPro versions earlier than 0.5.0, a category used only by rater 2 was placed at the end of the table, and the weighted kappa was calculated with a false order. Since 0.5.0, the category goes in its proper place. If you calculated a weighted kappa with an earlier version and one rater did not use some category, run it again. The simple kappa and the McNemar and Bowker tests do not depend on the order.

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4.15 Linear mixed models

A mixed model separates the **fixed** effects, those chosen on purpose and about which you want to draw conclusions (the hybrids, the rates), from the **random** effects, which represent a sample of many possible ones (blocks, locations, years, plants measured several times). With it, each source of variation carries its own error, missing plots do not break the analysis, and the information between blocks is not lost.

The six methods of the family run in the **R engine** inside the browser (section 4.6), with the packages `lme4`, `lmerTest` and `emmeans`; repeated measures use `nlme`. The engine is downloaded the first time; after that it works offline.

Method	Purpose
Linear mixed model (formula)	Any combination of fixed and random effects, written as a formula.
Randomized blocks with random blocks	One to three factors in complete blocks; with missing plots it recovers the inter-block information.
Split plot (mixed model)	Main plots, subplots and, if requested, sub-subplots, with their errors as random effects.
Lattice and alpha-lattice (inter-block recovery)	Random incomplete blocks within replicates: the means recover the inter-block information.
Trials at several locations	Fixed genotypes at locations with replicates; the location and the genotype × location interaction, random.
Repeated measures over time	The same unit measured on several dates: choose the covariance structure between dates.

The model written as a formula

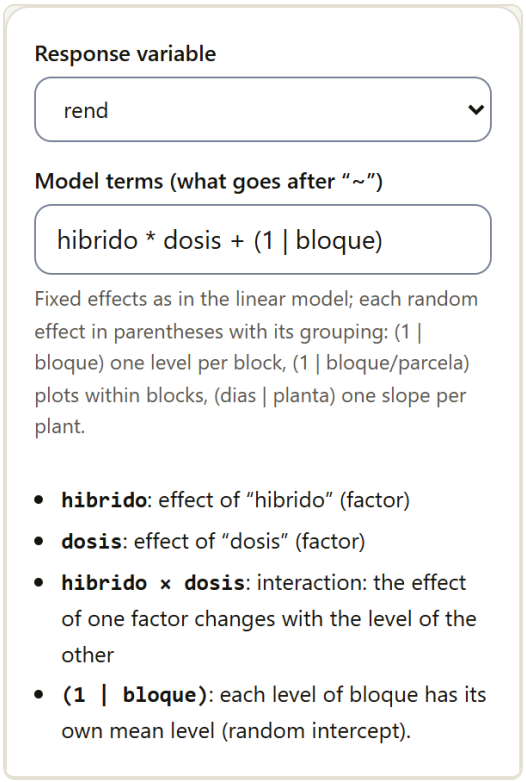


Figure 4.38. The mixed model formula and what the app understood from it, term by term.

The Model terms (what goes after "~") field takes the right-hand side of the formula. Fixed effects are written as in the general linear model (section 4.10); each random effect goes in parentheses with its grouping:

- `(1 | bloque)`: a level of its own for each block (random intercept).
- `(1 | bloque/parcela)`: plots within blocks.
- `(dias | planta)`: a slope of its own for each plant, correlated with its intercept; with `(dias || planta)`, without that correlation.

Below the field, the app explains each term it read, so you can confirm that the model is the one you intended. Numeric columns that are treatments (a rate) are checked in Numeric columns that are treatments (factor).

Next come Compare the means of (checkboxes with the fixed factors that enter as main effects; by default, the first three), Within each level of (with two or more factors, to compare one separately at each level of another when they interact), the Comparison adjustment (Tukey, Bonferroni, Holm, Šidák or No adjustment (LSD)), the Sums of squares (type III, II or I), the Degrees of freedom for the tests (Satterthwaite, or Kenward-Roger, more exact with few groups and slower) and the Estimation: REML, which gives unbiased variance components, or maximum likelihood, to compare models with different fixed effects. Kenward-Roger only works with REML. At the end come the Confidence level and the Significance level.

With the maize and random blocks (`hibrido * dosis + (1 | bloque)`), the tests of the hybrids, the rates and their interaction give the same F values as the block ANOVA (60.63, 292.15 and 1.47), because the data are balanced. What changes is the standard error of the means: it includes the block variance (43.9% of the random variation), so the hybrid means carry an SE of 0.1079 with 4.25 degrees of freedom, while their differences, in which the blocks cancel out, keep the 33 degrees of freedom of the error.

With the two maize factors, the result brings ten tables: the tests of the fixed effects, the **variance components** with their percentage, the **test of the random effects** (likelihood ratio, like `lmerTest::ranova`; its *p* is conservative, because a variance cannot be negative, and the true one is usually half of it), the marginal means of each factor with their letters and their pairwise comparisons, the coefficients, the predictions of the random effects (**BLUP**) and the model fit. When a factor has more than 45 pairs, its table of pairs is not shown, though the letters are; with more than 50 levels, only the means. If the BLUPs exceed 120 rows, their table does not appear either: a note below the coefficients says so, and the equivalent R code gives them with `ranef()`.

The plots are the marginal means of the first compared factor, with their interval and their letters (with Within each level of, one line for each level of the other factor), the residuals against the fitted values and their Q-Q plot. The traffic light checks the normality of the residuals (Shapiro-Wilk), the normality of the random effects (the BLUPs of the first grouping, if it has 5 levels or more), the equality of variances (Levene with the median, among the levels of the first fixed factor) and the optimizer warnings; a singular fit shows in amber.

Designs as mixed models

The other four methods ask for the roles of the design, build the formula themselves and show it below the form ("Model: ..."). Numeric columns that act as treatments are taken as factors, and the tests use type III sums of squares; the other options are those of the formula model. In randomized blocks with two or three factors and in split plots, the form also offers Within each level of.

Split plots ask for the Main-plot factor, the Subplot factor, if needed the Sub-subplot factor (optional), and the Blocks (random); they include the blocks and the main plots (and, with a third factor, the subplots) as random effects. If the main plots were randomized without blocks, leave — no blocks — and choose the Replicate (identifies each main plot). With Yates's oats (1935), the tests are those of the stratum analysis of section 4.10, but the mixed model remains exact with missing plots, when the stratum analysis no longer is. To see nitrogen separately within each variety, choose the variety in Within each level of: the means table gives the nitrogen letters within each variety, and the plot draws one line per variety.

Table 4.9. Variance components (REML) of the oats in split plots.

Group	Variance	SD	% of total
bloque:variedad (the main plots)	106.1	10.30	21.31
bloque	214.5	14.65	43.10
Residual	177.1	13.31	35.59

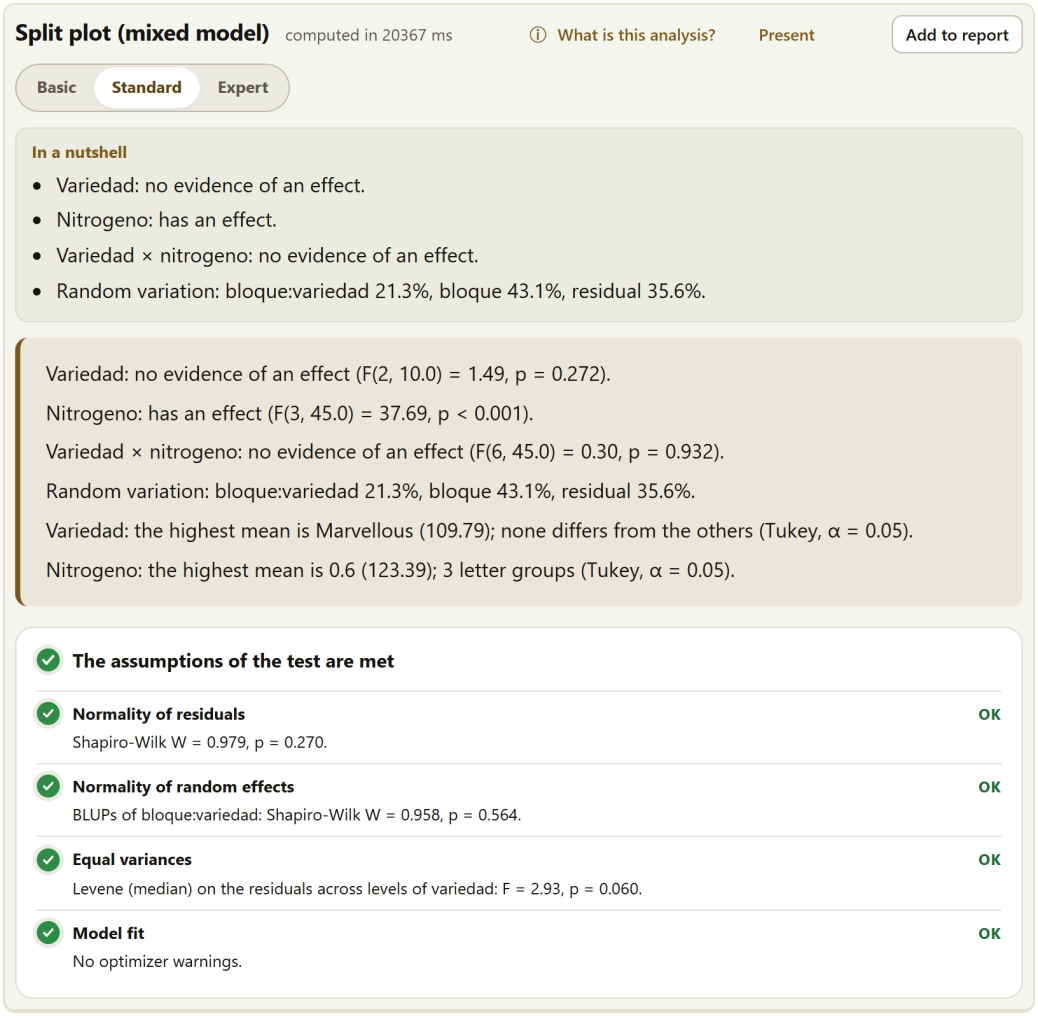


Figure 4.39. The oats in split plots as a mixed model: varieties are tested with 10 degrees of freedom (those of the main plots) and nitrogen with 45.

In the **lattice and alpha-lattice**, choose the Treatments (genotypes, varieties), the Replicate and the Incomplete block (within replicate). The incomplete blocks are random, and the genotype means **recover the inter-block information** that the intra-block analysis (section 4.10) wastes; the Replicates also random checkbox changes the model when the replicates are also a sample. With the alpha-lattice of 24 genotypes, the genotypes differ ($F(23, 34.7) = 5.45$, $p < 0.001$) and the blocks within replicates account for 42.1% of the random variation.

Trials at several locations ask for the Genotypes (fixed), the Location or environment and the Replicate (within each location); they set the location, the replicate within each location and the genotype × location interaction as random. The Fixed locations (only these locations are of interest) checkbox makes them a fixed effect. In the oats with 14 environments, the genotypes differ ($F(9, 117) = 5.41$) and the environment accounts for 79.7% of the variation, against 6.5% for the interaction. There the BLUPs add up to 196 rows (14 environments, 42 replicates and 140 genotype × environment combinations), so their table is not shown and the note below the coefficients says so. To study the interaction itself (stability, AMMI, GGE) there is the multi-environment family (section 4.20).

At the foot of randomized blocks, split plots and lattice as mixed models, the "Continue in the LABG suite" strip suggests AgriDesign, which generates the randomization, the field map and the field sheet of the design (chapter 10).

Repeated measures over time

When the same plant, plot or animal is measured on several dates, the measurements of one unit resemble each other, and all the more the closer the dates are. The method asks for the Time (measurement date) (numeric or factor), the Unit measured at each date (plant, plot, animal) (its number may repeat in each treatment: the unit is the combination of unit, treatment and block), the Treatment factors (up to 2; optional) and the Blocks. The Covariance structure between dates can be set or left at The best by the criterion (all are compared), which fits the candidates and chooses the one with the lowest value according to the Criterion to choose it (AIC or BIC). Continuous AR(1) enters only with numeric time, and the unstructured one with 8 dates or fewer (unless requested); one that cannot be fitted is marked "could not be fitted".

Table 4.10. The seven covariance structures compared with maize height on five dates (REML, the same fixed effects).

Structure	Parameters	AIC	BIC	
Independent	24	486.6	535.6	
Compound symmetry	25	474.3	525.4	
Heterogeneous compound symmetry	29	431.4	490.6	
AR(1)	25	451.8	502.8	
Heterogeneous AR(1)	29	420.1	479.4	selected
Continuous AR(1)	25	451.8	502.8	
Unstructured	38	427.6	505.3	

The conclusion says which structure won and tests the treatments, time and their interaction. AR(1) assumes that the correlation decreases with the distance between dates (φ, φ^2, φ^3...); the heterogeneous version also allows a variance of its own for each date, which is needed when the plant grows and so does its variation. Continuous AR(1) uses the actual distance between unequally spaced dates. In the maize, heterogeneous AR(1) wins, with $\varphi = 0.778$ between neighboring dates.

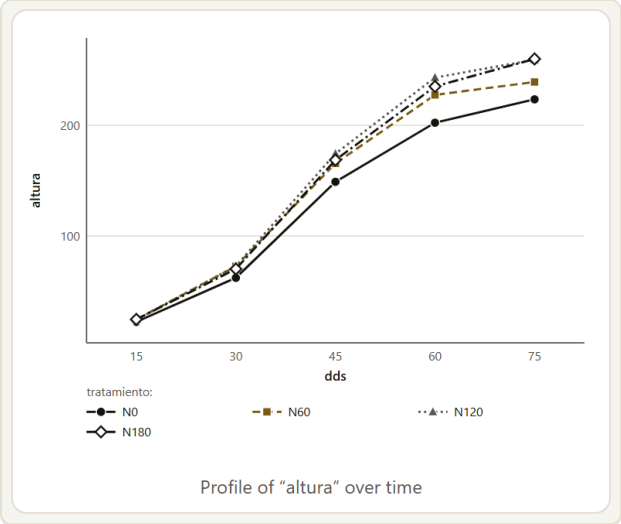


Figure 4.40. Maize height on five dates: the rates separate over time.

Treatment alone has no effect ($F(3, 57) = 1.39$, $p = 0.256$), but its interaction with time does ($F(12, 57) = 4.29$, $p < 0.001$): at 15 days the four rates measure the same, and from 30 days on the control without nitrogen falls behind. That is why the app proposes comparing the treatments `Within each level of` the date, and its means table gives the letters for each date separately. Another table gives the estimated matrix of covariances and correlations between dates.

The tests use the error degrees of freedom, $N - p$, for all terms, like `nlme`. The traffic light checks the normality of the normalized residuals (Shapiro-Wilk) and the equality of their variances between dates (Levene); if either fails, it proposes a heterogeneous structure or a transformation, and with a positive response `Try the alternative` runs the logarithm (section 4.7).

4.16 Variance components, heritability and repeatability

Three methods for quantitative genetics and breeding, which run in the browser without the R engine: they estimate the variance components by REML (with the average information algorithm) and were validated against `lme4`.

Variance components

They tell how much variation each random factor contributes. Choose the Random factors (up to 4), How they are related (Crossed; Nested (each within the previous ones), in table order; or Crossed, with their two-way interactions) and, if needed, the Fixed factors (optional, up to 3), which enter as main effects and are not tested: for their F tests there is the linear mixed model. In the alpha-lattice, with the genotypes fixed and the blocks nested in the replicates, the replicate contributes 43.6% of the variance ($\sigma^2 = 0.1139$), the blocks within the replicate 23.7% and the error 32.6%. Each component carries its SE and its likelihood ratio test, with the p of the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, because a variance cannot be negative: the replicate gives $\chi^2 = 6.41$ ($p = 0.006$) and the blocks, $\chi^2 = 5.01$ ($p = 0.013$). With only three replicates, the replicate variance is very imprecise: its SE, 0.128, is larger than the estimate. The plot shows the percentage of the total variance that each component contributes.

> **TWO *P* VALUES FOR THE SAME VARIANCE**
>
> For a random intercept, the test of the mixed models of section 4.15 (`ranova`) gives the p of χ^2 with 1 degree of freedom, which is conservative; these three methods give that of the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, which is half as large. With the same model (and REML), the variance and the χ^2 agree: only the way the p is calculated changes.

Heritability and genotype BLUPs

With the genotypes as random, the method estimates the genetic variance, the heritability and the genotypic value of each one. Choose the Genotypes (random), the Replicate or complete block (fixed), the Incomplete block within replicate (random), for the alpha-lattice and other designs with incomplete blocks, and the Environment or location (fixed); with several environments, the genotype × environment interaction enters as random and the conclusion compares its variance with the genetic variance. Below the roles, the app shows the model it built ("Model: ..."), and the title of the result states the design it understood. For the selection gain, choose the Selected proportion (from 5% to 50%) and, in Select the values that are, the highest (yield, weight, content) or the lowest (days to flowering, height, damage).

> **SPECIFY ALL THE COLUMNS OF THE DESIGN**
>
> If the replicate or the incomplete block is missing, its variation goes into the error, without warning, and the heritability usually drops: with the alpha-lattice, the mean-basis H^2 is 0.840 with the replicates and the blocks, 0.780 with the replicates only and 0.580 with neither (the title then says "completely randomized"). The app stops only when there is a single plot per genotype and environment, because then the genotype × environment interaction is confounded with the error.

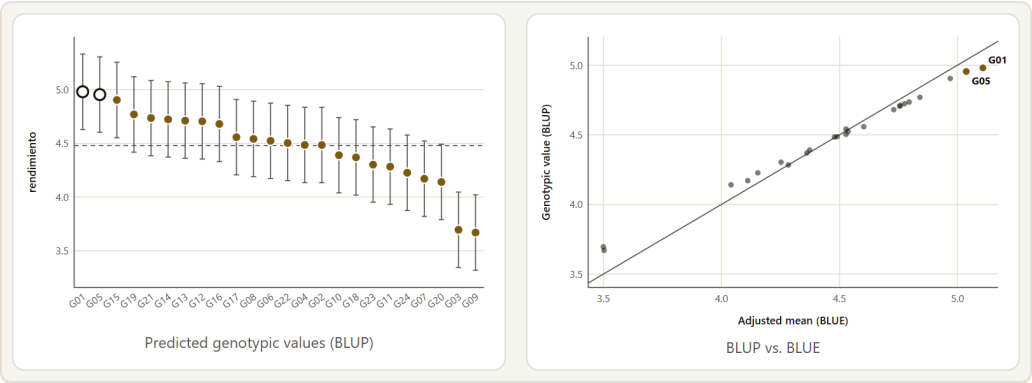


Figure 4.41. The BLUPs of the 24 genotypes of the alpha-lattice, from highest to lowest (hollow, the two selected), and their relation to the adjusted means (BLUE).

The conclusion begins with the genetic variance (σ^2_g = 0.1429, χ^2 = 23.62, $p < 0.001$). The mean-basis heritability is 0.840 ± 0.063: 84% of the variation among the genotype means is genetic. Since the trial has incomplete blocks, the heritability table also gives the **Cullis** (0.809) and **Piepho and Möhring** (0.803) heritabilities, the most accurate with unbalanced data or incomplete blocks (Schmidt et al., 2019), and the plot-level heritability (0.636). The genetic parameters include the genetic (8.44%) and error (6.38%) coefficients of variation; their ratio, 1.323, greater than 1, indicates favorable conditions for selection.

The **expected gain** from selecting 2 of the 24 genotypes (10% of 24, rounded; the table note gives the actual 8.3% and the selection intensity, i = 1.840) is calculated in three ways: with the mean of the selected BLUPs (0.4881, 10.90% of the mean), with the breeder's equation $i \cdot H^2 \cdot \sigma_F$ (0.6374) and with the selection differential of the BLUEs times H^2 (0.4981). The last table ranks the genotypes with their BLUP, their SE, their **reliability** and their adjusted mean (BLUE), and marks the selected ones (G01 and G05). The BLUPs shrink toward the overall mean in proportion to how little is known about each genotype; that is why, in the plot, the cloud of BLUPs against BLUEs is flatter than the diagonal.

The traffic light of the three methods checks the normality of the residuals and that of the random effects of the first term (Shapiro-Wilk of their BLUPs, with 5 levels or more), outlying observations (residuals of more than 3.5 standard deviations) and the REML fit: whether it converged and whether any component ended at zero, on the boundary and without an SE. With the alpha-lattice, the normality of the residuals shows in amber (p = 0.011 in the heritability): it is a warning, not an error, and it is worth checking for unusual plots.

Repeatability

When the same individual is measured several times (the harvests of a tree, the lactations of an animal), repeatability is the correlation between two measurements of the same individual. Choose the Individual measured several times (tree, plant, animal) and, if needed, the Fixed factors (optional: harvest, year, rater). With 20 guava trees over four harvests (harvest as a fixed factor), $r = 0.532 \pm 0.116$: 53% of the variation between measurements is due to permanent differences between trees. With 3.69 measurements per tree (the harmonic mean), the determination of the permanent value is $R^2 = 0.807$, and the conclusion adds that a determination of 90% would require 8 measurements per tree.

Table 4.11. Measurements needed per guava tree for each desired determination (Cruz, Regazzi and Carneiro, 2012).

Desired R^2	Measurements (η)	Rounded
0.80	3.53	4
0.85	4.99	5
0.90	7.93	8
0.95	16.74	17
0.99	87.25	88

The last table gives the predicted **permanent value** of each tree (the overall mean plus its BLUP), with its SE and its reliability. The plots are that value for each tree, from highest to lowest with its interval, and the determination according to the number of measurements, from 1 to 10, which shows how much is gained with each additional harvest.

4.17 Generalized linear models

Counts (insects per shoot, lesions per leaf), proportions (seeds germinated out of those sown) and times or concentrations do not have the constant variance or the normal distribution that ANOVA requires. Instead of transforming them, a generalized linear model uses the distribution that suits them and a **link** that connects the mean with the treatments, and it presents the results on the original scale: rates, probabilities and means.

Method	Purpose
Counts (Poisson and negative binomial)	Poisson, quasi-Poisson or negative binomial, chosen by the overdispersion; treatments in blocks, covariates and exposure.
Proportions (binomial)	Binomial or quasibinomial: successes out of *n* trials or 0/1 responses; logit, probit or complementary log-log link; Firth correction for separation.
Skewed positive responses (gamma)	Gamma or inverse Gaussian, for times, concentrations or weights whose variance grows with the mean.

Method	Purpose
Generalized linear model (formula)	Any of the seven families from a written formula.
Counts or proportions in random blocks (mixed)	Generalized mixed model with random blocks and, if needed, an observation-level effect.
Generalized linear mixed model (formula)	Binomial, Poisson, negative binomial or gamma with random effects written as in `lme4`.

The first four run in the browser (iteratively reweighted least squares, validated against R's `glm` and `glm.nb`); the two mixed ones, in the R engine.

Counts

Choose Response: the count, the Treatment factors (up to 3, crossed), the Blocks, the Covariates (optional, up to 2) and the Exposure (optional: area or time sampled), when the units were not sampled equally: with it, the means are rates per unit. The default Family is Automatic (based on overdispersion): the Poisson is fitted and, if Pearson's test finds **overdispersion** (more variation than the Poisson allows: $p < 0.05$ and χ^2/df greater than 1.1), the analysis switches to the negative binomial. You can also set the Poisson, the quasi-Poisson or the negative binomial.

The Link is the log; the Poisson and the quasi-Poisson also allow the identity, the square root and the power (μ^{λ}, with its Link exponent λ (μ^λ)), and the negative binomial, the identity and the square root. With the automatic family, the list offers only the links that the negative binomial also has (log, identity and square root), because the analysis switches to it if there is overdispersion. Next come the Analysis of deviance (type II by default, type III with sum-to-zero contrasts or sequential type I), the Coefficient intervals (see below) and, as in the mixed models, Compare the means of, Within each level of and the Comparison adjustment.

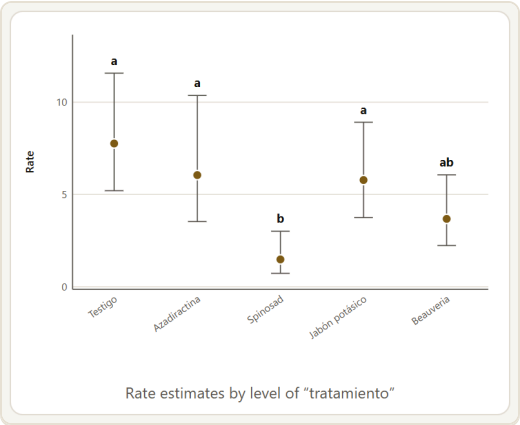


Figure 4.42. Avocado thrips per shoot with the negative binomial: rates, intervals and Tukey letters.

With the avocado thrips, the Poisson had overdispersion ($\chi^2/df = 2.23$, $p = 0.002$) and the app switched to the negative binomial ($\theta = 8.390$). The block has no effect ($p = 0.468$) and the treatment does ($\chi^2 = 21.34$, 4 df, $p < 0.001$): spinosad leaves 1.483 thrips per shoot, against 7.761 in the control.

The rates are compared on their own scale, with asymmetric intervals (calculated on the link scale and transformed back) and Tukey letters. The tables are the analysis of deviance, the coefficients on the link scale, the rate of each treatment, their pairwise comparisons (as rate ratios) and the model fit. Besides the plot of the rates, the result includes the plot of the Pearson residuals against the fitted values. The traffic light checks overdispersion, Pearson residuals greater than 3 and convergence.

Proportions and skewed responses

For **proportions**, the Response: successes (germinated, diseased) or 0/1 holds the successes, and you choose the column of Trials in each row (seeds sown, plants assessed), or — the response is 0/1 —. The automatic family starts with the binomial and switches to the quasibinomial if there is overdispersion (with 0/1 it stays with the binomial); the link is logit, probit or complementary log-log. With the beans, the binomial is enough (Pearson $\chi^2/df = 1.18$, $p = 0.281$): the treatment changes germination ($\chi^2 = 241.95$, 4 df), from 0.2800 in the control to 0.9450 with scarification.

Skewed positive responses are analyzed with the **gamma**, for times and concentrations whose variance grows with the square of the mean, or with the **inverse Gaussian**, more skewed, whose variance grows with the cube. The method uses the log by default; the gamma also allows the inverse, the identity and the power, and the inverse Gaussian, $1/\mu^2$ (its canonical link), the inverse, the identity and the power. With Box and Cox's (1964) poisons and the gamma, poison ($F = 48.92$) and treatment ($F = 23.68$) have an effect and their interaction does not ($p = 0.192$), without transforming the survival time as in section 2.9. With the free-dispersion families (gamma, inverse Gaussian, quasi-Poisson and quasibinomial), the dispersion is estimated by Pearson and the tests are F tests; the mean comparisons use t in the gamma and z in the others, like `emmeans`.

Profile intervals and Firth correction

With the log and logit links, each coefficient carries its ratio (rate ratio, odds ratio or ratio of means) with its interval. In Coefficient intervals, choose Wald (estimate ± z·SE), symmetric on the link scale, or Profile likelihood, like R's `confint()` but with the exact root of the profile: it can be asymmetric and is more exact with few data or probabilities near 0 or 1. With the thrips, the rate ratio of spinosad against the control is 0.191; its 95% interval goes from 0.085 to 0.430 with Wald and from 0.080 to 0.418 with the profile. With other links, the profile interval is given on the link scale. In the free-dispersion families the profile uses the Pearson dispersion, and in the negative binomial it keeps θ fixed; if the likelihood does not drop enough toward one side, that limit is shown as "—" and a warning says so.

With 0/1 responses or few trials, **separation** can occur: a group with 0% or 100% successes. Maximum likelihood drives its coefficient to infinity, and its SE is no longer useful. The Firth correction (for separation: groups with 0% or 100%) checkbox, which appears with the binomial family (or the automatic one) and the logit link, penalizes the likelihood: it gives finite coefficients, tests each coefficient and each term with the penalized likelihood ratio and gives the intervals from the penalized profile, like the `logistf` package; there is no AIC. When the ordinary fit has separation, its warning ends with "Use Firth's correction." and the fit row of the traffic light shows in red.

With the example "Bean: days to germination" (120 seeds from three treatments and three lots), the response `germino` (0/1) and the formula `lote + tratamiento + peso_semilla`, all 40 scarified seeds germinated. The table below compares the two fits: without Firth, the scarification coefficient blows up and the Tukey letters do not separate any treatment; with Firth, the treatment has an effect ($\chi^2 = 15.15$, 2 df, $p < 0.001$), with no effect of the lot ($p = 0.373$) or of seed weight ($p = 0.317$).

Table 4.12. Scarification against the control in the beans, with the ordinary fit and with the Firth correction.

Fit	Coefficient (SE)	Test of the coefficient	Probability of germinating	Letters (control, soaked, scarified)
Ordinary	30.69 (1.06×10^7)	$z = 0.00$, $p = 1.000$	1.0000	a, a, a
Firth	3.505 (1.413)	$\chi^2 = 15.07$, $p < 0.001$	0.9876	b, ab, a

With a formula and with random effects

The **generalized linear model with a formula** accepts in Model terms (what goes after "~") the same terms as the general linear model, with the Numeric columns that are factors (rate, numbered block) checkboxes, and the seven families: binomial, quasibinomial, Poisson, quasi-Poisson, negative binomial, gamma and inverse Gaussian. With the binomial or the quasibinomial, indicate the trials column (or the response is 0/1); with the count families, the exposure if needed.

Generalized mixed models run in the R engine (`lme4`, `car` and `emmeans`). Counts or proportions in random blocks (mixed) asks for the Treatment factors (up to 3, crossed) and the Blocks (random); Generalized linear mixed model (formula) asks for the Model terms including the random effects (what goes after "~"), such as `tratamiento + (1 | bloque)` or `(1 | localidad/bloque)`. Both accept the trials column (with the binomial) and the exposure (with the Poisson or the negative binomial). The families are Poisson, binomial, negative binomial and gamma; the gamma allows only random intercepts, such as `(1 | bloque)`, and estimates its dispersion. The model builder (section 4.4) sends a count response (Poisson) or a positive skewed response (gamma) with random effects to the formula model.

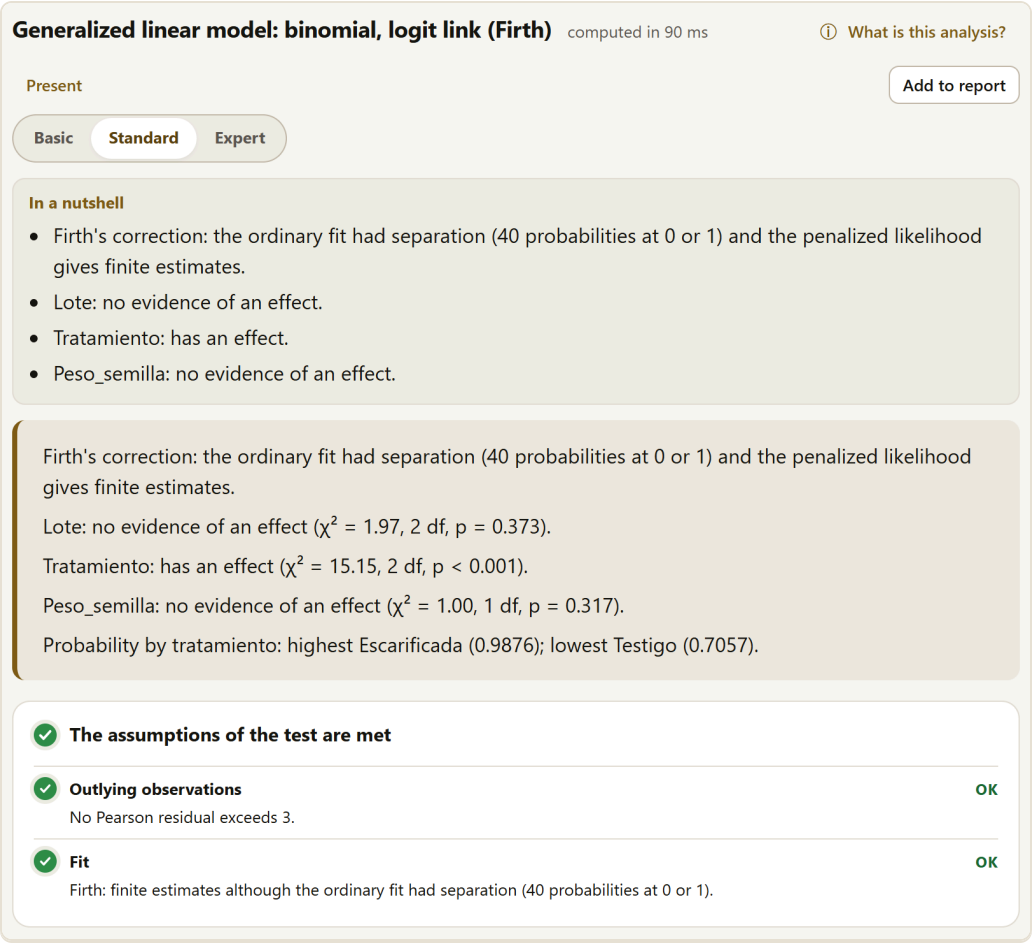


Figure 4.43. Bean germination with the Firth correction: the conclusion warns of the separation in the ordinary fit and tests each term with the penalized likelihood.

The result includes the Wald tests (their note states the approximation used), the coefficients (with the log and logit links, also their ratios with Wald intervals), the variances of the random effects on the link scale, the test of each random effect, the means on the response scale (with the random effects at zero), their pairwise comparisons and the fit, with the θ of the negative binomial or the dispersion of the gamma. The traffic light checks overdispersion (or dispersion), Pearson residuals greater than 3 and the optimizer warnings.

Table 4.13. The avocado thrips with three models: the conclusion is the same, with different strength.

Model	Blocks	Test of the treatment	Control	Spinosad
Negative binomial (automatic)	fixed	$\chi^2 = 21.34$, 4 df, $p < 0.001$	7.761	1.483
Quasi-Poisson (formula, type I)	fixed	$F = 3.69$, $p = 0.023$	7.584	1.452
Mixed Poisson with observation-level effect	random	Wald $\chi^2 = 16.07$, 4 df, $p = 0.003$	7.283	1.385

In the mixed model, the block variance ends up at zero and the app says so in the traffic light ("Singular fit: some variance component is zero"): in this trial the blocks contributed nothing, as their test in the fixed-effects model already indicated ($p = 0.468$).

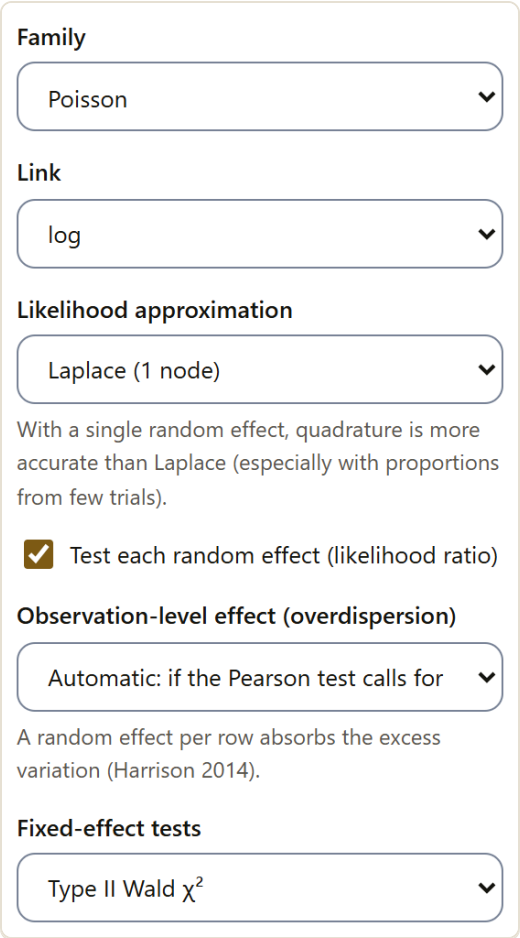


Figure 4.44. The options of the generalized mixed model: the likelihood approximation, the test of the random effects and the observation-level effect.

- Likelihood approximation: Laplace (1 node), by default, or adaptive Gauss-Hermite quadrature with 5, 10 or 25 nodes, more exact, especially with proportions from few trials. It appears with the binomial or the Poisson and a single random effect, an intercept per group (in the random-blocks method, always). The observation-level effect is a second random effect: with Yes, always, the app asks you to use Laplace; if the automatic mode adds it, the app returns to Laplace on its own and says so.
- Test each random effect (likelihood ratio), checked by default: compares the model with the one without that effect (with a model without random effects if none is left); its p is conservative. With the gamma it is not offered, because its Laplace likelihood cannot be compared with that of the model without random effects.
- Observation-level effect (overdispersion), with the Poisson or the binomial with trials: Automatic: if the Pearson test calls for it ($p < 0.05$ and χ^2/df greater than 1.1), Yes, always or No. It adds a random effect per row that absorbs the extra variation (Harrison, 2014).
- Fixed-effect tests: Type II Wald χ^2 or Type III Wald χ^2 (sum-to-zero).

4.18 Nonlinear models

Growth that slows down as it approaches its final size, a herbicide response that falls in an S shape, or a yield that stops rising with more fertilizer are neither straight lines nor parabolas: they have parameters with biological meaning (the asymptote, the dose for 50% effect, the optimum rate) that only a nonlinear model estimates directly. The family includes three methods.

Method	Purpose
Growth curve	Logistic, Gompertz or Weibull over time: the asymptote (the final size) and the inflection point.
Dose-response (log-logistic)	4- or 3-parameter log-logistic, with the ED10, ED50 and ED90, and 5- or 4-parameter Brain-Cousens for hormesis (a stimulus at low doses).
Plateau response to rate	Linear-plateau, quadratic-plateau, Mitscherlich and Michaelis-Menten (with and without a response at zero rate): the optimum rate, the maximum yield and, with the prices, the economic rate.

The three share the form: the Response variable, the Time (days, weeks) or the Dose, the Groups to compare (optional) (varieties, products), the Model, whose equation appears below it, and the confidence and significance levels. They run in the browser, by nonlinear least squares with an analytic gradient, validated against R's `nls`. With groups, one curve is fitted per group, and the extra-sum-of-squares F tests whether a common curve is enough and whether each parameter can be common. The result includes the fitted curve over the data, the residuals against the fitted values and a two-row traffic light: whether the fit reached the optimum and whether any residual exceeds 3σ. The other options of the form are these:

Option	What it does
Lowest AIC (fits all and compares), in Model	Fits all the models of the method to the same data (with the same groups and weights) and keeps the one with the lowest AIC among those that converge. The **Model comparison by AIC** table gives, for each one, its parameters, the AIC, the difference from the best (ΔAIC), the Akaike weight, σ and R^2.
Weights (optional)	A column with the weight of each row, for example the number of plants in a mean: the row with more weight counts more, like `weights` in `nls`. Rows with zero weight are left out; a negative weight stops the calculation.
Parameter intervals	Wald (estimate ± t·SE), symmetric, or Profile likelihood, which refits the other parameters at each value: they can be asymmetric and are more exact when the curve is strongly nonlinear, but they take longer. If the data do not bound a parameter on one side, that limit is shown as "—". The linear-plateau stays with Wald: its breakpoint does not allow profiling.
Dose on a logarithmic scale in the plot	In dose-response and in the plateau response. The dose is drawn on a log scale, and the zero dose goes one decade below the lowest dose, with the mark "0". It does not change the fit.
Product price (per unit of response) and Input price (per unit of rate)	Only in the plateau response, optional: with both, the economic optimum rate (below).

Growth curves

Three models: the **logistic** (symmetric around its inflection point), the **Gompertz** (which slows down more gradually) and the **Weibull** (the most flexible). With maize height on five dates and one logistic curve per nitrogen rate, the model explains 99.1% of the variation and the curves differ (F = 12.40 with 9 and 68 df). With Lowest AIC (fits all and compares) the logistic also wins: the Weibull, with one more parameter per curve, is at ΔAIC = 3.73, and the Gompertz, at 41.22.

B4

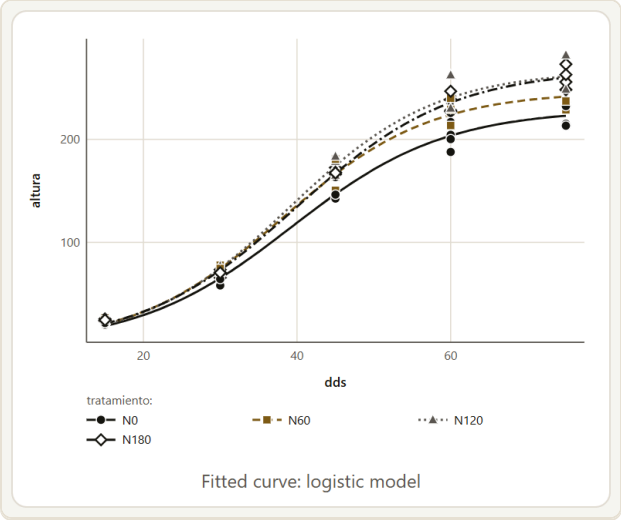


Figure 4.45. One logistic curve per rate: nitrogen changes the final height, not the timing of growth.

The comparison by parameter tells what changed: the **asymptote** (the final height) differs between rates ($F = 9.46$, $p < 0.001$), from 229.3 without nitrogen (N0) to 268.0 with N180; the **inflection point** (around 39 days in all four) and the **scale** do not differ ($p = 0.414$ and 0.895).

The parameter table gives each estimate with its SE, its interval and its meaning, and the residual plot completes the diagnosis.

B4

Dose-response

The 4-parameter **log-logistic** model (upper limit d and lower limit c, slope b and the **ED50** e) is the standard for herbicides, fungicides and toxicants (Seefeldt et al., 1995); the 3-parameter model fixes the lower limit at zero. The derived values are the doses for 10, 50 and 90% effect, with their SE by the delta method.

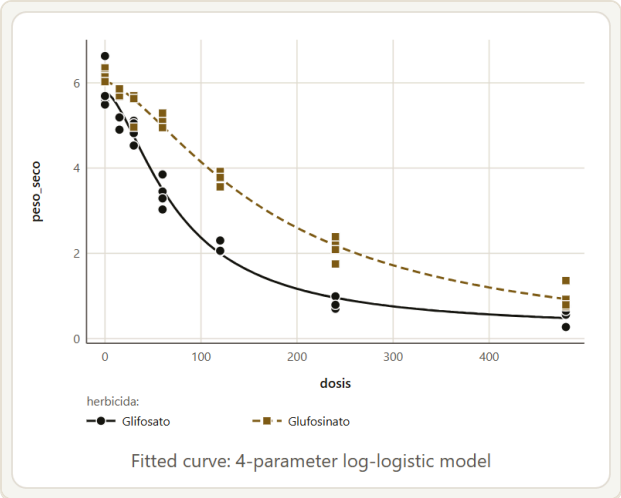


Figure 4.46. Weed dry weight with two herbicides: glyphosate reaches half the effect with less product.

The ED50 of glyphosate is 75.13 (95% CI: 63.32 to 86.94) and that of glufosinate 174.2 (121.6 to 226.7): it takes 2.3 times more glufosinate for the same effect. The curves differ ($F = 52.14$), but only in the ED50 ($F = 29.45$); the slope and the two limits can be common ($p = 0.732$, 0.508 and 0.178).

The ED90 of glufosinate (752.1) falls outside the doses tested, and its interval is very wide (237.9 to 1266): extrapolating the curve costs precision.

The **Brain-Cousens** models (the 5-parameter one, and the 4-parameter one with the lower limit at zero) add a parameter *f* for **hormesis**: a stimulus at low doses before the decline, frequent with herbicides at sublethal doses. With $f > 0$ and $b > 1$, the derived values are the **dose of maximum response**, the maximum response and the ED50, which here is the dose, after the maximum, at which the response falls halfway between *d* and *c*; their SEs come from the implicit derivative, like `ED` and `MAX` in `drc`. In these models the parameter *e* is no longer the ED50, and the meaning column says so. If there is no stimulus, the conclusion points it out: in the herbicides, *f* comes out negative (−0.027 for glyphosate and −0.017 for glufosinate), there is no dose of maximum response, and a log-logistic is enough.

> **THE SIMPLEST MODEL AND THE MOST HONEST INTERVAL**
>
> With `Lowest AIC (fits all and compares)`, in the herbicides the 3-parameter log-logistic wins (AIC = 22.56): the 4-parameter one is at ΔAIC = 2.85, because its lower limits do not depart from zero ($p = 0.318$ and 0.748 in the parameter table). With it, the ED50s are 79.41 (95% CI: 70.51 to 88.32) and 166.6 (150.4 to 182.8).
>
> With `Profile likelihood`, the interval of the parameter *e* for glufosinate (its ED50 in the log-logistic) goes from 121.6 to 226.7 (Wald) to 137.2 to 294.2: asymmetric, longer toward the high doses, where there is less information. The table of derived values keeps Wald intervals by the delta method, which are symmetric and, with poorly bounded curves, can reach negative values.

Plateau response to rate

For fertilization and irrigation, five models that stop at a maximum: the **linear-plateau** (rises in a straight line up to the rate x_0 and then stays flat), the **quadratic-plateau** (a parabola up to its maximum and flat afterward), and the **Mitscherlich** and the **Michaelis-Menten**, which approach their asymptote without reaching it; the Michaelis-Menten starts from zero at zero rate, and the variant `Michaelis-Menten with response at zero rate` also estimates that response. With the maize and one curve per hybrid, the linear-plateau explains 93.7% of the variation and the curves differ ($F = 12.38$ with 6 and 39 df).

Table 4.14. The optimum rate (start of the plateau) and the plateau yield of each hybrid, linear-plateau model.

Hybrid	Optimum rate x_0	95% CI	Plateau (t/ha)	95% CI
H-40	102.1	74.15 – 130.1	6.250	6.051 – 6.449
H-52	84.39	66.81 – 102.0	6.631	6.432 – 6.831
H-66	95.83	75.25 – 116.4	7.191	6.992 – 7.391

The comparison by parameter pinpoints the difference: between hybrids, the yield without nitrogen, a, changes (F = 3.93, p = 0.028); the slope and the optimum rate can be common (p = 0.357 and 0.498). With Lowest AIC (fits all and compares) the linear-plateau wins (Akaike weight 0.75), followed by the quadratic-plateau (ΔAIC = 2.28, weight 0.24), the Mitscherlich (9.13) and the Michaelis-Menten with response at zero rate (12.23). The Michaelis-Menten without it comes last, with a negative R^2: it passes through zero at zero rate, and the maize yields more than 4 t/ha without nitrogen.

The rate that gives the most yield is not always the most profitable one: each kilogram of nitrogen has a cost. With the two prices, the app calculates the **economic optimum rate**, where the slope of the curve equals the price ratio (the input price divided by the product price): there, the last unit of input still pays for itself with what it adds. With grain at 5000 per ton and nitrogen at 25 per kilogram, in the same currency, the ratio is 0.005: each additional kilogram must give at least 0.005 t (5 kg) of grain. It is calculated by the quadratic-plateau, the linear-plateau, the Mitscherlich and the two Michaelis-Menten models, with its SE by the delta method and the response at that rate.

B4

Table 4.15. The optimum rate and the economic rate of each hybrid, quadratic-plateau model, with grain at 5000 per ton and nitrogen at 25 per kilogram.

Hybrid	Optimum rate	Plateau (t/ha)	Economic rate	95% CI	Response (t/ha)
H-40	147.1	6.246	120.8	85.74 – 155.9	6.180
H-52	127.2	6.628	108.8	78.88 – 138.7	6.582
H-66	154.0	7.241	130.1	98.17 – 162.1	7.181

With the quadratic-plateau, the economic rate is 18 to 26 kg/ha below the optimum and sacrifices less than 0.07 t/ha of grain. In the linear-plateau the slope is constant: as long as the input pays (here, from 0.020 to 0.026 t per kilogram, much more than 0.005), the economic rate is the start of the plateau, the same x_0 as in table 4.14. If the curve starts flatter than the price ratio, the economic rate is 0 and the conclusion says that at those prices the first unit of input no longer pays for itself.

With the parabola (section 4.11), the optimum came out at 147.7 kg/ha; with the plateau, between 84 and 102. The difference is well known: the parabola forces the curve to drop after the maximum and pushes the maximum toward higher rates. When the data level off, as here between 120 and 180 kg, the plateau model usually recommends less fertilizer; the AIC comparison and the residual plot help decide which shape best describes the data.

4.19 Response surfaces

When two or more quantitative factors act together (nitrogen and potassium, temperature and time), response surface methodology looks for the combination that gives the best response. The factors (numeric, from 2 to 6) are **coded** (center at 0 and a step of 1) so that the coefficients are comparable, and the app converts everything back to the original units. The family includes five methods.

Method	Purpose
Generate a response surface or mixture design	Without data: the runs of a central composite, a Box-Behnken, a 2^k factorial with center points or a simplex for mixtures, in random order and ready to open in Data.
Response surface (second order)	Central composite or Box-Behnken: lack of fit, stationary point, canonical analysis and ridge.
First order and steepest ascent	Factorial with center points: the curvature and the path toward the best response.
Mixtures (Scheffé models)	Components that add up to the same total (substrates, rations, formulations): the best mixture.
Desirability of several responses	Several responses at once (high yield, low cost, a quality at a certain value): the most desirable combination.

All of them run in the browser; the models are validated against R's `lm` and `eigen`, and the designs, against their construction in R.

Generating the design

The generator needs no data. Choose the Design (Central composite, Box-Behnken, 2^k factorial with center points, Simplex-lattice (mixtures) or Simplex-centroid (mixtures)), the number of Factors or Components (from 2 to 6; from 3 to 5 in the Box-Behnken) and the Center points or the Centroid replicates (empty: the design's defaults). In the central composite, α sets the distance of the axial points: Rotatable (α = F^¼), with the same precision at the same distance from the center; Orthogonal; Face-centered (α = 1), with only three levels per factor; or Two orthogonal blocks, with the factorial and axial parts on different days or batches. From five factors on, it uses the half fraction of the factorial. In the simplex-lattice you also choose the Lattice degree (2, 3 or 4).

Table 4.16. The size of each design with three factors or components and the default center points.

Design	Runs	How they are formed
Rotatable central composite	20	8 factorial, 6 axial with α = 1.6818 and 6 center points
Orthogonal central composite	20	The same, with α = 1.5246
Face-centered central composite	20	The same, with α = 1

Table 4.16. The size of each design with three factors or components and the default center points. *(continued)*

Design	Runs	How they are formed
Central composite in two orthogonal blocks	20	The same, with α = 1.8091 and in two blocks
Box-Behnken	15	3 at the center; each factor at −1, 0 and +1
2^3 factorial with center points	12	8 factorial and 4 center points: for the first order and the curvature
Simplex-lattice of degree 2	6	The vertices and the midpoints of the edges
Simplex-lattice of degree 3	10	The mixtures in thirds, one of them at the centroid
Simplex-centroid	7	The vertices, the midpoints of the edges and the centroid

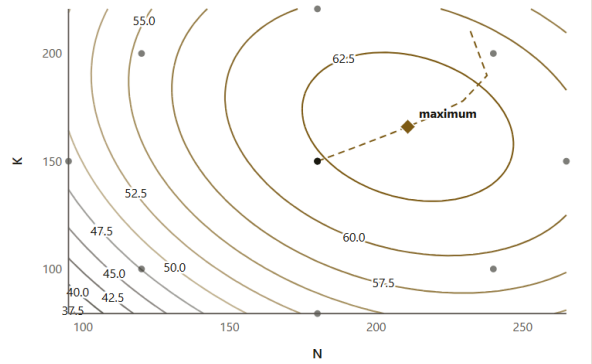
The conclusion says how many coefficients the second-order model has (or the Scheffé quadratic, in mixtures) and how many degrees of freedom remain for the error; if none remain, it asks you to add center points in surface designs and to replicate some mixtures (for example the centroid) in mixture designs, such as the simplex-lattice of degree 2 with three components: 6 mixtures for 6 coefficients.

With the Factor names (optional), the Low level (−1) of each factor (optional) and the High level (+1) of each factor (optional), separated by commas, the **Design runs** table gives each run coded and in original units; if an axial point falls at a negative level, the conclusion warns about it. In mixtures, the Mixture total (1 for proportions, 100 for percentages, or the total mass) converts the proportions into amounts. With a Seed for the random order (optional), the Run column gives the random field order, within each block, and the Standard column, the design order; the same seed always gives the same order. For example, a rotatable central composite with two factors, with the names "N, K", the low levels "120, 100" and the high levels "240, 200", has 13 runs (4 factorial, 4 axial with α = 1.4142 and 5 center points), with the axial points at N = 95.15 and 264.85 and at K = 79.29 and 220.71: the design of the tomato example.

The Open the design in Data (with a column for the response) button takes the runs to a new worksheet (chapter 2) with the columns "corrida" and "estandar", the block (if there are two), the factors (in original units if the levels or the total were given; otherwise, coded) and an empty "respuesta" column to record the result of each run. The plot shows the runs on the first two coded factors or, in a three-component mixture, in the triangle.

Second-order surface

For central composite or Box-Behnken designs. Choose the response, the Design factors (numeric, 2 to 6), the Blocks (optional), the Model (Second order (with quadratic terms) or First order with interactions) and, in Find, Maximum response or Minimum response. With the tomato in a central composite of nitrogen and potassium, the model explains 97.5% of the variation and there is no lack of fit (F = 0.04 with 3 and 4 df, p = 0.989).



Contours of "rendimiento": N × K

Figure 4.47. The contours of tomato yield: the maximum (diamond) and the ridge (dashed line) that starts at the center of the design.

The **stationary point** is a maximum at N = 210.8 and K = 166.0, with an estimated yield of 63.75 (95% CI: 62.61 to 64.89). It lies within the experimental region: 0.61 coded units from the center, while the data reach 1.42.

The **canonical analysis** confirms it: the two eigenvalues are negative (−2.360 and −4.086); with different signs it would be a saddle point, and one near zero would indicate a ridge, a line of almost equally good combinations.

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The tables are the ANOVA by groups of terms (with the lack of fit and the pure error), the coefficients coded and in original units, the coding, the stationary point, the canonical analysis and the **ridge**: the best combination at each distance from the center, useful if the optimum falls outside the region or is a saddle point. The plots are the contour plots of each pair of factors (up to six pairs), the **3D surface** of the first two factors, the response along the ridge with its confidence band, and the residuals. The Rotate the 3D surface: buttons (↺, ↻, ↑ and ↓) turn it sideways and show it from higher up or lower down. With more than two factors, those that are not in a plot are held at the stationary point if it is inside the region or, if not, at the center.

The Factor coding is, by default, From the design (center and step of its levels): with five symmetric levels (central composite), the step goes from the center to the factorial point and the axial points lie at ±α. With Manual: the −1 and +1 levels of each factor, two fields per factor appear (N: level −1, N: level +1...) to use another coding, for example that of the trial protocol. The predictions, the stationary point and the ridge in original units do not change; the coded coefficients and the distances do. In the tomato, with N from 100 (−1) to 260 (+1), the coefficient of N goes from 4.243 to 5.657 and the stationary point lies 0.50 units from the center instead of 0.61, but it remains at N = 210.8 and K = 166.0, with 63.75.

THE TRAFFIC LIGHT AND WHAT YOU ARE LOOKING FOR

The traffic light has three rows: the lack of fit (gray if there are no replicated points), the residuals beyond 3σ and the stationary point, in amber if it is a saddle point or falls outside the region. That last row does not compare the type of point with what you are looking for: with Minimum response, the tomato maximum leaves it green. What matters then is in the conclusion, which searches along the ridge: the lowest estimated response is 45.66, with N = 102.1 and K = 112.4.

First order and steepest ascent

Far from the optimum, a factorial with center points and a first-order model are enough: the **path of steepest ascent** tells you which way to move. With the tomato, the flat model has lack of fit (F = 11.32, p = 0.017): the response curves, and the app advises moving to the second order. The path rises 51.53 kg of N and 25.62 of K per coded unit; its table gives the combinations and the predicted response.

Mixtures

In a mixture, the proportions matter, not the amounts: if one component goes up, another goes down. **Scheffé** models have no intercept and are fitted on the proportions: each row is divided by its sum, so proportions, percentages or amounts all work, as long as every row adds up to the same total (if not, the app warns about it). Choose the response, the Mixture components (2 to 6; each row sums to the same total), the Scheffé model (Linear (each component alone), Quadratic (with binary blends) or Special cubic (with ternary blends)) and, in Find, Mixture with the highest response or Mixture with the lowest response.

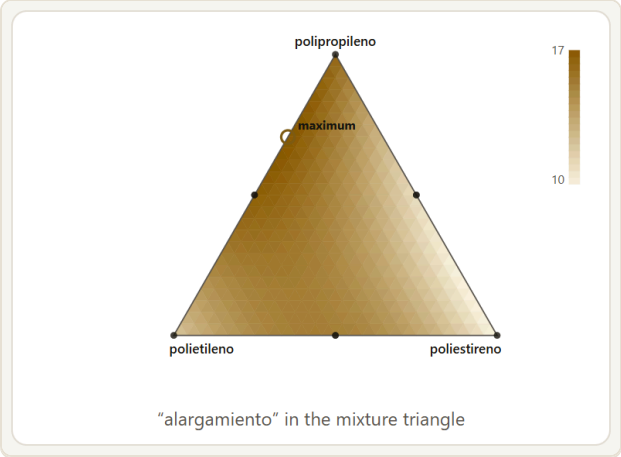


Figure 4.48. Yarn elongation in the mixture triangle: the best mixture (circle) combines polyethylene and polypropylene, without polystyrene.

With Cornell's (2002) yarn elongation, a simplex-lattice of polyethylene, polystyrene and polypropylene with replicates, the quadratic model explains 95.1% of the variation. The coefficient of each component is the response of the pure component (11.7, 9.4 and 16.4), and that of each pair, how far the blend departs from the sum of its parts: polyethylene with polystyrene (β = 19.0, p < 0.001) and with polypropylene (β = 11.4, p = 0.002) are **synergies**; polystyrene with polypropylene, an **antagonism** (β = −9.6, p = 0.005).

The best mixture of the model, searched over the whole triangle, has 29.4% polyethylene and 70.6% polypropylene, without polystyrene: an estimated elongation of 17.38 (95% CI: 16.36 to 18.41), more than any pure component.

The ANOVA tests the model against the mean (the proportions add up to 1, so the mean is already in the model) and the **lack of fit** against the pure error of the replicated mixtures. In the yarn, the quadratic has as many coefficients as distinct mixtures (6), and that row of the traffic light stays gray: there is nothing to test it with. The linear model does show lack of fit ($F = 32.32$ with 3 and 9 df, $p < 0.001$) and explains only 42.7%. The special cubic cannot be estimated with this design: the app says "The mixtures do not allow the special cubic model to be estimated" and explains that it needs the centroid of each three-component face. Besides the triangle (with three components), the **Cox trace** moves each component from 0 to 1 starting at the centroid and divides the rest equally among the others: the slope of each line tells how much weight that component carries. The method has no residual plot.

Desirability of several responses

A single response is rarely all that matters: you want high yield, low cost and a quality at a certain value. Derringer and Suich's **desirability** fits each response with its own second-order model (with the same design and the same coding), converts each prediction into a d between 0 (unacceptable) and 1 (what is sought) and combines them into an **overall desirability** D, their geometric mean: a single response outside its limits is enough for D to be 0. The combination with the highest D is sought with the Nelder-Mead method from each design point and from the center.

Choose the `Design factors (numeric, 2 to 6)` and the `Responses to combine (1 to 6)`. Each response has its own box: `Find` (`Highest`, `Lowest` or `A target value`), the `Lower limit` and the `Upper limit` (with the `Target value` between them, if you look for a value), the `Exponent` (1: linear; greater than 1 requires getting closer to the target; with a target value, the `Exponent (above the target)` governs the other side) and the `Importance`, which weights the response in the geometric mean. When you select a response, it looks for the highest value, with the limits at its minimum and maximum: you have to adjust them to what you require of each one. The `Search region` is `The cube of the design levels` or `The design sphere (rotatable central composite)`.

The method has no traffic light: if the model of any response has an R^2 below 0.7, the conclusion warns that its prediction, and with it the desirability, is not very reliable.

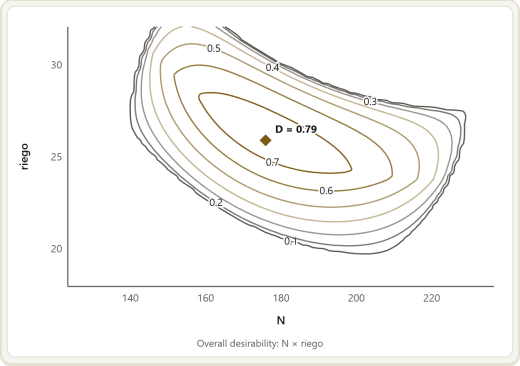


Figure 4.49. The overall desirability for the fruit: the diamond marks the combination of nitrogen and irrigation that best balances yield, cost and pH.

In the fruit example (a central composite of nitrogen and irrigation depth), with the highest yield between 70 and 86, the lowest cost between 11 and 15 and a pH target of 6.5 between 6 and 7, the most desirable combination is N = 175.9 and riego = 25.89, with D = 0.786. There the estimated yield is 79.50 (d = 0.593), the cost 11.72 (d = 0.819) and the pH 6.500 (d = 1.000). With the sphere, the same point comes out, because it lies close to the center.

The tables give the combination (coded and original), each response at that point with its d and the R^2 of its model (0.995, 0.992 and 0.976), and the D of each run, with the predicted responses: the eight runs on the edge get D = 0, because some response falls outside its limits, and the five at the center, 0.743.

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4.20 Multi-environment trials

The same genotypes evaluated at several locations or in several years respond differently to each environment: this is the genotype × environment interaction (G × E). The five methods of the family measure it, describe it and help choose genotypes that are stable or adapted to each region.

Method	Purpose
Combined analysis of several environments	The combined ANOVA with the G × E interaction, the analysis of each environment and the homogeneity of their errors.
Two-stage analysis (one variance per environment)	The adjusted means of each environment and the weighted combined analysis: for errors that differ between environments or genotypes that are missing.
Stability	The regression on the environmental index and six more measures: Wricke, Shukla, Lin and Binns, Huehn, Kang and Annicchiarico.
AMMI	The interaction axes with their tests, the ASV, the WAAS and the AMMI1 and AMMI2 biplots.
GGE biplot	Which won where, mean vs. stability, the ideal genotype and environment and the comparison of two genotypes.

All of them ask for the Response variable, the Genotypes (varieties, hybrids, lines), the Environments (locations, years, dates) and, if there are any, the Replicates or blocks (optional); without them (— one mean per cell —), each row is the mean of one genotype in one environment. They also ask whether Best is The highest response or The lowest response, and the significance level. At least 3 genotypes and 3 environments are needed (2 in the GGE). They run in the browser and were validated against R's `lm` and `svd`. At the foot of the combined analysis and the stability, the "Continue in the LABG suite" strip suggests ClusteringPro, to group genotypes or environments; at the foot of the AMMI and the GGE, PCAPro as well (chapter 10).

> **WHEN A GENOTYPE IS MISSING IN AN ENVIRONMENT**
>
> Except for the two-stage analysis, the methods need the complete genotype × environment table. The Impute empty genotype × environment cells (EM-AMMI) checkbox, checked by default, fills each empty cell with the additive model plus the interaction axes requested in Imputation axes (from 0, additive effects only, to 3; 1 by default) and repeats until the value stops changing. If G2 were missing in E5 and G7 in E12 in the oats, 4.173 and 1.660 would be imputed in 31 iterations: the conclusion and the **Cells imputed by EM-AMMI** table list them, the interaction loses one degree of freedom per cell (115 instead of 117), and the traffic light marks it in amber (in red if more than 10% of the cells are imputed). Without the checkbox, the app stops and says which combinations are missing.

Combined analysis

With the oats of Olivoto and Lúcio (2020), 10 genotypes in 14 environments with three replicates, the combined ANOVA finds the interaction (F = 2.76, p < 0.001): the ranking of the genotypes changes between environments. The genotypes are tested in two ways: against the error (F = 14.93) and against the interaction, as is appropriate if the environments are a sample of many possible ones (F = 5.41, the same as in the mixed model of section 4.15). G8 has the highest overall mean (3.004).

The method also analyzes each environment separately (mean, MSE, CV and the genotype F) and checks the **homogeneity of the errors**: in the oats, the largest MSE is 6.71 times the smallest (Bartlett p < 0.001), so the tests of the combined analysis are approximate, and the traffic light says so. With 12 genotypes or fewer, the **Each genotype across environments** plot joins the means of each genotype in the environments ordered from the lowest to the highest mean: the lines that cross are the interaction.

Two-stage analysis

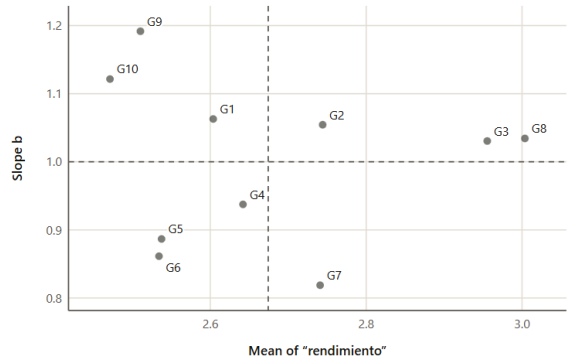
When the errors of the environments differ as much as in the oats, or genotypes are missing in some environments, the two-stage analysis is the better choice. In the first stage, each environment is analyzed on its own and gives the adjusted mean of each genotype with its variance; in the second, those means are combined by generalized least squares, each weighted according to its precision, and the interaction variance is estimated by REML. It does not impute: it uses the means that exist. Each environment needs its own error, so replicates or several plots per genotype are required.

In the oats, the genotypes differ (Wald: $F = 5.99$ with 9 and 117 df, $p < 0.001$), G8 has the highest adjusted mean (3.016 ± 0.07732), and the interaction variance, now without the error of each environment, is 0.05407. The traffic light row for the errors shows in amber (maximum/minimum ratio of 6.71): this is exactly the case the method is meant for. The tables are the first stage (mean, MSE, CV and genotypes of each environment), the adjusted means next to the simple mean, and the test of the genotypes; the plot gives each adjusted mean with its 95% interval. When there are replicates, the assistant (section 4.2) includes it among the next steps of any analysis in this family.

Stability

There are several ways to measure it. The **regression on the environmental index** (Finlay and Wilkinson; Eberhart and Russell) gives each genotype a slope b (1: it responds like the average; greater: it takes advantage of good environments; smaller: it is adapted to poor ones) and a variance of the deviations S^2_d (0: its response is predictable). **Wricke's ecovalence** and **Shukla's variance** tell how much each genotype contributes to the interaction, and **Lin and Binns' superiority** (Pi) how far it is from the best genotype in each environment. **Huehn's nonparametric statistics** work with ranks: S1 and S2 measure how much each genotype changes place between environments (with their Z test), and S3 and S6 combine response and stability. **Kang's YS** gives each genotype points for its yield rank, subtracts points for its Shukla variance and selects those above the average; it needs replicates. **Annicchiarico's reliability index** (Wi) expresses the response as a % of the mean of each environment and penalizes instability (Wi = mean − z·SD), across all environments, in the favorable ones and in the unfavorable ones; it is calculated only when the best is the highest response.

The options are the Standard error of b, From each genotype's regression or From the pooled error of the environments, and the Risk level of the Annicchiarico index (0.25, 0.10 or 0.05): a lower risk penalizes instability more.



Mean vs. slope (Eberhart and Russell)

Figure 4.50. Mean against slope: on the right and close to the $b = 1$ line are the genotypes that do well everywhere.

Table 4.17. Three oat genotypes according to each stability measure.

	G8	G3	G10
Mean	3.004	2.955	2.471
b	1.034	1.031	1.122
p of $S^2_d = 0$	0.013	0.840	< 0.001
Wricke (% of G × E)	8.23	2.32	25.48
Lin and Binns Pi	0.04117	0.04099	0.3437
Huehn S1	2.97	2.70	4.57
Kang YS	5	12	−9
Annicchiarico Wi (%)	107.41	107.98	77.54
ASV (AMMI)	0.574	0.113	1.225
WAAS (AMMI)	0.327	0.106	0.652

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G8 has the highest mean but significant deviations ($S^2_d > 0$): it yields a lot, though not always what is expected. G3, almost as productive, meets $b = 1$ and $S^2_d = 0$, is the most stable by Wricke and the best by Lin and Binns: the ideal genotype of Eberhart and Russell. The other measures agree: G3 is the one that changes place least between environments (Huehn), the one with the highest Kang YS and the one with the highest Annicchiarico Wi, just above G8. Kang selects seven genotypes (G1 to G6 and G8). G10, the one with the lowest mean, is also the one that contributes most to the interaction. This result does not repeat the combined ANOVA: whether there is interaction is seen in the combined analysis.

GGE biplot

The GGE biplot combines the genotype effect and its interaction (G + G × E), which is what matters for choosing. Choose the Table (Environment-centered or Centered and divided by the standard deviation of each environment) and the Singular value partitioning: Symmetric, Genotype-focused (to compare genotypes) or Environment-focused (to compare environments). In the oats, the first two axes explain 64.34% of G + G × E.

Two more plots place an **ideal genotype** (high and stable) and an **ideal environment** (one that discriminates strongly among genotypes and represents the others) on the average-environment axis, with distance rings, and their tables rank genotypes and environments by their distance to them. In the oats, the genotype closest to the ideal is G8 and the environment, E13, with any of the three partitionings (the table notes recommend the genotype-focused partitioning for the first and the environment-focused one for the second). With Compare two genotypes (optional) and ... with, a plot draws the line that joins them and the perpendicular that divides the environments, and a table compares, in each environment, the difference predicted by the biplot with the observed one: G8 outperforms G3 in 13 of the 14 environments according to the biplot (all but E9), but in only 9 according to the means. The biplot summarizes two axes; fine decisions are made with the data.

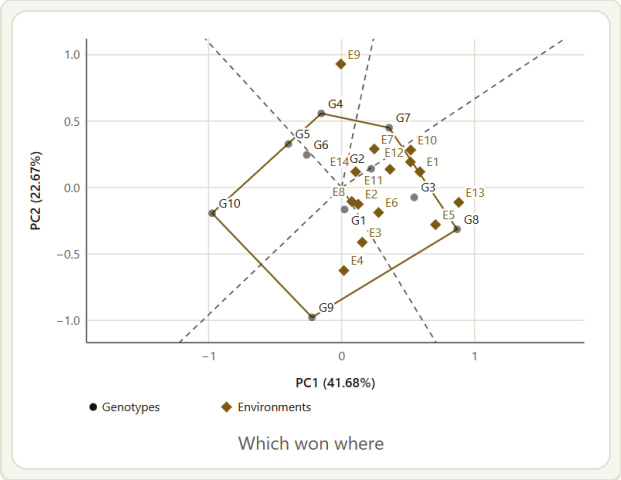


Figure 4.51. Which won where: the genotype at the vertex of each sector is the best in the environments that fall in it.

In **which won where**, the polygon joins the genotypes farthest from the origin, and the rays divide the plane into sectors. In the oats there are 4 sectors, which the conclusion calls "possible mega-environments": G8 wins in nine environments, G9 in E3 and E4, G7 in E7 and E14, and G4 in E9. With data from a single season they are no more than that: to declare them mega-environments, the pattern has to repeat across years (Yan et al., 2000).

The second plot, **mean vs. stability**, orders the genotypes along the average-environment axis: G8 has the best performance and, among the top three, G3 is the most stable.

AMMI

The additive main effects and multiplicative interaction model decomposes the interaction into **axes** (principal components of the table of interactions). The app tests each axis with Gollob's F and, with Cornelius's F_R, the interaction that remains after n axes, which is more reliable: in the oats, Cornelius retains 3 axes and Gollob 4; the first two explain 66.22% of the interaction. The tables give the genotype and environment scores and Purchase's **ASV** (lower, more stable: G3, with 0.11), and the plots are the AMMI1 (mean against axis 1) and AMMI2 (axis 1 against axis 2) biplots.

The **WAAS** averages the absolute scores of each genotype over several axes, weighted by the % of the interaction of each axis (lower, more stable), and the **WAASY** combines it with the response, both rescaled from 0 (the worst) to 100 (the best). The options are the WAAS axes (The significant ones (Gollob's F), or from 1 to 5) and the Response weight in the WAASY (%) (0: stability only; 100: response only; 50 by default). In the oats, with the 4 significant axes, G3 is the most stable (WAAS = 0.106) and the best by WAASY (95.46), followed by G8 (79.81). The **Response vs. stability (WAAS)** plot crosses the overall mean with the mean WAAS: the lower right quadrant brings together high response and stability.

> **TWO WAYS OF COUNTING THE AXES**
>
> The AMMI conclusion counts the axes that are significant by Gollob consecutively from the first one; the WAAS counts all the axes with a p below the significance level and uses that number of axes, starting from the first (without replicates there is no error to test them, and it uses all of them). If a non-significant axis fell between two significant ones, the two numbers would differ; in the oats they agree (4). To set another number, choose the axes in WAAS axes.

4.21 Multivariate analysis

When several variables are measured at once on each unit (the length, width, weight and wall of a fruit), analyzing them one by one repeats tests and wastes the relationship among them. MANOVA tests the treatments on the set as a whole, and discriminant analysis looks for the combination of variables that best separates the groups. Both run in the browser, validated against R's `manova` and `summary.manova` and against `lda` from the MASS package.

MANOVA

Choose the Response variables (2 to 10), the Treatment factors (up to 3, crossed), the Blocks (optional), the Main test and the Significance level. The main test is the one the conclusion uses: Pillai, the most robust and the default; Wilks; Hotelling-Lawley; or Roy, which gives an upper bound for F and an optimistic p. The MANOVA: the four tests table includes all four. The sums of squares are sequential (type I), as in `summary.manova`: the block enters first and then the factors, in the order of their columns in the table (not in the order in which you check them); with unbalanced data, that order changes the test of each term.

With four chili pepper varieties in three blocks, five fruits per plot, the variety changes the set of four measurements (Pillai = 2.668, $F(12, 159) = 106.40$, $p < 0.001$). The block is borderline: Pillai gives $p = 0.063$, and Roy, with its optimistic p, 0.005; that is why you should decide with the main test and not with the most favorable one. The traffic light checks that there are enough error degrees of freedom for the number of responses (green with at least three times as many; here, 54 for 4) and the equality of the covariance matrices with **Box's M test** among the treatment combinations, without the blocks ($\chi^2(30) = 30.13$, $p = 0.459$). The other tables give the ANOVA of each response separately, Box's M test group by group and the means of each factor; the plot shows the profile of the standardized means (with 12 levels or fewer).

> **! THE FRUITS OF A PLOT ARE NOT REPLICATES**
>
> The example has the five fruits of each plot in separate rows, and the MANOVA takes each fruit as a replicate: the error ends up with 54 degrees of freedom when the experiment has only 12 plots. This is the pseudoreplication that the method's fact sheet warns about (chapter 9), and it makes the tests too optimistic. To test the varieties with the correct error, average the fruits of each plot in your spreadsheet and import the table of means (SigmaPro has no tool for averaging rows). The table below compares the two analyses.

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Table 4.18. The chili pepper MANOVA with each fruit as a replicate and with the plot means (Pillai test).

	Each fruit as a replicate (60 rows)	Plot means (12 rows)
Error	54 df	6 df
Variety	Pillai = 2.668, $F(12, 159) = 106.40$, $p < 0.001$	Pillai = 2.969, $F(12, 15) = 120.26$, $p < 0.001$
Block	Pillai = 0.258, $F(8, 104) = 1.93$, $p = 0.063$	Pillai = 1.106, $F(8, 8) = 1.24$, $p = 0.386$
Traffic light: degrees of freedom	Green: 54 df for 4 responses	Amber: 6 df for 4 responses
Traffic light: Box's M	Green: $\chi^2(30) = 30.13$, $p = 0.459$	Gray: with 3 plots per variety and 4 responses, no group has more units than variables, and the test is not calculated (its table does not appear either)

The varieties differ anyway, but with the means the block is no longer borderline, and the traffic light calls for caution: with 6 error degrees of freedom, four responses are too many. With few plots, it is better to analyze fewer responses, the most important or the least redundant ones.

Below the plot, two buttons, CSV for PCAPro (principal components) and CSV for ClusteringPro (clustering), download only the variables of the analysis with their group (in the MANOVA, with the block and the factors), ready to open in those suite apps from their data block. At the foot of the result, the Continue in the LABG suite strip suggests the same two apps (section 10.2); its Send data to … button, in contrast, downloads the full table of the Data block (section 10.4).

Linear discriminant analysis

Choose the Groups to separate and the Variables that describe them (2 to 15). The result gives the discriminant functions with the proportion of the separation that each one explains, their coefficients, the group centroids and the **cross-validated classification**: each unit is classified with the function fitted without it, which gives an honest success rate (the one that uses the same data as the fit is optimistic). The traffic light checks that the smallest group has more units than variables, and the equality of the covariances among the groups (Box's M), which also comes with its table.

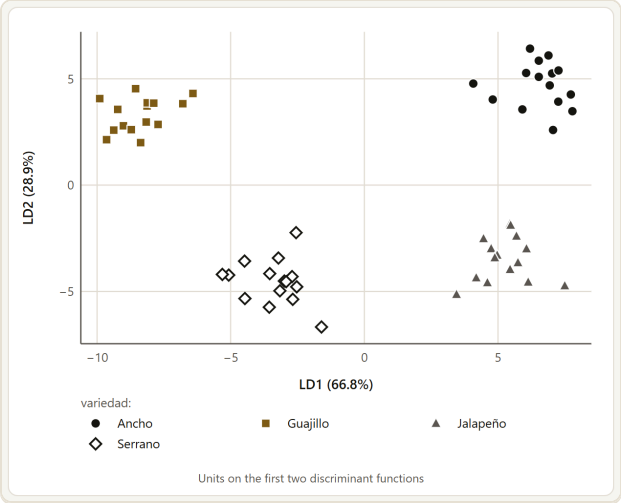


Figure 4.52. The 60 fruits on the first two discriminant functions: four clouds with no overlap.

With the chili peppers there are three functions; the first explains 66.8% of the separation and the second 28.9%. With cross-validation, all 60 fruits are assigned to their own variety (100%). Here each fruit is indeed a unit: what you want is to classify fruits.

The coefficients are not standardized: compare them only among variables on similar scales. The app warns about this when it names the variables with the most weight in the first function.

The suite strip proposes ClusteringPro, to look first for the natural groups in the data, and PCAPro; the two CSV buttons are the same as in the MANOVA.

B4

4.22 Power, replicates and sampling

Before the trial: how many replicates, how many plants or how many seeds you need to detect a difference that matters or to estimate with an acceptable error. Six of the eight methods of the Sample size, power and sampling family need no data. When data are loaded, they are in the catalog like any other method, with the No data needed mark; with an empty worksheet, the Calculate a sample size or power button of the Explore block enables them, and the catalog of the Analysis block then shows only this family.

The power calculations ask What do you want to calculate?: the sample size, the power of a given size or the effect that can be detected (the difference between means, the variation among the ANOVA means or the correlation; for two proportions, only the size or the power). They use the same formulas as R (`power.t.test`, `power.anova.test`, `power.prop.test` and `pwr.r.test` from the pwr package) and plot power against sample size. The Alternative hypothesis can be two-sided or one-sided; with a one-sided one, the app tests the positive direction, so the difference or the correlation must be written as positive numbers (with a negative number, the power comes out as 0% and the sample size makes no sense).

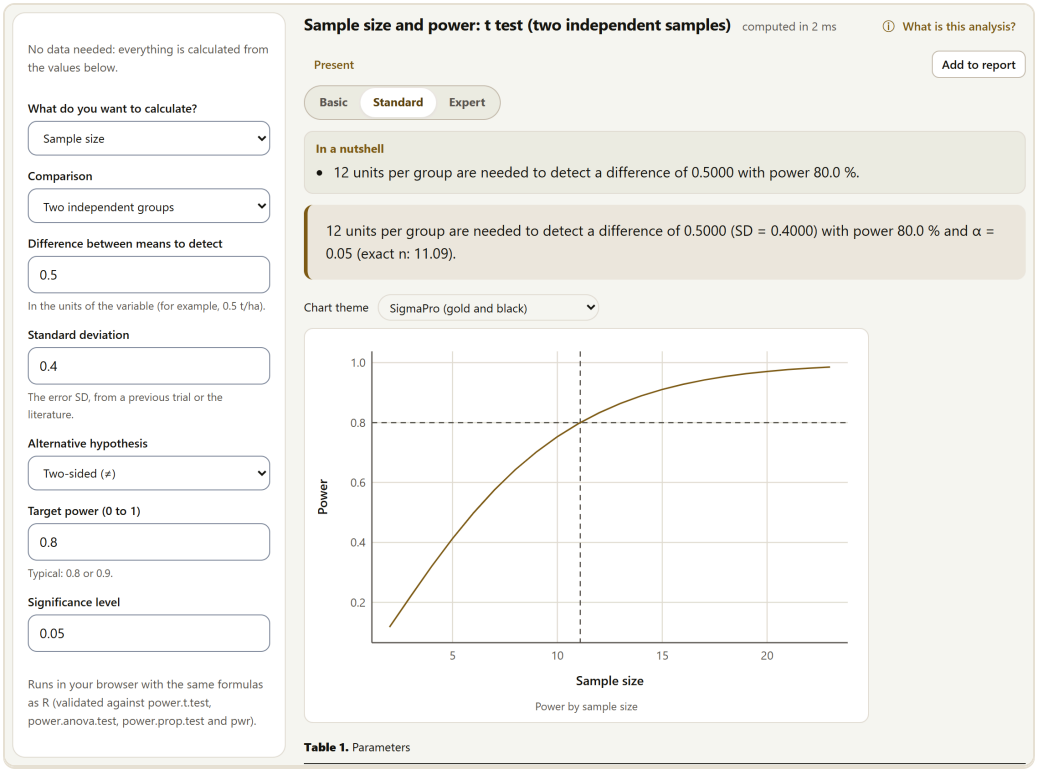


Figure 4.53. To detect a difference of 0.5 with a standard deviation of 0.4, you need 12 units per group.

Table 4.19. The six calculations without data, with an example of each.

Method	With	Result
Means (t test)	Difference 0.5, SD 0.4, power 0.8	12 units per group (11.09)
ANOVA of several treatments	Means 4.2, 4.8, 5.1 and 5.6; error SD 0.6	5 replicates per treatment (4.915)
Two proportions	65% versus 80%	138 units per group (137.9)
Correlation	40 pairs, power 0.8	A correlation of 0.4265 can be detected
Replicates for an experiment	CV 12%, difference of 15% of the mean, 6 treatments in blocks	11 replicates
Sample size to estimate	Proportion 0.3 with error ± 0.05, population of 900	238 units (323 without the finite population)

In the **ANOVA**, the `Expected treatment means (optional)`, separated by commas, replace the number of treatments and the variance of the means. To solve for `Detectable variation among means`, leave the means empty and enter the `Treatments`: in that mode the means are not used. With 4 treatments, 5 replicates, an error SD of 0.6 and a power of 0.8, a variance among means of 0.3349 can be detected (Cohen's $f = 0.8352$); the R code leaves as `NULL` whatever is being solved for.

The **replicates for an experiment** use the formula of Cochran and Cox (1957) with the Expected coefficient of variation (%) from previous trials, the Difference to detect (% of the mean), the Treatments and the Design (randomized complete blocks or completely randomized). The error degrees of freedom depend on the replicates, so the app searches for them until the formula holds: the Search for the number of replicates table shows each step, and the plot compares the required replicates with the ones plugged in. In the ANOVA and in the replicates, the Continue in the LABG suite strip proposes AgriDesign to randomize the trial and draw its field layout.

The **sample size to estimate** asks What do you want to estimate? (a mean or a proportion), the expected variation, the acceptable error and the Population size (0 if very large). It uses Cochran's (1977) formulas: $n_0 = (z \cdot S/e)^2$ for the mean and $n_0 = z^2 \cdot p(1-p)/e^2$ for the proportion. With a finite population, the mean is corrected with $n = n_0/(1 + n_0/N)$ and the proportion with $n = n_0/[1 + (n_0 - 1)/N]$, because the variance of a proportion carries the factor $(N - n)/(N - 1)$. In the example of the table, $n_0 = 322.7$ and $n = 237.7$, which is rounded up to 238. The plot shows how many units are needed for each error: halving the error quadruples the sample when the population is large.

B4

Estimating from a sample

The other two methods analyze a sample already taken, in the Variable measured in the sample. **Simple random sampling** estimates the mean or the proportion (Estimate: automatic, which gives the proportion if the column is 0/1) and, with the Population size (0 if very large or unknown), the total, with the finite population correction; its plot shows how the interval narrows as the sample grows. **Stratified sampling** also asks for the Strata (orchards, fields, regions, soil types) and the Population size of each stratum (N_h), one field per stratum with all the units in it, not only those sampled. With Allocate a new sample of (optional), it proposes how to allocate it among the strata: proportional and Neyman allocation.

With the avocado from three orchards (1,580 trees), the simple estimate gives a mean of 51.89 kg per tree (SE = 1.975) and the stratified one 54.32 (95% CI: 52.00 to 56.65), with a considerably smaller standard error (1.142) and 32.29 effective degrees of freedom (Satterthwaite). The difference has an explanation: the sample took 16 trees from Planicie, which has 57% of the trees and the most productive ones, and 20 from the other two orchards, which have 43%; the simple estimate weighs them equally and is pulled down, while the stratified one weighs each orchard by its size. That is why this example serves to compare the two methods, but the correct estimate for these data is the stratified one.

For a new sample of 60, Neyman takes 17 in Ladera, 29 in Planicie and 16 in Vega: more in the large strata and in the variable ones. Each stratum is rounded up, so the sum can exceed the total; when it does, the conclusion says so ("Rounded up, they add up to 62"). The How to allocate a sample of 60 table gives the unrounded values of the two allocations (proportional: 15.95, 34.18 and 9.87; Neyman: 16.14, 28.22 and 15.63) so you can adjust them by hand. If all the strata were measured completely (each N_h equal to the number sampled), it is a census: there is no sampling error, the SE is 0, the interval reduces to the estimated value, and the conclusion explains this.

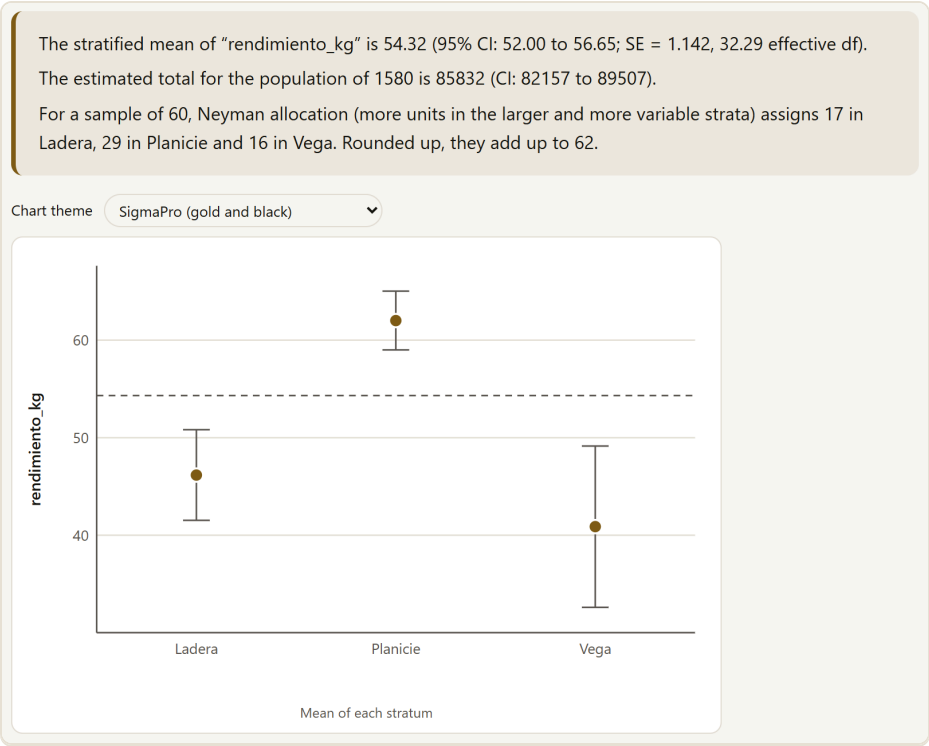


Figure 4.54. The avocado by orchard: the stratified mean weighs each orchard by its size, and the Neyman allocation warns when rounding exceeds the requested total.

B4

4.23 Survival analysis

When the response is the time until something happens (germinating, flowering, dying) and in some units it did not happen before the end of the trial, those units are **censored**: you know that their time is longer than the one observed, but not by how much. Averaging the days without them, or with them as if they had germinated on the last day, biases the result; survival analysis uses them correctly. Both methods were validated against R's `survival` package.

Both ask for the Time to event (or to censoring) and the Event (1 = occurred, 0 = censored): 0 if the unit finished the trial without the event or was lost. The Strata (optional) (lots, sites, sowing dates) give each stratum its own baseline hazard, and the groups are compared within them. The Confidence level changes the intervals; the conclusions and the traffic light always decide with $\alpha = 0.05$.

Kaplan-Meier curves

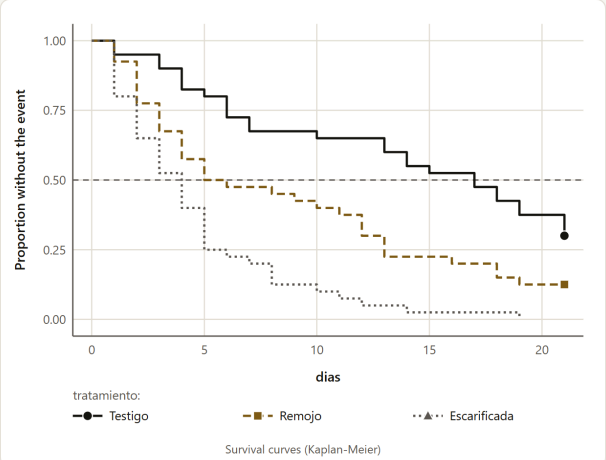


Figure 4.55. The proportion of seeds not yet germinated over time: scarification speeds up germination.

With 120 bean seeds in three treatments, checked every day until day 21, the median number of days to germination is 17 in the control, 5.5 with soaking and 4 with scarification, and the curves differ (log-rank $\chi^2(2) = 37.89$, $p < 0.001$). In all, 17 seeds (14.2%) did not germinate: they are censored, and the traffic light reports it (it turns amber if they are more than half).

The dashed line marks 0.5: where each curve crosses it lies its median. The marks at the end of the curves are the censored units.

B4

The Groups to compare (optional) can be up to 12; without them, a single curve comes out, with its band. The options are the Curve interval (log, the one in `survival`; log-log; or linear) and, with groups, the Test to compare the curves: log-rank, with the same weight for all times, or Peto-Peto, which gives more weight to early times (with the bean, $\chi^2(2) = 31.47$); the strata come into play when you choose groups (stratified by lot, $\chi^2(2) = 32.81$). The tables give the median of each group with its interval and the 25% and 75% quartiles (a "—" when the curve does not go down that far, like the 75% of the control, which stays at 0.30), the table of the test (observed against expected events) and the survival table at each time with events.

Cox model

Cox's proportional hazards model measures the effect of the Factors (treatments, varieties) and of the Numeric covariates on the instantaneous hazard of the event, as a **hazard ratio** (HR): when it is greater than 1, the event comes sooner. In the factors, each level is compared with the first. Tied times are handled with Efron (the one in `survival`, better with whole days) or with Breslow.

With the bean, the treatment and the seed weight, stratified by lot: scarification multiplies the hazard of germinating by 4.459 (95% CI: 2.628 to 7.568) and soaking by 2.096, relative to the control. The HR of the weight is 0.02058 **per gram**; since the seeds weigh about 0.4 g, it is better to read it per tenth of a gram: $0.02058^{0.1} = 0.678$, a third less hazard for each additional 0.1 g. The traffic light checks the **proportionality of the hazards** (like `cox.zph`: $p = 0.753$, no evidence that the effects change over time), that there are at least 10 events per coefficient (here, 103 events for 3 coefficients: 34.33) and that the fit converged. The tables include the three global tests and Harrell's concordance C (0.690); the plot shows each HR with its interval on a logarithmic scale, with the line at HR = 1.

4.24 Time series

A time series is a variable measured at equal intervals (every month, every week, every year), where nearby values resemble each other. The four methods ask for the Variable measured over time and the Time column (to order the series), and they were validated against R (`acf`, `pacf`, `Box.test`, `decompose`, `arima` and the `trend` package). The example is a weather station with the rainfall and temperature of each month from 2017 to 2024.

The time column is numeric (1, 2, 3..., or a period number) or of date type, and each time must appear only once; if the rows are already in order, — rows are already in order — is enough. The column only orders the series: the calculations assume equal intervals, so with dates there must be one per period (for example, one per month). Be careful with a years × months table converted to long format with Wide ⇄ long... (section 2.11): it ends up sorted by month and then by year, so "rows are already in order" would analyze a scrambled series. First create a continuous time column and use it as the time column; since the year usually ends up as a factor, the expression converts it to a number: `(as.numeric(anio) - 2017) * 12 + mes` gives 1 in January 2017 and 96 in December 2024. Autocorrelation and decomposition need the complete series, with no missing values; Mann-Kendall omits them and ARIMA keeps them in their place.

- **Autocorrelation**: the ACF and the PACF with their band, and the Ljung-Box and Box-Pierce tests. With monthly data, it is best to set the Lags for the Ljung-Box and Box-Pierce tests (0 = 10) to 12 or 24; with 12, monthly rainfall is strongly autocorrelated (Ljung-Box $Q(12) = 427.37$, $p < 0.001$), with the annual cycle in plain view: positive at 1 month, negative at 6. The conclusion names only the first eight lags outside the band; lag 12 (0.76), that of the annual cycle, is in the table and in the plot. To check the residuals of an ARMA model, the Previously fitted parameters (fitdf) discount their degrees of freedom.
- **Decomposition**: it separates the trend (moving average over one cycle), the seasonal component and the residual, additive or multiplicative (Type), with the Observations per cycle (frequency): 12 for months. In temperature, seasonality explains 96.01% of the variation without the trend. The plots show the series with its trend, the seasonal component and the residual.
- **Monotonic trend**: the Mann-Kendall test (it does not assume normality) and Sen's slope, resistant to extreme values, with the Slope confidence level. Temperature shows no trend over the eight years ($\tau = 0.058$, $p = 0.403$; Sen's slope of 0.008333 per month, 95% CI: −0.01364 to 0.02881).

As in survival analysis, the autocorrelation conclusions decide with $\alpha = 0.05$; the chosen level changes the band.

ARIMA and forecasting

The ARIMA model describes the series with its past values (autoregressive, p), its past errors (moving averages, q) and the differences that make it stable (d), with its seasonal version (P, D, Q and the Seasonal period (1 = no seasonal part)). In Model order, you choose Set it manually and enter p and q, or ask to Search for the lowest AIC (p, q = 0…2; P, Q = 0…1), which uses the d and D you set and, with a period, tries 36 models in a few seconds. Without differences, the Series mean is estimated or omitted. Last come the Steps to forecast (up to 120) and the Forecast interval level.

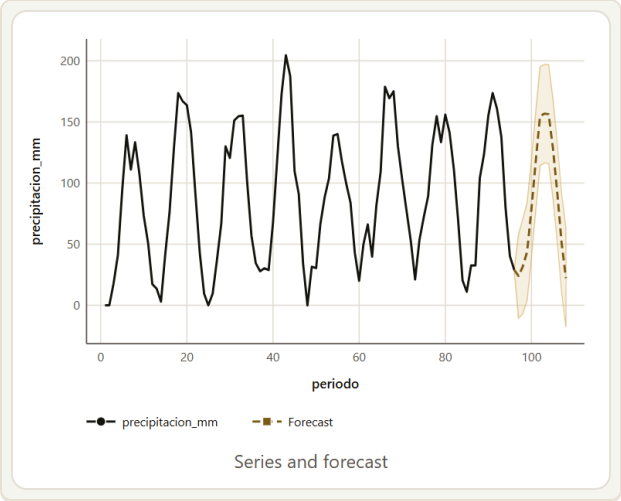


Figure 4.56. Monthly rainfall and its twelve-month forecast with ARIMA(1, 0, 0)(0, 1, 1)$_{12}$.

With the rainfall, ARIMA(1, 0, 0)(0, 1, 1)$_{12}$ leaves residuals without autocorrelation (Ljung-Box $Q(22) = 15.46$, $p = 0.842$): the traffic light is green. It is also the model the AIC search chooses with $d = 0$ and $D = 1$ (AIC = 739.14, the lowest of the 36). The forecast repeats the summer peak, close to 157 mm.

The lower limits of some dry months are negative (–10.33 mm in the first): the model does not know that rainfall cannot go below zero. To respect that, it is better to model a transformation of the variable (section 2.9).

The tables give the coefficients with their SE, the fit (σ^2, log-likelihood, AIC, BIC and the Ljung-Box test of the residuals, with 2 periods of lags if there is a seasonal part and 10 if not), the step-by-step forecast with its interval and, if the order was searched, the models compared by AIC. A second plot shows the autocorrelation of the residuals: if the model captured the structure, its bars stay inside the band.

5

BLOCK 5 · RESULTS

B5

Results: the LABG Report

Every analysis worth keeping goes to the report, where it joins the others in order, with tables and figures numbered consecutively, its assumptions, a paragraph for your thesis or paper, the technical details and the R code. The report does not store fixed results: it stores the requests, and it is recalculated with whatever data are there, so correcting a value corrects the whole report.

5.1 Adding to the report

5.2 The header and the index

5.3 Each analysis in the report

5.4 The paragraph for your thesis or paper

5.5 Technical details and R code

5.6 A living report

5.1 Adding to the report

In Explore and in Analysis, every result has in its header, to the right of ⓘ What is this analysis?, the Add to report button. When you click it, a notice confirms how many analyses the report holds (“Added to the report (3 in total). See it in the Results block.”) and the button changes to In the report ✓; if you click it again, it takes you to the Results block instead of adding the analysis twice.

What gets added is the **request**: the method, the variables with their roles, and the options. If you then change an option (another comparison test, another confidence level), that is a different request: the button says Add to report again and, if you click it, both stay in the report.

When the report is empty, the block offers to take you to Analysis or Explore, or to Open a .labg project: a saved project brings the data and the complete report (section 7.6).

5.2 The header and the index

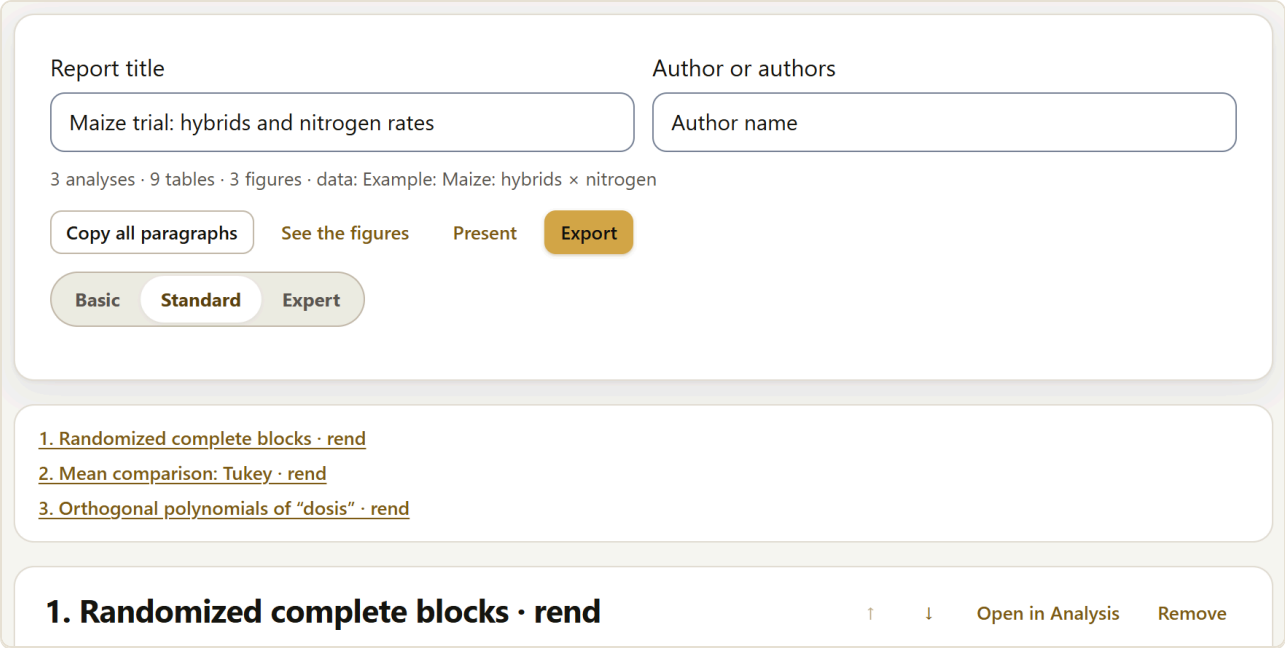


Figure 5.1. The header of a report with three maize analyses, its index and the beginning of the first analysis.

- Report title and Author or authors: they go on the cover of the exported document. The browser remembers the author for your next report.
- The **summary**: how many analyses, tables and figures the report has and which data they come from (and how many analyses could not be computed, if there are any).
- Copy all paragraphs (section 5.4), See the figures (chapter 6), Present and Export (chapter 7). Present shows the whole report as full-screen slides, one analysis after another, with its name in the top bar, so you can present it at a meeting or at a thesis defense (section 4.5).
- The **level of detail** (section 4.5): in Basic, each analysis shows the reading of its tables instead of the tables and hides the technical details and the R code; in Expert, it opens them. The exported document is the same at all three levels.
- The **index**: a link to each analysis, with its number.

B5

5.3 Each analysis in the report

Each analysis takes up a card with its number and its title (the method and the response: "Randomized complete blocks · rend"). The bar on the right has four buttons:

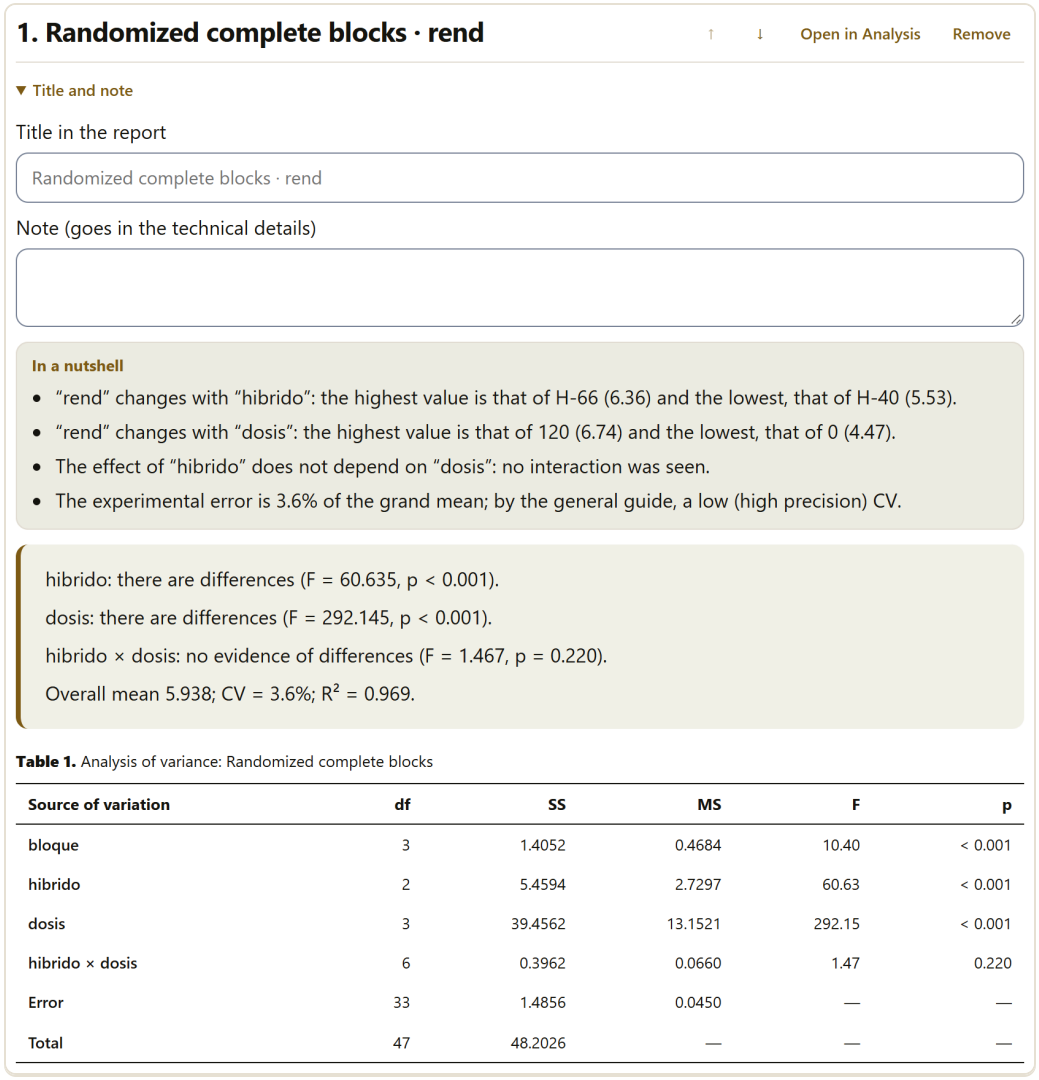
Action	What it does
↑ and ↓	Move the analysis up or down in the report; the tables and figures are renumbered.
Open in Analysis	For analyses that come from the Analysis block: takes the request to that block with its variables and its options and computes it again, so you can review or change it. Anything you change there is a new request: you add it to the report and remove the earlier one if it is no longer useful.
Open in Explore	For summaries that come from Explore, the same button takes you to that block, but it does not put the variables back: a notice reminds you that the summary is adjusted there and added again.
Remove	Takes it out of the report. Ctrl+Z does not bring it back: you add it again from its block.

B5

Below the bar, Title and note expands so you can change the **title** under which the analysis appears in the report (the suggested one stays as a guide, in gray) and write a **note**, which goes in the technical details: where the data came from, what was removed, what was decided and why.

Then comes the same content as in the source block, in this order:

- In a nutshell (in analyses from the Analysis block) and the **conclusion** (section 4.5).
- The **tables**, each with its Copy table button carrying its number (Copy table 1, Copy table 2...), which copies it tab-separated so you can paste it into a spreadsheet or into a table in a word processor. Download as CSV is not in the report: that button is in the result in Analysis or Explore (section 4.5), and the spreadsheet of chapter 7 contains every table with all its digits. Below each table, collapsed, is How to read this table (section 4.5): it is an on-screen aid; in its place, the exported document has a panel with gauges for the key statistics of each analysis (section 7.2).
- The **figures**, also numbered consecutively, with the Chart theme selector above them. It is the same one as in Analysis: it changes the theme of every chart in the app.
- The **Assumptions**: the analysis's traffic light, if it has one (section 4.5), with the What to do box and its recommendation. Completely randomized and randomized complete block designs have no traffic light in Analysis; in the report, SigmaPro checks the assumptions of the same model on its own (normality of residuals, equal variances among treatments, block × treatment additivity in block designs, independence of residuals, and outliers) and adds its traffic light. With the maize data, all five lights are green and What to do says: "The assumptions are reasonably met: the analysis of variance and the mean comparisons will be reliable."
- For your thesis or paper, with How to report it (APA) collapsed at the end (section 5.4), and, collapsed, the Technical details and the Equivalent R code (section 5.5).



1. Randomized complete blocks · rend ↑ ↓ Open in Analysis Remove

▼ Title and note

Title in the report

Randomized complete blocks · rend

Note (goes in the technical details)

In a nutshell

- "rend" changes with "hibrido": the highest value is that of H-66 (6.36) and the lowest, that of H-40 (5.53).
- "rend" changes with "dosis": the highest value is that of 120 (6.74) and the lowest, that of 0 (4.47).
- The effect of "hibrido" does not depend on "dosis": no interaction was seen.
- The experimental error is 3.6% of the grand mean; by the general guide, a low (high precision) CV.

hibrido: there are differences (F = 60.635, p < 0.001).

dosis: there are differences (F = 292.145, p < 0.001).

hibrido × dosis: no evidence of differences (F = 1.467, p = 0.220).

Overall mean 5.938; CV = 3.6%; R^2 = 0.969.

Table 1. Analysis of variance: Randomized complete blocks

Source of variation	df	SS	MS	F	p
bloque	3	1.4052	0.4684	10.40	< 0.001
hibrido	2	5.4594	2.7297	60.63	< 0.001
dosis	3	39.4562	13.1521	292.15	< 0.001
hibrido × dosis	6	0.3962	0.0660	1.47	0.220
Error	33	1.4856	0.0450	—	—
Total	47	48.2026	—	—	—

Figure 5.2. An analysis in the report with its title and its note expanded, In a nutshell, the conclusion and the first table. The report's tables are numbered consecutively, from the first analysis to the last.

5.4 The paragraph for your thesis or paper

In the For your thesis or paper section, each analysis has a **Methods** paragraph and a **Results** paragraph, written with its numbers and with the citation for the method when there is one. They are a starting point: the app says so next to the button, and you should review the wording and add the agronomic interpretation.

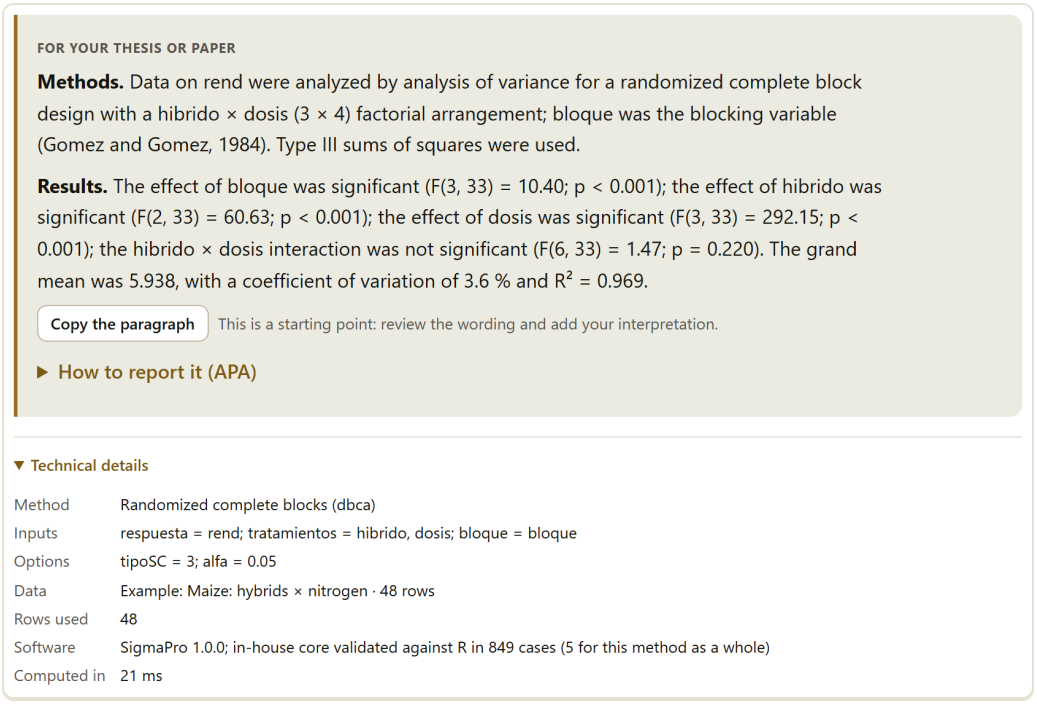


Figure 5.3. The Methods and Results paragraph of the maize analysis of variance, with How to report it (APA) collapsed and, below, its technical details.

Copy the paragraph copies the two paragraphs of one analysis. Copy all paragraphs, in the header, gathers those of the whole report into one text with two sections: **Methods**, with the paragraph of each analysis and, at the end, a sentence that cites SigmaPro and R with their versions and the number of validation cases (if any analysis ran in the R engine, another sentence says so, with the version of R used inside the browser); and **Results**, with the paragraph of each analysis, numbered. With the maize data, the results of the first two analyses read as follows:

RESULTS (TEXT COPIED FROM SIGMAPRO)

1. The effect of bloque was significant (F(3, 33) = 10.40; p < 0.001); the effect of hibrido was significant (F(2, 33) = 60.63; p < 0.001); the effect of dosis was significant (F(3, 33) = 292.15; p < 0.001); the hibrido × dosis interaction was not significant (F(6, 33) = 1.47; p = 0.220). The grand mean was 5.938, with a coefficient of variation of 3.6 % and R² = 0.969.

2. The means, from highest to lowest, were H-66 (6.359 a), H-52 (5.922 b) and H-40 (5.533 c); means sharing a letter do not differ.

The paragraphs use the column names exactly as they are in the table ("rend", "hibrido"): replace them with the full names when you paste them, or give the columns clear names in the Data block (section 2.7). The references cited in the paragraphs appear in full in the exported report.

The paragraphs, like the tables, come out in the language of the interface: with the selector set to ES (section I.5) they are written and copied in Spanish, with the same numbers.

Below the button, How to report it (APA) gives the same two paragraphs with APA 7 conventions (italics, *p* without a leading zero, intervals in square brackets) and in both languages, without changing the app's language (section 4.5). In APA style, the first maize result begins: "The effect of bloque was significant ($F(3, 33) = 10.40$, $p < .001$)".

5.5 Technical details and R code

At the end of each analysis, two collapsed sections hold what another person needs to repeat it.

The **Technical details** (figure 5.3) give the method with its internal name; the inputs (which column played each role); all the options, including those left as they came; the data (their source, the number of rows and the filter, if there was one); the rows that actually entered the computation; the software: the SigmaPro version, how many validation cases against R the app has and how many of them belong to that method (if it ran in the R engine, the versions of webR, R and its packages); the computation time; and the note, if one was written.

The **Equivalent R code** is the same as in section 4.5: a script that, with the exported data, gives the same numbers in R. Copy the code copies it. The .labg project (section 7.6) gathers the code of the whole report into a single script, together with the data.

5.6 A living report

Because the report stores requests and not results, each analysis is computed with the current data every time it is displayed or exported.

- If you correct a value in the worksheet, add a row or turn on a **filter** (section 2.10), every analysis in the report is recalculated, and the technical details state the filter it was computed with.
- If an analysis can no longer be computed (a column it used was deleted or renamed), its card says so with the box from section 4.5: what happened and what to do. The summary in the header counts it, and the others are unaffected. When the column comes back, so does the analysis.
- Analyses in the R engine show "Computing in the R engine…" while they run; the export waits for them.

> **CLOSING DOES NOT LOSE THE REPORT**
> The report, with the data, is saved automatically in the browser while you work (section 7.7), and you can save it in a .labg project to take it to another computer or share it (section 7.6).

B5

6

BLOCK 6 · FIGURES

Figures: ready to publish

The charts of the analyses in the report come together in this block, numbered and with their captions, in the order of the report. Here you choose the error bars, the chart theme and the width, download them as SVG, vector PDF, PNG or TIFF at 600 dpi, touch them up with the ✎ editor and, for the exact size a journal asks for, open them in the Figure Studio.

6.1 The figures in the report

6.2 Error bars, theme and width

6.3 The downloads

6.4 The ✎ editor

6.5 The Figure Studio

6.1 The figures in the report

The block shows all the figures of the analyses in the report that have charts (designs, comparisons, regression, correlation, contingency and almost all the others), in a gallery of cards. If the report has none, the block says so and offers to take you to Results.



Figure 6.1. The figures of the maize report. Each card says which analysis it comes from and has its numbered caption and its downloads.

Each card says which analysis it comes from ("From analysis 2: Mean comparison: Tukey · rend") and shows the figure with its **caption**: the number (the same one it has in the report and in the exported document), the title and a description of what it shows, which is also the figure's alternative text for screen readers. In mean plots, the caption states the error bars ("with ± one standard error"), as journals require.

6.2 Error bars, theme and width

Above the gallery, three settings apply to every figure in the report: on screen, in the exported document and in the downloads. The app remembers them in the browser.

- **Error bars**: ± 1 SE (the standard error of each mean, the default) or 95% CI (the confidence interval). When you change them, the mean plots are redrawn and their captions change.
- Chart theme: the look of the figures, among the six in table 6.1. The theme applies to every chart in the app: the same selector appears above the charts of each analysis in Analysis and in Results (also in Check the model assumptions), and changing any one of them changes them all.
- Width in the document and in the downloads: 8.5 cm (one column), 12 cm, 16 cm (page width, the default) or 17.5 cm (two columns). The height follows from each figure's proportions.

Table 6.1. The chart themes.

Theme	How the figures look
SigmaPro (gold and black)	The default: text and lines in black (in white with the app in dark mode), gold accents and a faint grid.
Minimal (grays, no grid)	Grays only, no grid and thin axes.
Grid (gray panel)	The plot area in gray with the grid in white and no axis lines; accents in blue.
Journal (black and white, serif)	Black and white, a serif typeface (Times New Roman), no grid and black axes: the style many print journals ask for.
Color-blind safe (Okabe-Ito palette)	The series in the Okabe and Ito palette (blue, orange, green, pink and sky blue), whose colors can be told apart with any type of color blindness.
Presentation (large text)	Larger text and thicker lines, for projecting in a room.

The theme changes the colors, the text and the lines, not the numbers: each figure shows the same data with any of the six. Whatever you touched up with the ✎ editor (section 6.4) is kept when you change the theme.

6.3 The downloads

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Each card downloads its figure at the chosen width in four formats. All of them come out with the chosen chart theme, in its light version even if the app is in dark mode, and with whatever you changed with the ✎ editor or the studio.

Format	What it is and what it is for
SVG	Vector, with the size in millimeters. It opens and can be edited in any vector drawing program, and it scales up without losing sharpness.
PDF	Vector, with lines and text in one of the standard PDF fonts (Helvetica; Times with the Journal theme): it is small, it scales up without losing sharpness and its text is searchable. It is the format many journals prefer.
PNG 600 dpi	An image at 600 dots per inch, with the resolution written inside the file: it goes into a word processor at its correct size.
TIFF 600 dpi	The same image in TIFF, the format some journals and publishers require.

To download all the figures at once, in a ZIP with the four formats, use the Export block (chapter 7).

6.4 The ✎ editor

Every SigmaPro figure, here, in Analysis and in Results, has a ✎ button in its corner that opens a floating panel to touch up **that** figure, as in a charting program.

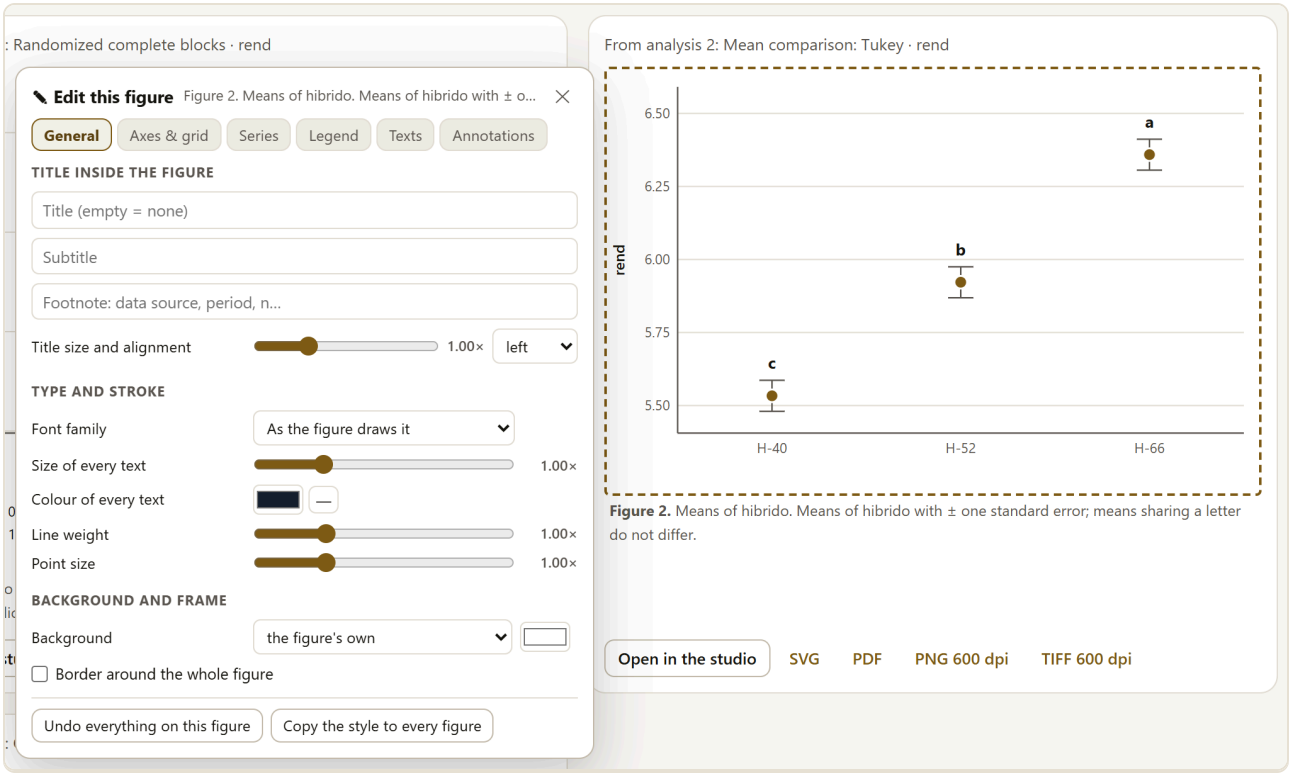


Figure 6.2. The ✎ editor with the figure of the hybrid means: the figure being edited is marked with a dashed border.

Tab	What you change
General	A title, a subtitle and a footnote inside the figure; the font family, the size and the color of all the text; the line weight and the point size; the background and a frame.
Axes & grid	The axis titles, the tick labels, the color and weight of the axes, the grid and the box around the plot area.
Series	Each color the figure uses, with the name of its legend entry: color, weight, stroke, opacity, or hiding it.
Legend	Show or hide it, eight positions, as it is, in a row or in a column, its size and its box; you can also drag it on the figure.
Texts	Each text in the figure, one by one: its wording, size, bold, italics, color, or hiding it.
Annotations	Notes, arrows, reference lines at a value of any of the axes, shaded bands and the panel letter (A, B...); notes and arrows can be dragged.

The changes are written into the figure itself, so what you see is what gets exported. The browser remembers them for each figure and reapplies them every time the figure is redrawn (when the data, the error bars or the theme change). Undo everything on this figure returns it to how SigmaPro drew it, and Copy the style to every figure carries the general settings to the others: the font, the sizes, the text color, the background, the frame and the style of the axes and the grid.

6.5 The Figure Studio

To give a figure the exact size a journal asks for, Open in the studio takes it to the LABG Figure Studio, the same one in every app of the suite: a split screen with the figure on the left, to scale and with the outline of its output size, and the inspector on the right.

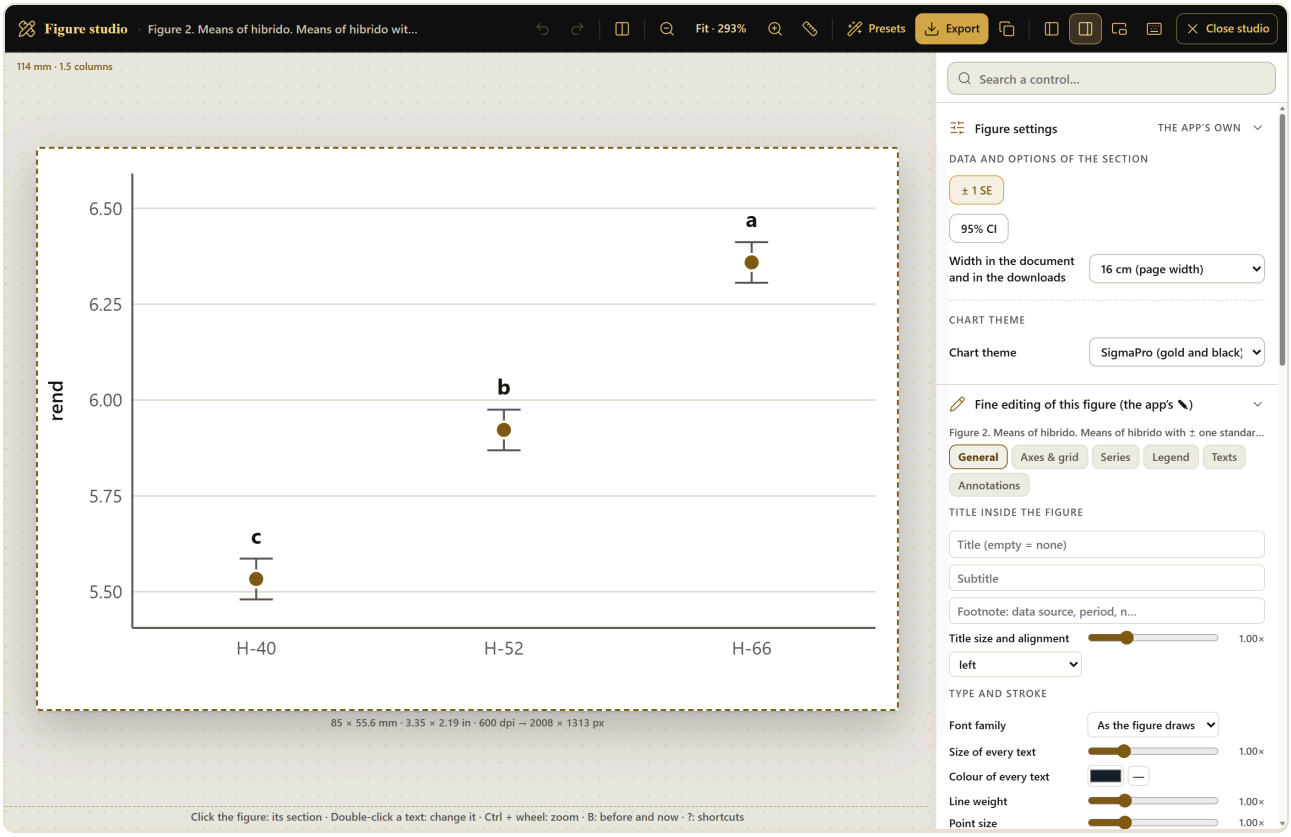


Figure 6.3. The Figure Studio: the figure at output scale on the left and the inspector on the right, with the settings the figure has in SigmaPro and the ✎ editor docked.

- **Journal sizes**: one, one and a half or two columns, in millimeters or inches, with text size and line weight designed for the output size. Below the figure, the final size and the number of pixels it will have.
- **Scientific palettes**, with a simulation of how a person with color blindness would see the figure.
- **The legend** inside the chart (eight places), outside on the right, below in a row or hidden, and you can drag it on the figure.
- **SigmaPro's settings, inside**: the error bars, the width and the chart theme from the Figures block appear in the inspector, under Figure settings (each change is made in the app), and the ✎ editor is docked as "Fine editing of this figure".

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- **Comfortable work**: undo and redo, before-and-after comparison with a curtain (B key), zoom, double-click on a text to change it, a control search, one-click presets (and your own, saved as JSON) and the ? key to see the shortcuts.
- **Export**: PNG, SVG, PDF and TIFF, at 300, 600 or 1200 dpi, with a transparent background if you want one. For the PDF, the studio asks SigmaPro for the vector figure instead of an image.

The studio remembers its settings in the browser and changes nothing in the app: when you close it, the figure goes back to its place with whatever was done to it. If for some reason it cannot open, the app keeps working as usual, with the ✎ editor and the block's downloads.

7

BLOCK 7 · EXPORT

Exporting and saving your work

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The report leaves SigmaPro as a document to edit or print, also in Markdown and as a class worksheet; its tables as a spreadsheet or LaTeX; and its figures in a single compressed file. The .labg project stores all your work so you can reopen it, share it or repeat it in R, and autosave protects it while you work. Everything is built in the browser: nothing leaves your computer.

7.1 The Export block

7.2 The full report

7.3 The class worksheet

7.4 The tables

7.5 The figures

7.6 The .labg project

7.7 Autosave

7.1 The Export block

A notice says how many analyses the report has; below it there are six cards. Each button builds its file in the browser and downloads it; if analyses in the R engine are still computing, it waits for them.

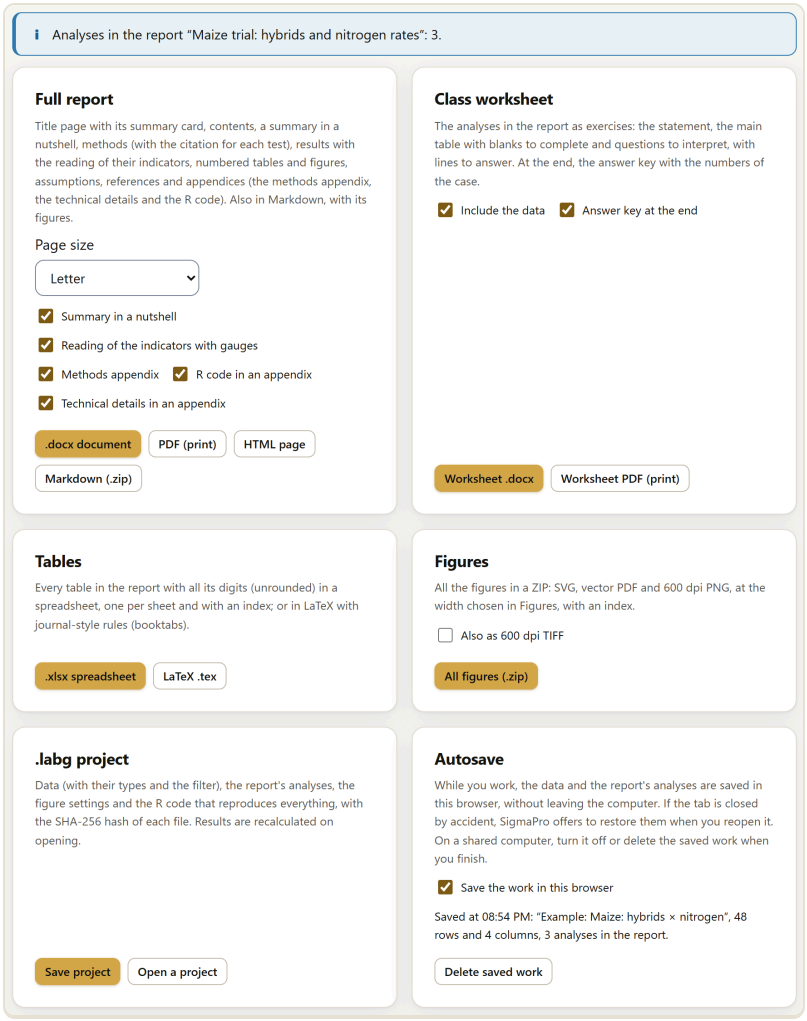


Figure 7.1. The Export block with the maize report.

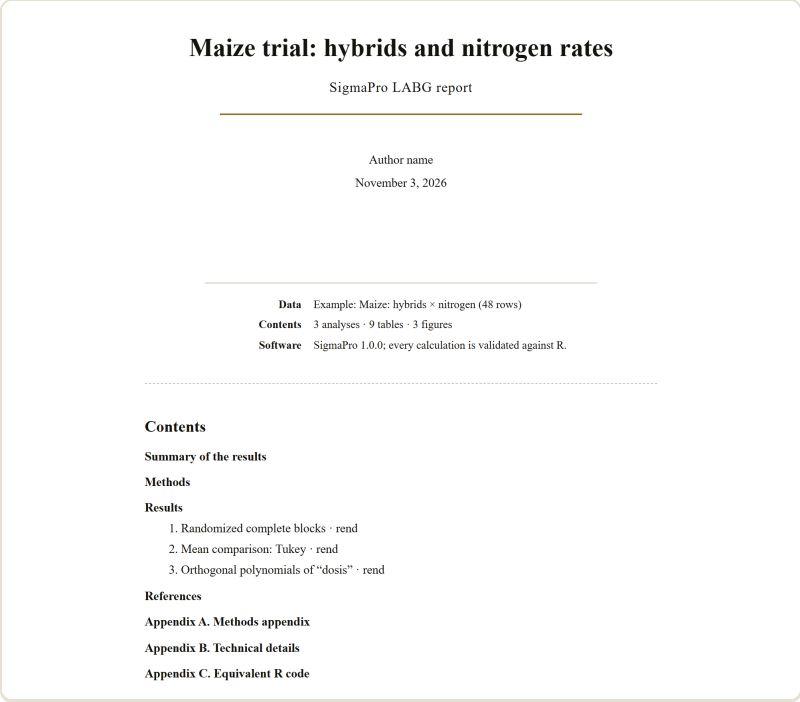
7.2 The full report

The report comes out in four formats with the same content: .docx document, to edit it in a word processor; PDF (print), which opens the browser's print dialog with the report paginated (there you choose "Save as PDF"); HTML page, a single file that opens in any browser, with vector figures; and Markdown (.zip), formatted text (see below).

You choose the Page size (letter or A4) and which parts it includes: the Summary in a nutshell, the Reading of the indicators with gauges and, in appendices, the Methods appendix, the R code and the Technical details of each analysis. The five checkboxes come checked, and the app remembers them. The document has:

1. A **cover** with the title, the author and the date, and an information box: the data (with their number of rows), the contents (how many analyses, tables and figures) and the SigmaPro version.
2. The **table of contents**, on a page of its own: the sections and, under Results, each analysis. In the HTML page and in the PDF each line is a link to its section; in the .docx it is a word-processor table of contents, which picks up the page numbers when its fields are updated (Word offers to do this when you open the document).
3. The **Summary of the results**: the In a nutshell box of each analysis (section 4.5), with its number and its title, for readers who want the essentials before the detail.
4. **Methods**: the methods paragraph of each analysis, with its citations, and the sentence that cites SigmaPro and R (section 5.4).
5. **Results**: each analysis with its number and its title, the conclusion, the results paragraph, the **reading of its indicators** (figure 7.3), its tables and its figures numbered consecutively and, if it has them, its assumptions with the traffic light.
6. **References**: the works cited by the methods, in full.
7. **Appendices**: the methods appendix (figure 7.4), the technical details and the R code of each analysis. They take the letters A, B and C in that order; if you leave one out, the others move up.

The **reading of the indicators** is the document version of How to read this table (section 4.5): one gauge per key statistic of the analysis, up to eight, in the order of its tables (the *p* of each effect, the CV, the R^2, the effect sizes…), each with its value and its zone, and at the foot the sources of the guides. The values of the pairwise comparisons and that of the intercept stay in their tables. In the .docx the panel goes in as a 300 dpi image with its alternative text; in the HTML page and in the PDF, as vector graphics.



Maize trial: hybrids and nitrogen rates

SigmaPro LABG report

Author name

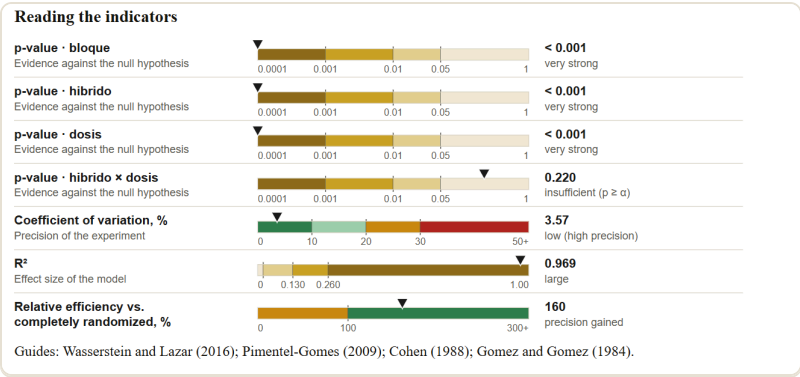
November 3, 2026

Data Example: Maize: hybrids × nitrogen (48 rows)
Contents 3 analyses · 9 tables · 3 figures
Software SigmaPro 1.0.0; every calculation is validated against R.

Contents

Summary of the results
Methods
Results
1. Randomized complete blocks · rend
2. Mean comparison: Tukey · rend
3. Orthogonal polynomials of "dosis" · rend
References
Appendix A. Methods appendix
Appendix B. Technical details
Appendix C. Equivalent R code

Figure 7.2. The beginning of the maize report exported as an HTML page: the cover with its information box and the table of contents. On screen, a dotted line marks where each page begins when printed.



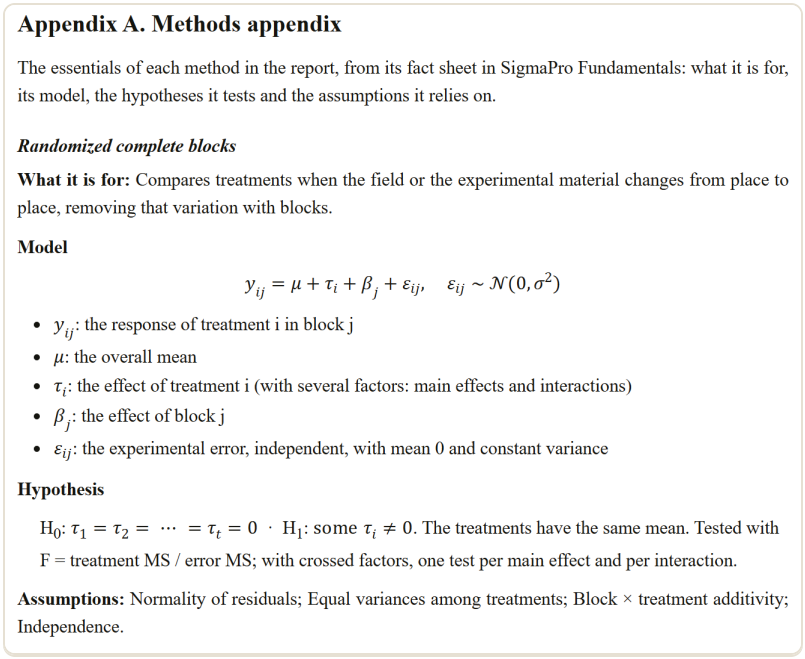
Reading the indicators

p-value · bloque Evidence against the null hypothesis	0.0001 0.001 0.01 0.05 1	**< 0.001** very strong
p-value · hibrido Evidence against the null hypothesis	0.0001 0.001 0.01 0.05 1	**< 0.001** very strong
p-value · dosis Evidence against the null hypothesis	0.0001 0.001 0.01 0.05 1	**< 0.001** very strong
p-value · hibrido × dosis Evidence against the null hypothesis	0.0001 0.001 0.01 0.05 1	**0.220** insufficient (p ≥ α)
Coefficient of variation, % Precision of the experiment	0 10 20 30 50+	**3.57** low (high precision)
R² Effect size of the model	0 0.130 0.260 1.00	**0.969** large
Relative efficiency vs. completely randomized, %	0 100 300+	**160** precision gained

Guides: Wasserstein and Lazar (2016); Pimentel-Gomes (2009); Cohen (1988); Gomez and Gomez (1984).

Figure 7.3. The reading of the indicators of the maize block analysis, in the exported HTML page.

The **methods appendix** gives, for each method in the report, the essentials of its fact sheet in Fundamentals (chapter 9): what it is for, its model with the meaning of each term, the hypotheses it tests, its assumptions and further reading. Each method appears once, even if the report uses it in several analyses. The formulas are in MathML in the page and in the PDF, and in the .docx they are **word-processor equations**, which you edit with its equation editor.



Appendix A. Methods appendix

The essentials of each method in the report, from its fact sheet in SigmaPro Fundamentals: what it is for, its model, the hypotheses it tests and the assumptions it relies on.

Randomized complete blocks

What it is for: Compares treatments when the field or the experimental material changes from place to place, removing that variation with blocks.

Model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

- y_{ij}: the response of treatment i in block j
- μ: the overall mean
- τ_i: the effect of treatment i (with several factors: main effects and interactions)
- β_j: the effect of block j
- ε_{ij}: the experimental error, independent, with mean 0 and constant variance

Hypothesis

H_0: $\tau_1 = \tau_2 = \cdots = \tau_t = 0$ · H_1: some $\tau_i \neq 0$. The treatments have the same mean. Tested with F = treatment MS / error MS; with crossed factors, one test per main effect and per interaction.

Assumptions: Normality of residuals; Equal variances among treatments; Block × treatment additivity; Independence.

Figure 7.4. The methods appendix of the maize report: the randomized complete blocks method, with its model and its hypotheses written as formulas.

Markdown (.zip) downloads a ZIP with `reporte.md` and the `figuras` folder. It is the same document as formatted text: each table as a table, with its numeric columns right-aligned and its notes; the table of contents with links to each section; the R code in code blocks; and the appendix formulas in TeX, between dollar signs. The figures and the reading panels go as 300 dpi PNG images in `figuras`. It is useful for taking the report to a code repository, a notebook or a document converter.

The tables in the .docx are word-processor tables, with the horizontal rules journals use, and you can edit them; the figures go in as 300 dpi images at the width chosen in Figures, with the current chart theme (table 6.1). The test battery opens each .docx with an independent reader and checks that its parts are valid and that the text of each table is exactly the app's, and it reads the Markdown with an independent CommonMark parser to check the same in its tables (chapter 8).

The document comes out in the language of the interface: with the selector set to ES (section I.5), the cover, the headings, the conclusions, the tables and the paragraphs come out in Spanish, with the same numbers; the comments in the R code are in Spanish in either language.

7.3 The class worksheet

For teaching with the analyses in the report: Worksheet .docx and Worksheet PDF (print) build an exercise sheet with the same data and, at the end, the answer key for the instructor.

The worksheet begins with lines for the name, the group and the date, some instructions and, with Include the data, the data table (up to 60 rows, with the Data filter if there is one). Each analysis in the report is an **exercise** with its statement, the report's methods paragraph (section 5.4), and numbered questions with lines to answer them (figure 7.5):

- **Complete the table**: in an analysis of variance, the mean squares and the F values are left blank; the sums of squares, the degrees of freedom and the *p* values are shown. In analyses without that table, the main table is given in full, to answer from it.
- **Decide**, with the *p* values and the analysis's α, which effects are significant.
- **Interpret** up to two indicators of the result, the single-valued ones: the CV, the R^2, an effect size.
- State which **assumptions** the method relies on, according to its fact sheet in Fundamentals (chapter 9).
- Write the **conclusion** in two or three sentences, without statistical symbols.

Exercise 1. Randomized complete blocks · rend

Data on rend were analyzed by analysis of variance for a randomized complete block design with a hibrido × dosis (3 × 4) factorial arrangement; bloque was the blocking variable (Gomez and Gomez, 1984). Type III sums of squares were used.

1. Complete the table: compute each mean square (MS = SS / df) and each F (the MS of the effect divided by the MS of the error it is tested against).

Table 2. Analysis of variance: Randomized complete blocks (to complete)

Source of variation	df	SS	MS	F	p
bloque	3	1.4052	_____	_____	< 0.001
hibrido	2	5.4594	_____	_____	< 0.001
dosis	3	39.4562	_____	_____	< 0.001
hibrido × dosis	6	0.3962	_____	_____	0.220
Error	33	1.4856	_____	—	—
Total	47	48.2026	—	—	—

2. With the p values in the table and α = 0.05, which effects are significant? Justify your answer.

3. Coefficient of variation, % = 3.57. What does this value say about the result?

Figure 7.5. The first exercise of the maize worksheet, in the print page: the ANOVA table to complete and the first questions.

With Answer key at the end, a separate page has the complete table, with the same number, and the answer to each question with the numbers of the case: the conclusion of the analysis, the interpretation of each indicator with its guide, the assumptions from the fact sheet and the In a nutshell box. To hand out the worksheet without the key, uncheck that box; you can edit the .docx to change or add questions.

7.4 The tables

- .xlsx spreadsheet: each table of the report on its own sheet, with **all its digits**, without the on-screen rounding, and a first sheet with the index. It is useful for further calculations, or for rebuilding a table in another format without retyping a single number. To download a single table as CSV, without building the spreadsheet, use the Download as CSV button of the result in Analysis or Explore (section 4.5).
- LaTeX .tex: the tables as LaTeX tables with the three journal rules of the `booktabs` package, one per `table` environment, with its title and its notes, ready to paste into a thesis or paper written in LaTeX.

7.5 The figures

All figures (.zip) downloads a compressed file with each figure of the report in SVG, vector PDF and PNG at 600 dpi, at the width chosen in the Figures block and with the changes made with the ✎ editor, plus an index with the number and the caption of each one. With Also as 600 dpi TIFF, it adds the TIFF. The figures come out with the current chart theme. A single figure is downloaded from its card in the Figures block (section 6.3).

7.6 The .labg project

The project stores **all your work** in a single file: the data, the report and the figure settings. Use it to continue another day, to take your work to another computer, to send it to an advisor or a reviewer, or to repeat all the analyses in R.

Save project downloads a file with the .labg extension, which is a ZIP with these files:

File	What it stores
`manifiesto.json`	The format, the SigmaPro version and the SHA-256 hash of each file.
`datos/hoja.json`	The complete worksheet: the column types, the factor levels, the computed columns and the filter.
`datos/datos.csv`	The rows the analyses use, as text, for R or any other program.
`analisis.json`	The report: each analysis with its inputs, its options, its title and its note.

File	What it stores
`estilos.json`	The figure settings.
`codigo/analisis.R`	The R code that repeats everything: unzip the project and run `source("codigo/analisis.R")` from its folder.
`LEAME.txt`	What the file is and how to open it.

Results are not stored: they are recalculated when the project is opened, so they come out exactly the same (the test battery checks this, table by table) and a project never carries outdated numbers.

Opening a project

You open a project with Open a project in Export, or with Open a .labg project in Results when the report is empty. The app checks the SHA-256 hash of each file against the manifest. If they match and there is no other work in the app, it opens the project directly; if data or analyses are loaded, it asks first, because opening the project replaces them.

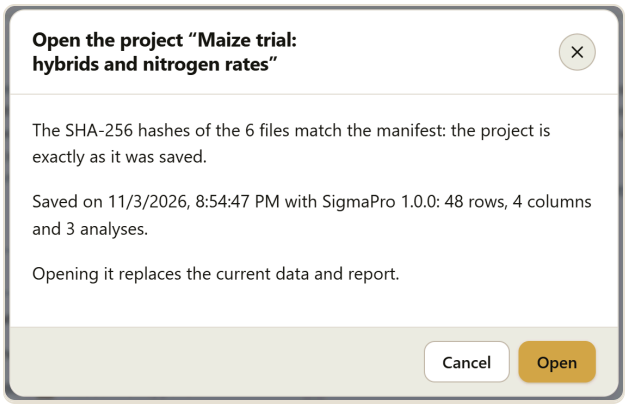


Figure 7.6. Opening a project with other work in the app: the hashes match and the app asks you to confirm.

If any file changed after it was saved (someone edited the CSV or the report by hand), its hash does not match: the dialog says so, with the name of each changed file, and the button changes to Open anyway. A file that is not a SigmaPro project, or that comes from another app of the suite or from a newer version of SigmaPro, does not open, and a notice says why. Once the project is open, the app goes to Results with the report already recalculated.

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7.7 Autosave

While you work, the data (with their types, their computed columns and their filter), the analyses in the report and the figure settings are saved in the browser's own storage, two seconds after the last change and whenever you switch tabs or close the tab. Nothing leaves your computer. Section 1.4 explains how to recover your work when you open SigmaPro again.

The Autosave card of the Export block says when the work was last saved and what was saved ("Saved at 12:28 PM: ..., 48 rows and 4 columns, 3 analyses in the report"). It has two controls:

- Save the work in this browser: checked by default. Unchecking it **deletes** what was saved and stops saving; checking it again saves right away.
- Delete saved work: deletes it now; the work is saved again at the next change.

If the browser does not allow saving (a private window, blocked or full storage), the app keeps working the same and the card says so: in that case, save the .labg project. If SigmaPro is open in two tabs, the card warns you when the storage is in use by the other one.

> **! ON A SHARED COMPUTER**
>
> The saved work stays in that browser, and the next person who opens SigmaPro will see the notice offering to recover it. On a lab computer or in a computer center, uncheck autosave or use Delete saved work when you finish, and take your work with you in the .labg project.

8

BLOCK 8 · VALIDATION

Validation: every number, against R

SigmaPro does not ask you to take it on trust: every result of the program is compared against R, with published or classic data, and the report of that comparison comes inside the app. This chapter explains the criterion, how to read the block, what else is checked besides the numbers, and how to repeat the whole validation on another computer.

8.1 The criterion

8.2 The Validation block

8.3 Five classes of results

8.4 Beyond the numbers

8.5 The release rule

8.6 How to repeat the validation

8.1 The criterion

A validation **case** is a dataset (from a book, a paper or an R package), a SigmaPro computation and R's answer to the same computation. The case passes if **every number** in R's answer matches SigmaPro's:

$$|SigmaPro - R| \le 10^{-6} \cdot max(1, |R|)$$

That is, six digits: absolute for numbers smaller than 1 and relative for larger ones. A few rules refine it: a tiny p value (smaller than 10^{-6}) must also match in its relative digits, so that a p of 10^{-20} does not pass just for being "almost zero"; R's missing values and infinities are expected as missing values and infinities; and text is compared exactly. Every key in R's answer is compared, and every one of them must exist in SigmaPro's result.

R's answers are computed by R scripts kept alongside the app (`tests/r/`), with R 4.4.2 and the reference packages for each method (`car`, `emmeans`, `lme4`, `survival`, `agricolae` and others). SigmaPro does not use R to compute, except in the R engine methods (section 4.6): its core is its own, and R is the judge.

8.2 The Validation block

The block shows the latest validation report, the same one that is generated when the tests are run, with its date at the top right. So that the app opens quickly, the full report is loaded the first time you open the block: the summary appears immediately and, for a moment, the table says "Loading the cases…".

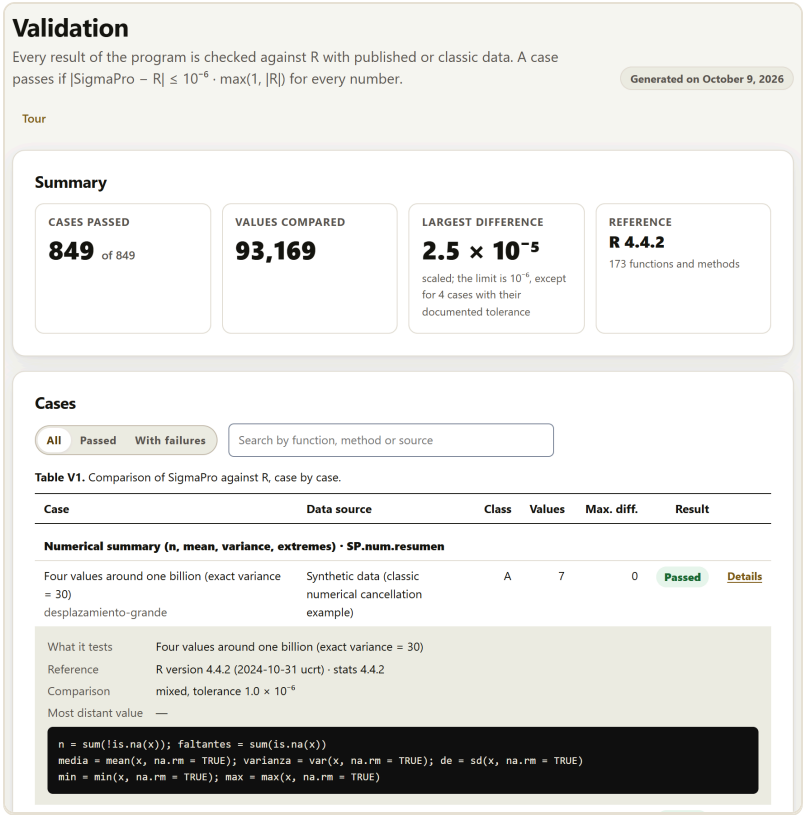


Figure 8.1. The Validation block: the summary, the filters and the first cases, with the details of one of them open.

- The **summary**: how many cases passed out of how many there are, how many numbers were compared, the largest scaled difference and the R version of the references, with how many functions and methods the cases cover. If the largest difference exceeds 10^{-6}, the tile says how many cases have their own documented tolerance (section 8.3).
- The **filters** All, Passed and With failures, and a search box by function, method or data source.
- The **table of cases**, grouped by the function or method they test. Each case says what it tests, where its data come from, its class (section 8.3), how many numbers it compares, the largest difference and whether it passed. The table shows the first 120 cases and brings in the next ones as you scroll down through it, or with Show more cases, which says how many are shown out of how many; the filters and the search box see all the cases, including those not shown yet.
- Details opens what lies behind each case: the R function and the versions of its packages, the comparison rule (with its tolerance; if the case has its own tolerances on some paths, it says so), the value that differed the most, and the R code that computed the reference, so you can repeat it.

Below the table, a card summarizes the **other checks** (section 8.4), each with its date, its result and its detail.

8.3 Five classes of results

Not all numbers can be compared in the same way. Each case carries a class that says what kind of computation it is and how it is validated.

Class	What it is	How it is validated
A	Closed formula or exact algorithm: sums of squares, F, t, estimators, their *p* values.	Against R, with 10^{-6}.
B	R uses a published approximation: the Shapiro-Wilk *p*, the quantiles of the studentized range.	The same published algorithm is programmed. If R itself is less precise by design, the reference is computed in R with more precision and the finding is documented.
C	R uses randomness: Dunnett by default, the simulated Fisher test.	A reference without randomness, by high-precision numerical integration; R's default value, separately, with the tolerance of its own error.
D	Iterative optimization: Box-Cox, REML, generalized models, nonlinear models, ARIMA.	Both programs with tightened convergence criteria; 10^{-6} on the converged result, or the documented tolerance of the path when R itself gives no more digits.
E	Tables from books and papers, printed with 2 to 4 decimals.	Half a unit of the last printed digit. The book or paper is cited, not the program it was computed with.

B8

When the tolerance of a path is not 10^{-6}, the reason is written in the case and appears in the published validation report; a tolerance is never loosened without saying why. The class B findings (R functions that, by design or by rounding, give fewer digits than requested) are described one by one in the validation document that comes with the app.

8.4 Beyond the numbers

Besides the cases against R, the test battery checks fifteen other things, each with its own independent program. Each one leaves its result, with its date, in the other checks card of the Validation block.

Check	What is checked
The equivalent R code	The code the app shows (section 4.5) is run in R, scenario by scenario, and must give the same numbers as the app.
The exports	The spreadsheet is read with R, cell by cell; the .docx, with an XML reader; the PDFs, PNGs and TIFFs, with image readers; the project, with its hashes, its R code run in full and reopened with the same tables; and autosave, recovered in a new page.
Performance	With about 100,000 rows, each common analysis within its time target.
Privacy and offline use	With data, analyses and exports, no data leave the browser; with no network, the app opens, computes and exports.
Accessibility	Every block and the analyses of every family, at desktop and phone widths, in light and dark mode: AA contrast of the text, names and labels for screen readers, no horizontal overflow, and keyboard use. In addition, an independent automated checker applies the WCAG 2.1 rules, levels A and AA, to each scenario; it is used only in the tests and is not part of the app.
The English interface	Every text of the interface, the results and the examples has its translation; each validation case gives the same numbers in Spanish and in English; and no result built in English leaves untranslated text.
Regression of results and reports	The fingerprint of each result (all its numbers) and of its report (tables, conclusions, traffic light and notes) is compared with the stored one: a number or a text that changes unintentionally is caught before a version is released.
Opening sequence	The opening screen (section 1.6) appears before the interface, ends on Start, is dismissed with Esc, respects the reduced-motion preference, fits on a phone and runs smoothly.
Fundamentals of each analysis	Each analysis has its fact sheet (chapter 9) with its nine sections in Spanish and in English; its related analyses, its example dataset and its columns exist in the app; the example runs with the method; and every formula is converted for display.
Glossary and "See it in action"	Each glossary term (section I.6) opens its example, and the highlighted number exists in the table it names, in Spanish and in English.
What guides you	The three paths of Home, the history, the empty states, the decision tree, the help, the tours of the ten blocks, presentation mode, the errors in plain words, and Validation opened from a direct link.

Check	What is checked
Quality of what you read	Each reading scale (section 4.5) is tested at each of its boundaries: just below, at the limit and just above. And every number in the conclusions, in In a nutshell and in the paragraphs of all the cases must be in the result or its tables, at the rounding with which it is written, or follow from them by a calculation the text states; a control with altered numbers measures whether the comparison detects them.
Links and icons of the app	Every file the page names exists; every link to a block, a fact sheet, a glossary term or an analysis in the tree leads to something that exists, including those the app builds on the fly; the icons measure what they declare; and the external addresses the app shows respond (this part needs internet).
Speed and best practices	A web page auditor measures the loading of Home, Fundamentals, the Suite and Validation on a phone with a slow network and on desktop, serving the app as on its published page: performance, accessibility, best practices and search engine visibility, with 90 or more out of 100 in each. It is not part of the app either.
Links and logos of the LABG Suite	Each app of the suite (chapter 10) has its logo and its data, or a declared "PENDIENTE" (pending); the suggestion rules name apps that exist; and the page, the repository and the manual of each published app respond. It needs internet: it checks the pages of the other apps, something the app never does.

8.5 The release rule

A method **does not appear in the app** until it has validation cases and all of them pass. The app checks this on its own when it opens, with the validation report it carries: if a case of a method failed, the method would disappear from the catalog and, if the assistant suggested it, the app would say that its validation has not passed yet instead of computing it. That is why everything this manual describes is validated.

8.6 How to repeat the validation

The validation does not depend on whoever did it: with R and node installed, anyone can regenerate the references with their own R and compare again.

You need R (4.4 or later) with the packages each case requires, and node (20 or later; nothing needs to be installed with npm). From the SigmaPro folder:

```
Rscript tests/r/generar-todo.R      # regenerates the references with your R
node tests/motor-r.mjs              # the R engine methods, in the browser
node tests/correr.mjs               # compares and writes the reports
node tests/navegador.mjs            # the same in the browser, via file://
node tests/codigo-r.mjs             # the app's R code gives the same in R
node tests/exportar.mjs             # the exports, with independent readers
```

B8

`motor-r.mjs` opens the app in the browser and downloads the R engine, so it needs internet; it runs before `correr.mjs`, which merges its cases into the report. `correr.mjs` writes the report that the Validation block shows and a version for publishing, in HTML and in CSV. The other checks of section 8.4 have their own programs in `tests/`: `idioma.mjs`, `regresion.mjs`, `calidad.mjs`, `intro.mjs`, `fundamentos.mjs`, `glosario.mjs`, `orientacion.mjs`, `enlaces.mjs`, `suite-enlaces.mjs`, `sin-conexion.mjs`, `rendimiento/medir.mjs`, `rendimiento/lighthouse.mjs` and `accesibilidad/auditar.mjs`; the two link checks need internet. The two external tools, the accessibility checker and the speed auditor, are installed separately, and an environment variable tells the tests where to find them: without the checker, the accessibility audit runs everything else and warns that that part was not done; without the auditor, the speed measurement does not run.

On Windows, `tests/pruebas.ps1` finds R, node and the browser and runs the comparison against R, the equivalent R code, the English interface, the regression, the fact sheets, the quality of what you read, the browser test, the glossary, what guides you and the opening sequence. With `-R`, it first regenerates the references; with `-Completo` it adds the exports, privacy, performance and accessibility (about 15 minutes) and, if the speed auditor is installed, its measurement (about as long again); and with `-Red`, the checks that need internet: the R engine and the links of the app and of the suite.

9

BLOCK 9 · FUNDAMENTALS

Fundamentals: what each analysis does and what data it needs

Each of SigmaPro's 78 analyses has a fact sheet: what it is for, the model and its hypotheses, the assumptions and how to check them in the app, the data it needs, how to read the result, its limits and where it comes from. This chapter explains how to reach the fact sheets, how to search them and what each one contains.

9.1 Three ways to reach a fact sheet

9.2 The catalog and its filters

9.3 The method map and the decision tree

9.4 A fact sheet, section by section

9.5 Data needed and the templates

9.6 Formulas, language and offline use

9.7 How the fact sheets were reviewed

9.1 Three ways to reach a fact sheet

The fact sheets live in block 9, but they are one click away from where you need them: when you choose what to analyze, when you have a result in front of you and when you are getting started.

From	How
The block bar	Fundamentals opens the catalog of the 78 fact sheets, with its search box and its filters (9.2).
An analysis	In the catalog of the Analysis block, each analysis carries the ⓘ icon (*What is this analysis?*); with an analysis open, the ⓘ What is this analysis? button is in the header of its result, also in the four summaries of the Explore block. Both open its fact sheet.
The Home block	The What you can analyze card lists the 78 analyses by family; each name opens its fact sheet.

From the fact sheet you go back to work with Open this analysis, which opens the method in the Analysis block (or, for descriptives, frequencies, proportions and goodness of fit, in the Explore block), and with Load the example data, which also loads the fact sheet's example dataset and fills in the form with its columns. That way, every fact sheet can be checked with one click.

9.2 The catalog and its filters

The catalog groups the fact sheets by family, like the catalog of the Analysis block. Each card says what the analysis is for and carries two tags: the type of response it measures and how its groups or factors are organized.

Control	What it does
Search	Searches the name, the summary, the family, the "what it is for" and the other names of each analysis (for example, "RCBD", "Tukey", "heritability"). If you type several words, all of them must appear; it ignores accents and capitalization.
Family	The 18 families: the 17 of the Analysis block and *Descriptives and intervals*, the family of the four summaries in Explore.
Response type	Continuous measurement, counts, yes / no (0/1), proportions, categories (tables), time to an event, time series, several responses or no data (planning).
Groups and factors	One sample, two groups, three or more groups, crossed factors, blocks, split plot, continuous predictors, random effects, several environments or no factors.
Design	The experimental design in which it is used: completely randomized, randomized complete blocks, Latin square, split plots and strips, lattice, incomplete blocks, repeated measures, several environments, sampling, response surfaces or no experimental design.
Glossary	Opens the glossary (also with Ctrl G): the statistics and concepts the fact sheets use, each with its definition and "See it in action" (section I.6).

The filters combine, and the count at the top says how many analyses remain; if none qualifies, Clear the filters resets them to All. At the foot of some families, the Related tools strip shows the LABG suite apps that continue that work (chapter 10). Fundamentals is also shown in full in the bar's Focused mode (section I.6): the catalog, the map, the tree and each fact sheet have their own guide.

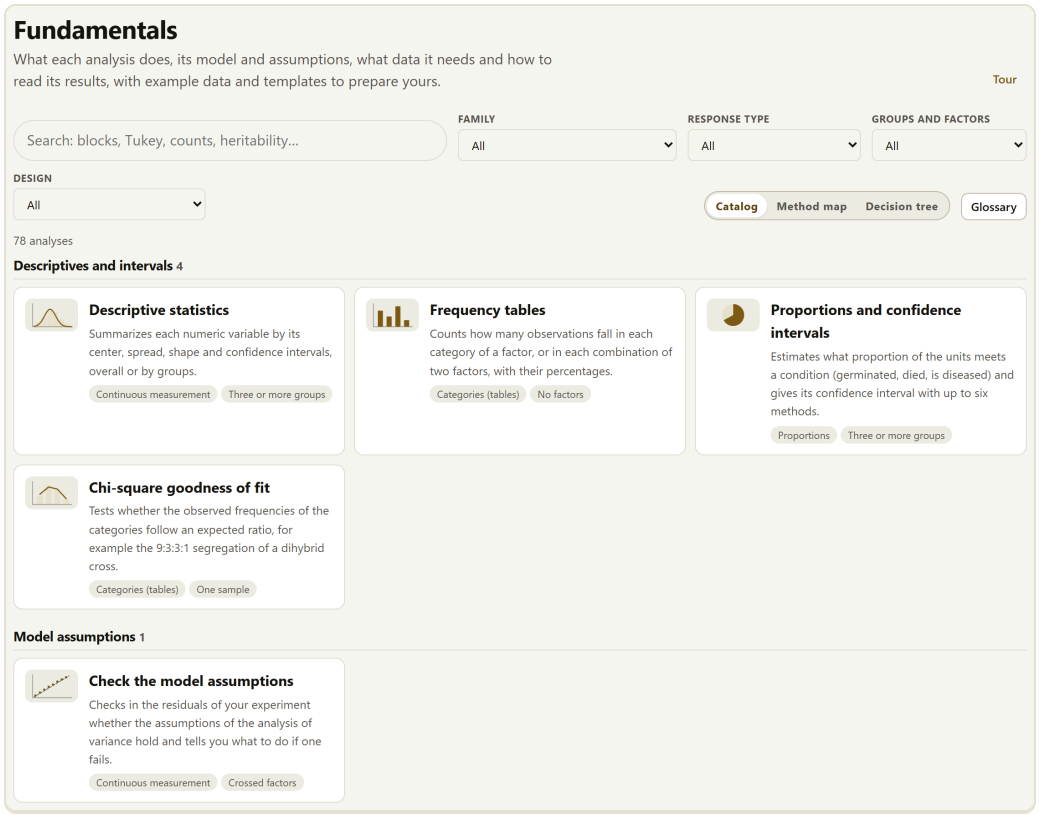


Figure 9.1. The Fundamentals catalog: the search box, the four filters and the first families.

9.3 The method map and the decision tree

Method map shows the same analyses in a table: the type of response in the rows and the structure of the groups and factors in the columns. It helps you find the analysis starting from the question: "I measure counts in blocks", "I compare two groups with a yes / no response".

Each name is a link to its fact sheet: it opens with a click or with the keyboard (Tab and Enter). The catalog filters also apply to the map, and the map remembers the chosen view while the tab stays open. On narrow screens, the table scrolls sideways inside its frame.

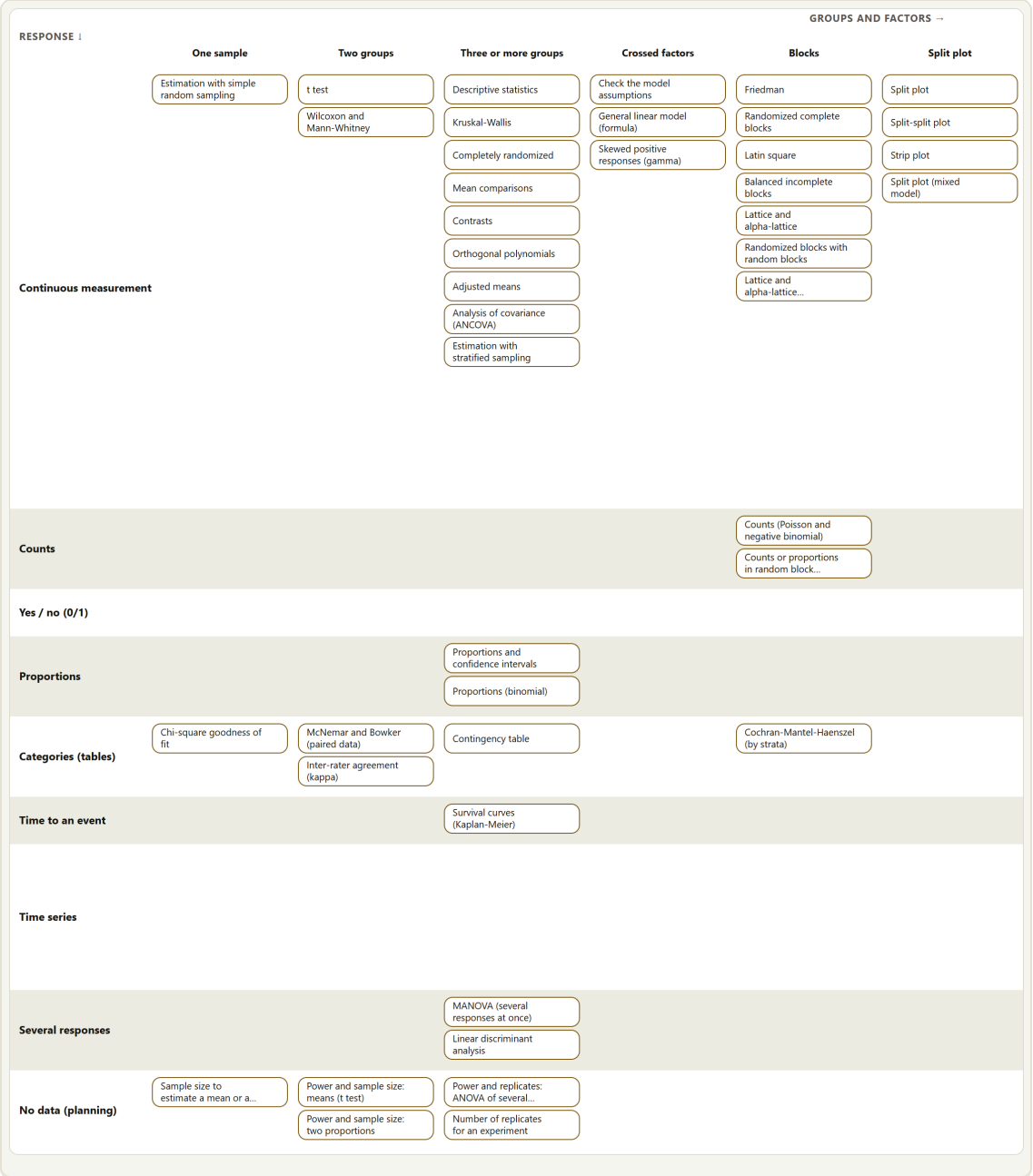


Figure 9.2. The method map (the columns on the right, such as continuous predictors and random effects, come into view when you scroll the table sideways).

The decision tree

Decision tree takes you from the research question to the analysis through a few questions in field language, without data (figure 9.3). The first column asks what you want to do with the data: describe them, compare treatments, relate numeric variables, associate categories, compare genotypes across several environments, estimate genetic variances, analyze the time to an event, a time series or several responses at once, estimate from a sample or plan a trial. Each answer opens the next question on the right (what did you measure?, how was the trial laid out?, what shape does the relationship have?), and a gold line joins each answer to its question.

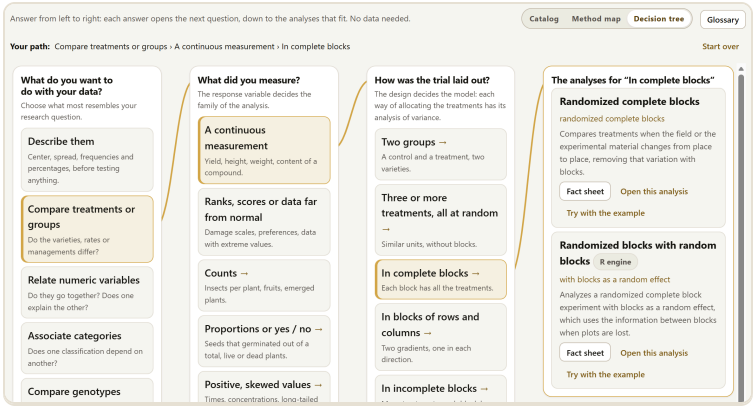


Figure 9.3. The decision tree along the path of a trial in complete blocks; the question columns continue further down. The leaf shows the two analyses that fit.

The last column shows the analyses that fit, each with when to use it and what it is for, and three buttons: Fact sheet, Open this analysis and, if its fact sheet has an example, Try with the example, which loads the example data and computes it. Every one of the 78 analyses is in some leaf; the sample size and power analyses open without data. Start over goes back to the first question, and the path is remembered while the tab stays open. The tree has no search box or filters; on a narrow screen, the columns are stacked one below the other.

You can also reach the tree from the third path of Home (section 1.1) and from the assistant's recommendation: See it in the decision tree opens it on the path of that analysis, with the analysis highlighted (section 4.2).

9.4 A fact sheet, section by section

All fact sheets have the same nine sections, in the same order, and an index on the left to jump between them, which marks the one you are reading. Each section folds or unfolds with a click on its title.

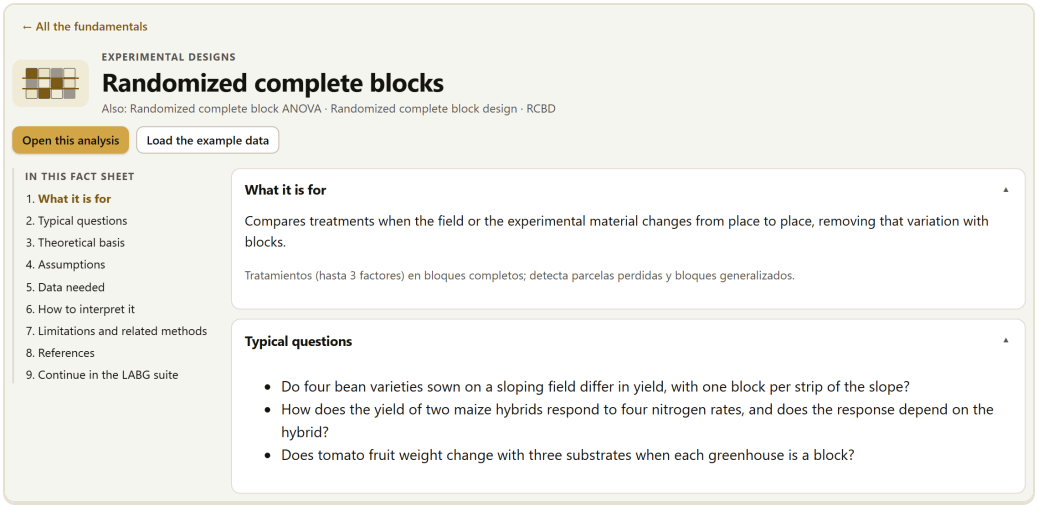


Figure 9.4. The randomized complete blocks fact sheet: the header, the index and the first two sections.

Table 9.1. The nine sections of a fact sheet.

#	Section	What it contains
1	What it is for	In one sentence, what question it answers; the other names it is known by and its family.
2	Typical questions	Two or three agronomic or biological questions that the analysis answers.
3	Theoretical basis	The idea of the method in 150 to 300 words, the model with its terms explained, the hypotheses H_0 and H_1 and the statistic that tests them.
4	Assumptions	Each assumption, how it is checked in SigmaPro (the traffic light, Check the model assumptions, a plot) and what to do if it does not hold, with buttons that open the check or the alternative.
5	Data needed	The gold box of section 9.5.
6	How to interpret it	What to look at in the result and in what order, a sample write-up for the thesis or the paper, and the guides for reading each statistic.
7	Limitations and related methods	When not to use it and which SigmaPro analysis to use instead, with links to their fact sheets.
8	References	3 to 5 references in APA format: the original paper or book and the reference texts.
9	Continue in the LABG suite	The suite apps that carry on the work of that analysis, with the reason (chapter 10). If none applies, it says so.

The theoretical basis ends with the model, each of its terms and the hypotheses being tested, with the statistic that decides them (figure 9.5).

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

y_{ij}	the response of treatment i in block j
μ	the overall mean
τ_i	the effect of treatment i (with several factors: main effects and interactions)
β_j	the effect of block j
ε_{ij}	the experimental error, independent, with mean 0 and constant variance

H_0: $\tau_1 = \tau_2 = \cdots = \tau_t = 0$

H_1: some $\tau_i \neq 0$

The treatments have the same mean. Tested with F = treatment MS / error MS; with crossed factors, one test per main effect and per interaction.

Figure 9.5. The model, its terms and the hypotheses in the randomized complete blocks fact sheet.

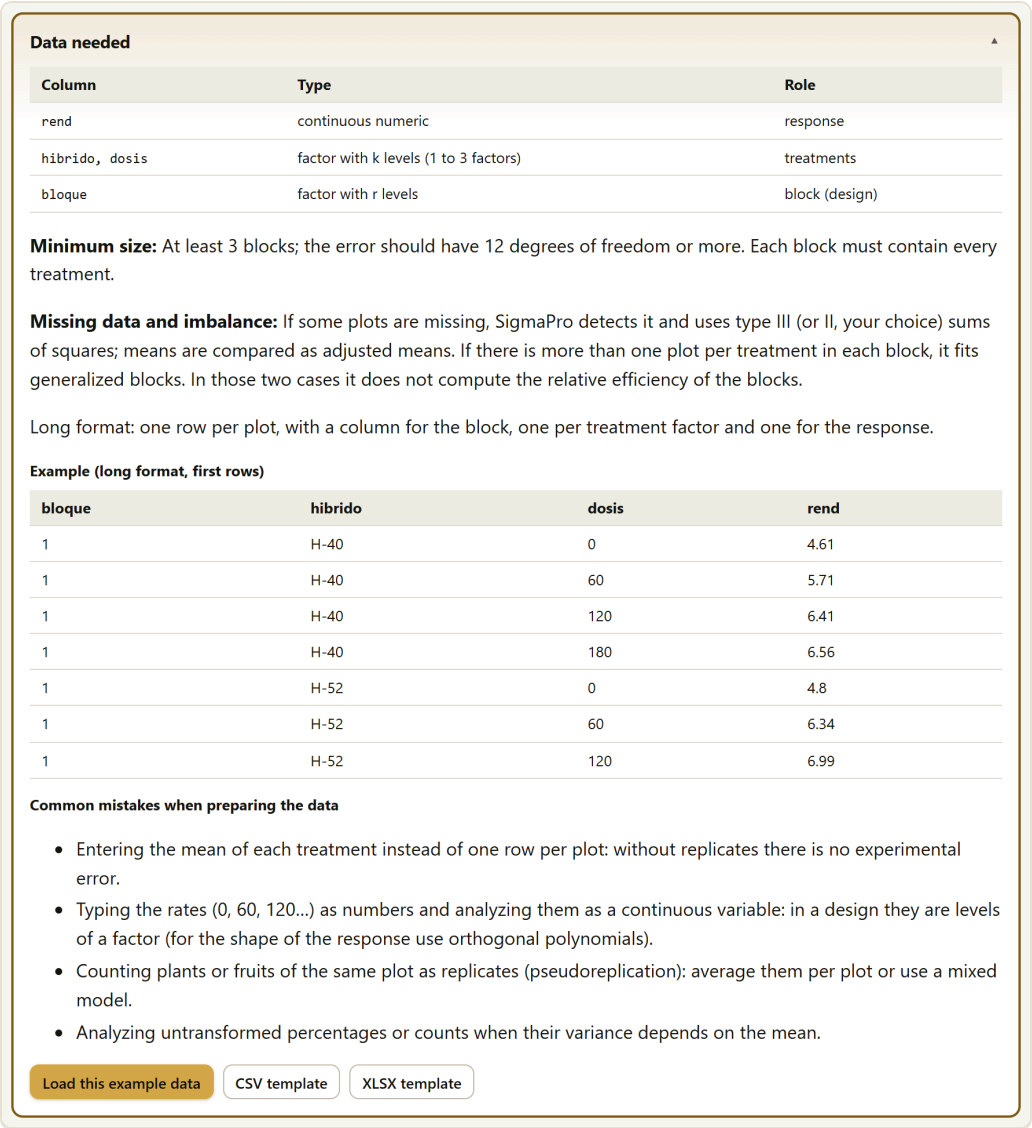
EXAMPLE

The sample write-ups in each fact sheet use the figures the app gives with its example dataset; when the fact sheet has none, the figures are presented as illustrative. Before copying them, replace the figures with those of your own result.

9.5 Data needed and the templates

Section 5 sits in a gold box because it answers the question that most often holds up an analysis: how should my data be laid out? It has six parts (figure 9.6).

- **Columns it needs:** each column with its type (continuous numeric, factor with k levels, count, 0/1, date) and its role in the analysis (response, treatments, block…). The names are those of the example dataset.
- **Minimum size:** what the method needs to run and what is advisable for the test to make sense.
- **Missing data and imbalance:** what SigmaPro does if a value is missing or the groups are not the same size.
- **Format:** almost always long (one row per plot, plant or date). When the wide format is a common source of confusion, the fact sheet shows the same data in wide format, marked "not like this"; the Data block converts them to long with Wide ⇄ long (section 2.11).
- **Example:** the first actual rows of the example dataset, up to 8.
- **Common mistakes when preparing the data:** those that make the analysis run but answer a different question (averages instead of plots, numeric doses treated as if they were continuous, the fruits of one plot as replicates).



Data needed

Column	Type	Role
`rend`	continuous numeric	response
`hibrido, dosis`	factor with k levels (1 to 3 factors)	treatments
`bloque`	factor with r levels	block (design)

Minimum size: At least 3 blocks; the error should have 12 degrees of freedom or more. Each block must contain every treatment.

Missing data and imbalance: If some plots are missing, SigmaPro detects it and uses type III (or II, your choice) sums of squares; means are compared as adjusted means. If there is more than one plot per treatment in each block, it fits generalized blocks. In those two cases it does not compute the relative efficiency of the blocks.

Long format: one row per plot, with a column for the block, one per treatment factor and one for the response.

Example (long format, first rows)

bloque	hibrido	dosis	rend
1	H-40	0	4.61
1	H-40	60	5.71
1	H-40	120	6.41
1	H-40	180	6.56
1	H-52	0	4.8
1	H-52	60	6.34
1	H-52	120	6.99

Common mistakes when preparing the data

- Entering the mean of each treatment instead of one row per plot: without replicates there is no experimental error.
- Typing the rates (0, 60, 120...) as numbers and analyzing them as a continuous variable: in a design they are levels of a factor (for the shape of the response use orthogonal polynomials).
- Counting plants or fruits of the same plot as replicates (pseudoreplication): average them per plot or use a mixed model.
- Analyzing untransformed percentages or counts when their variance depends on the mean.

Load this example data | CSV template | XLSX template

Figure 9.6. "Data needed" in the randomized complete blocks fact sheet.

B9

`CSV template` and `XLSX template` download the fact sheet's example table (`plantilla-<análisis>.csv` or `.xlsx`) so that you can fill it in with your own data and then import it in the Data block. `Load this example data` loads the complete dataset and opens the analysis with its columns already chosen. The planning fact sheets (power, number of replicates, sample size and response-surface design) use no data: instead, their table of columns lists the values the form asks for.

9.6 Formulas, language and offline use

The formulas in the fact sheets are mathematical text, not images: they can be read by a screen reader, enlarged without losing sharpness and copied.

SigmaPro writes them with its own converter to MathML, the formula language that browsers understand, without downloading any library. Each fact sheet is in Spanish and in English, with the same figures in both; you switch with the `ES · EN` selector like the rest of the app, and in English the notation follows the convention of that language (SE, SS and MS instead of EE, SC and CM).

The 78 fact sheets are app files (`data/fundamentos/`) that are loaded when they are opened. They work when you double-click `index.html`, in the published version and offline: the installed app stores them along with everything else. They also make up the methodological appendix of the exported report (section 7.2).

9.7 How the fact sheets were reviewed

A fact sheet that describes something the app does not do is worse than no fact sheet at all. That is why each one was checked against the code and against the running app, in six independent reviews of 13 fact sheets each.

Each review checked the model, the hypotheses, the assumptions (and that the check and the alternative it proposes exist), the data and the example table, that the example runs and gives the figures in the write-up, the interpretation and the references (author, year, title, journal, pages and DOI). Of the 78 fact sheets, 49 changed in the review. Along the way, 18 errors in the app itself also turned up, from a calculation (weighted kappa when a rater does not use some category) to labels and messages; they were fixed, and the calculation errors have their new test case against R. The details are in `docs/revision-fundamentos.md`, and the list of analyses with the status of their fact sheet, in `docs/inventario-analisis.md`.

In addition, an automated test (`node tests/fundamentos.mjs`, part of the release battery of chapter 8) checks in every version that each analysis has its fact sheet with the nine sections in both languages, that the examples run, that the formulas convert and that no fact sheet names an option or a column that does not exist.

10

BLOCK 10 · LABG SUITE

LABG Suite: where the work continues

SigmaPro is part of the LABG Suite, a set of free apps that run in the browser. Some work begins or ends outside SigmaPro: planning and randomizing the design, exploring many variables at once, reducing dimensions, forming groups; and the suite reaches beyond statistics, from phenology and forest inventory to population genetics and phylogeny. This chapter explains the LABG Suite block, the suggestions that appear within your work and how to take the data from one app to another.

10.1 The LABG Suite block

10.2 Suggestions within your work

10.3 Where each app appears

10.4 Sending the data to another app

10.5 Hiding a suggestion

10.6 How to cite the apps

10.1 The LABG Suite block

The block presents all of SigmaPro's sister apps, 22 of them, grouped by area as on the suite portal (table 10.1). At the end of the Home block, the same box, "Explore the LABG suite", shows only the four that most often continue an agronomic analysis: PCAPro, ClusteringPro, AgriDesign and CorrelationPro.

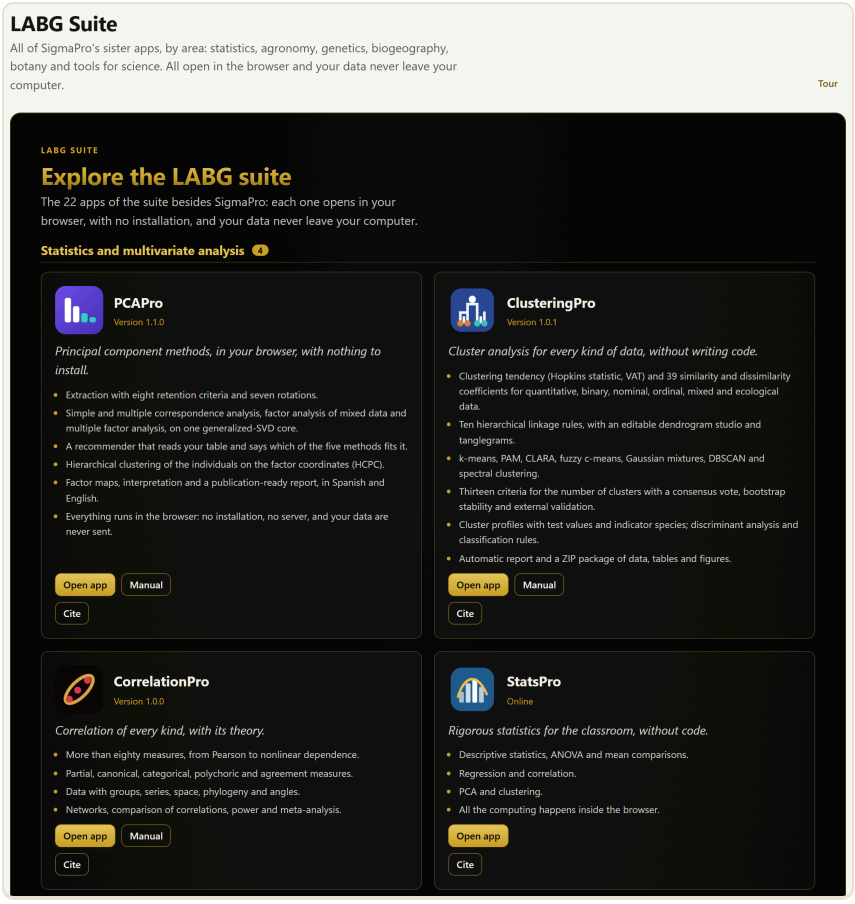


Figure 10.1. The LABG Suite block: the start, with the first of its six areas.

Each card has:

- The app's **official logo**, which is also a link to it, its name and its published **version** (or Online, when the app does not publish a version number).
- Its **tagline** and four to six **features**, taken from each app's published documentation.
- Open app and Manual, which open the app or its manual in a new browser tab; an app without an online manual has only the first.
- Cite, which unfolds the citation in APA format and in BibTeX, with buttons to copy it (section 10.6).

Table 10.1. The 22 apps of the LABG Suite block, by area, with their version (November 2026).

Area	Apps
Statistics and multivariate analysis	**PCAPro** 1.1.0, **ClusteringPro** 1.0.1, **CorrelationPro** 1.0.0, **StatsPro** (online)
Experimentation and agronomy	**AgriDesign** 1.0, **PhenologyPro** 1.2.0, **EconomicsPro** 1.0.7, **ForestPro** 1.0.0, **PostHarvestPro** 1.0.0
Genetics, breeding and genetic resources	**BreedingPro** 1.0.1, **PopGeneticsPro** 1.0.1, **GermplasmPro** 1.0.0, **CytogeneticsPro** 1.0.0
Biogeography and phylogeny	**BioModelling Pro** 1.1.1, **PollinationPro** 1.2.5, **PhylogenyPro** 1.4.0, **CladisticsPro** 1.0.1, **PalynoPro** 1.0.0
Botany for the classroom	**FloralPro** 1.0.0, **LeafPro** 1.2.0
Tools for science	**SciMetricsPro** 1.0.1, **ReviewPro** 1.0.0

At the foot, Open the LABG suite portal opens the suite portal (on Home, the link reads See the 23 apps of the suite, counting SigmaPro). An app without a public release yet would appear with its provisional mark, its details marked **PENDING** and the Coming soon button instead of the links; today all of them are published.

> **NOTE**
> The suite's apps open in your browser, with nothing to install, and process the data on your computer, like SigmaPro. Opening one of them does not send it anything: SigmaPro does not send your data anywhere.

10.2 Suggestions within your work

Outside block 10, the suite's apps appear only where they continue what you are doing, in a discreet strip titled Continue in the LABG suite. Never in pop-up windows and never as advertising.

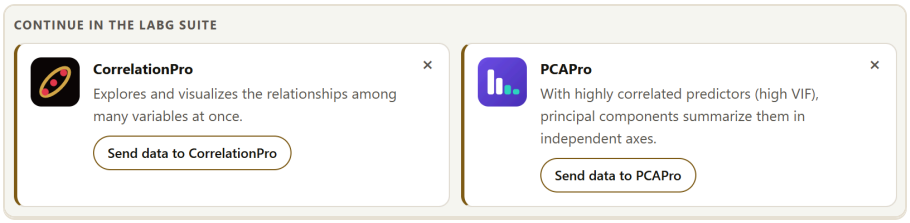


Figure 10.2. At the foot of a linear regression with the coffee data: the two apps that continue that work.

Each card carries the app's logo as a link (it opens in a new tab), a sentence that says why the app is useful there, Send data to … when the app is already published (10.4) and a × button to hide the card (10.5). The strips appear in three places:

- **At the foot of the result** of an analysis, in the Analysis block, and in section 9 of its Fundamentals fact sheet.
- **At the foot of a family** in the Fundamentals catalog, as Related tools.
- **In the Data block**, when the loaded table has a structure that another app can make use of: a design with treatments and blocks, many numeric variables, coordinates or dates. The card says what it saw in the table ("The table has 8 numeric variables").

10.3 Where each app appears

The rules are written in an app file (`data/suite-rules.js`) and are the same for everyone: they do not depend on what you have used before.

Table 10.2. Which app is suggested, and where.

App	At the foot of these analyses	In the Data block, if the table…
AgriDesign	The experimental designs (completely randomized, blocks, Latin square, split plots, split-split plots and strip plots, incomplete blocks, lattice) and their mixed versions: to randomize the design and produce the field map and the field book. The number of replicates and the power of the ANOVA: to plan the trial.	has treatments and blocks or replicates
CorrelationPro	Correlation of two variables, correlation matrix, partial correlation, linear regression, stepwise selection and best subsets.	has 8 or more numeric variables
PCAPro	The same correlations and regressions (in regression, because of collinearity), MANOVA, discriminant analysis, AMMI and GGE.	has 8 or more numeric variables
ClusteringPro	MANOVA, discriminant analysis, multi-environment trials, stability, AMMI and GGE: to classify genotypes, sites or individuals into groups.	has 8 or more numeric variables
BioModellingPro	—	has latitude and longitude columns
PhenologyPro	—	has dates

In the Fundamentals catalog, the Related tools strip follows the same logic by family: AgriDesign at the foot of the designs, the mean comparisons and power; CorrelationPro and PCAPro at the foot of correlation and regression; PCAPro and ClusteringPro at the foot of multivariate analysis and multi-environment trials. An analysis with no related app says so in section 9 of its fact sheet and refers you to this block. The other apps of the suite have no suggestion rules: they are in Block 10.

10.4 Sending the data to another app

Send data to … sends nothing over the internet: it saves the Data block's table as a CSV file on your computer and opens the other app in a new tab.

1. With the table loaded, press Send data to PCAPro (or to the corresponding app).
2. The browser downloads `sigmapro-para-pcapro.csv` and opens PCAPro; a notice reminds you of the file name.
3. In the other app, open that file from its data block.

The file contains the table SigmaPro is using (with the active filter, if there is one): the numbers with all their digits, each factor with the names of its levels and missing values as empty cells, in UTF-8 so that accented characters arrive intact. For apps that work with numeric variables, such as PCAPro, the file has a label column (the first non-numeric one) and all the numeric ones. If no data are loaded, the button says so and downloads nothing.

10.5 Hiding a suggestion

The × button on a card hides it in that place (for example, PCAPro at the foot of the regression), and SigmaPro remembers this in this browser. The same app keeps appearing in the other places.

To bring them all back, the LABG Suite block has Show again the suggestions you hid below the cards. If all the cards of a strip are hidden, the whole strip disappears.

10.6 How to cite the apps

If you used another app of the suite in your work, cite it the way you cite SigmaPro (Home block and appendix E): with its version and its DOI.

Cite unfolds the app's citation in APA format and in BibTeX, built from the app's published data (its citation file and its metadata), so it changes with each version; Copy APA and Copy BibTeX copy it to the clipboard. An app without a DOI is cited with its web address, and one that does not publish a version number, without it. For example, for PCAPro:

EXAMPLE

Barrera-Guzmán, L. Á. (2026). PCAPro: an offline, browser-based platform for guided principal component analysis (Version 1.1.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.22649709

A

APPENDICES

Appendices

What is worth having at hand and did not fit in any chapter: which files the app reads and writes, what to do when something does not turn out as expected, a glossary of the terms the manual uses, the references for the methods and the data, and the program's license, with how to cite it.

A The files the app reads and writes

B When something goes wrong

C Glossary

D References

E The license and how to cite SigmaPro

A The files the app reads and writes

SigmaPro runs in the browser and sends nothing to any server: files come in from your computer and go out to the downloads folder. Work in progress is saved only in the browser's own storage (section 7.7).

A.1 · What the app reads

Format	Extension	What it takes
Delimited text	`.csv`, `.tsv`, `.txt`	Separated by commas, semicolons or tabs, with a decimal point or decimal comma and the missing-value codes you specify (section 2.3).
Spreadsheet workbook	`.xlsx`, `.xlsm`, `.ods`	One sheet at a time. Workbooks in the old format (`.xls`) do not open: save them first as `.xlsx` or `.csv`.
Pasted table	—	What you copied from a spreadsheet or a document (section 2.4).
Project	`.labg`	The data, the report and the figure settings, with verification of their hashes (section 7.6).

A.2 · What the app writes

Block	File	What it contains
2 4 9	`.csv`	Send data to … in the suite strip: the table in use, with the active filter, as `sigmapro-para-pcapro.csv` (or the corresponding app), to open it in another suite app (section 10.4).
3 4	`.csv`	Download as CSV: a table from a result, with unrounded values (sections 3.2 and 4.5).
4	`.csv`	From MANOVA and discriminant analysis, the variables with their group, to open them in PCAPro or ClusteringPro (section 4.21).
6	`.svg` · `.pdf` · `.png` · `.tif`	A figure, at the chosen width; PNG and TIFF at 600 dpi (section 6.3).
7	`.docx` · `.html` · PDF	The complete report; the PDF is saved from the browser's print dialog (section 7.2).
7	`.zip` (Markdown)	The report in `reporte.md`, with its figures and reading panels in PNG (section 7.2).
7	`.docx` · PDF	The class worksheet, with its answer key if requested (section 7.3).
7	`.xlsx` · `.tex`	The report tables with all their digits, or in LaTeX with booktabs (section 7.4).
7	`.zip`	All the figures in SVG, PDF and PNG (and TIFF, if requested), with an index (section 7.5).
7	`.labg`	The project: data, report, figure settings and the R code that reproduces everything (section 7.6).

Ap

B When something goes wrong

The app's messages say what happened and, almost always, how to fix it. These are the most common cases.

What happens	What to do
The numbers were read as text, or many missing values appeared on import	Check the Decimal separator (comma or point) and the column separator. If the column contains stray text ("ND", "s/d"), add it to Missing-value codes; the app says how many values it could not convert (section 2.3).
A column of block numbers or doses is not taken as a factor	Change its type to *factor* in the list of variables or in its ⋮ menu (section 2.7). In the analyses, numeric columns with few values appear as "(numeric)" and can be used as a factor.
The measurements from several dates are in separate columns	Convert the table to long form with Wide ⇄ long… (section 2.11).
"Column '…' does not exist" when writing a formula	Check how the name is written (capitals, accents); names with spaces go between backticks: `` `peso seco` ``.
"The R engine could not be loaded" or "The R package '…' could not be installed"	When the app is opened with a double-click, the R engine is downloaded from the internet the first time (about 25 MB): check the connection and try again. After that it stays stored in the browser (section 4.6).
A mixed model says "Singular fit"	Some variance component ended up at zero: the data are not enough to estimate it. Remove that random effect or simplify the model (sections 4.15 and 4.17).
"This analysis does not yet have an approved validation against R"	The method did not pass its validation and the app does not compute it (section 8.5). It should never appear in a published version.
An analysis in the report says it could not be computed	A column it used was deleted or renamed, or the filter left too few rows. Bring the column back or adjust the filter, and the analysis returns (section 5.6).
When opening a project, "Some files changed after being saved"	Someone edited the .labg file outside the app. Open it anyway only if you know what was changed (section 7.6).
The autosave card says the browser does not allow saving	It is a private window, or storage is blocked or full: save the .labg project so you do not lose your work (section 7.7).
"The Figure studio is still loading"	Wait a moment and press Open in the studio again.

C Glossary

Block — A group of experimental units that are similar to one another (a stretch of the field, a greenhouse, a working day) and that receives all the treatments; the variation between blocks is removed from the error.

BLUE — Best linear unbiased estimator: the adjusted mean of a genotype taken as a fixed effect.

BLUP — Best linear unbiased predictor: the predicted value of a random effect (a genotype, a block), shrunk toward the mean in proportion to how little is known about it.

Censoring — In survival analysis, a unit that reached the end of the trial without the event: its time is known to be longer than the one observed.

Coefficient of variation (CV) — The error standard deviation as a percentage of the mean; it measures the precision of a trial. In the Explore descriptives, it is the standard deviation of the variable as a percentage of its mean.

Comparison letters — In mean comparisons, two means that share a letter do not differ at the chosen α level; the letter a goes with the highest mean.

Confidence interval (CI)	The range of values for a parameter that are compatible with the data, at the chosen level (for example, 95%).
Covariate	A numeric variable that is not a treatment but explains part of the response (the number of plants, the altitude).
Degrees of freedom (df)	The number of independent pieces of information with which something is estimated; Satterthwaite and Kenward-Roger approximate them in mixed models.
ED50	In a dose-response curve, the dose that gives half of the effect between the limits of the curve; in the log-logistic model it is the parameter e. ED10 and ED90 give 10% and 90%.
Error stratum	Each unit size in a design with plots of several sizes (main plots, subplots), with its own error.
Factor and level	A classification variable (the hybrid, the dose as a treatment) and each of its values.
Fixed and random effect	Fixed: the levels were chosen on purpose and you want to draw conclusions about them. Random: the levels are a sample of many possible ones (blocks, locations, years).
Heritability	The proportion of the variation (among genotype means or among plots) that is genetic.
Interaction	When the effect of one factor depends on the level of another (one hybrid responds more to nitrogen than another). If it is significant, it is advisable to compare one factor within each level of the other.
Lack of fit	The part of the variation of the means of the replicated points that the model does not explain; if it is significant, the model does not have the right shape.
Marginal mean	The mean the model predicts for a level, with equal weights over the other factors; with covariates or unbalanced data, it is the one that gets published (adjusted mean).
Mega-environment	A group of environments in which the same genotype wins, repeatedly across years. With data from a single season, the sectors of the GGE biplot are possible mega-environments.
Odds ratio and hazard ratio	How many times more (or less) favorable an outcome is in one group than in another (odds), or how many times greater the instantaneous risk of an event is (hazards, in the Cox model).
Overdispersion	More variation than the assumed distribution allows (Poisson, binomial).
Power	The probability of detecting a difference that exists; it is set before the trial (usually 80%).
***p*-value**	The probability of obtaining a result as extreme as the one observed, or more so, if there were no effect.
Relative risk	The ratio of the proportions of an outcome in two groups (for example, the germination of the treated seed divided by that of the untreated seed).
REML	Restricted maximum likelihood: it estimates the variance components after discounting the degrees of freedom spent on the fixed effects, and so corrects the downward bias of maximum likelihood.
Residual	What is left of each observation after subtracting what the model predicts.

Standard error (SE)	The standard deviation of an estimator: how much it would vary if the trial were repeated.
Traffic light	The box that says whether the assumptions of an analysis hold: green, fine; amber, check; red, problem; gray, not applicable or not enough data to check it. Not every analysis has one (section 4.5).
Type I, II and III sums of squares	Three ways of testing each term with unbalanced data: in sequence, adjusted for the terms that do not contain it, or adjusted for all the others (section 4.10).
Variance component	The part of the total variation contributed by a random factor.

D References

The works for the methods the manual explains, for the R packages SigmaPro was validated against and for the published data of the examples. The exported report has, in its own list, the works cited by its analyses.

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E The license and how to cite SigmaPro

E.1 · What is inside the app

SigmaPro is written in HTML, CSS and JavaScript, with its own computing core: file reading, model algebra, distributions, methods, plots, exports (.docx, .xlsx, PDF, ZIP) and validation. To match R's numbers, some algorithms were translated from R's free source code; each is declared, with its publication and license, in `LICENSES-TERCEROS.md`. The icons of the suite's shared modules are from Lucide (ISC license).

The R engine methods use webR (R compiled for the browser) and R packages under their own free licenses, downloaded the first time or served from `motor-r/`, not bundled with the program.

E.2 · License

The program is distributed under the **GNU General Public License, version 3 or later** (GPL-3.0-or-later): it may be freely used, copied, studied and modified if modified versions keep the license and the attribution. Full text: `LICENSE`. This manual is © 2026 Luis Ángel Barrera-Guzmán, all rights reserved.

E.3 · How to cite it

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